APPLIED MATHEMATICS FOR TODAY

Pupil's book five

BY UTHMAN BALIJUKA

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ACKNOWLEDGEMENT

I am very grateful to almighty Allah who has enabled me to accomplish this mission to produce this book. Similarly, I wish to express and convey my gratitude to all teachers who contributed materially, professionally and financially to the production of this book, thank you very much.

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AUTHOR

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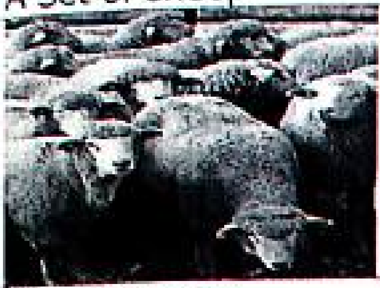
TOPIC 1: SET CONCEPTS

A Set is a collection of clearly or well defined members.

Primary five visited a school farm and discovered the following sets of animals and items.

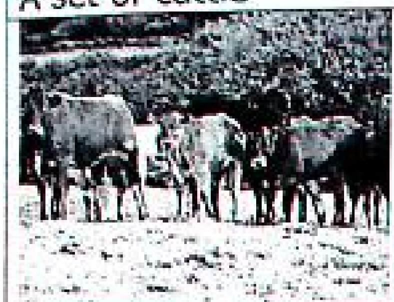
A Set of sheep



A Set of goats

A set of garden tools

A set of cattle



A set of pigs

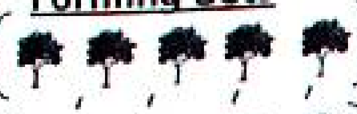
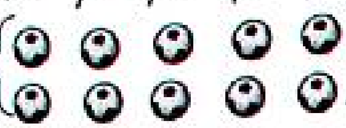
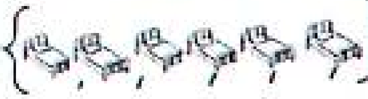
A set of rabbits

1.1 Description of Set

Activity 1:1 Describe the following sets by listing the elements

1. A set of vowel letters.
2. A set of counting numbers.
3. A set of months of the year.
4. A set of days of the week.
5. A set of city buildings in Kampala.
6. A set of sugar factories in Uganda.
7. A set of National parks in Uganda.
8. A set of pupils in primary five.
9. A set the first 8 even numbers.
10. A set cash crops in Uganda.
11. A set first 9 odd numbers.
12. A set of fruit trees in the school compound.
13. A set of multiples of 2 less than 10.
14. A set of first 6 prime numbers.
15. A set of square numbers less than 50.

1.2 Forming Sets

1. {  } A set of 5 trees
2. {  } A set of ______
3. {  } A set of ______
4. { a, e, i, o, u } ______
5.
6. {  } ______
7. { Mirinda, Pepsi, Novida Cocacola Fanta } ______
8. { 1,2,3,4,5,6,7,8,9 } ______
9. { a,b,c,d,e,f,g,h..... } ______

5. {△, ○, ▭, ▽} ______ 10. {Maths, science, SST English, R.E} ______

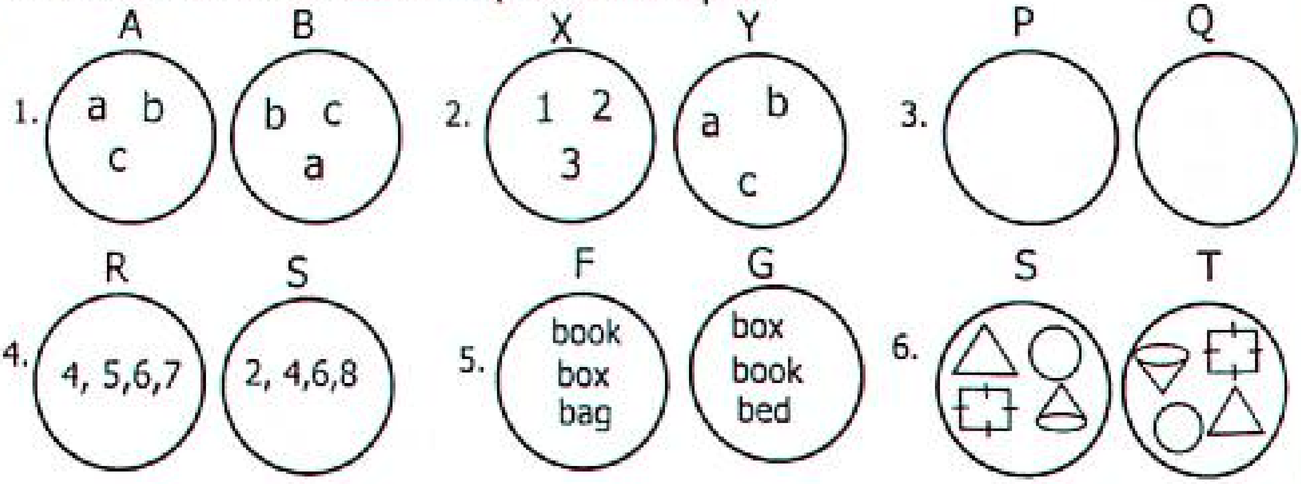
1.3 Equal and not equal sets

Equal sets are sets with same members and have equal number of elements or members. The order or arrangement does not matter.

Example 1 Given set A= {a, b, c, d} B= {d, a, c, b} Set A has 4 members → n (A) =4. Set B has 4 members → n (B) = 4. The two sets have the same number of elements and exactly the same. Therefore Set A is equal to set to set B. Set A = set B.	**Example 2** Given set P={knife,panga,fork,spoon,tray} Q={fork,knife,tray,panga,spoon} Set P and set Q have the same number of members which are exactly the same. Therefore set P is equal to set Q Set P = Set Q
Example 3 X={bag, book, pen, box} Y={Jumper, book, radio, pen} Set X and Y have the same number of members but elements are quite different, therefore, set X is not equal to set Y. Set X ≠ set Y	**Example 4** R = {boy, girl, man} S = {boy, woman, girl} Set R has 3 elements Set S has also 3 elements They don't have exactly the same elements. ∴ Set R ≠ set S

Activity 1:3

State whether the sets are equal or not equal.



7. Given that A = {2, 4, 6, 8, 10}   B = {6, 8, 4, 2,10}
8. B = {a, b, c, d, e}   D = {a, e, i, o, u}
9. F = {chair}   D = {a, e, i, o, u}
10. H= {Monday, Tuesday, Wednesday}   K= {January, February, March}
11. S = {p, o, t}   T = {t, o, p}
12. X = {24, 26, 28, 30}   Y = {30, 26, 24, 28}

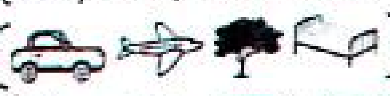
1.4 <u>Equivalent and Non-Equivalent sets</u>

Equivalent sets are sets with the same number of elements or members though the elements may not be exactly the same.

Example 1	**Example 2**
Set M = {bag, book, pen, box} Set N = {Jane, Joan, Jolly, John} Set M and N have the same number of members but the members or elements are different, therefore set M is equivalent to set N. Set M ↔ Set N	Set S = {Philly, Josse, Adam} Set T = {Abdallah, Uthman, Imran} n(S) = 3 n(T) = 3 Therefore set S and set T are equivalent. Set S ↔ Set T

Activity 1:4

Write equivalent and non-equivalent

1. K = {a, e, i, o, u}   L = {b, c, d, e, f}
2. W = {2, 4, 6, 8}   X = {1, 2, 3, 4}
3. A = {1, 2, 3, 4, 5, 6}   b = {s, m, a, r, t}
4. R = {6, 7, 8, 9}   S = {b, a, n, d}
5. P = {box, bed, bag}   Q = {bag, ball, box, bell}
6. S = {Sun, Mon, Tue, Wed}   T = {Kato, Wasswa, Kizza, Kigongo}
7. L = {}   M = {}
8. H = {}   K = {Circle, Cylinder, Triangle}
9. A = {p, o, t}   B = {k, y, z}
10. R = {b, r, e, a, d}   T = {b, o, a, r, d}

1.5 <u>Empty set</u>

An empty set is a set with no members or elements in it. We use symbol Ø, an empty set is sometimes called a **<u>null set</u>**.

Example 1 Set R = {A lady who has ever been a Vice president of Uganda} Uganda had a Vice president in the names of Specious Kazibwe. Set R is not empty set.	**Example 2** A = {People who walk on two legs} Set A is not an empty set
Example 3 B = {English alphabet with 30 letters} English alphabet has 26 letters, *Therefore it's an empty set.*	**Example 4** Set S = {Pupils in P.5 with 5 eyes} All pupils have 2 eyes. Therefore set S is an empty set.

Activity 1:5

State empty set or not empty set.

1. Q ={Goats which eat meat}
2. K = {A boy with metallic legs}
3. M = {The first 8 odd numbers}
4. Z = {The sun rises from the south}
5. T = {Women who are doctors}
6. T = {Nursery pupils who teach primary five}
7. K = {Rabbits without hair}
8. X = {First 6 odd numbers}
9. F = {Daughters who are as old as their mothers}
10. D = {A set of letters of alphabet}
11. Q = {Cars which can fly like helicopters}
12. J = {Male teachers who are pregnant}
13. L = {Bulls which produce milk}
14. R = {Even numbers between zero and ten}

1.6 Union of sets

Union of a set is a set of all members made up of two or more sets.

- ✓ Union of set is a set of all members made from two or more sets without repeating the members that appear in both sets.
- ✓ Members that appear in both sets are written once in the union sets. Symbol of union set "∪"

TOPIC 1: Set Concepts

Example 1	Example 2
Given that, T = {b, r, e, a, d} S = {b, l, a, c, k} Find T ∪ S. In this example set T and set S can be fused in one set without repeating the members. Set T ∪ S = {b, r, e, a, d, l, c, k}	If P ={b,l,a,n,k,e,t} Q ={b,u,c,k,e,t} What is P ∪ Q. P ∪ Q= {b, l, a, n, k, e, t, u, c}

Activity 1:6

Find the Union of sets in the numbers below,

1. L = {Allan peter, Bosco, Ben} K = {John, Bosco, Simon, Allan}.
   Find K ∪ L.
2. B ={1, 2, 3, 4, 5, 6} D ={3, 5, 7, 6}
3. F ={b, l, a, c, k, b, o, r, d} G = {b, r, o, a, d}
4. H ={Elgon, Moroto, Rwenzori} L={Muhavura, Moroto, Napak}
5. Q ={Hoima, Lira, Kampala, Mbale} R ={Jinja, Mbale, Gulu, Kampala}
6. B ={Brenda,Allan,Betty,Suzie,Fatuma} E={Molly,Bridget,Moureen,Allen}
7. K ={James,John,Joan,Joseph} L={Joseph,Jossy,Joan,Jackson,Jim}
8. Z={Bananas,Oranges,Pineapples,Mango} Y={Oranges,Pawpaws,Mango}

1.7 Intersection of sets

An intersection of sets is a set obtained from two or more sets but mostly considering the common elements or members. We use the symbol "∩" as to show intersection of sets.

Example 1	Example 2
Set A ={2, 4, 6, 8, 10} B ={1,2,3,4,5,6} What is A ∪ B. Common element = {2, 4, 6} Therefore A ∩ B = {2, 4, 6}	P = {Plate, spoon, folk, saucepan} Q = {Cup, Plate, ladle, spoon, folk} Find P ∩ Q. Common members = {plate, spoon, folk} Therefore P ∩ Q = {plate, spoon, folk}

Activity 1:7

List the intersection sets in the numbers below.

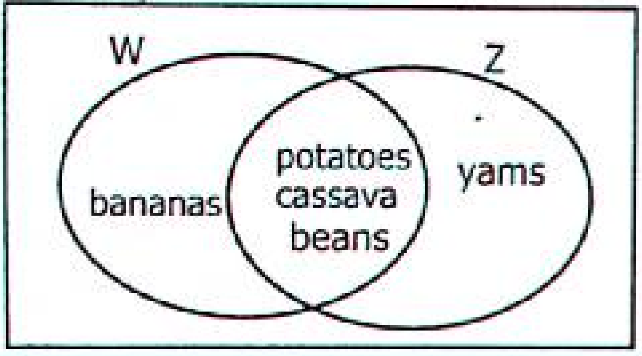
1. Y= {2, 4, 6, 7, 8} Z = {3, 5, 6, 7, 9} Find Y ∩ Z.
2. F = {a, b, c, d, e, f} G = {a, e, I, o, u} Find F ∩ Z.
3. R = {book, box, bag, bed} S = {book, bed, bench}. Find R ∩ S.

4. A = {b, o, a, r, d}   B = {b, o, a, t}. Find $A \cap B$.
5. X = {red,blue,green,yellow}   Y={blue,pink,Indigo,green} what is $X \cap Y$
6. H = {ruler, pen, pencil} J = {book, rubber, ruler}   Find $H \cap J$.
7. T = {b, e, a, n, s}   S = {p, e, a, s}. Find $T \cap S$.
8. K = {1, 2, 3, 4, 6, 12}   L = {1, 2, 3, 6, 9, 18}. Find $K \cap L$.

1.8 Representing sets on the venn diagram

Example 1

W = {bananas, cassava, potatoes, beans}
Z ={Yams, potatoes, beans, cassava}
Represent the two sets on the venn diagram.
Common members= {potatoes, beans, cassava}



a) Find $P \cup Q$.

Here we write all elements in the venn diagram.

$P \cup Q$= {bananas, potatoes, beans, cassava, yams}

b) Find $P \cap Q$.

$P \cap Q$= {potatoes, beans, cassava}

Activity 1:8

Represent the given sets on the venn diagram.

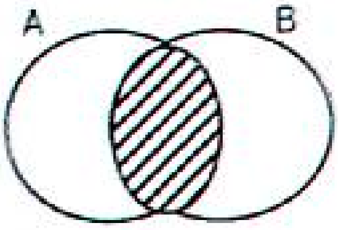
1. Given that M = {April,May,March,July}   N ={June,March,August,April}
   a) Represent the information on the venn diagram.
   b) Find $M \cup N$ c) Find $M \cap N$
2. If A ={hat, helmet, cup}   B = {shirt, trousers, helmet}
   a) Represent the above information on the venn diagram.
   b) What is $A \cup B$?   c) What is $A \cap B$?
3. Set L ={b, r, e, a, d}   K = {b, r, e, a, k}
   a) Represent the information on the venn diagram.
   b) From the venn diagram, find $L \cup K$.
4. Given A ={2, 4, 6, 8, 10}   B = {1, 2, 4, 5, 6}
   a) Show the above sets on the venn diagram.
   b) What is $A \cup B$ c) Find $A \cap B$.

5. Set R ={odd numbers less than 15} S = {counting numbers less than 10}
   a) List elements in set R. b) List all the elements in set S.
   c) Represent the above sets on the venn diagram.
   d) Find $R \cup S$ e) Find $R \cap S$

6. P ={Multiples of 2 less than 18} Q={Multiples of 3 less than 20}
   a) List the elements in set P. b) Write down all the members in set Q.
   c) Find $P \cup Q$. d) Find $P \cap Q$.

6. Given that, F = {all prime numbers less than 15}
   G = {odd numbers less than 10}
   a) Represent set F and set G on the venn diagram.
   b) Find $F \cup G$. c) Find $F \cap G$

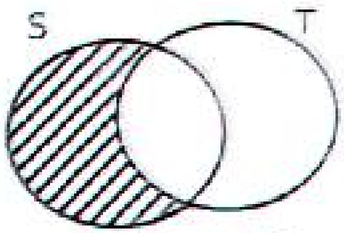
1.9 Describing Shaded Regions of Venn Diagrams

1.


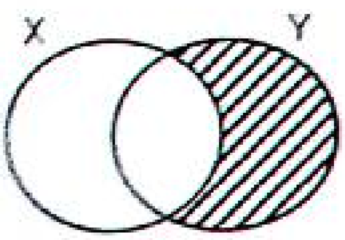
Set $A \cap set\ B$

2.

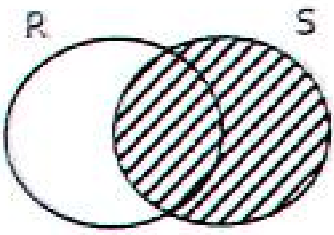

Set $P \cup set\ Q$

3.


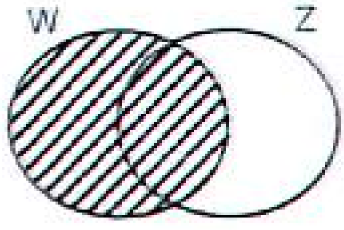
Set S – set T

4.


Set $Y - set\ X$

5.


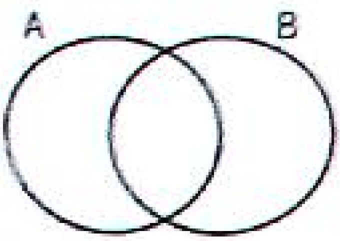
Set S

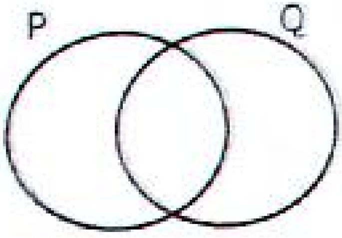
6.


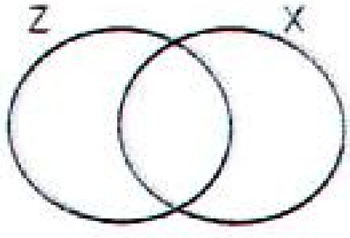
Set W

Activity 1:9

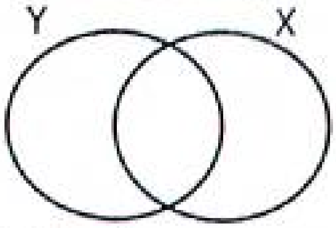
Shade the regions as described in the question.

1. Shade $A \cup B$


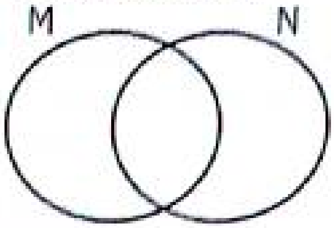
2. Shade $P \cap Q$


3. Shade Z – X


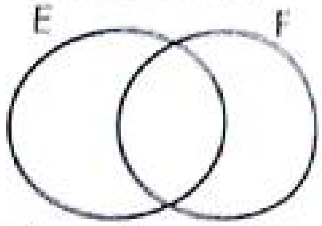
4. Shade $X - Y$



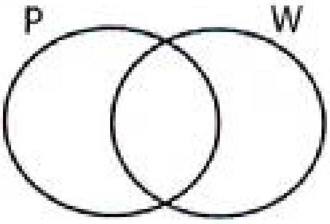
5. Shade set M



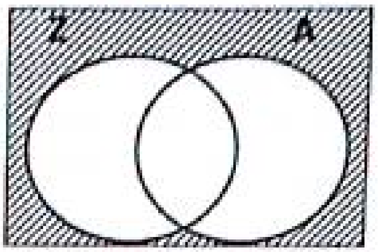
6. Shade set F



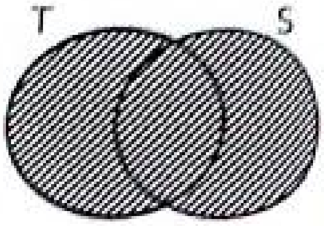
7. Shade set W



8. Name the shaded part

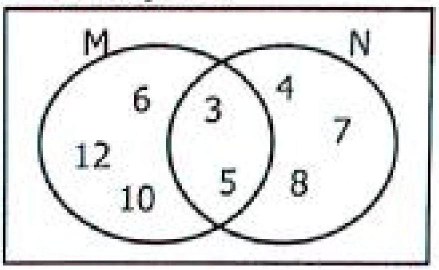


9. Name the shaded part



1.10 Forming Sets from the Venn Diagrams

Example 1



a) List the elements in set M.
Set M = {3, 5, 6, 10, 12}

b) List all elements in set N.
Set N = {3, 4, 5, 7, 8}

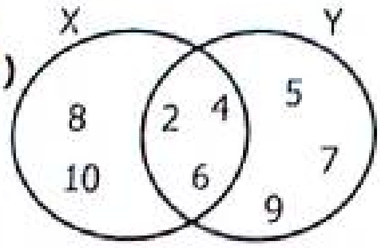
c) Find $M \cup N$.
$M \cup N = \{3, 4, 5, 6, 7, 8, 10, 12\}$

d) Find $M \cap N$
$M \cap N = \{3, 5\}$

Activity 1:10

Form sets from the venn diagram below.

1)



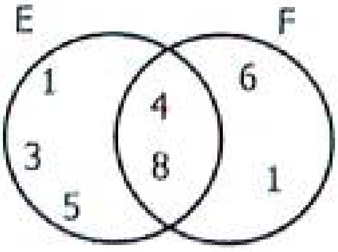
a) List elements in set X.

b) List elements in set Y.

c) Find $X \cap Y$.

d) Find $X \cup Y$.

2)



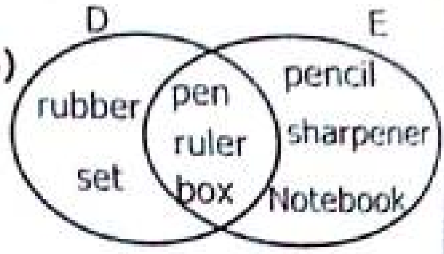
a) List elements in set E.

b) List elements in set F.

c) Find $E \cup F$.

d) Find $E \cap F$.

3)



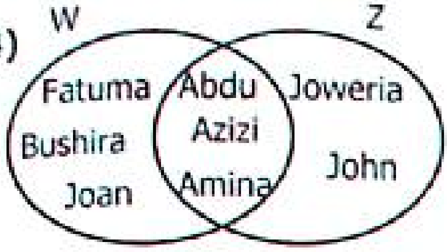
a) List elements in set D.

b) List elements in set E.

c) Find $D \cup E$.

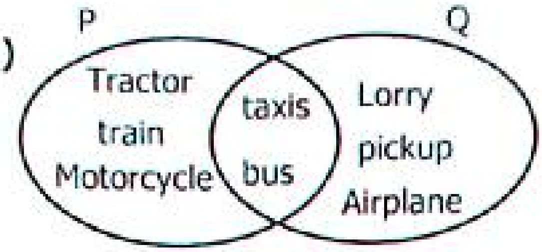
d) Find $D \cap E$.

4)



a) List elements in set Z.
b) List elements in set W.
c) Find $W \cup Z$.
d) Find $W \cap Z$.

5)



a) List elements inset P.
b) List elements in set Q.
c) Find $P \cup Q$.
d) Find $P \cap Q$.

1.11 Difference of Sets

Example 1

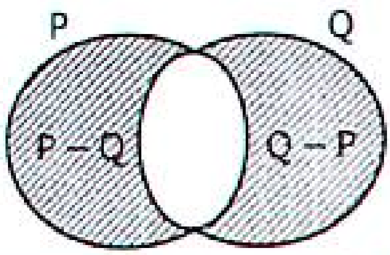
Given that A = {~~box~~, bag, ~~table~~, chair} B={stool, bed, ~~box~~, form, ~~table~~}

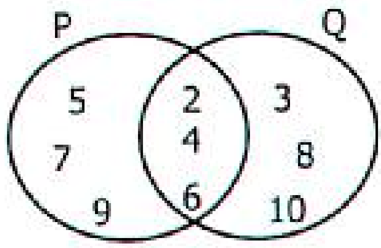
a) Find set A – set B.
Note the common members = {box, table}
Therefore set A – set B = {bag, chair}

b) What is set B – set A?
Set B – set A = {stool,bed,form}

Example 2

Use the venn diagram to answer questions that follow.





a) Find set P – set Q.
P – Q={5, 7, 9}

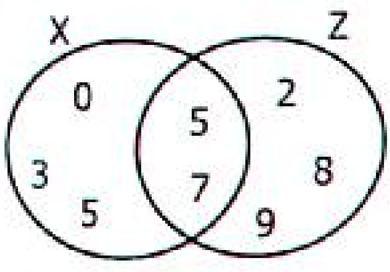
b) What is Q – P?
Q – P={3,8,10}

c) Find $P \cup Q$.
PUQ={2,3,4,5,6,7,9,10}

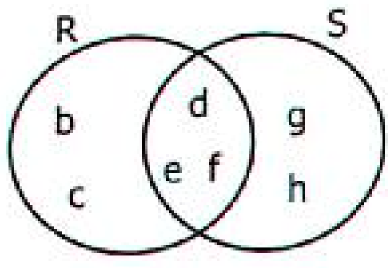
d) Find $P \cap Q$.
$P \cup Q$ ={2,4,6}

Activity 1:11

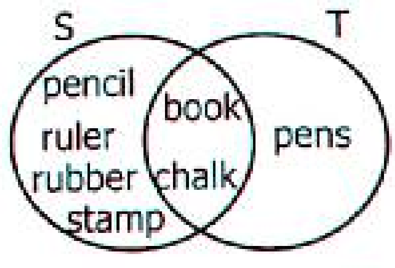
1. Use the venn diagram below to answer questions that follow.



a) Find X – Z.
b) Find Z – X.



a) List set R – set S.
b) List set S – set R.



a) List set S – set T.
b) List set T – set S.

2. Given Set R = {cup,plate,knife,spoon} Set T={tin,plate,fork, knife,flask}
   a) Find set R – set T b) List the elements of set T – set S.
3. Set P = {a, b, c, d, e}, Q ={a, e, i, o, u}
   i) List the elements of set P – set Q. ii)List the elements of set Q – set P.
4. If set H = {1, 2, 3, 4, 5} Set K = {2, 4, 6, 8, 10}
   a) Find H – K. b) Find K – H.
5. If set A = {factors of 12} B = {factors of 18}
   a) List all elements in set A
   b) Write down all elements in Set B.
   c) Find set A – set B
   d) Find set B – set A
6. Given that set G ={Multiples of 2 less than 15}.
   Set H={Multiple of 3 less than 15}
   a) List members in set G.
   b) List members in set H.
   c) Find set G – set H.
   d) Find set H – set G.
7. If set X ={Prime numbers less than 20} Y={odd numbers less than 20}
   a) List the elements in set X.
   b) Write down all elements in set Y.
   c) Find set $X \cup$ set Y.
   d) Find set $X \cap$ set Y

1.12 Finding Number of Elements in Given Sets

Example 1

Given that K = {1, 2, 3, 4, 12, 24}

Find n(K) count the number of elements.

n(K) =7

Example 2

If R = {a, b, c, d, e, f, g, h}

Find n(R)

n(R)=8

Example 3

A ={2,4,7,9,12,15} B = {3, 6, 9, 12, 15, 18}

a) Find n(A)
n(A)= 6

b) Find n(B)
n(B) = 6

c) Find n(A – B)
Find A – B
A – B ={2, 4, 7}
n(A – B)=3

d) Find n(B – A)
B – A = {3, 6, 18}
n(B – A) = 3

Example 4

Given that;

P={cassava,bananas,potatoes,rice}

Q={rice, potatoes,wheat,Irish,cassava}

a) Find n$(P \cap Q)$

$P \cap Q$ ={rice, potatoes, cassava}

n$(P \cap Q)$ = 3

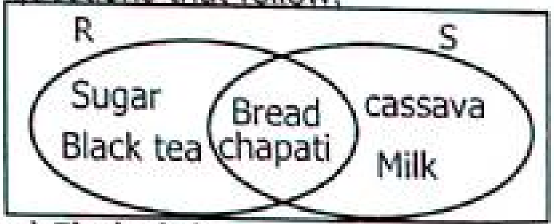
b) Find n$(P \cup Q)$

$P \cup Q$={rice, potatoes, bananas, wheat cassava Irish}

n$(P \cup Q)$ = 6

Example 5

Use the venn diagram to answer questions that follow.



a) Find n(R).

R = {black tea, sugar, bread, chapati}

n(R) = 4

b) Find n$(R \cap S)$ = 2

$R \cap S$ ={bread, chapati}

n$(R \cap S)$ = 2

Activity 1:12

Draw venn diagrams to represent sets below and answer questions that follow;

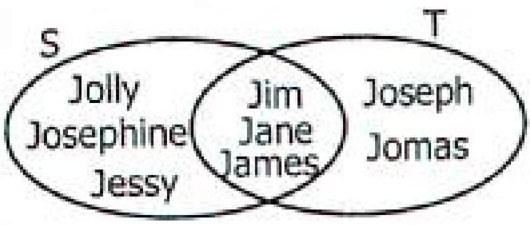
1. W={March,April,October,May,June} X={July,April,August,May,Dec,March}
   a) Find n(X) b) Find n(W) c) Find $n(W \cap X)$
   d) List $n(W \cup X)$ e) What is n(W – X) f) What is n(X – W)

2. E = {k, e, t, l, b} F ={b, e, a, t}
   a) Find $n(E \cap F)$ b) Find $n(E \cup F)$ c) What is n(E – F) d) What is n(F-E)

3. A = {Mellane,Mable,Madrine,Margret}
   B={Molly,Mabella,Mable,Mercy,Mabimbo}
   a) Find $n(A \cup B)$ b) Find $n(A \cap B)$ c) Find n(A – B) d) Find n(B – A)

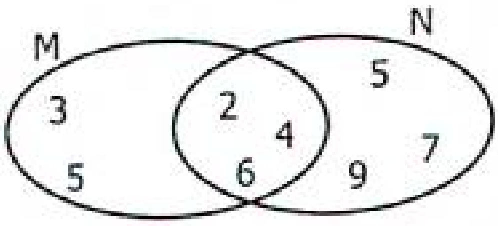
4. P = {Opio, Okello, Obala} Q={Omollo, Opida, Okello, Owino, Obala}
   a) Find n(Q) b) Find $n(P \cup Q)$ c) Find $n(P \cap Q)$ d) Find n(Q – P)

5.



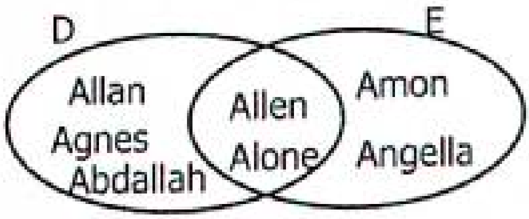
a) Find n(S – T). c) Find $n(S \cup T)$.
b) Find n(T – S). d) Find $n(S \cap T)$.

6.



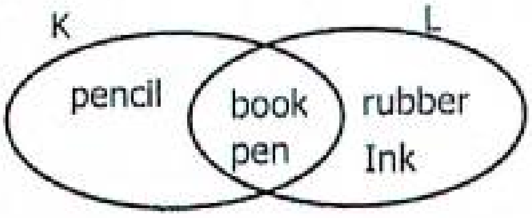
a) List elements inset M.
b) List elements in set N.
c) Find n(M – N).
d) What is n(N – M).

7.



a) Find n(E).
b) Find n(D).
c) What is $n((D \cup E)$?
d) Find $n(D \cap E)$.

8.



a) What is n(K – L)
b) Find n(L – K)
c) Find $n(K \cup L)$
d) Find $n(K \cap L)$

TOPIC 2: WHOLE NUMBERS

Numeration system is a system of presenting numbers or ideas. The numeration system used in most countries in the world is Hindu-Arabic system.

A digit is a single symbol that represents a quantity for example 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. These digits are put together to form different numbers.

2.1 Forming Numbers From Digits

Example 1

Write down all 3digit numbers formed by the digits 4, 8, 5.

Formation of different numbers by the help of a table.

4	8	5
485	845	584
458	854	548

The numbers are 485, 845, 584, 458, 854 and 548

Example 2

Find the sum of the biggest and smallest number from digits 6,4 and 7.

6	4	7
647	467	746
674	476	764

Sum

$$\begin{array}{lr} \text{smallest} & 4\,6\,7 \\ \text{largest} & +7\,6\,4 \\ \hline & 1\,2\,3\,1 \end{array}$$

Example 3

Use the digits 6, 4, 9 and 8 to find the difference between the highest and lowest number formed.

Highest number = 9864

Lowest number = 4689

Difference

$$\begin{array}{r} 9\,8\,6\,4 \\ -\,4\,6\,8\,9 \\ \hline 5\,1\,7\,5 \end{array}$$

Example 3

Use the digits 6, 8 and 0 to find the difference between the highest and lowest number formed.

$$\begin{array}{lr} \text{Highest number} = & 8\,6\,0 \\ \text{Lowest number} = & -\,6\,0\,8 \\ \hline & 2\,5\,2 \end{array}$$

Activity 2:1

List all numbers formed from the digits below.

1) 8,4,6 2) 5,2,7 3) 3,0,9 4) 7,5,0 5) 3,7,4
6) 9,5,6 7) 1,3,2,4 8) 8,4,7,9 9) 1,0,4,5 10) 5,8,6
11) 1,5,7,0 12) 9,3,7,5 13) 4,7,8 14) 8,0,3 15) 1,2,3,4

13) Write down numbers which can be formed using 7, 5 and 9.
14) Use the digits 3, 5, 8 and 2 to form 24 different numbers.
15) Write down all even numbers formed from 5347.

16) Given the digits 9, 5 and 4.

a) Find the difference between the largest and the smallest numbers formed from the digits above.

b) Work out the sum of the largest and lowest numbers formed by the digits above.

17) Write down the smallest number than can be formed using 5, 0, 7 and 3.

18) Write down the biggest number that can be formed from 7130.

19) Use the digits 8, 3, 9, 7 and to the find the difference between the largest and lowest number formed from the digits.

20) Work out the sum of the largest number and smallest number formed from the given digits 4, 7, 5 and 3.

2.2 Place Values of Whole Numbers

Place values of whole numbers from the smallest place values.

- Ones	-Thousands	-Millions
-Tens	- Ten thousands	-Ten millions
-Hundreds	- Hundred thousands	-Hundred millions

Example 1

Write down the place value of each digit in the number 8 7 5 6.

T.Th	Th	H	T	O
	8	7	5	6

Ones
Tens
Hundreds
Thousands

Example 2

What is the place value of 5 in 675631?

6 7 5 6 3 1

Thousands

The place value of 5 is thousands.

Activity 2:2

Write down the place value of each digit in the numbers below.

1) 438	2) 580	3) 902	4) 279	5) 4861	6) 8073
7) 1234	8) 9876	9) 14281	10) 84,578	11) 10205	12) 84347

13) What is the value of 9 in the number 193,000?

14) Write down the place value of 6 in 2,463.

15) What is the place value of 2 in 425,637?

16) What is the place value of 5 in 74567?

2.3 Finding Values of Whole Numbers

- ✓ Place values are key points in finding values.
- ✓ Finding the values of different digits, one should be able to compute with the help of place values.

Example 1

Write down the value of each digit in the number 38572.

3 8 5 7 2

Place value	Method	Value
Ones	2×1	2
Tens	7×10	70
Hundreds	5×100	500
Thousands	8×1000	8000
Ten thousands	3×10000	30000

Example 2

Write down the value of 3 in the number 86341.

8 6 3 4 1

Hundreds = $(3\times100) = 300$

The value of 3 is 300.

Example 3

What is the value of 7 in the number 37629?

3 7 6 2 9

Thousands = $(7\times1000) = 7000$

The value of 7 is 7000.

Example 4

Work out the sum of 5 and 4 in the number 14052.

1 4 0 5 2

Tens $\rightarrow 5\times10=50$

Thousands $\rightarrow 4\times1000=4000$

Sum

$$\begin{array}{r} 4000 \\ +\ 50 \\ \hline 4{,}050 \end{array}$$

Example 5

Find the difference between the values of 2 and 9 in the number 2796.

2 7 9 6

Ones $\rightarrow 6\times1=6$

Thousands $\rightarrow 2\times1000=2000$

Difference

$$\begin{array}{r} 2000 \\ -\ \ \ 6 \\ \hline 1{,}994 \end{array}$$

Example 6

Calculate the product of the values of 6 and 2 in the number 6423.

6 4 2 3

Tens $\rightarrow 2\times10 = 20$

Ten thousands $\rightarrow 6\times1000 = 6000$

$$\begin{array}{r} 6000 \\ \times\ 20 \\ \hline 12000 \end{array}$$

Example 7

Calculate the quotient of the values of 6 and 3 in 1603.

1 6 0 3

Ones $\rightarrow 3\times1=3$

Hundreds $\rightarrow 6\times100=600$

Quotient

$$\left(\frac{600}{3}\right) = 200$$

Activity 2:3

1. Find the value of each digit in the numbers below.
   a) 4 3 6 7 b) 4 9 1 c) 7 6 5 d) 3 2 9 e) 1 5 3 6 f) 9 5 7 4
   g) 4 5 6 3 h) 6 8 9 3 i) 76329 j) 80302 k) 47482 L) 75630
2. Work out the value of 5 in the numbers 35201.
3. Work out the sum of the value of 9 and 7 in the number 9675.
4. What is the sum of the value of 3 and 4 in the number 2374?
5. Find the difference between the value of 5 and 2 in the number 65729.
6. What is the product of the value of 6 and 4 in 9634?
7. Calculate the product of the value of 6 and place Value of 3 in the number 23681.
8. Calculate the quotient of the value of 6 and 5 in 1625.
9. What is the quotient of the values of 3 and 2 in the number 83962?
10. Find the quotient the value of 4 and 8 in the number 78247.

2.4 <u>Expanding Whole Numbers Using Values</u>

✓ Place values are still important here in this lesson.

Example 1 Expand 7 5 8 9 using values. 7thousands+5hundreds+8tens+9ones $7 \times 1000 + 5 \times 100 + 8 \times 10 + 9 \times 1$ $7000 + 500 + 80 + 9$	**Example 2** Expand 4 0 9 6 using values. 4thousands+0hundreds+9tens+6ones $4\times1000+0\times100+9\times10+6\times1$ $4000+90+6$
Example 3 Expand 4 8 3 2 6 using values 4ten thousands+8thousands+ 3hundreds+2tens+6ones $4\times10,000+8\times1000+3\times100+2\times10+6\times1$ $40,000+8000+300+20+6$	**Example 4** Expand 6 7 3 5 9 using values. 6tenthousands+7thousands+3hundred s+5tens+9ones $6\times10000+7\times1000+3\times100+5\times10+9\times1$ $60000+7000+300+50+9$

Activity 2:4

Expand the following using values.

1) 549 2) 632 3) 409 4) 7020 5) 5408
6) 14567 7) 90,908 8) 97,438 9) 12,345 10) 38901
11) 10307 12) 84,237 13) 134628 14) 283967 15) 903476

2.5 <u>Expanding Whole Numbers Using Exponents (Using powers of ten)</u>

Example 1
Expand 2 3 7 9 using place values.
2thousands + 3hundreds + 7tens + 9ones
$2 \times 1000 + 3 \times 100 + 7 \times 10 + 9 \times 1$

Example 2

Expand 35,987 using factors of ten.

2ten thousands + 5thousands + 9hundreds + 8tens + 7ones

$3 \times 10000 + 5 \times 1000 + 9 \times 100 + 8 \times 10 + 7 \times 1$

$3 \times 10 \times 10 \times 10 \times 10 + 5 \times 10 \times 10 \times 10 + 9 \times 10 \times 10 + 8 \times 10 + 7 \times 1$

$(3 \times 10^4) + (5 \times 10^3) + (9 \times 10^2) + (8 \times 10^1) + (7 \times 10^0)$

Activity 2:5

Expand the following using powers of ten (Exponents)

1) 940	2) 278	3) 1637	4) 5469	5) 206	6) 980
7) 90140	8) 6097	9) 63745	10) 91072	11) 32198	12) 27027

Expand the following using places values

1) 444	2) 536	3) 162	4) 890	5) 9435	6) 8402
7) 7543	8) 6787	9) 24256	10) 98987	11) 28256	12) 74568

2.6 Finding Expanded numbers

Example 1

Find the expanded number.

$7\times1000\times3\times100+5\times10+8\times1$

7×1000	=	7 0 0 0
3×100	=	3 0 0
5×10	+	5 0
8×1		8
		7,3 5 8

Example 3

4thousands+6hundreds+1tens+8ones

4thousands	=	4 0 0 0
6hundreds	=	3 0 0
1tens	=	+ 1 0
8ones	=	8
		4 3 1 8

Example 4

Find the expand number.

6000 + 400 + 90 + 3

6 0 0 0 0
4 0 0
+ 9 0
3
6 0,4 9 3

Example 4

$(8\times10^4)+(5\times10^3)+(2\times10^2)+(0\times10^1)+(7\times10^0)$

8×10^4	=	8 0 0 0 0
5×10^3	=	5 0 0 0
2×10^2	=	+ 2 0 0
0×10^1	=	0
7×10^0	=	7
		8 5 2 0 7

Activity 2:6

Find the expanded numbers (write the expanded numbers in short)

1) 8thousands+5hundreds+3tens+7ones
2) 80000+300+70+4
3) 6thousands+1hundreds+9tens+3ones
4) 6000+8000+500+40+1
5) $5\times10000+1\times1000+3\times100+7\times10+2\times1$
6) 40000+3000+900+80+2
7) 5thousands+7hundreds+2tens+4ones
8) $7\times1000+2\times10+8\times1$

9) $(3\times10^4)+(8\times10^3)+(4\times10^2)+(6\times10^1)$ 10) $(7\times10^4)+(2\times10^2)+(6\times10^0)$

11) $(2\times10^3)+(3\times10^2)+(4\times10^1)+(5\times10^0)$ 12) $(8\times10^3)+(9\times10^2)+(2\times10^1)+(5\times10^0)$

2.7 Writing Figures in Words

Example 1

Write 4 2 3 6 2 in words.

Thousands	Units
42	3 6 2

Forty two thousand, three hundred sixty two.

Example 2

Write 1 0 1 0 1 1 in words.

Thousands	Units
1 0 1	0 1 1

One hundred one thousand, eleven.

Activity 2:7

Write the following in words.

1) 346 2) 407 3) 179 4) 908 5) 7,456 6) 4,895

7) 90,478 8) 340307 9) 11,289 10) 40,786 11) 107060 12) 80706

13) $(8\times10^3)+(6\times10^1)+(5\times10^0)$. 14) $(4\times10^3)+(5\times10^1)+(8\times10^0)$

15) $(8\times10^4)+(4\times10^2)+(2\times10^0)$ 16) $(3\times10^4)+(5\times10^2)+(6\times10^0)$

2.8 Writing in Figures

Example 1

Write one hundred thirty nine thousand, eighty hundred four in figures.

139 thousand 804 units

$$\begin{array}{lr} \text{139 thousand} = & 139{,}000 \\ \text{804units} & \underline{+\ 804} \\ & \underline{139{,}804} \end{array}$$

Example 2

Eighty hundred forty thousand, nine hundred fifty three.

840thousand 953 units

$$\begin{array}{lr} \text{840thousand} = & 840{,}000 \\ \text{953 units} = & \underline{+\ 953} \\ & \underline{840{,}953} \end{array}$$

Activity 2:9

Write the following in figures.

1. Sixty three thousand, nine hundred forty three.
2. Seventy three thousand, seven.
3. Forty six thousand, four hundred nine.
4. Two hundred two thousand, four hundred seventy seven.
5. Seventy four thousand, seven hundred ninety one.
6. Nine hundred three thousand, six hundred fourteen.
7. One hundred four thousand, one hundred eleven.
8. Five hundred twelve thousand, seven hundred eleven.
9. Eighty hundred eighty thousand three hundred nine.
10. Two hundred thirty four thousand, six hundred seventy eighty.
11. Three hundred thirty three thousand, four hundred forty four.
12. Six hundred thousand, four hundred two.

Rounding off Whole Numbers

- ✓ Rounding off whole or decimal number we consider the place values.
- ✓ When rounding off, we maintain or increase the value of the place value stated by either adding **Zero** or **One**, ten, hundred or thousand.

2.9 Round off the Whole Numbers to the Nearest tens

Example 1

Round off 384 to the nearest tens.

384 → 380 + 4

4 is nearer to 0 than to 10 ∴ we add 0

$$\begin{array}{r} 380 \\ +00 \\ \hline 3.80 \\ \hline \end{array}$$

Example 2

Round off 7843 to the nearest tens.

7 8 4 3 → 7 8 4 0 + 3

3 is nearer to 0 than 10 ∴ we add 0

$$\begin{array}{r} 7840 \\ +00 \\ \hline 7840 \\ \hline \end{array}$$

Example 3

Round off 547 to the nearest tens.

547 → 540 + 7

7 is nearer to 10 than 0 ∴ we add 10

$$\begin{array}{r} 540 \\ +10 \\ \hline 550 \\ \hline \end{array}$$

Example 4

Round off 14899 to the nearest tens.

1 4 8 9 0 + 9

9 is nearer to 10 than 0 ∴ we add 10

$$\begin{array}{r} 14890 \\ +10 \\ \hline 14900 \\ \hline \end{array}$$

Activity 2:9

Round the following numbers to the nearest tens.

1) 42	2) 53	3) 741	4) 444	5) 1764	6) 8793
7) 12322	8) 7381	9) 1768	10) 8795	11) 12329	12) 7386
13) 8767	14) 5798	15)3285	16) 79	17) 7598	18) 48327

2:10 Round off to the Nearest Hundreds

Example 1

Round off 9845 to the nearest hundreds.

9 8 4 5 → 9 8 0 0 + 4 5

45 is nearer to 0 than 100 ∴ we add 000

$$\begin{array}{r} 9800 \\ +000 \\ \hline 9800 \\ \hline \end{array}$$

Example 2

Round off 428 to the nearest hundreds.

4 2 8 → 4 0 0 + 2 8

$$\begin{array}{r} 400 \\ +00 \\ \hline 400 \\ \hline \end{array}$$

Example 3

Round off 479 to the nearest hundreds.

4 7 9 → 4 0 0 + 7 9

79 is nearer to 100 than 0 ∴ we add 100

$$\begin{array}{r} 400 \\ 100 \\ \hline 500 \\ \hline \end{array}$$

Example 4

Round off 1863 to the nearest hundreds.

1 8 6 3 → 1 8 0 0 + 6 3

$$\begin{array}{r} 1800 \\ +100 \\ \hline 1900 \\ \hline \end{array}$$

Activity 2:10

Round the following to the nearest hundreds.

1) 542	2) 438	3) 1945	4) 121	5) 1783	6) 1925
7) 7899	8) 999	9) 8076	10) 1111	11) 3351	12) 8239
13) 4037	14) 7304	15) 3074	16) 7034	17) 209	18) 447

2.11 Round off the Nearest Thousands

Example 1

Round 7432 to the nearest thousands.

$7432 \rightarrow 7000 + 32$

432 is near 0 than 1000, $\therefore$ we add 0

$$\begin{array}{r} 7000 \\ +0000 \\ \hline 7000 \\ \hline \end{array}$$

Example 2

Round off 8298 to the nearest thousands.

$8298 \rightarrow 8000 + 298$

298 is near 0 than 1000, $\therefore$ we add 0

$$\begin{array}{r} 8000 \\ +0000 \\ \hline 8000 \end{array}$$

Example 3

Round 9748 to the nearest thousands

$9748 \rightarrow 9000 + 748$

748 is nearer to 1000,

therefore we add 1000.

$$\begin{array}{r} 9000 \\ +1000 \\ \hline 10000 \\ \hline \end{array}$$

Example 4

Round off 6500 to the nearest thousands.

$6500 \rightarrow 6000 + 500$

500 is nearer 1000 than 0,

therefore we add 1000

$$\begin{array}{r} 6000 \\ +1000 \\ \hline 7000 \\ \hline \end{array}$$

Activity 2:11A

Round off the following to the nearest thousands.

1) 7200	2) 9497	3) 3262	4) 8438	5) 34562	6) 6734
7) 14993	8) 3789	9) 23456	10) 62567	11) 46382	12) 12642
13) 65432	14) 46500	15) 12600	16) 62100	17) 59011	18) 9124

Activity 2:11B

Round off the following according to the instructions.

1. Round off two hundred sixty nine to the nearest tens.
2. Round off three hundred fifty seven to the nearest tens.
3. Round off seven hundred sixty three to the nearest tens.
4. Round off seven hundred forty nine to the nearest hundreds.
5. Round off three thousand four hundred sixty five to the nearest hundreds.
6. Round off twenty thousand nine hundred twenty two to the nearest hundreds.

7. Round off fourteen thousand six hundred ninety three to the nearest thousands.
8. Round off ninety three thousand four hundred forty nine to the nearest thousands.
9. Round off forty seven thousand three hundred sixty eight to the nearest thousands.
10. Round off ten thousand seven hundred ninety three to the nearest thousands.

2.12 Changing Hindu – Arabic to Roman Numerals

Hindu – Arabic is a system which uses the ordinary arithmetic symbols for example 0, 1, 2, 3, 4, 5, 6, 7, 8 to 9.
Roman numerals system mainly uses the additive and subtractive number system.

Hindu – Arabic	1	2	3	4	5	6	7	8	9	10
Roman Numerals	I	II	III	IV	V	VI	VII	VIII	IX	X

20	30	40	50	60	70	80	90	100	200
XX	XXX	XL	L	LX	LXX	LXXX	XC	C	CC

There are various categories (groups) of Roman numerals.
i) Numbers 2 and 3 are repeated Roman numerals.

2 → 1 + 1 = II
3 → 1 + 1 + 1 = III

20 → 10 + 10 = XX
30 → 10 + 10 + 10 = XXX

ii) Numbers 4 and 9 are obtained by subtraction.

4 → 1 subtracted from 5 = IV
9 → 1 subtracted from 10 = IX
40 →10 subtracted from 50 = XL

60 → 50 + 10 = LX
70 → 50 + 20 = LXX
80 → 50 + 30= LXXX

iii) 6, 7 and 8 are obtained by addition.

6 → 5 + 1 = VI
7 → 5 + 2 = VII
8 → 5 + 3 = VIII

90 → 100 – X = XC
100 = C

Expressing Hindu-Arabic to Roman Numerals

Example 1
Write 28 in Roman numerals.
Expand 28
28 = 20 + 8
XX VIII
28 → XXVIII

Example 2
Write 43 in Roma numerals.
Expand 43
43 = 40 + 3
XL III
43 → XLIII

Example 3
Express 86 in Roman numerals

8 6 = 8 0 + 6
LXXX VI
8 6 → LXXXVI

Example 4
Express 32 in Roman numerals.

3 2 = 3 0 + 2
X X X I I
3 2 → XXXII

Activity 2:12

1. Write the following to Roman numbers.

a) 22 b) 25 c) 37 d) 33 e) 42 f) 47 g) 49
h) 50 i) 64 j) 73 k) 79 L) 88 m) 97 n) 95
o) 96 p) 63 q) 145 r) 193 s) 167 t) 128 u) 180

2.13 Expressing Roman Numerals to Hindu – Arabic Numerals

Example 1
Express XXXIII to Hindu – Arabic numerals.

X X X → 3 0
I I I + 3
XXXIII → 33

Example 2
Change XLIV to Hindu Arabic numerals.

X L → 4 0
I X → + 4
X L I X 4 4

Example 3
Express LXXIX in Arabic numerals.

L X X → 7 0
I X → 9
LXXIX → 79

Example 2
Convert LXXVI to Hindu Arabic.

L X X → 70
V I → 6
LXXVI → 76

Activity 2:13

1. Express LXXXIII in Arabic numerals.
2. Change XLVIII in Hindu-Arabic.
3. Convert XXIV in Hindu-Arabic numerals.
4. Change XLIV in Hindu-Arabic.
5. Okware is XXVII years. How old is he in Hindu – Arabic numerals?
6. Its XXXVIII years since the president came to power. Convert it in Hindu-Arabic numerals.
7. Byaruhanga has XXIX goats, write the number in Hindu-Arabic numerals.
8. Express LXXII in Arabic numerals.
9. A shopkeeper has LXXXVIII kilograms of sugar remaining in his shop. Convert that figure in Arabic numerals.
10. There are XCVI pupils in primary five. How many pupils are in the class in Arabic numerals?
11. Ogala had XLVII chicken. Write this number in Hindu-Arabic numerals.
12. Our teacher is XXXIV years old. How old is he in Hindu-Arabic numerals?

TOPIC : 3. OPERATION ON WHOLE NUMBERS

3:1 Addition of Large Numbers

Procedure

- ✓ Arrange the numbers vertically according to the place values.
- ✓ Add downwards starting from the place value on your right, re-group where applicable.
- ✓ The number where the sum is a two digit number, write the ones and carry the tens to the next digit on the left.

Addition of Large Numbers and Re-Grouping

Example 1	Example 2	Example 3
Without re-grouping. Add: 347869 + 450120 347867 $+450120$ 797989	Find the sum of 123456 and 645321. 123456 $+645321$ 768777	45678 $+58726$ 104404

Activity 3:1

Add the following numbers.

1) 43002 + 54974	2) 74342 + 42517	3) 23456 + 65432	4) 57368 + 42531
5) 748395 + 936765	6) 483674 + 193814	7) 747693 + 203894	8) 930746 + 194894
9) 634893 + 178636 + 234724	10) 43624 + 49733 + 18267	11) 84956 + 45687 + 60408	12) 95678 + 68690 + 12345

3:2 Addition of Word Problems

Example 1

A piece of land measures 24,974 square metres and another one measures 78,647 square meters. Find the size of the two plots of land.

24,974 square metres
\+ 78,647 square metres
103621 square metres

Example 2

Katono bought a sack of rice at sh.285700 and a sack of posho at sh.248500. Calculate his total expenditure.

Sh. 285,700
Sh. + 248500
Sh. 534,200

He spent sh.534,200

Activity 3:2

1. Angellina bookshop Kampala sold 42879 text books in 2021 and 36943 textbooks in 2022. How many books did they sell in the two years?
2. The population in Amratar district is 276347 and 307894 in Gulu district. Calculate the total population in the two districts.
3. Find the sum of 784621 and 942767.
4. In the year 2021 Uganda exported 159340 tonnes of coffee and 347286 tonnes in 2022. Find the total export for the two years.
5. Kirihura district produces 27872 litres of milk a day and Mbarara produces 42647 litres. Calculate the total amount of milk produced by the two districts in a day.
6. Uganda Baati factory produced 347004 iron sheets in April and 674973 iron sheets were made in May. How many iron sheets were produced in the two months/
7. Mr. Masaba bought three goats at sh. 584000 and later bought two more goats at sh. 327000. How much did he pay altogether?
8. A fruit factory packed 984340 litres of Mango juice and 147630 litres of apple juice per month. How many litres of juice are produced in the two months?
9. Okello a shopkeeper in Kitigum town recorded the following sales; study the table below and answer questions that follow.

Day	Sales
Monday	Sh. 246700
Tuesday	Sh. 954300
Wednesday	Sh. 650490
Thursday	Sh. 790800
Friday	Sh. 687200

a) What was the value of goods sold on Monday and Tuesday?
b) Workout the total sales on Wednesday and Thursday.
c) Find the total sales on Monday and Thursday.
d) Calculate the total sales for all the five days.

3:3 Subtraction up to 6 digit Numbers

- ✓ Note the words used in subtraction are minus, take away, difference, decrease, how much more, what must be added to and range.
- ✓ Read and arrange the numbers vertically according to the place values, then subtract from the ones position.
- ✓ Re-group where applicable, the difference got gives the answer.

Example 1

Subtract 334674 – 179930.

Thousands	Units
2 12 13	16
3 3 4	6 7 4
- 1 7 9	9 3 0
1 5 4	7 4 4

Example 2

Subtract 131498 from 881467.

Thousand	Units
7 10	13 15 17
8 8 1	4 6 7
- 1 3 1	4 9 8
7 4 9	9 6 9

Activity

Workout the following numbers.

1) 349638 – 143504   2) 934674 – 387657   3) 367490 – 141959
4) 930000 – 831407   5) 777404 – 119870   6) Subtract 486231 – 296539
7) Subtract 248673 – 99782   8) Subtract 17864 from 43426
9) Subtract 358750 from 609200   10) 493240 minus 298721
11) Sh. 24950 minus 97,200   12) Subtract 149630 from 190,000
13) Subtract 284000 from 300,000   14) 700,000 take away 358050
15) 48,242 mangoes take away 29782 mangoes.
16) What number must be added to 90862 to get 120,000?
17) What number must be added to 32493 to get 40000?
18) Find the difference between 79840 and 52084.

3:4 More About subtraction of Whole Numbers

Example 1

Mr. Okwalinga planted 248000 banana seedlings and only 228473 germinated. How many seedlings did not germinate?

Planted seedlings	2 4 8 0 0 0 seedlings
Germinated seedlings	– 2 2 8 4 7 3 seedlings
Failed to germinate	1 9 5 2 7 seedlings

Activity 3:4

1. By how much is sh.764239 greater than 454264?
2. By how much is 100438 greater than 94600?
3. Ashopkeeper bought items for his shop on credit at sh.960,000. After a week he paid sh.486,000. How much..........
4. What must be added to 468396 to get 930189?
5. A shopkeeper bought 45900kg of maize floor and sold 36950kg. How many kilogrammes of maize remained?
6. Okot's salary was decreased from sh.755000 by sh.197000. How much does he earn now?
7. A farmer planted 29361 seedlings of tomatoes, 12346 seedlings did not grow. How many seedlings grew?
8. A tank holds 20000 litres of water. If 1824 litres are used. How many litres of water is left in the tank?
9. Mabongo had 2720 animals on his farm. 932 animals were sold off during Easter days. How many animals remained?

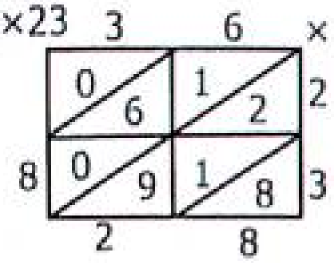
3:5 Multiplication of Two Digit Numbers

Words used in multiplication; Multiply, product, times and of

Multiplication using lattice method is the easiest method

Example 1

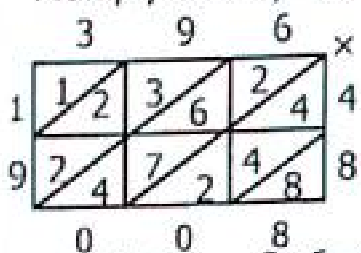
Multiply 36×23



1) 2×6=12 3) 3×6=18
2) 2×3=06 4) 3×3=09
36 × 23 = 828

Example 2

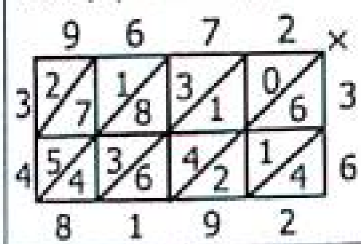
Multiply 396 by 48.



4×6=24 8×6=48
4×9=36 8×9=72
4×3=12 8×3=24
396×48=19008

Example 3

Multiply 9672 by 36.



3 × 2 = 06 6 × 2 = 12
3 × 7 = 21 6 × 7 = 42
3 × 6 = 18 6 × 6 = 36
3 × 9 = 27 6 × 9 = 54
9672×36=348192

Activity 3:5

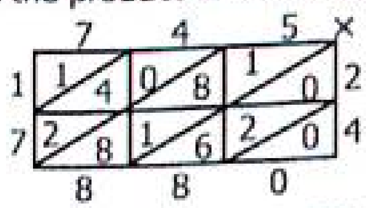
Multiply the following numbers;

1) 68×14 2) 36×23 3)49×28 4) 248×24 5) 672×25 6) 847×36
7) 346×35 8) 438×46 9) 359×17 10) 3579×28 11) 2468×32 12) 642×25

3:6 Multiplication of Word Problems

Example 1

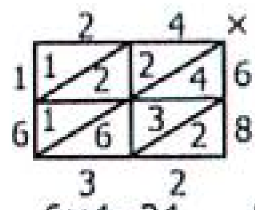
Find the product of 745 and 24.



2×5=10 4×5=20
2×4=08 4×4=16
2×7=14 4×7=28
∴745×24=17880

Example 2

There are 24 classrooms in Green Hill Academy. If each classroom has 68 pupils. How many pupils are in the school?



6×4=24 8×4=32
6×2=12 8×2=16
∴ 24×68=1632pupils

Activity 3:6

Workout the following numbers;

1. Multiply 385 by 36.
2. Multiply 542 by 54.
3. What is the product of 39 and 42?
4. What is the product of 24 and 34?
5. Find the product of 496 and 68?
6. Find the product of 284 and 18.
7. A Pick-up can carry 360 crates of soda. If each crate contains 24 bottles of soda,
   a) How many bottles does it carry in one trip?
   b) If it makes 3trips, how many bottles it would have carried?
8. Find the area of a rectangular piece of timber whose sides are 84cm by 36cm.
9. A rectangular playground measures 120metres by 64 metres. Calculate......
10. How many pupils are in 18 classrooms if each classroom has 72pupils?
11. A mathematics examination has 32 questions, if 48 pupils did the examination. How many questions did they answer altogether?
12. There are 256 books on each shelf in the library. How many books are in the library if there are 14 shelves?
13. Universal chalk factory produces 144 cartons of dairy. Each carton contains 24 boxes of chalk. How many boxes of chalk does the factory produce daily?
14. Nabaasa went fishing and caught 86 fish. He sold each fish at sh.7500. How much money did he get from all the sales?

3:7 Division of Whole Numbers

- ✓ To divide by one or two digit numbers, we use the concept of multiples of the divisor.
- ✓ We look for multiplier that gives the multiple nearest to the dividend.

Example 1

Divide 5664 by 6.

Write the multiples of 6.

M_6 = 6,12,18,24,30,36,42,48,54,60

```
              944
            6|5664
6×9   =      -54↓
               26
6×4   =       -24↓
              -24
6×4   =        24
    5664÷6= 944
```

Example 2

Share 2466 by 9.

M_9 = 9,18,27,36,45,54,63,72,81,90

```
             274
           9|2466
9×2   =     -18↓
              66
9×7   =      -63↓
              36
9×4   =      -36
2466÷9 = 274
```

Example 3	Example 4
Divide 748 by 4 Write the multiples of 4. M_4 = 4,8,12,16,20,24,28,32,36,40	What is the quotient of 6538 and 7? M_7 = 7,14,21,28,35,42,49,56, 63, 70
$\begin{array}{lr} & 187 \\ & 4\overline{)748} \\ 4\times1 = & -4\downarrow \\ & 34 \\ 4\times8 = & -32\downarrow \\ & 28 \\ 4\times7 = & -28 \\ & \overline{} \end{array}$ 748 ÷ 4 = 187	$\begin{array}{lr} & 934 \\ & 7\overline{)6538} \\ 7\times9 = & -63\downarrow \\ & 23 \\ 7\times3 = & -21\downarrow \\ & 28 \\ 7\times4 = & -28 \\ & \overline{} \end{array}$ 6538÷7 = 934

Activity 3:7

Workout the following numbers;

1. Divide 1456 by 4
2. Divide 3816 by 4.
3. Divide 9540 by 5.
4. Divide 8745 by 5.
5. Find the quotient of 1536 and 6.
6. Workout the quotient of 4098 and 6.
7. Divide 3192 by 7.
8. Divide 6909 by 7.
9. Divide 2384 by 8.
10. Divide 3648 by 8.
11. Find the quotient of 2385 and 9.
12. What is the quotient of 1701 and 9.

3:8 Division by Two Digit Numbers

Example 1	Example 2
Divide 4860 by 10. M_{10}=10,20,30, 40,50, 60,70, 80,90,100	Workout the quotient of 7884 by 12. M_{12}=12,24,36,48, 60, 72,84,96,108
$\begin{array}{lr} & 486 \\ & 10\overline{)4860} \\ 10\times4 = & -40\downarrow \\ & 86 \\ 10\times8 = & -80\downarrow \\ & 60 \\ 10\times6 = & -60 \\ & \overline{} \end{array}$ 4860÷10 = 486	$\begin{array}{lr} & 657 \\ & 12\overline{)7884} \\ 12\times6 = & -72\downarrow \\ & 68 \\ 12\times5 = & -60\downarrow \\ & 84 \\ 12\times7 = & -84 \\ & \overline{} \end{array}$ 7884÷12 = 657

Activity 3:8

1. Divide 1480 by 10.
2. Divide 78930 by 10.
3. Divide 6875 by 11
4. Divide 5720 by 11.
5. Divide 6240 by 12.
6. Divide 2976 by 12
7. Divide 3720 by 15.
8. Divide 2750 by 15.
9. Divide 12544 by 14.
10. Divide 3472 by 14.
11. Divide 4624 by 16.
12. Divide 46830 by 16.

3:9 Solving Problems Involving Division

Example 1	Example 2
A school of 35 classrooms with equal numbers of pupils has a population of 1680. How many pupils are in each class?	In a biscuit factory, a man packs 24 biscuits in each packet. If he has 5280 biscuits to pack, how many packets will he pack?
M_{35}=35,70,105,140,175,210, 245, 280, 315,350	M_{24} = 24,48,72,96,120,144,168,192, 216,240
$\begin{array}{lr} & 48 \\ & 35\overline{)1680} \\ 35\times4 = & -140\downarrow \\ & 280 \\ 35\times8 = & -280 \end{array}$	$\begin{array}{lr} & 220 \\ & 24\overline{)5280} \\ 24\times2 = & 48\downarrow \\ & 48 \\ 24\times2 = & -48\downarrow \\ & 0 \\ & 0 \end{array}$
Each class has 48pupils	220 packets

Activity 3:9

1. Find the quotient of 1806 and 21. 2. Divide sh.65250 among 25 people.
3. Share sh.246400 among 16 workers.
4. A butcher supplies 20kg of meat to each hotel in Kampala city. If he has 8740kg of meat. How many hotels will he supply?
5. A man had 2400 acres of land. He divided it among his 12children. How many acres did each get?
6. A fruit seller packs 24 oranges in each box. How many boxes are needed to pack 12576 oranges?
7. The District Education Officer Maracha had 13440 books to be shared among 12 schools. How many books did each school get?
8. There are 1932 prisoners in Mbale Central Prison. 21 prisoners occupy each cell. How many cells are in the prison?
9. Kotido District registered 68640 candidates for Primary Leaving Examinations (P.L.E). If each centre is required to seat 55 candidates, how many examination centres are required?
10. A lorry carried 18000 eggs from Uganda to Burundi. Given each tray contains 30 eggs, how many trays of eggs did it carry?

3:10 Order of Operation (BODMAS)

Brackets (B), Order (O), Division (D), Multiplication (M), Addition (A), Subtraction (S)

TOPIC 3: Operation on Whole Numbers

Example 1	Example 2
$8 \div (4 \times 2)$ BODMAS <u>Workout the brackets first</u> $8 \div (4 \times 2)$ $8 \div 8$ $= 1$	$\frac{2}{3} \times 12 + 12 \div 6$ <u>Order of working</u> Divide, multiply and finally add. $\frac{2}{\cancel{3}_1} \times \cancel{12}^4 + 12 \div 6$ $2 \times 4 + 2$ $8 + 2 = 10$

Activity 3:11

Workout the following;

1) $8 \div (4 \times 3)$ 2) $(8 \div 2) \times 3$ 3) $12 - (16 \div 8)$ 4) $4 + (12 \div 2)$

5) $18 \div (3 \times 2)$ 6) $3 + (3 \times 2) - 7$ 7) $(8 \times 6) \div (12 \div 4)$ 8) $(6 \times 12) - (24 \div 8)$

9) $9 + 15 - 10$ 10) $36 \div (3 \times 2) - 4$ 11) $42 \div (7 \times 3) + 1$ 12) $(8 + 9) \times 20$

3:12 <u>Base Five and Base Ten or Denary Base</u>

- ✓ Decimal system means grouping numbers in tens. The decimal system can also be called the base ten system.
- ✓ Other number systems are called non-decimal system for example

GROUPINGS	BASE	BASE NAME	DIGITS USED
Grouping in twos	Base two	Binary	0,1
Grouping in threes	Base three	Ternary	0, 1, 2
Grouping in fours	Base four	Quarternary	0, 1, 2, 3
Grouping fives	Base five	Quinary	0, 1, 2, 3, 4
Grouping sixs	Base six	Senary	0, 1, 2, 3, 4, 5
Grouping sevens	Base seven	Septenary	0, 1, 2, 3, 4, 5, 6
Grouping eights	Base eight	Octal base	0, 1, 2, 3, 4, 5, 6, 7,
Grouping nines	Base nine	Nonary	0, 1, 2, 3, 4, 5, 6, 7, 8,

The table below shows the difference between the decimal system and the base five system (Quinary base).

Base ten	1	2	3	4	5	6	7	8	9
Representation	X	XX	XXX	XXXX	XXXXX	XXXXX X	XXXXX XX	XXXXX XXX	XXXXX XXXX
Base five	1	2	3	4	10	11	12	13	14

10	11	12	13	15	16	17	18	19
XXXXX XXXXX	XXXXX XXXXX X	XXXXX XXXXX XX	XXXXX XXXXX XXX	XXXXX XXXXX XXXXX	XXXXX XXXXX XXXXX X	XXXXX XXXXX XXXXX XX	XXXXX XXXXX XXXXX XXX	XXXXX XXXXX XXXXX XXXX
20	21	22	23	30	31	32	33	34

Counting in base five

0	1	2	1	4
10	11	12	13	14
20	21	22	23	24
30	31	32	33	34
40	41	42	43	44

3.13 Place Values in Base Five

Example 1

Write down the place value of each dig in 24_{five}

$2\ 4_{five}$

4 – Ones

2 – fives

Example 2

Write down the place values in the number $1\ 4\ 3_{five}$

$1\ 4\ 3_{five}$

3 – Ones

4 – Fives

1 – Five fives

No.	Place value	Read as
24_{five}	1 group of fives, 4 ones	Two four base five.
143_{five}	1 group of five fives, 4 groups of fives, 3 ones	One four three base five.
201_{five}	2 group of five fives, 1 ones	Two zero one base five.

Activity 3:13

1. Write the following in words.

a) 13_{five} b) 44_{five} c) 132_{five} d) 102_{five}

e) 120_{five} f) 203_{five} g) 304_{five} h) 1234_{five}

2. Write the place value of each digit in the numbers below.

a) 14_{five} b) 23_{five} c) 142_{five} d) 101_{five}

e) 100_{five} f) 244_{five} g) 441_{five} h) 321_{five}

3:14 Expanding in Base five (A)

Example 1

Expand 123_{five}

(1group of five fives)+2groups of fives) +3ones

$(1\times5\times5) + (2\times5) + (3\times1)$ OR

$(1\times5^2) + (2\times5^1) + (3\times5^0)$

Changing Base Five to Base Ten (B)

Example 1	Example 2
Express 32_{five} to base ten. 32five = (3×fives)+(2×ones) (3×5)+(2×1) 15 + 2 17_{ten}	Change 203_{five} to base ten. (2×five fives)+(0×fives)+(3×ones) (2×5×5)+(0×5)+(3×1) 50+0+3 53_{ten}

Activity 3:14

1. Write in expanded form.

a) 21_{five} b) 30_{five} c) 14_{five} d) 33_{five}
e) 111_{five} f) 201_{five} g) 234_{five} h) 421_{five}

2. Change the following to base ten.

a) 22_{five} b) 14_{five} c) 33_{five} d) 10_{five}
e) 101_{five} f) 104_{five} g) 212_{five} h) 432_{five}
i) 321_{five} j) 231_{five} k) 312_{five} L) 123_{five}
m) 213_{five} n) 414_{five} o) 44_{five} p) 204_{five}

3:15 Changing from Base Ten to Base Five

Remember; The basic digits for base five are 0, 1, 2, 3 and 4.

Example 1

Change 23_{ten} to base five.

Base	No	Rem
5	23	
	4	3

$23_{ten} = 43_{five}$

23÷5=4 rem 3
4 is less than 5 we don't need to divide again

Example 2

Express 43ten to base five.

Base	No	Rem
5	43	
5	8	3
	1	3

43÷5=8 rem,3
8÷5=1rem,3

$43_{ten} = 133_{five}$

Activity 3:15

Change the following to base five.

1) 13_{ten} 2) 20_{ten} 3) 24_{ten} 4) 8_{ten} 5) 42_{ten} 6) 34_{ten}
7) 21_{ten} 8) 44_{ten} 9) 60_{ten} 10) 67_{ten} 11) 78_{ten} 12) 83_{ten}
13) 141_{ten} 14) 100_{ten} 15) 55_{ten} 16) 38_{ten} 17) 40_{ten} 18) 52_{ten}

3:16 Addition in Base Five

- ✓ Arrange the numbers vertically.
- ✓ Add each column correctly.
- ✓ Don't forget that the numeral written should be less than 5 i.e {0,1,2,3,4}

Example 1

Add $13_{five} + 24_{five}$

$$\begin{array}{rr} & 1\ 3_{five} \\ + & 2\ 4_{five} \\ \hline & 4\ 2_{five} \end{array} \qquad 7 \div 5 = 1 \text{ rem } 2$$

Example 2

Add $32_{five} + 44_{five}$

$$\begin{array}{rr} & 3\ 2_{five} \\ + & 4\ 4_{five} \\ \hline & 1\ 3\ 1_{five} \end{array} \qquad \begin{array}{l} 6 \div 5 = 1\text{rem}1 \\ 8 \div 5 = 1\text{rem}3 \end{array}$$

Example 3

Find the sum of 241_{five} and 24_{five}.

$$\begin{array}{rr} & 2\ 4\ 1_{five} \\ + & 2\ 4_{five} \\ \hline & 3\ 2\ 0_{five} \end{array} \qquad \begin{array}{l} 5 \div 5 = 1r0 \\ 7 \div 5 = 1r2 \end{array}$$

Example 4

Work out $333_{five} + 134_{five}$

$$\begin{array}{rr} & 3\ 3\ 3five \\ + & 1\ 3\ 4five \\ \hline & 1\ 0\ 2\ 2five \end{array} \qquad \begin{array}{l} 7 \div 5 = 1r2 \\ 7 \div 5 = 1r2 \\ 5 \div 5 = 1r0 \end{array}$$

Activity 3:16

1. Add $32_{five}+14_{five}$
2. Add $13_{five}+44_{five}$
3. Find the sum of 24_{five} and 33_{five}
4. Work out $34_{five}+42_{five}$
5. Add$11_{five}+44_{five}$
6. $221_{five}+34_{five}$
7. Add$44_{five}+32_{five}$
8. Add $113_{five} +34_{five}$
9. Find the sum of $132_{five}+43_{five}$
10. Add $121_{five}+141_{five}$
11. Add $234_{five}+213_{five}$
12. What is the sum of 332_{five} and 124_{five}

3.17 Subtraction in Base Five

Example 1

Subtract $32_{five} - 14_{five}$

$$\begin{array}{rr} & 3\ \ 2_{five} \\ - & 1\ \ 4_{five} \\ \hline & 1\ \ 3 \end{array} \qquad \begin{array}{l} 7 - 4 = 3 \\ 2 - 1 = 1 \end{array}$$

Example 2

Subtract $431_{five} - 34_{five}$

$$\begin{array}{rr} & 4\ 3\ 1_{five} \\ - & 3\ 4_{five} \\ \hline & 3\ 4\ 2_{five} \end{array} \qquad \begin{array}{l} 6 - 4 = 2 \\ 7 - 3 = 4 \end{array}$$

Example 3

$$\begin{array}{rr} & \overset{2}{\cancel{3}}\ \overset{4}{\cancel{0}}\ \overset{5}{2}_{five} \\ - & 4\ 4_{five} \\ \hline & 2\ 0\ 3_{five} \end{array} \qquad \begin{array}{l} 7 - 4 = 3 \\ 4 - 4 = 0 \\ 2 - 0 = 2 \end{array}$$

Example 4

$$\begin{array}{rr} & \overset{1}{\cancel{2}}\ \overset{5}{\cancel{1}}\ \overset{5}{0}_{five} \\ - & 4\ 3_{five} \\ \hline & 1\ 2\ 2_{five} \end{array} \qquad \begin{array}{l} 5 - 3 = 2 \\ 5 - 4 = 1 \end{array}$$

Activity 3:17

Subtract the following numbers;

1. $44_{five} - 13_{five}$
2. $40_{five} - 23_{five}$
3. $30_{five} - 12_{five}$
4. $43_{five} - 34_{five}$
5. $103_{five} - 24_{five}$
6. $122_{five} - 104_{five}$
7. $134_{five} - 44_{five}$
8. $321_{five} - 243_{five}$
9. $332_{five} - 233_{five}$
10. $114_{five} - 31_{five}$
11. $111_{five} - 34_{five}$
12. $330_{five} - 243_{five}$
13. $100_{five} - 23_{five}$
14. $103_{five} - 43_{five}$
15. $222_{five} - 33_{five}$
16. $204_{five} - 123_{five}$

TOPIC. 4 : NUMBER PATTERNS AND SEQUENCES

In this topic, we shall be looking at different types of numbers. i.e

-Counting numbers	-Composite numbers	-Multiple numbers
- Prime numbers	-Odd numbers	-Triangular numbers
-Square numbers	- Even numbers	-Rectangular numbers

1) **Whole numbers** are numbers that begin with zero with no fractions. These are {0, 1, 2, 3, 4, 5, 6, 7..........}

2) **Counting numbers**: These are numbers we use when counting, sometimes they are called natural numbers for example; {1,2,3,4,5,6,7,8.......}

3) **Even numbers** are numbers exactly divisible by 2 and give no remainder i.e 0,2,4,6, 8.......... The first even number is zero.

4) **Odd numbers** are numbers that are not exactly divisible by 2. When divided by 2, it gives 1 as a remainder e.g 0, 1, 3, 5, 7, 9, 11.............

5) **Prime numbers** are numbers with only two factors, 1 and itself. Examples are {2, 3, 5, 7, 11, 13, 17, 19............}

6) **Composite numbers** are numbers which have more than two factors. Examples of composite numbers are 4, 6, 8, 9, 10, 12, 14, 15, 16, 18..................

7) **Square numbers** are numbers got by multiplying two equal numbers 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 125..............

8) **Triangular numbers** are numbers formed by adding counting numbers.

1 = 1
1 + 2 = 3
1 + 2 + 3 = 6
1 + 2 + 3 + 4 = 10
1 + 2 + 3 + 4 + 5 = 15
1 + 2 + 3 + 4 + 5 + 6 = 21
1 + 2 + 3 + 4 + 5 + 6 + 7 = 28

The first seven triangular numbers are 1, 3, 6, 10, 15, 21 and 28

4.1 Finding the Next Number in the Sequences By Increasing

Example 1

What is the next number in the sequence?

12, 14, 16, 18, 20, 22, 24, 26, 28

12 + 2 = 14
14 + 2 = 16
16 + 2 = 18
18 + 2 = 20
20 + 2 = 20
22 + 2 = 24
24 + 2 = 26
26 + 2 = 28

Example 2

Find the missing numbers 1, 2, 4, 7, 11, 16, 21, 28

1 + 1 = 2
2 + 2 = 4
4 + 3 = 7
7 + 4 = 11
11 + 5 = 16
16 + 6 = 21
21 + 7 = 28

Example 3

Find the next number in the sequence.

1, 2, 3, 4, 5, 6, <u>9</u>, <u>10</u>

$1 + 1 = 2$ $\quad$ $6 + 1 = 7$

$1 + 2 = 3$ $\quad$ $7 + 1 = 8$

$1 + 4 = 5$ $\quad$ $8 + 1 = 9$

$5 + 1 = 6$ $\quad$ $9 + 1 = 10$

Example 4

Find the missing numbers 2, 3, 5, 7, <u>11</u>, 13, <u>17</u>...............

These are prime numbers, with only two factors 1 and itself.

Activity 4:1

Find the missing two numbers in the sequences below;

1) 1, 3, 5, 7, 9__ __
2) 0, 2, 4, 6, 8, 10, __ __
3) 2, 4, 7,11, 15, 21,__ __
4) 3, 7, 11, 15, 19, 23, __ __
5) 5, 7, 10, 14, 19, 25, __ __
6) 1, 5, 9, 13, __ __
7) 2, 2, 3, 5, 8, 12, __ __
8) 5, 10, 15, 20, 25, __ __
9) 1, 3, 6, 10, 15, 21, 28, 36, __ __
10) 7, 14, 21, 28, 35, 42, __ __
11) 115, 215, 315, 415, __ __
12) 1, 4, 9, 16, 25, 36, 49, 64, __ __
13) Find the missing numbers

Pair of shoes	2	__	5	__	8	10	12	__	15
No. of shoes	4	6	__	8	__	__	__	18	__

14) Fill the missing numbers in the table below.

Stools	1	3	5	__	9	__	__	12	__
Number of legs	3	9	__	21	__	36	18	__	15

4.2 Finding the Next Numbers in the Reducing Order

Example 1

Find the next three numbers in the sequence below;

100, 81, 64, 49, <u>36</u>, <u>25</u>, <u>16</u>

These are sequence numbers

$10 \times 10 = 100$ $\quad$ $7 \times 7 = 49$

$9 \times 9 = 81$ $\quad$ $6 \times 6 = 36$

$8 \times 8 = 64$ $\quad$ $5 \times 5 = 25$

$4 \times 4 = 16$

Example 2

What is the next 3 numbers in the sequence below;

27, 24, 21,18, 15,<u>12</u>, <u>9</u>, <u>6</u>

The common difference is 3, therefore keep on subtracting 3.

$27-3 = 24$ $\quad$ $18-3 = 15$ $\quad$ $9-3 = 6$

$24-3 = 21$ $\quad$ $15-3 = 12$

$21-3 = 18$ $\quad$ $12-3 = 9$

Activity 4:2

Find the missing numbers in the sequences below;

1) 18, 16, 14, 12, __,__,__
2) 30, 26, 22, 18, __,__
3) 23, 21,19,17,__,__,__
4) 40, 39, 37, 34, 30,__,__,__

5) 72, 64, 56,48,___,___,___

6) 80,40,20,10,___,___,___

7) 23, 22, 20,17,___,___,___

8) 31, 29,23,___,___,___

9) 95,90,85,80,___,___,___

10) 64, 49,___ ,25,___,9, 4

4.3 Multiples of Numbers

A multiple is the product of multiplying a number by a whole number. For example multiples of 4.

$4\times1=4$ $\quad 4\times2=8$ $\quad 4\times3=12$ $\quad 4\times4=16$ $\quad 4\times5=20$ $\quad 4\times6=24$

Therefore multiples of 4 are 4, 8, 12, 16, 20, 24..............

Example 1

List the multiple of 6 less than 40.

$6\times1=6$ $\quad 6\times2=12$ $\quad 6\times3=18$

$6\times4=24$ $\quad 6\times5=30$ $\quad 6\times6=36$

$6\times7=42$ $\quad 6\times8=48$

$M_6=\{6, 12, 18, 24, 30, 36\}$

Example 2

Write down the multiples of 8 between 10 and 50.

$8\times1=8,$ $\quad 8\times2=16,$ $\quad 8\times3=24$

$8\times4=32,$ $\quad 8\times5=40,$ $\quad 8\times6=48$

$8\times7=56,$ $\quad 8\times8=64,$ $\quad 8\times9=72$

$M_8=\{16, 24, 32, 40, 48\}$

Activity 4:3

1. List the multiple of 4 less than 30.
2. Write down the multiples of 5 less than 40.
3. Write down multiples of 2 less than 15.
4. Find the multiples of 7 less than 40.
5. Find the multiples of 10 between 9 and 80.
6. List the multiples of 9 between 20 and 60.
7. Write down the multiples of 2 between 10 and 40.
8. List the multiples of 7 between 40 and 60.
9. List all the multiples of 12 between 40 and 100.

4.4 Lowest Common Multiples (L.C.M)

Example 2

Find the L.C.M of 16 and 24.

$M_{16} = \{16,32,48,64, 80,............\}$

$M_{24} = \{24,48,72,96, 120...........\}$

The L.C.M = 48

Method 2

2	16	24
2	8	12
2	4	6
2	2	3
3	1	3
	1	1

$L.C.M=2\times2\times2\times2\times3$

$L.C.M=4\times4\times3$

$L.C.M=48$

Method 3

Prime factorise 16 and 24 separately

2	16
2	8
2	4
2	2
	1

2	24
2	12
2	6
3	3
	1

$2_1,2_2,2_3,2_4$ $\quad 2_1,2_2,2_3,3_1$

$F_{16}UF_{24} = \{2_1,2_2,2_3,2_4,3_1\}$

$L.C.M = 2\times2\times2\times2\times3$

$L.C.M = 48$

Example 2
Find the L.C.M of 4 and 6.
(use multiple)
M_4 = 4, 8, 12, 16, 20, 24............
M_6 = 6, 12, 18, 24, 30, 36
Common multiples are 12 and 24
L.C.M = 12

Method 2
(Prime factorise)

2	4	6
2	2	3
3	1	3
	1	1

L.C.M = 2×2×3
L.C.M = 12.

Method 3
Prime factorise each number.

2	4
2	2
	1

2	6
3	3
	1

$F_4=\{2_1, 2_2\}$ $F_6=\{2_1, 3_1\}$
$F_4 \cup F_6=\{2_1, 2_2, 3_1\}$
L.C.M = 2×2×3
L.C.M = 12

Activity 4:4

Find the lowest common multiple (L.C.M) of the following numbers.

1) 6 and 8 2) 12 and 9 3) 12 and 18 4) 12 and 20 5) 18 and 15
6) 18 and 24 7) 9 and 10 8) 20 and 16 9) 6 and 9 10) 30 and 20
11) 10 and 15 12) 8 and 7 13) 16 and 15 14) 12 and 15 15) 15 and 9

4.5 Finding Factors

A factor is a number which divides another number exactly number of times.

Example 1 *(Method 1)*
Write down all the factors of 24
1 × 24 = 24
2 × 12 = 24
3 × 8 = 24
4 × 6 = 24
24 = {1, 2, 3, 4, 6, 8, 12, 24}
(Method 2)
24 ÷ 1 = 24
24 ÷ 2 = 12
24 ÷ 3 = 8
24 ÷ 6 = 4
24 ÷ 8 = 3
24 ÷ 12 = 2
24 ÷ 24 = 1
24 = {1, 2, 3, 4, 6, 8, 12, 24}

Example 2
Find the factors of 42.
(Method 1)
1 × 42 = 42
2 × 21 = 42
3 × 14 = 42
6 × 7 = 42
42 = {1, 2, 3, 6, 7, 14, 21, 42}

(Method 2)
43 ÷ 1 = 42 42÷7=6
42 ÷ 2 = 21 42÷14=3
42 ÷ 3 = 14 42÷21=2
42 ÷ 6 = 7 42÷42=1
42 = {1, 2, 3, 6, 7, 14, 21, 42}

Activity 4:5

Write down the factors of each of the following numbers;

a) 12 b) 16 c) 8 d) 20 e) 24 f) 42 g) 32 h) 30
i) 25 j) 40 k) 15 L) 21 m) 50 n) 80 o) 100 p) 80

4.6 Finding Common Factors and Greatest Common Factors (G.C.F)

Abbreviations

Lowest Common Factor (L.C.F)
Highest Common Factor (H.C.F)
Greatest Common Factor (G.C.F)

Example 1

Find the common factors of 12and 18

F_{12}	F_{18}
1 × 12	1 × 18
2 × 12	2 × 9
3 × 4	3 × 6

$F_{12} = \{1, 2, 3, 4, 6, 12\}$
$F_{18} = \{1, 2, 3, 6, 9, 18\}$
Common factors of 12 and 18
$F_{12} \cap F_{18} = \{1, 2, 3, 6\}$

Example 2

List the common factors of 24 and 30.

F_{24}	F_{30}
1 × 24	1 × 30
2 × 12	2 × 15
3 × 8	3 × 10
4 × 6	5 × 6

$F_{24} = \{1, 2, 3, 4, 6, 8, 12, 24\}$
$F_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$
Common factors of 24 and 30 {1, 2, 3, 6}

Example 3

Find the GCF of 18 and 24.

F_{18}	F_{24}
1 × 18	1 × 24
2 × 9	2 × 12
3 × 6	4 × 6

$F_{18} = \{1, 2, 3, 6, 9, 18\}$
$F_{24} = \{1, 2, 3, 6, 8, 12\}$
Common factors = {1, 2, 3, 6}
GCF = 6

Example 4

What is the GCF of 12 and 20?

F_{12}	F_{20}
1 × 12	1 × 20
2 × 6	2 × 10
3 × 4	4 × 5

$F_{12} = \{1, 2, 3, 4, 6, 12\}$
$F_{20} = \{1, 2, 4, 5, 10, 20\}$
Common factors = {1,2,4}
GCF = 4

Activity 4:6

Find the GCF/greatest common factor, HCF (Highest Common Factor)

1) 18 and 9
2) 12 and 24
3) 30 and 20
4) 24 and 28
5) 21 and 28
6) 40 and 30
7) 14 and 21
8) 28 and 36
9) 20 and 40
10) 36 and 30
11) 6 and 12
12) 6 and 9
13) 10 and 20
14) 24 and 32
15) 28 and 32
16) 20 and 15

4.7 Prime Numbers and Composite Numbers

A prime number is a number which has only two factors 1 and itself.
A composite number is one which has more than two factors.

Number	Factors	Number of factors	Type of number
2	1, 2	2 factors	Prime numbers
3	1, 3	2 factors	Prime numbers
4	12, 4	3 factors	Composite numbers
5	1, 5	2 factors	Prime numbers
6	1, 2, 3, 6	4 factors	Composite numbers
7	1, 7	2 factors	Prime numbers
8	1, 2, 4, 8	4 factors	Composite numbers
9	1, 3, 9	3 factors	Composite numbers
10	1, 2, 5, 10	4 factors	Composite numbers

- ✓ Therefore 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31................are prime numbers.
- ✓ Numbers like 4, 6, 9, 10, 12, 14, 15, 16, 18, 20, 21, 22.........are composite numbers because they have more than 2 factors.

4.8 Prime Factorisation of Numbers

When prime factorizing any given number, prime numbers are used to divide that is to say 2, 3, 5, 7, 11, 13, 17, 19, 23..................
The prime number used should be exactly without leaving any remainder.
There are two main methods used to prime factorise numbers namely;

i) Prime factorizing using the ladder
ii) Prime factorizing using a factor tree

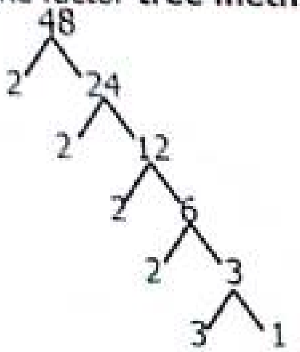
Prime factors are represented in three ways
Subscription form/Set notation, Multiplication form & Power or exponent form

Example 1

Prime factorise 48 using the **ladder method.**

2	48
2	24
2	12
2	6
3	3
	1

Using the factor **tree method**



(i) Set notation; $12 = \{2_1, 2_2, 2_3, 2_1, 3_1\}$ (ii) Multiplication form; $48=2\times2\times2\times2\times3$

(iii) Power/exponent form; $48=2\times2\times2\times2\times3$

$= 2^4\times3^1$

Activity 4:8 A

Use the ladder method to prime factorise the following;

1) 12 (in subscript form) 2) 24 (in subscript form) 3) 30 (in subscript form)
4) 20 (in multiplication) 5) 32 (in multiplication) 6) 45 (in multiplication)
7) 28(in power form) 8) 16 (in power form) 9) 72 (in power form)
10) 80(in power form) 11)100 (power form) 12) 63 (set notation)
13) 84(set notation) 14) 200(set notation) 15)96(subscription form)

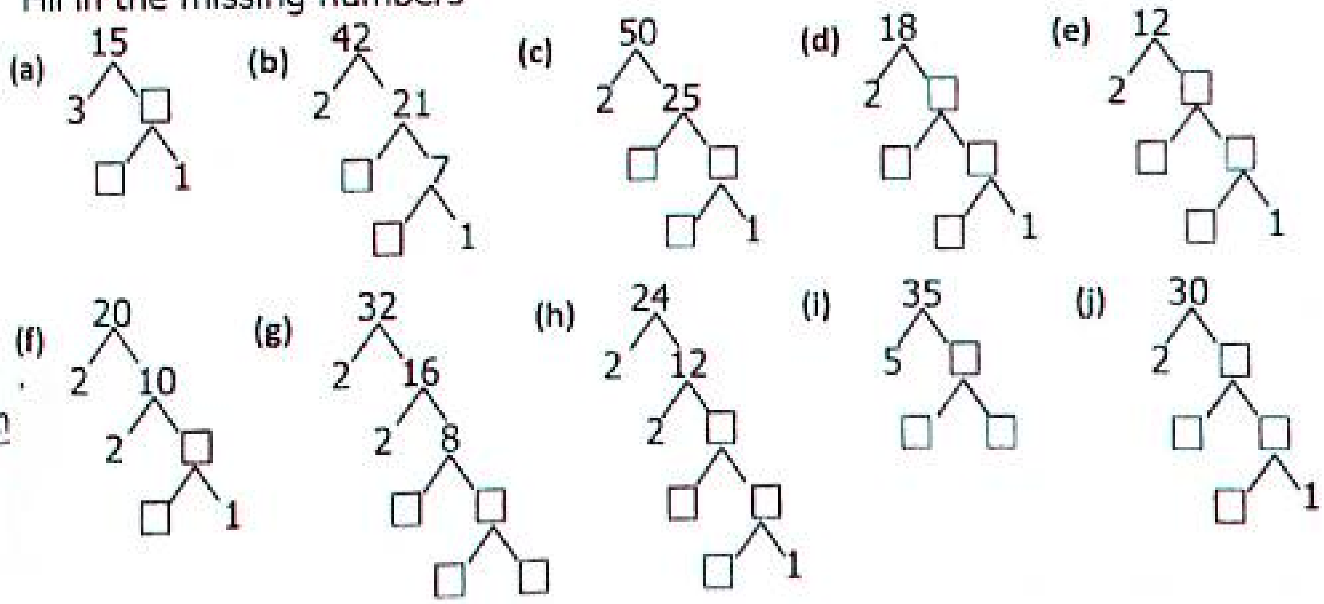
Activity 4:8 B

Use a factor tree to prime factorise the following and give your answer as instructed;

1) 12 (multiplication) 2) 24 (in power form) 3) 28 (multiplication form)
4) 40 (in exponent form) 5) 48 (set notation) 6) 63 (multiplication)
7)72(subscription form) 8) 144 (power form) 9)128(multiplication form)

Activity 4:8C

Fill in the missing numbers



4.9 Square Numbers

- ✓ A square number is a product number obtained when a number is multiplied by itself.
- ✓ Square numbers can also be formed by adding **odd** numbers or they can be formed by multiplying **consecutive counting** numbers by themselves.

Examples of square numbers

Number	Method	Square numbers
1	1×1	1
2	2×2	4
3	3×3	9
4	4×4	16
5	5×5	25
6	6×6	36
7	7×7	49
8	8×8	64
9	9×9	81
10	10×10	100
11	11×11	121
12	12×12	144
13	13×13	169
14	14×14	196
15	15×15	225

Example 1

Find the next number in the square 1, 4, 9, 16, 25, 36, 49

Method 1

$1 + 3 = 4$ $\quad 16 + 9 = 25$

$4 + 5 = 9$ $\quad 25 + 11 = 36$

$9 + 7 = 16$ $\quad 36 + 13 = 49$

Method 2

$1 \times 1 = 1$ $\quad 4 \times 4 = 16$ $\quad 7 \times 7 = 49$

$2 \times 2 = 4$ $\quad 5 \times 5 = 25$

$3 \times 3 = 9$ $\quad 6 \times 6 = 36$

Activity 4:9

Find the missing number.

1) 100, 81, 64, 49,___ 2) 1, 4, 9, 16,___, ___ 3) 16, 25, 36, 49, 64, ___

4) $\frac{1}{2}, \frac{1}{4}, \frac{1}{16}, \frac{1}{25}$, ___, ___ 5) $\frac{2}{3}, \frac{4}{6}, \frac{9}{16}, \frac{16}{25}, \frac{25}{36}$, ___, ___ 6) 225, 196,169,___,___

7) 100,121,144,___,___ 8) 49,64,81,___,___ 9) 49,36, 25, 16,___,___

4.10 Finding Square root of Whole Numbers

- ✓ A square root is a number that is multiplied by itself to get a square number.
- ✓ To achieve a square root of a number, you prime factorise the given square number.
- ✓ The symbol for square root is √
- ✓ Divide using prime numbers to find square roots.

Example 1

Find the square root of 36.

2	36
2	18
3	9
3	3
	1

$\sqrt{36} = \sqrt{(2\times2)\times(3\times3)}$

$\sqrt{36} = 2\times3$

$\sqrt{36} = 6$

Example 2

What is the square root of 100?

2	100
2	50
5	25
5	5
	1

$\sqrt{100} = \sqrt{2\times2\times5\times5)}$

$\sqrt{100} = 2\times5$

$\sqrt{100} = 10$

Activity 4:10

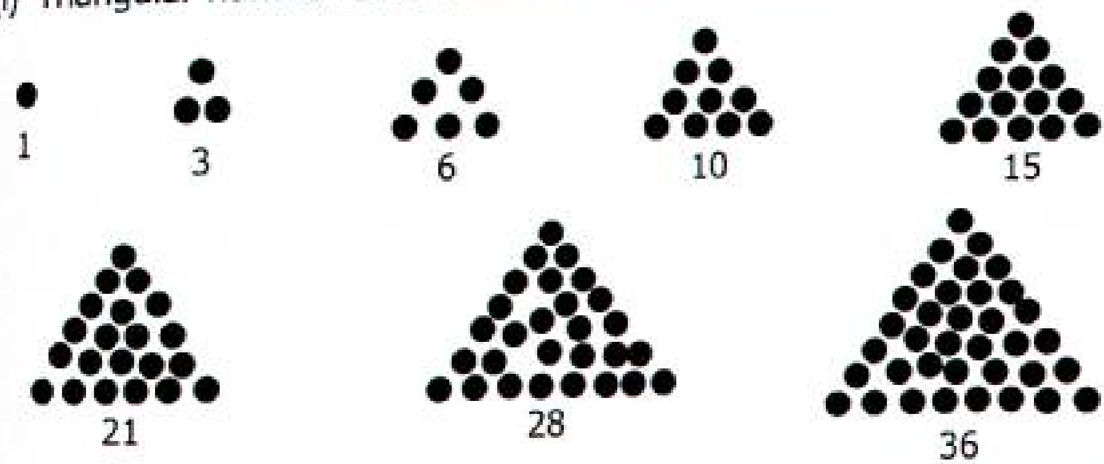
Find the square root of the following numbers;

a) 4 b) 16 c) 25 d) 81 e) 144 f) 64

g) 169 h) 196 i) 121 j) 49 k) 9 L) 36

4.11 Triangular Numbers

(i) Triangular numbers are numbers that can make a triangular pattern.



(ii) Triangular numbers can also be obtained by adding consecutive numbers for example

1	=	1
1 + 2	=	3
1 + 2 + 3	=	6
1 + 2 + 3 + 4	=	10
1 + 2 + 3 + 4 + 5	=	15
1 + 2 + 3 + 4 + 5 + 6	=	21
1 + 2 + 3 + 4 + 5 + 5 + 7	=	28
1 + 2 + 3 + 4 + 5 + 6 + 7 + 8	=	36 e.t.c

(iii) We can still find triangular numbers with the help of a formula.

$Triangular\ number = \frac{n(n+1)}{2}$

Example 1 Calculate the 6th triangular number. <u>Method 1</u> $1 + 2 + 3 + 4 + 5 + 6$ $(1 + 2 + 3) + (4 + 5 + 6)$ $6 + 15$ 21	<u>Method 2</u> $\Delta\ number = \frac{n(n+1)}{2}$ $= \frac{6(6+1)}{2}$ $\frac{6 \times 6 + 6 \times 1}{2} = \frac{36+6}{2} = \frac{42}{2}$ $3 \times 7 = 21$
Example 2 What is the 8th triangular number? *<u>Method 1</u>* 1+2+3+4+5+6+7+8 6+9+21 15+21 36 The 8th triangular number is 36	<u>Method 2</u> Triangular no. $= \frac{n(n+1)}{2}$ $= \frac{8(8+1)}{2}$ $= \frac{8 \times 9}{2}$ $= \frac{72}{2} = 36$ The 8th triangular number is 36

Activity 4:11

1. Find the 9th triangular number.
2. What is the 7th triangular number?
3. Calculate the 12th triangular number.
4. Work out the 10th triangular number.
5. Work out the 14th triangular number.
6. Calculate 15th triangular number.
7. If the nth triangular number is 15. Find the value of n.
8. Given that, the nth triangular number is 90. Find the value of n.
9. What is the 20th triangular number?
10. If the nth triangular number is 171. Find the value of n.

TOPIC . 5: FRACTIONS

- ✓ A fraction is a part of a whole.
- ✓ Fractions are grouped into three categories namely; Vulgar fraction or common fractions and decimal fractions.
- ✓ There are 5 types of fractions.

i) Proper fractions ii) Improper fractions iii) Mixed fractions
iv) Equivalent fractions v) Decimal fractions

Proper fractions are fractions whose numerator is less than the denominator.

Examples of proper fractions are $\frac{1}{2}, \frac{3}{4}, \frac{2}{3}, \frac{9}{10}, \frac{12}{15}, \frac{19}{20}, \frac{23}{50}, \frac{79}{100}, \frac{120}{137}$ $e.t.c$

An **improper fraction** is a fraction where the numerator is greater than the denominator.

Examples include; $\frac{6}{5}, \frac{7}{2}, \frac{12}{7}, \frac{24}{21}, \frac{40}{37}, \frac{50}{24}, \frac{100}{92}, \frac{260}{132}, \frac{300}{180}$ $e.t.c$

A **mixed number** is a number having both whole number and a proper fraction.

Examples include; $1\frac{3}{2}, 2\frac{4}{13}, 1\frac{7}{12}, 1\frac{12}{11}, 1\frac{23}{28}, 4\frac{7}{12}, 8\frac{7}{8}, 10\frac{11}{12}, 12\frac{8}{15}$ $e.t.c$

5.1 Expressing Mixed Numbers to Improper Fractions

Example 1

Express $5\frac{2}{3}$ as an improper fraction

$$5\frac{2}{3} = \frac{D \times W + N}{D}$$

$$5\frac{2}{3} = \frac{3 \times 5 + 2}{3}$$

$$= \frac{15 + 2}{3}$$

$$5\frac{2}{3} = \frac{17}{3}$$

Example 2

Express $7\frac{7}{12}$ as an improper fraction.

$$7\frac{7}{12} = \frac{D \times W + N}{D}$$

$$7\frac{1}{12} = \frac{(12 \times 7) + 7}{12}$$

$$= \frac{84 + 7}{12}$$

$$7\frac{7}{12} = \frac{91}{12}$$

Activity 5:1

Express each of the following fractions as improper fractions;

1) $3\frac{1}{2}$ 2) $4\frac{2}{3}$ 3) $5\frac{4}{5}$ 4) $6\frac{7}{12}$ 5) $1\frac{1}{2}$ 6) $2\frac{5}{10}$
7) $8\frac{2}{7}$ 8) $9\frac{3}{8}$ 9) $\frac{3}{8}$ 10) $5\frac{7}{10}$ 11) $13\frac{1}{5}$ 12) $3\frac{5}{12}$
13) $9\frac{1}{8}$ 14) $5\frac{7}{11}$ 15) $9\frac{2}{3}$ 16) $2\frac{5}{15}$ 17) $2\frac{1}{3}$ 18) $12\frac{3}{4}$

5:2 Expressing Improper Fractions to Mixed Numbers

Example 1

Change $\frac{13}{5}$ to a mixed fraction.

$$\begin{array}{r|l} & 2\text{rem } 3 \\ 5 & 13 \\ & -10 \\ & 3 \end{array}$$

$\frac{13}{5} = 2\frac{3}{5}$

Example 2

Express $\frac{75}{9}$ as a fraction.

$$\begin{array}{r|l} & 11\text{rem } 7 \\ 8 & 95 \\ & -8 \\ & 15 \\ & -8 \\ & 7 \end{array}$$

$\frac{75}{9} = 11\frac{7}{15}$

Activity 5:2

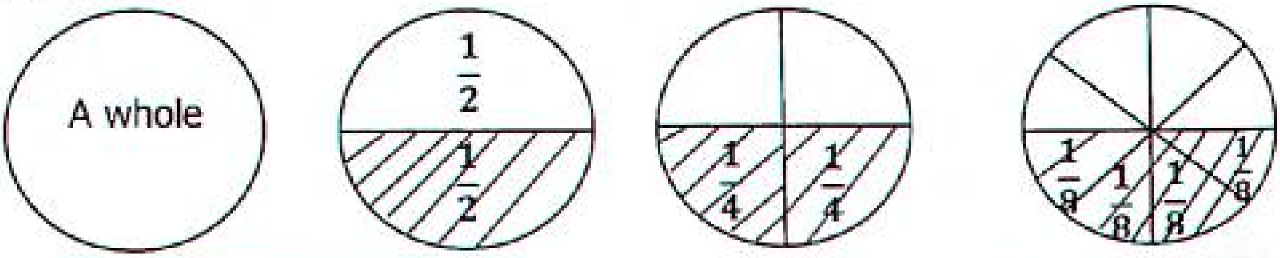
Change the following fractions as mixed fractions

a) $\frac{7}{2}$ b) $\frac{9}{4}$ c) $\frac{17}{4}$ d) $\frac{23}{7}$ e) $\frac{48}{5}$ f) $\frac{37}{6}$ g) $\frac{29}{30}$ h) $\frac{42}{8}$

i) $\frac{48}{9}$ j) $\frac{15}{6}$ k) $\frac{13}{9}$ L) $\frac{12}{7}$ m) $\frac{38}{12}$ n) $\frac{25}{12}$ o) $\frac{77}{12}$ p) $\frac{67}{8}$

5.3 Finding Equivalent Fractions

Example 1



ONE							
$\frac{1}{2}$				$\frac{1}{2}$			
$\frac{1}{4}$		$\frac{1}{4}$		$\frac{1}{4}$		$\frac{1}{4}$	
$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$

$\frac{1}{2} = \left(\frac{1}{4} + \frac{1}{4}\right)$

$\frac{1}{2} = \frac{2}{4}$

$\frac{1}{2} = \frac{2}{4} = \frac{4}{8}$

$\frac{1}{2} = \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right)$

$\frac{1}{2} = \frac{4}{8}$

Example 1

Write four equivalent fractions to $\frac{2}{3}$.

$\left(\frac{2}{3} \times \frac{2}{2}\right), \left(\frac{2}{3} \times \frac{3}{3}\right), \left(\frac{2}{3} \times \frac{4}{4}\right), \left(\frac{2}{3} \times \frac{5}{5}\right)$

$\frac{2}{3} = \frac{4}{6}, \frac{6}{9}, \frac{8}{12}, \frac{10}{15}$

Example 3

Write four equivalent fractions to $\frac{7}{10}$.

$\left(\frac{7}{10} \times \frac{2}{2}\right), \left(\frac{7}{10} \times \frac{3}{3}\right), \left(\frac{7}{10} \times \frac{4}{4}\right), \left(\frac{7}{10} \times \frac{5}{5}\right)$

$\frac{7}{10} = \frac{14}{20}, \frac{21}{30}, \frac{28}{40}, \frac{35}{50}$

Activity 5:3

Write the next 4 equivalent fractions for each of the following;

1) $\frac{3}{5}$ 1) $\frac{2}{5}$ 2) $\frac{4}{5}$ 4) $\frac{3}{10}$ 5) $\frac{3}{11}$ 6) $\frac{5}{12}$

7) $\frac{7}{8}$ 8) $\frac{3}{5}$ 9) $\frac{1}{7}$ 10) $\frac{6}{12}$ 11) $\frac{7}{8}$ 12) $\frac{8}{9}$

13) $\frac{1}{4}$ 14 $\frac{4}{7}$ 15) $\frac{11}{12}$ 16) $\frac{12}{15}$ 17) $\frac{7}{20}$ 18) $\frac{7}{30}$

19. Find the next equivalent fraction to $\frac{3}{8}$.

20. What are the three equivalent fractions to $\frac{4}{7}$.

21. Workout the fourth equivalent fraction to $\frac{2}{5}$

22. Find the missing fractions in the given sequences.

a) $\frac{3}{4}, \frac{6}{8}, \frac{9}{12}, \frac{12}{16}$ ____, ____ b) $\frac{5}{6}, \frac{10}{12}, \frac{15}{18}, \frac{20}{24}$ ____, ____ c) $\frac{4}{5}, \frac{8}{10}, \frac{12}{15}, \frac{16}{20}$, ____, ____

5.4 Finding Missing Number in Equivalent Fractions

Example 1 (Method 1)

Find the missing number.

$$\frac{2}{3} = \frac{\square}{12}$$

$$\left(\frac{2}{3} \times \frac{2}{2}\right), \left(\frac{2}{3} \times \frac{3}{3}\right), \left(\frac{2}{3} \times \frac{4}{4}\right)$$

$$\frac{2}{3} = \frac{8}{12}$$

$\square = 8$

(Method 2)

$$\frac{2}{3} = \frac{\square}{12}$$

$$3 \times \square = 2 \times 12$$

$$\frac{\cancel{3}^1 \square}{\cancel{3}_1} = \frac{24}{3}$$

$\square = 8$

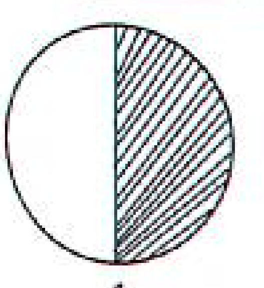
Activity 5:4

Find the missing numbers.

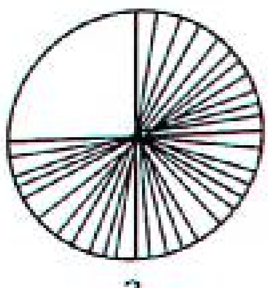
1) $\frac{2}{7} = \frac{\square}{28}$ 2) $\frac{1}{5} = \frac{\square}{20}$ 3) $\frac{3}{4} = \frac{9}{\square}$ 4) $\frac{3}{5} = \frac{\square}{30}$ 5) $\frac{7}{10} = \frac{\square}{30}$ 6) $\frac{3}{8} = \frac{\square}{32}$

7) $\frac{4}{5} = \frac{8}{\square}$ 8) $\frac{5}{8} = \frac{\square}{24}$ 9) $\frac{3}{4} = \frac{\square}{20}$ 10) $\frac{3}{8} = \frac{18}{\square}$ 11) $\frac{1}{2} = \frac{\square}{8}$ 12) $\frac{7}{8} = \frac{\square}{40}$

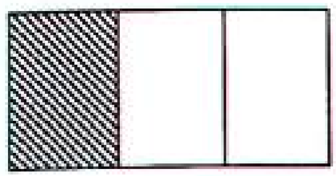
5.5 Equivalent Fractions by Shading



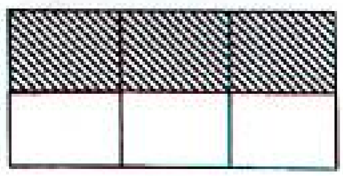
$\frac{1}{2}$



$\frac{3}{4}$



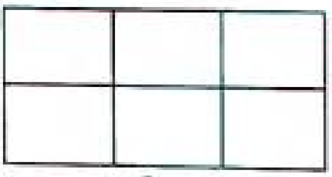
$\frac{1}{3}$



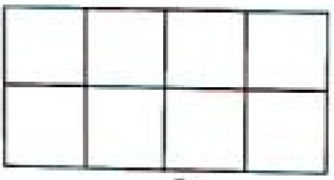
$\frac{3}{6}$

Activity 5:5

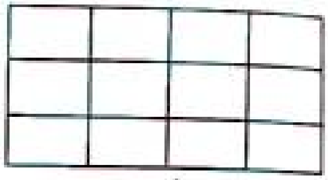
Shade the fractions

(a) 

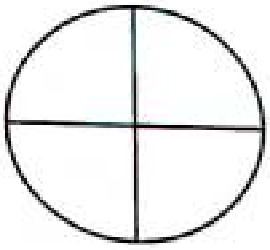
$\frac{2}{3}$

(b) 

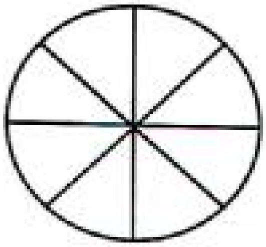
$\frac{3}{4}$

(c) 

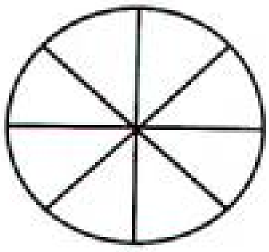
$\frac{1}{3}$

(d) 

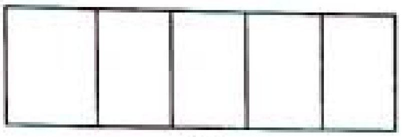
$\frac{1}{2}$

(e) 

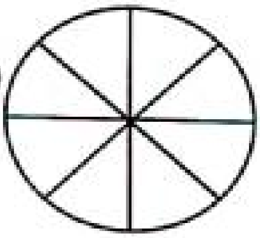
$\frac{1}{4}$

(f) 

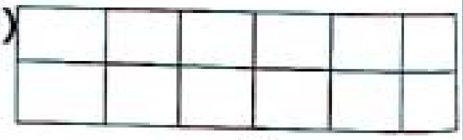
$\frac{3}{4}$

(g) 

$\frac{2}{5}$

(h) 

$\frac{5}{8}$

(i) 

$\frac{2}{3}$

5:6 <u>Reducing Fractions to the Lowest Terms</u>

Two methods are used to reduce fractions;

(i) Using factors

(ii) Prime factorization

Example 1

Reduce $\frac{12}{18}$ to the lowest term.

<u>Method 1</u>

Use of factors

F_{12} 1×12, 2×6, 3×4

F_{18} 1×18, 2×9, 3×6

$F_{18} = \{1,2,3,6,9,18\}$

$F_{12} = \{1,2,3,4,6,12\}$

$$\frac{12}{18} \div \frac{6}{6} = \frac{2}{3}$$

<u>Method 2</u>

Use prime factorization method.

1	12
2	6
3	3
	1

1	18
3	9
3	3
	1

$$\frac{12}{18} = \frac{\cancel{2} \times 2 \times \cancel{3}}{\cancel{2} \times \cancel{3} \times 3}$$

$$\frac{12}{18} = \frac{2}{3}$$

Example 2

Reduce $\frac{30}{72}$ to the lowest term using G.C.F.

Method 1

F_{30} 1 × 30
2 × 15
3 × 10
5 × 6

$F_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$

1 × 72
2 × 36
4 × 18
6 × 12
8 × 9

$F_{12} = \{1, 2, 3, 4, 6, 8, 9, 18, 24, 36, 72\}$

$\frac{30}{72} \div \frac{6}{6} = \frac{5}{12}$

Prime factorise 30 and 72.

Method 2

2	30
3	15
5	5
	1

2	72
3	36
5	18
3	9
3	3
	1

$$\frac{30}{72} = \frac{\cancel{2} \times \cancel{3} \times 5}{\cancel{2} \times 2 \times 2 \times \cancel{3} \times 3}$$

$$\frac{30}{72} = \frac{5}{12}$$

Activity 5:6

Reduce the fractions below to their lowest term.

1) $\frac{18}{24}$ 2) $\frac{24}{30}$ 3) $\frac{10}{12}$ 4) $\frac{16}{20}$ 5) $\frac{8}{12}$

6) $\frac{24}{36}$ 7) $\frac{20}{40}$ 8) $\frac{6}{24}$ 9) $\frac{21}{42}$ 10) $\frac{6}{8}$

11) $\frac{32}{48}$ 12) $\frac{12}{48}$ 13) $\frac{35}{40}$ 14) $\frac{16}{64}$ 15) $\frac{25}{100}$

5.7 Ordering Fractions

- ✓ Ordering fractions is done in two ways ascending and descending order.
- ✓ Ascending order means arranging from the smallest to the highest.
- ✓ Descending order means arranging from biggest to smallest.

Using Lowest Common Multiples (L.C.M) is the easiest method to determine the size of the fraction in natural numbers.

Example 1

Arrange $\frac{2}{3}, \frac{7}{12}, \frac{3}{4}$ *and* $\frac{1}{2}$ in ascending order.

The L.C.M of 3, 12, 4 and 2 is 12.

$\frac{2}{3_1} \times \cancel{12}^4$ $\frac{7}{\cancel{12}_1} \times \cancel{12}^1$ $\frac{3}{4_1} \times \cancel{12}^3$ $\frac{1}{2_1} \times \cancel{12}^6$

2×4=8 7×1=7 3×3=9 1×6=6

Ascending order $\frac{1}{2}, \frac{7}{12}, \frac{2}{3}, \frac{3}{4}$.

Example 2

Arrange $\frac{5}{8}, \frac{5}{24}, \frac{5}{6}, \frac{5}{12}$ in descending order.

The L.C.M of 8, 24, 6 and 12 is 24.

$\frac{5}{8_1} \times \cancel{24}^3$ $\frac{5}{\cancel{24}_1} \times \cancel{24}^1$ $\frac{5}{6_1} \times \cancel{24}^4$ $\frac{5}{\cancel{12}_1} \times \cancel{24}^2$

5×3=15 5×1=5 5×4=20 5×2=10

Descending order $\frac{5}{6}, \frac{5}{8}, \frac{5}{12}, \frac{5}{24}$.

Activity 5:7

Arrange the fractions below as instructed.

1) Arrange $\frac{5}{6}, \frac{11}{12}, \frac{5}{8}$ *(in ascending order)*.
2) Arrange $\frac{2}{3}, \frac{1}{2}, \frac{3}{4}$ *(in ascending order)*.
3) Arrange $\frac{4}{5}, \frac{9}{10}, \frac{15}{20}$ *(in ascending order)*.
4) Arrange $\frac{2}{3}, \frac{5}{9}, \frac{3}{4}$ *(from the smallest)*.
5) Arrange $\frac{5}{12}, \frac{5}{6}, \frac{3}{4}$ *(from the smallest)*.
6) Arrange $\frac{3}{5}, \frac{4}{10}, \frac{15}{20}$ *(from the smallest)*.
7) Arrange $\frac{1}{3}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}$ *(in descending order)*.
8) Arrange $\frac{2}{3}, \frac{2}{4}, \frac{2}{6}, \frac{2}{8}$ *(in descending order)*.
9) Arrange $\frac{11}{12}, \frac{5}{6}, \frac{23}{24}, \frac{5}{8}$ *(in descending order)*.
10) Arrange $\frac{7}{10}, \frac{7}{20}, \frac{1}{2}, \frac{3}{4}$ *(from the biggest)*.
11) Arrange $\frac{3}{4}, \frac{5}{6}, \frac{1}{2}$ *(from the biggest)*.
12) Arrange $\frac{3}{8}, \frac{3}{12}, \frac{3}{6}, \frac{3}{8}$ *(from the biggest)*.

5.8 Comparing Fractions Using 'Greater than', 'Less than' or 'Equal to'

Use < or > or = to compare fractions

Example 1

$\frac{3}{4} > \frac{7}{12}$

L.C.M of 4 and 12 is 12.

Multiply each side by12

$\cancel{12}^{3} \times \frac{3}{\cancel{4}_{1}}$ —— $\frac{7}{\cancel{12}_{1}} \times \cancel{12}^{1}$

3×3 $\quad 7 \times 1$

$9 \ \underline{>} \ 7$

Example 2

$\frac{17}{20} < \frac{9}{10}$

L.C.M of 20 and 10 is 20.

Multiply each side by 20.

$\cancel{20}^{1} \times \frac{17}{\cancel{20}_{1}}$ —— $\frac{9}{\cancel{10}_{1}} \times \cancel{20}^{2}$

1×17 —— 9×2

$17 \ \underline{<} \ 18$

Activity 5:8

Use < or > or = to compare fractions below;

1) $\frac{3}{4}$ —— $\frac{5}{6}$
2) $\frac{1}{2}$ —— $\frac{3}{8}$
3) $\frac{7}{12}$ —— $\frac{5}{8}$
4) $\frac{1}{2}$ —— $\frac{1}{3}$
5) $\frac{2}{3}$ —— $\frac{2}{5}$
6) $\frac{7}{15}$ $\quad \frac{1}{12}$
7) $\frac{14}{15}$ —— $\frac{11}{12}$
8) $\frac{2}{9}$ —— $\frac{7}{9}$
9) $\frac{7}{12}$ —— $\frac{7}{10}$
10) $\frac{1}{2}$ —— $\frac{3}{9}$
11) $\frac{7}{12}$ —— $\frac{5}{6}$
12) $\frac{4}{12}$ —— $\frac{1}{3}$
13) $\frac{9}{16}$ —— $\frac{3}{4}$
14) $\frac{3}{6}$ —— $\frac{6}{12}$
15) $\frac{5}{8}$ —— $\frac{5}{6}$

5.9 Addition of Fraction with Different Denominators

To add fractions with different denominator, find the lowest common denominator (L.C.D)

TOPIC 5: Fractions

Example 1

Add $\frac{3}{4}+\frac{1}{7}$

The L.C.D of 4 and 7 is 28.

$\frac{7\times3+4\times1}{28}$

$\frac{21+4}{28}=\frac{25}{28}$

Example 3

$\frac{7}{12}+\frac{2}{3}$ L.C.D=12

$\frac{1\times7+4\times2}{12}$

$\frac{7+8}{12}=\frac{15}{12}$

$1\frac{\cancel{3}^{1}}{\cancel{12}_{4}} = 1\frac{1}{4}$

Example 4

$\frac{3}{4}+\frac{5}{6}$ L.C.D = 12

$\frac{3\times3+2\times10}{12}$

$\frac{9+10}{12}=\frac{19}{12}$

$\frac{19}{12} = 1\frac{7}{12}$

Activity 5:9

1) $\frac{1}{5}+\frac{1}{2}$ 2) $\frac{2}{3}+\frac{1}{5}$ 3) $\frac{5}{6}+\frac{3}{8}$ 4) $\frac{5}{6}+\frac{1}{8}$ 5) $\frac{1}{3}+\frac{2}{5}$

6) $\frac{3}{20}+\frac{2}{5}$ 7) $\frac{7}{8}+\frac{2}{3}$ 8) $\frac{1}{5}+\frac{3}{4}$ 9) $\frac{7}{15}+\frac{1}{6}$ 10) $\frac{2}{7}+\frac{3}{4}$

11) $\frac{2}{5}+\frac{1}{6}$ 12) $\frac{7}{12}+\frac{5}{6}$ 13) $\frac{1}{3}+\frac{1}{4}+\frac{1}{2}$ 14) $\frac{2}{3}+\frac{1}{2}+\frac{5}{6}$ 15) $\frac{3}{4}+\frac{2}{3}+\frac{3}{8}$

5.10 Addition of Mixed Numbers

Example 1

Add $4\frac{2}{3}+\frac{5}{12}$.

L.C.D=12

$4+\frac{(4\times2+1\times5)}{12}$

$4+\frac{8+5}{12}$

$4+\frac{13}{12}$

$4+1+\frac{1}{2}$

$5\frac{1}{12}$

Example 2

Add $\frac{3}{4}+2\frac{3}{8}$.

L.C.D=8

$2+\left(\frac{2\times3+1\times3}{8}\right)$

$2+\frac{6+3}{8}$

$2+\frac{9}{8}=2+1\frac{1}{8}$

$=3\frac{1}{8}$

Example 3

Add $5\frac{2}{3}+7\frac{5}{6}$

$5+7+\left(\frac{2}{3}+\frac{5}{6}\right)$ L.C.D=6

$12+\left(\frac{2\times2+1\times5}{6}\right)$

$12+\left(\frac{4+5}{6}\right)$

$12+\frac{9}{6}$

$12+1\frac{3^{1}}{6_{2}}=12\frac{1}{2}$

Example 4

Add $4\frac{7}{12}+\frac{3}{4}+8\frac{3}{8}$

$4+8+\left(\frac{7}{12}+\frac{3}{4}+\frac{3}{8}\right)$

L.C.D=24

$12+\left(\frac{2\times7+6\times3+3\times3}{24}\right)$

$12+\left(\frac{14+18+9}{24}\right)$

$12+\frac{41}{24}=12+1\frac{17}{24}$

$=13\frac{17}{24}$

Activity 5:10

1) $4\frac{2}{7}+\frac{3}{4}$ 2) $12\frac{11}{12}+\frac{5}{6}$ 3) $8\frac{7}{8}+\frac{3}{4}$ 4) $\frac{7}{12}+3\frac{12}{15}$

5) $\frac{13}{15}+2\frac{2}{3}$ 6) $\frac{17}{24}+7\frac{3}{4}$ 7) $4\frac{11}{12}+5\frac{2}{3}$ 8) $9\frac{1}{4}+12$

9) $2\frac{1}{15}+1\frac{3}{5}+1\frac{1}{5}$ 10) $12\frac{1}{2}+3\frac{1}{3}+\frac{1}{8}$ 11) $6\frac{1}{3}+9\frac{1}{4}+4\frac{1}{4}$ 12) $3\frac{1}{12}+\frac{1}{6}+6\frac{2}{3}$

13) $8+1\frac{3}{4}+\frac{5}{8}$ 14) $7\frac{1}{2}+\frac{5}{6}+3\frac{1}{4}$ 15) $6\frac{2}{3}+\frac{5}{6}+4\frac{11}{12}$ 16) $2\frac{1}{2}+\frac{3}{4}+1\frac{5}{6}$

17) $4\frac{1}{2}+3$ 18) $12\frac{7}{8}+3$ 19) $2+3\frac{3}{8}$ 20) $6+\frac{5}{9}$

5.11 Addition of Fractions (Word problems)

Example 1

Mr. Odur spent $\frac{2}{5}$ of his salary on food and $\frac{3}{8}$ on rent. What fraction did he spend on the two items?

$\frac{8\times2+5\times3}{40}$ L.C.D =40

$\frac{16+15}{40}=\frac{31}{40}$

He spent $\frac{31}{41}$ of his salary

Example 2

A primary five pupil ate $\frac{2}{3}$ of the sugarcane during breakfast and $\frac{1}{4}$ at lunch. What part of the sugarcane did the pupil eat?

$\frac{2}{3}+\frac{1}{4}$ L.C.D =12

$\frac{4\times2+3\times1}{12}$

$\frac{8+3}{12}=\frac{11}{12}$

He ate $\frac{11}{12}$

$\frac{2}{3}=\frac{4}{6},\frac{6}{9},\frac{8}{12},\frac{10}{15}$

$\frac{1}{4}=\frac{2}{8},\frac{3}{12},\frac{5}{20},\frac{6}{24}$

$\frac{8}{12}+\frac{3}{12}=\frac{8+3}{12}=\frac{11}{12}$

Activity 5:11

1. Jafar spent $\frac{7}{12}$ on rent and $\frac{3}{18}$ on fees. What fraction did he spend altogether?
2. At Buganda primary school, $\frac{5}{12}$ is spent in class, $\frac{1}{3}$ in music and $\frac{1}{4}$ in athletics. What fraction do the pupils spend at school?
3. Obangaina ate $\frac{1}{2}$ of the cake in the morning and $\frac{5}{12}$ in the evening. What fraction of the cake did he eat?
4. Motherwell Junior School collected $\frac{7}{20}$ of the school fess in the first month of the term and $\frac{5}{8}$ in the second month. Workout the fraction of the school fees collected in the two months?
5. A house maid filled $\frac{2}{5}$ of a tank with water in the morning and $\frac{1}{2}$ in the evening. What fraction of the tank did he fill with water?
6. Kusiima painted his building with different colour. $\frac{2}{9}$ with blue, $\frac{1}{3}$ red and $\frac{3}{18}$ yellow. What fraction is painted altogether?
7. A primary five teacher of mathematics spent $\frac{1}{2}$ of an hour giving example, $\frac{1}{6}$ writing an activity and $\frac{1}{3}$ marking an activity. How long was the lesson.
8. In a school library, $\frac{7}{18}$ of the books are science $\frac{1}{3}$ are mathematics and $\frac{2}{9}$ are social studies. What fraction do the three type of books represent.

5.12 Subtraction of Fractions with Common Fractions

Example 1

Subtract; $\frac{5}{6}-\frac{7}{18}$. L.C.D= 18

$\frac{3\times5-1\times7}{18}$

$\frac{15-7}{18}=\frac{8}{18}$

$\frac{8}{18}\div\frac{2}{2}=\frac{4}{9}$

Example 2

Subtract $\frac{3}{4}$ from $\frac{7}{8}$. L.C.D = 8

$\frac{7}{8}-\frac{3}{4}$

$\frac{1\times7-2\times3}{8}$

$\frac{7-6}{8}=\frac{1}{8}$

Activity 5:12

Subtract the following fractions.

1) $\frac{3}{4}-\frac{7}{12}$ 2) $\frac{8}{9}-\frac{3}{4}$ 3) $\frac{11}{12}-\frac{5}{6}$ 4) $\frac{5}{6}-\frac{1}{3}$ 5) $\frac{5}{7}-\frac{1}{2}$ 6) $\frac{4}{5}-\frac{2}{3}$

7) $\frac{7}{5}-\frac{3}{4}$ 8) $\frac{5}{8}-\frac{7}{12}$ 9) $\frac{6}{7}-\frac{2}{3}$ 10) $\frac{14}{15}-\frac{3}{5}$ 11) $\frac{17}{24}-\frac{5}{8}$ 12) $\frac{19}{24}-\frac{5}{8}$

5.13 Addition and Subtraction of Fractions

Example 1

Work out $\frac{3}{4}+\frac{7}{12}-\frac{5}{6}$

L.C.D of 4, 12 and 6 is 12.

$$\frac{3\times 3+1\times 7-2\times 5}{12}$$

$$\frac{9+7-10}{12}$$

$$\frac{16-10}{12}=\frac{6^1}{\cancel{12}_2}$$

$$=\frac{1}{2}$$

Example 2

$$\frac{2}{3}-\frac{3}{4}+\frac{5}{8}$$

Apply BODMAS

Re-arrange the fractions

$\frac{2}{3}+\frac{5}{8}+\frac{3}{4}$ L.C.D = 24

$$\frac{8\times 2+3\times 5-6\times 3}{24}$$

$$\frac{16+15-18}{24}$$

$$\frac{31-18}{24}=\frac{13}{24}$$

Activity 5:13

1) $\frac{2}{3}+\frac{1}{2}-\frac{3}{4}$ 2) $\frac{5}{6}+\frac{3}{8}-\frac{5}{8}$ 3) $\frac{3}{7}+\frac{2}{3}-\frac{11}{21}$ 4) $\frac{3}{4}+\frac{5}{6}-\frac{2}{3}$

5) $\frac{4}{5}+\frac{1}{3}-\frac{8}{15}$ 6) $\frac{1}{4}+\frac{1}{2}-\frac{1}{3}$ 7) $\frac{2}{3}-\frac{7}{10}+\frac{4}{5}$ 8) $\frac{1}{4}-\frac{1}{2}+\frac{3}{8}$

9) $\frac{5}{8}-\frac{3}{4}+\frac{1}{2}$ 10) $\frac{4}{5}-\frac{2}{3}+\frac{7}{15}$ 11) $1\frac{1}{2}+3\frac{2}{3}-2\frac{1}{8}$ 12) $2\frac{3}{5}+\frac{4}{5}-1\frac{2}{3}$

13) $5\frac{3}{8}-1\frac{1}{4}+2\frac{1}{2}$ 14) $3\frac{1}{2}-2\frac{5}{7}+1\frac{3}{4}$ 15) $1\frac{2}{5}-1\frac{1}{4}+2\frac{1}{3}$ 16) $5\frac{1}{6}-4\frac{2}{3}+1\frac{7}{12}$

5.14 Multiplication of Fractions by Whole Numbers

Example 1

What is $\frac{3}{4}\times 240$?

M_4 = 4, 8, 12, 16, 20, 24, 28

$\frac{3}{4_1}\times \cancel{240}^{60}$

(3×60)

= 180

Example 2

Find $\frac{5}{6}$ *of* 180 pupils.

M_6 = 6, 12, 18, 24, 30

$\frac{5}{6_1}\times \cancel{180}^{30}$pupils

(5×30)pupils

150pupils

Example 3

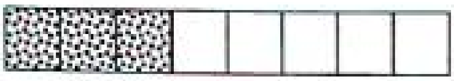
In a class of 40 pupils $\frac{3}{8}$ of the pupils are girls the rest are boys.

a) What is the fraction of the boys?

$\frac{1}{1}-\frac{3}{8}$ L.C.D = 8

$\frac{(8\times1-1\times3)}{8}$

$=\frac{8-3}{8}=\frac{5}{8}$



Girls Boys

b) How many girls are in the class?

$Girls = \left(\frac{3}{8}\times 40\right)$

$M_8 =$ 8,16,24,32,40,48

$\frac{3}{\cancel{8}_1}\times \cancel{40}^5$ girls

(3×5)girls

=15girls

Example 4

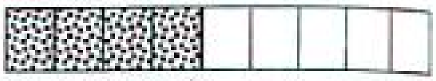
A tank of water contain 1800litres. If $\frac{4}{9}$ of water was used,

a)What fraction of water remained in the tank?

$\frac{1}{1}-\frac{4}{9}$ L.C.D = 9

$\frac{9\times1-1\times4}{9}$

$=\frac{9-4}{9}=\frac{5}{9}$



Used water Water in the tank

b) Calculate the volume of water which remained.

$\frac{5}{9}\times 1800 litres$

$M_9 =$ 9, 18, 27, 36

$\frac{5}{\cancel{9}_1}\times \cancel{1800}^{200}$litres

5×200litres = 1000litres

Activity 5:14

1) What is $\frac{7}{12}\times480$?
2) What is $\frac{5}{8}\times 56$?
3) Find $\frac{7}{15}$ *of* 3000.
4) What is $\frac{3}{7}\times 3500kg$?
5) What $\frac{3}{8}\times$ 32pupils?
6) Workout $\frac{5}{6}$ *of* 24000/=.
7) What is $\frac{5}{16}$ *of* 32?
8) Workout $\frac{2}{3}$ of 1hour.
9) What is $\frac{5}{12}$ of 48pupils?

10). What is $1\frac{1}{9}$ of 45 mangoes?

11) An Askari earns sh.120,000 monthly. He spends $\frac{3}{4}$ on fees and the rest on food.

(a) What fraction does he spend on food?

(b) How much does he spend on fees?

(c) How much does he spend on food?

12. A science text book contains 96pages. Olele read $\frac{5}{8}$ of the book. How many pages did he read?

13. The distance from Kampala to Kitgum is 630km. Komeko bus broke down after covering $\frac{7}{9}$ of the journey. What distance had it covered?

14. Mr. Owere had sh.200,000/=. He gave away $\frac{2}{5}$ of it to his wife.

(a) How much did he give to the wife? (b) How much did he remain with?

15. In a group of 420people, $\frac{5}{7}$ are females and the rest are male.
   (a) Find the fraction of males. (b) How many males are in the group?
16. During the last election, $\frac{7}{8}$ of the electorate voted for Museveni Kaguta Tibihaburwa and the rest to Kyagulanyi Robert.
   (a) What fraction did Kyagulanyi Robert get?
   (b) If together they collected 4,800,000 votes, how many voted Museveni Kaguta?
17. In a school of 800pupils, $\frac{3}{5}$ of them are boys and the rest are girls.
   (a) Find the fraction of the girls. (b) Find the number of girls.

5.15 <u>Finding the Reciprocal of Different Numbers or Fractions</u>

- ✓ A reciprocal is one divided by the quantity.
- ✓ The reciprocal of a fraction is the opposite of a given fraction.

Example 1	**Example 2**
What is the reciprocal of $\frac{2}{3}$? *Name the reciprocal* Let the reciprocal of $\frac{2}{3} = p$ $\frac{2}{3}\ of\ p = 1$ $\frac{2}{3} \times p = 1$ Multiply both sides by the L.C.D $3 \times \frac{2}{3}p = 1 \times 3$ $\frac{2}{2}p = \frac{3}{2}$ $p = \frac{3}{2}$ *The reciprocal of* $\frac{2}{3}$ *is* $\frac{3}{2}$ *or* $1\frac{1}{2}$	What is the reciprocal of 4? Let the reciprocal of $4 = Z$. $4 \times Z = 1$ $4Z = 1$ Divide both sides by 4 $\frac{4Z}{4} = \frac{1}{4}$ $Z = \frac{1}{4}$ The reciprocal of 4 is $\frac{1}{4}$

Activity 5:15

Workout the reciprocal of the following;

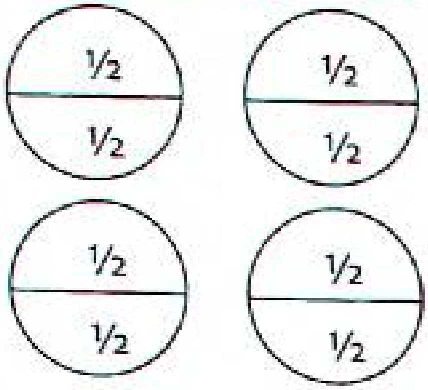
1) 2 2) 6 3) 8 4) 10 5) 12 6) $\frac{3}{4}$ 7) $\frac{7}{12}$ 8) $\frac{3}{5}$
9) $\frac{1}{2}$ 10) $\frac{3}{8}$ 11) $1\frac{1}{3}$ 12) $2\frac{2}{5}$ 13) $1\frac{1}{8}$ 14) $1\frac{1}{4}$ 15) $3\frac{3}{4}$ 16) $4\frac{3}{5}$

5.16 Dividing Whole Numbers by Fractions

Example 1

How many halves are in 4 oranges?

Method 1: Using diagrams



There are 8 halves.

Method 2: By calculations

$4\ oranges\ divide\ by\ \frac{1}{2}$.

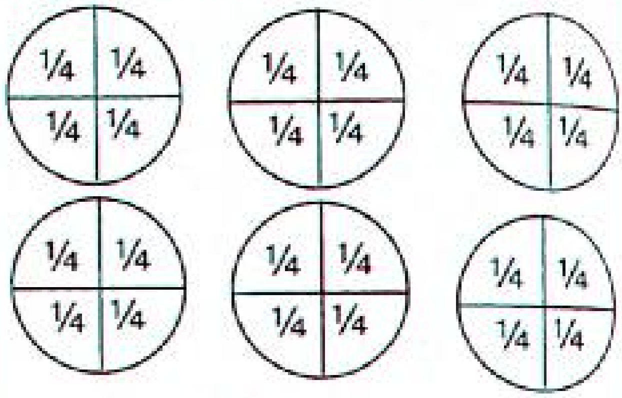
$4 \div \frac{1}{2} = (4 \times \frac{2}{1})halves$

8 halves

Example 2

Ambrose packed $\frac{1}{4}$ kg of rice in a paper bag.

If he had 6kg, how many did he pack?



He packed 24 bags.

Method 2

$(6 \div \frac{1}{4})bags$

$(6 \times \frac{4}{1})bags$

24 bags

Activity 5:16

1. How many $\frac{1}{2}$ kg of meat are in 8kg?
2. How many halves can be cut from 10books?
3. Mr. Mukasa bought 3cakes to be shared by his children, if each child got $\frac{1}{4}$ of the cake, how many children does Mr. Mukasa have?
4. How many $\frac{2}{3}$ kg packets of sugar can be packed from 18kg?
5. A bag of salt contain 20kg. How many packets of a half kilogram can be packed from it?
6. 15liters of milk were shared by children in P.5 class. If each child got $\frac{1}{4}$ litre, how many children are in the class?
7. How many $\frac{3}{4}$ litre bottles of cooking oil can be packed from 12litres?
8. How many half kg pack can be got from 5kg?
9. A shopkeeper prepared 10litres of mango juice. He packed it in $\frac{1}{2}$ litre pack bottles, how many bottles did he pack?
10. Jomayi had 15 acres of land to be divided into small plots of $1\frac{1}{2}$ acres each. How many plots of land did he get?

5.17 Dividing Fractions by Natural Numbers

Example 1

Workout $\frac{1}{2} \div 4$

Make 4 a fraction by putting it over 1

$\frac{1}{2} \div \frac{4}{1}$

Change $\frac{4}{1}$ to its multiplicative inverse

$\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$

Example 2

Work out $\frac{2}{3} \div \frac{12}{1}$

Change $\frac{12}{1}$ to its multiplicative inverse

$\frac{\cancel{2}^{1}}{3} \times \frac{1}{\cancel{12}_{6}}$

$\frac{1\times1}{3\times6} = \frac{1}{18}$

Example 3

Work out $3\frac{3}{8} \div 4$

Change the mixed number to improper fraction

$\frac{27}{8} \div \frac{4}{1}$

Find the reciprocal of $\frac{4}{1}$

$\frac{27}{8} \times \frac{1}{4} = \frac{27}{32}$

Example 4

Calculate $7\frac{1}{2} \div 3$

Change the mixed number to improper fraction.

Change $\frac{3}{1}$ to its reciprocal

$\frac{\cancel{15}^{5}}{2} \times \frac{1}{\cancel{3}_{1}}$

$\frac{5\times1}{2\times1} = \frac{5}{2}$

$2\frac{1}{2}$

Activity 5:17

Work out the following numbers

1) $\frac{1}{2} \div 3$ 2) $\frac{2}{3} \div 6$ 3) $\frac{5}{6} \div 24$ 4) $\frac{7}{12} \div 28$ 5) $\frac{11}{12} \div 88$

6) $\frac{7}{8} \div 21$ 7) $\frac{2}{3} \div 24$ 8) $\frac{8}{15} \div 72$ 9) $\frac{7}{9} \div 14$ 10) $1\frac{1}{2} \div 12$

11) $2\frac{4}{5} \div 28$ 12) $3\frac{1}{3} \div 30$ 13) $5\frac{2}{3} \div 10$ 14) $2\frac{1}{4} \div 180$ 15) $3\frac{2}{3} \div 44$

5.18 Dividing a Fraction by a Fraction

Example 1

$\frac{2}{3} \div 2\frac{1}{2}$

$\frac{2}{3} \div \frac{5}{2}$

$\frac{2}{3} \times \frac{2}{5} = \frac{4}{15}$

Example 2

$\frac{4}{5} \div \frac{2}{5}$

$\frac{\cancel{4}^{2}}{\cancel{5}_{1}} \times \frac{\cancel{5}^{1}}{\cancel{2}_{1}}$

$2 \times 1 = 2$

Example 3

$\frac{5}{6} \div \frac{2}{3}$

$\frac{5}{\cancel{6}_{2}} \times \frac{\cancel{3}^{1}}{2}$

$\frac{5}{4} = 1\frac{1}{4}$

Activity 5:18

Work out the following numbers.

1) $\frac{2}{3} \div \frac{1}{3}$ 2) $\frac{3}{4} \div \frac{5}{6}$ 3) $\frac{7}{8} \div \frac{1}{2}$ 4) $\frac{11}{12} \div \frac{6}{7}$ 5) $\frac{4}{5} \div \frac{2}{3}$ 6) $\frac{3}{4} \div \frac{1}{2}$

7) $\frac{5}{6} \div \frac{9}{20}$ 8) $\frac{2}{3} \div \frac{4}{5}$ 9) $\frac{1}{4} \div \frac{2}{3}$ 10) $5\frac{1}{3} \div \frac{2}{3}$ 11) $13\frac{1}{2} \div \frac{2}{3}$ 12) $6\frac{1}{2} \div 2\frac{1}{2}$

DECIMALS

5.19 Place Values of Decimals

All digits after the decimal point towards the right hand are decimals. The place values of decimals begin from tenths, hundredths, thousandths, ten thousandths etc.

Example 1

Write down the place values of 23678.

2 3 6 7 8
- 8 – Ones
- 7 – Tens
- 6 – Hundreds
- 3 – Thousands
- 2 – Ten thousands

Example 2

Write down the place value of each digit of 236.78.

2 3 6. 7 8
- 8 – Hundredths
- 7 – Tenths
- 6 – Ones
- 3 – Tens
- 2 – Hundreds

Example 3

Write the place value of each digit

0 .4 6 7
- 7 – Thousandths
- 6 – Hundredths
- 4 – Tenths
- 0 – Ones

Example 4

Write the place value of each digit below.

8 4. 0 3
- 3 – Hundredths
- 0 – Tenths
- 4 – Ones
- 8 – Tens

Activity 5:19

Write the place values of each of the following decimals

1) 0.6 2) 0.43 3) 0.24 4) 0.124 5) 0.420 6) 7.52

7) 12.036 8) 144.8 9) 0.478 10) 921.03 11) 39.87 12) 342.342

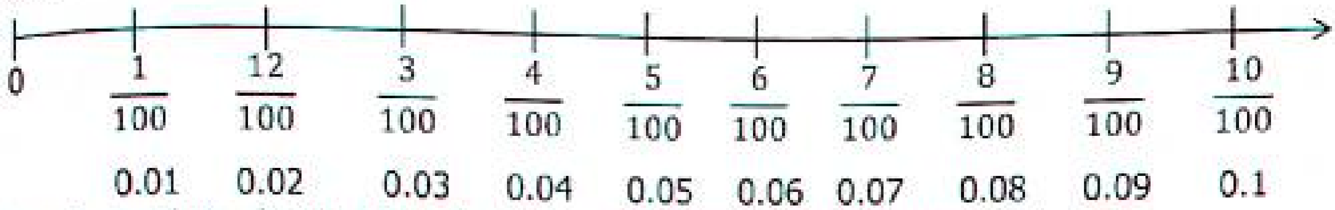
5.20 Decimals on a Number line

One whole is divided into ten equal parts, each part is equal to a tenth.

0	$\frac{1}{10}$	$\frac{12}{10}$	$\frac{3}{10}$	$\frac{4}{10}$	$\frac{5}{10}$	$\frac{6}{10}$	$\frac{7}{10}$	$\frac{8}{10}$	$\frac{9}{10}$	1
0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0

TOPIC 5: Fractions

If each tenth is divided into ten equal parts, the outcome will be one hundredth of a whole number.



If each tenth is divided into ten equal parts, the outcome will be one hundredth of a whole number.

Values of Decimals

Procedure

Decimal places are the number of digits after the decimal point.

$\frac{2}{1} = 2$ (2 is a whole number no decimal place)

$\frac{2}{10} = 0.2$ (one place of decimal)

$\frac{3}{100} = 0.02$ (two place of decimal)

$\frac{5}{1000} = 0.005$ (three places of decimals)

- ✓ The zero (0) before the decimal point is used to keep the place for whole numbers.
- ✓ The number of "0" of the denominator in a common fraction is equal to the number of decimal places when the common fraction is changed to a decimal fraction.

Example 1	**Example 2**	**Example 4**
Change $\frac{8}{10}$ to a decimal. $\frac{8}{10} = 0.8$	Convert $\frac{12}{100} = 0.12.$ $\frac{12}{100} = 0.12$	Change $\frac{48}{1000}$ to decimal. $\frac{48}{1000} = 0.048$

Activity 5:20

Express the following as decimal fractions

1) $\frac{8}{1}$	2) $\frac{12}{1}$	3) $\frac{184}{1}$	4) $\frac{8}{10}$	5) $\frac{9}{10}$	6) $\frac{3}{10}$
7) $\frac{12}{100}$	8) $\frac{53}{100}$	9) $\frac{72}{100}$	10) $\frac{97}{100}$	11) $\frac{148}{1000}$	12) $\frac{254}{1000}$
13) $\frac{107}{1000}$	14) $\frac{750}{1000}$	15) $1\frac{4}{10}$	16) $12\frac{4}{100}$	17) $72\frac{23}{100}$	18) $13\frac{7}{100}$

5.21 Expressing Decimals to Common Fractions

Example 1	Example 2
Express 0.8 as a fraction. $0.8 = \frac{8}{10} \textit{ reduce by } 2$ $\frac{8}{10} \div \frac{2}{2} = \frac{4}{5}$ $0.8 = \frac{4}{5}$	Change 0.12 as a common fraction. $0.12 = \frac{12}{100}$ (reduce by 4) $\frac{12}{100} \div \frac{4}{4} = \frac{3}{25}$ $0.12 = \frac{3}{25}$
Example 3 Change 5.32 as a common fraction. 5.32 = 5 + 0.32 $5.32 = 5 + \frac{32}{100}$ $5\frac{32}{100} \textit{ reduce the fraction by } 4$ $5 + \left(\frac{\cancel{32}^{8}}{\cancel{100}_{25}} \div \frac{4^{1}}{4_{1}}\right)$ $5 + \frac{8}{25} = 5\frac{8}{25}$	**Example 4** Change 15.7 as a common fraction. 15 + 0.7 $15 + \frac{7}{10}$ $15\frac{7}{10}$

Activity 5:21

Express the decimals as common fractions in their lowest terms.

1) 0.7	2) 0.1	3) 0.4	4) 0.9	5) 0.15
6) 0.24	7) 0.87	8) 0.42	9) 0.253	10) 0.537
11) 0.112	12) 0.345	13) 2.5	14) 8.7	15) 12.4
16) 150.3	17) 12.17	18) 182.28	19) 148.48	20) 720.072

5.22 Expressing Fractions to Decimals

Example 1	Example 2
Change $\frac{4}{5}$ 0.8 5⟌ 4 0 5 × 0 = -0 4 0 5 × 8 -4 0 - -	Express $\frac{1}{4}$ as a common fraction. Divide 1 by 4 0. 2 5 4⟌ 1 0 0 4 × 0 = -0 10 4 × 2 -8 20 4 × 5 = -20 - -

Activity 5:22

Express the following fractions to decimals.

1) $\frac{2}{5}$ 2) $\frac{3}{5}$ 3) $\frac{2}{10}$ 4) $\frac{3}{4}$ 5) $\frac{1}{8}$ 6) $\frac{3}{8}$ 7) $\frac{5}{8}$ 8) $\frac{3}{6}$ 9) $\frac{6}{12}$ 10) $\frac{4}{10}$

5.23 Ordering Decimal Fractions

Procedure

✓ Express the decimals to fractions

✓ Find the Lowest Common Denominator (L.C.D)

✓ Multiply each fraction by the Lowest Common Denominator.

✓ Arrange in ascending or descending order.

Example 1

Arrange 0.8, 0.28, 0.08, 0.88 starting from the smallest.

$\frac{8}{10}$	$\frac{28}{100}$	$\frac{8}{100}$	$\frac{88}{100}$
The L.C.D = 100			
$\frac{8}{\cancel{10}_1} \times \cancel{100}^{10}$	$\frac{28}{\cancel{100}_1} \times \cancel{100}^{1}$	$\frac{8}{\cancel{100}_1} \times \cancel{100}^{1}$	$\frac{88}{\cancel{100}_1} \times \cancel{100}^{1}$
8×10	28×1	8×1	88×1
80	28	8	88

Arrangement from the smallest

0.08, 0.28, 0.8, 0.88

Example 2

Arrange 0.22, 0.02, 0.9, 0.117 in descending order (from the biggest)

L.C.D = 1000

$\frac{22}{\cancel{100}_1} \times \cancel{1000}^{10}$	$\frac{2}{\cancel{100}_1} \times \cancel{1000}^{10}$	$\frac{9}{\cancel{10}_1} \times \cancel{1000}^{100}$	$\frac{117}{\cancel{100}_1} \times \cancel{1000}^{1}$
22×10	2×10	9×100	117×1
220	20	900	117

Arrangement in descending order

0.9, 0.22, 0.17, 0.02

Activity 5:23

Arrange the following fractions as instructed.

1. Arrange 0.2, 0.5, 0.35 (from the smallest)
2. Arrange 0.21, 0.12, 1.2, 0.02 (from the smallest)
3. Arrange 0.43, 0.34, 3.4, 4.3 (from the biggest)
4. Arrange 2.7, 0.27, 0.72, 7.2 (from the biggest)
5. Arrange 0.6, 0.26, 2.6,0.62 (from the biggest)
6. Arrange 0.8, 0.3, 0.5, 0.2 (from the biggest)
7. Arrange 1.2, 0.21, 2.1, 0.12 (in ascending order)
8. Arrange 0.48, 0.84, 0.048, 0.408 (in ascending order)

9. Arrange 0.11, 0.011, 0.101, 1.1 (In ascending order)
10. Arrange 0.3, 0.07, 0.7, 0.15 (in descending order)
11. Arrange 0.09, 0.009, 0.9, 0.19 (in descending order)
12. Arrange 0.24, 2.2, 0.02,0.42 (in descending order)

5.24 Addition and Subtraction of Decimal Fractions

Procedure

- ✓ Arrange the digits according to the place values
- ✓ Add accurately
- ✓ Re-group in tenths, hundredths or thousandths where necessary

Example 1

Add 2.4+9.7+0.42

$$\begin{array}{r} 2.4 \\ 9.7 \\ 0.42 \\ \hline 12.52 \\ \hline \end{array}$$

Example 2

Find the sum of 0.8+2+7.27

$$\begin{array}{r} 0.8 \\ 2.0 \\ +\ 7.27 \\ \hline 10.07 \\ \hline \end{array}$$

Example 3

Subtract 4.24 – 1.38.

$$\begin{array}{r} 4.24 \\ -1.38 \\ \hline 2.86 \\ \hline \end{array}$$

Example 4

Subtract 2 from 3.42

$$\begin{array}{r} 3.42 \\ -2.00 \\ \hline 1.42 \\ \hline \end{array}$$

Example 5

What number must be added to 12.5 to get 18.28?

$$\begin{array}{r} 18.28 \\ -12.5 \\ \hline 5.78 \\ \hline \end{array}$$

Example 6

What number must be subtracted from 0.78 to get 1?

1.00 – 0.78

$$\begin{array}{r} 1.00 \\ -0.78 \\ \hline 0.22 \\ \hline \end{array}$$

Activity 5:24

1) 2.41+2.8
2) 32.8+6.7
3) 1.8+0.43+0.08
4) 1.35+3.48+1.4
5) 1.84+0.83+0.25
6) 2.483+1.7-2
7) 6.1+1.45-5
8) 4+0.75-0.2
9) 8.013+2.95-4.520
10) 72.8+18.004-1521
11) 0.03+0.3+3
12) 0.49+0.36+5
13) 1.62+359-621
14) 0.49+3.9+0.85
15) 0.5+0.8-03
16) 0.08-0.006+0.4
17) 3.56-6.44+5.8
18) 1.7-3.42+4.32
19) 0.7-0.94+2.1
20) 0.005-0.46+0.5
21) 8.06-12.2+3.5

5.25 Combined Addition and Subtraction of Decimals

Example 1

Work out 6.4+0.83-2.9

$$\begin{array}{r} 6.4 \\ +\,0.83 \\ \hline 7.23 \end{array} \qquad \begin{array}{r} 7.23 \\ -\,2.90 \\ \hline 4.33 \end{array}$$

Example 3

Work out 8.02 – 13.45+7.9

Apply the rule of BODMAS (Add then subtract)

8.02 + 7.9 – 13.45

$$\begin{array}{r} 8.02 \\ +\,7.9 \\ \hline 15.92 \end{array} \qquad \begin{array}{r} 15.92 \\ -\,13.45 \\ \hline 2.47 \end{array}$$

Example 2

Work out 15.03+3-12.4

15.03+3.00-12.4

$$\begin{array}{r} 15.03 \\ +\,3.00 \\ \hline 18.30 \end{array} \qquad \begin{array}{r} 18.30 \\ -\,12.40 \\ \hline 5.90 \end{array}$$

Example 4

Work out 1.25 – 7.5+10

1.25 + 10.00 – 7.5

$$\begin{array}{r} 1.25 \\ +\,10.00 \\ \hline 11.25 \end{array} \qquad \begin{array}{r} 11.25 \\ -\,7.50 \\ \hline 3.75 \end{array}$$

Activity 5:25

Work out the following numbers

1) 3.75+6.2
2) 12.1+0.24
3) 7.93+5.5
4) 1.068+0.94
5) 10.68+9.4
6) 110+0.062
7) 18+6.31+0.3
8) 43.06+6+0.88
9) 9-8.72+5.6
10) 0.175+1.75-17.5
11) 4.8 – 3.07
12) 8 – 2.08
13) 12 – 0.457
14) 3.48 – 2
15) 17.24 – 9
16) 9.008 – 3
17) 8.62 – 0.498
18) 22 – 0.88
19) 9 – 8.72
20) 1.75 – 0.175

5.26 Word Problems Involving Decimals

Activity 5:26

1) A baby drank 1.litres of milk on Saturday and 1.4litres on Sunday. How many litres of milk did the baby drink altogether?
2) Catherine bought 2.5kg of sugar and Candy bought 1.8kg. How much sugar did they buy altogether?
3) Mr. Ssemonde bought 3.58metres of cloth and Mammonde bought 6.45metres of cloth. What length of cloth do they have altogether?
4) A rectangular flower garden measures 10.2metres by 4.8metres. Find the total distance round the garden.
5) Longido is 1.75metres and Bangido is 1.37metres. Find the difference of their height.
6) What is the difference between 43.4 and 18.8?
7) Find the difference between 12.5 and 7.7.
8) Dramati bought 8.2kg of salt and gave Amandu 3.42kg. How much salt did he remain with?

TOPIC 6: LINE, ANGLES AND GEOMETRICAL FIGURES

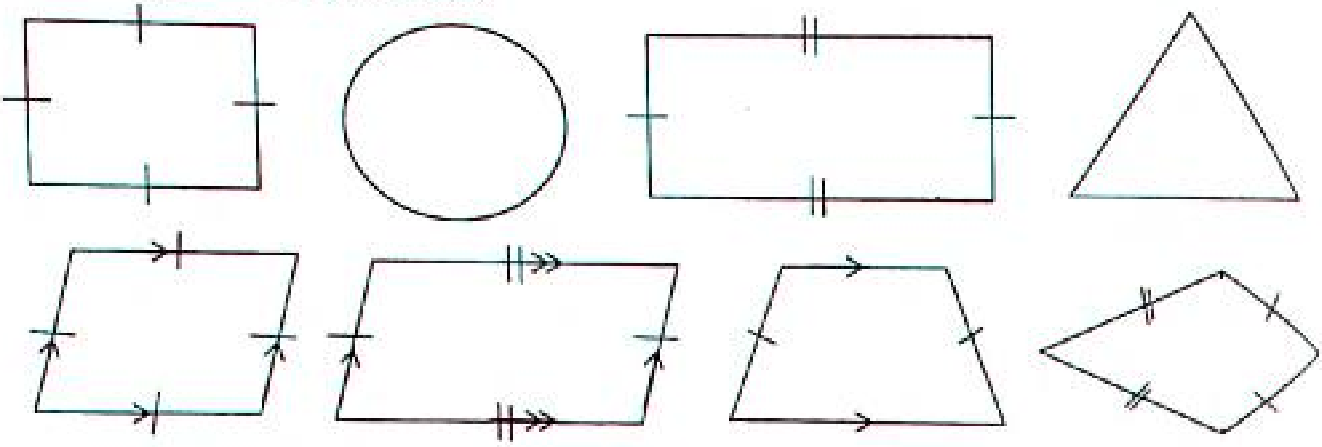
Geometry is a branch of mathematics that deals with the measurements, properties, lines, angles, surfaces and solids.

Or geometry is the type of mathematics that deals with points, lines, shapes and surfaces.

In geometry we study about size, shapes, positions, angles and dimension of things.

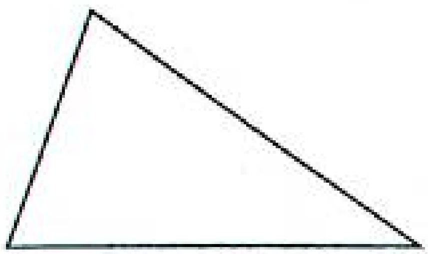
6.1 Two Dimension Shapes in Geometry

These are flat shaped like square, circle, rectangles, triangles, kite, trapezium, rhombus, parallelogram etc..

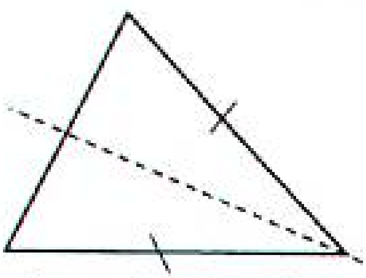


Lines of Symmetry in Polygons

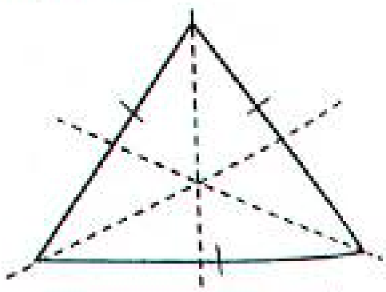
A line of symmetry is the line that divides a shape or an object into two equal and symmetrical parts. This means that if you were to fold the shape along the line, both halves would match exactly like the examples below.



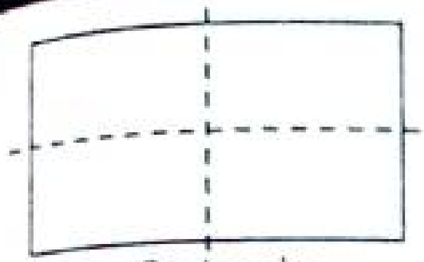
Scalene triangle
No line of symmetry



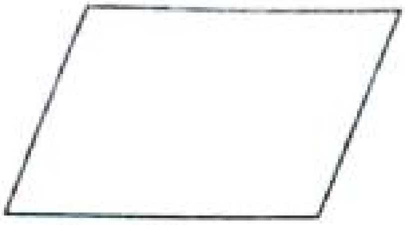
Isosceles triangle
1 line of symmetry



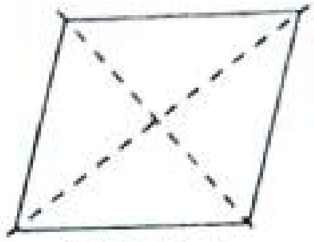
Equilateral triangle
3 lines of symmetry



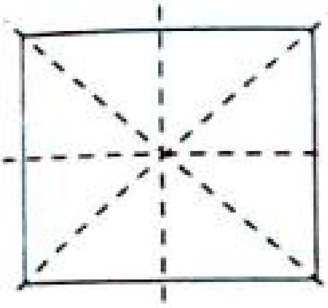
Rectangle
2 lines of symmetry



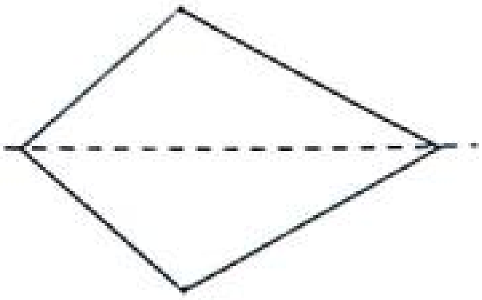
Parallelogram
No line of symmetry



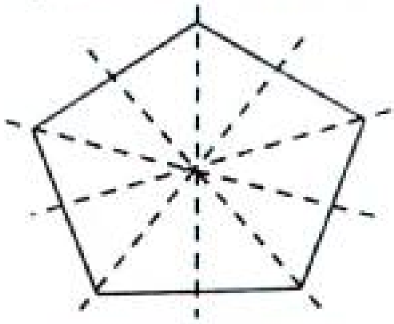
Rhombus
2 lines of symmetry



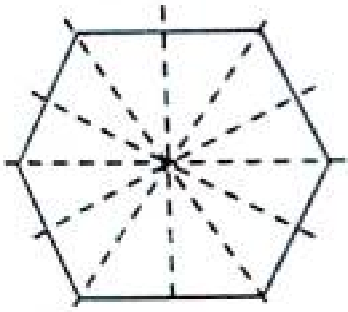
Square
4 lines of symmetry



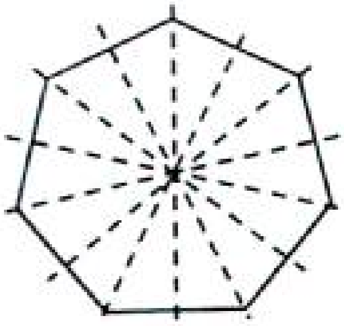
Kite
1 line of symmetry



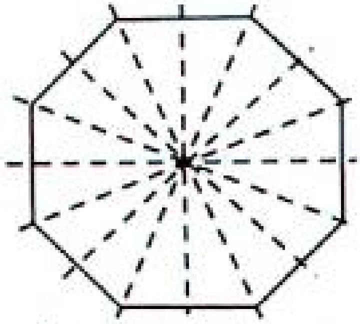
Regular Pentagon
5 lines of symmetry



Regular Hexagon
6 lines of symmetry



Regular Heptagon
7 lines of symmetry

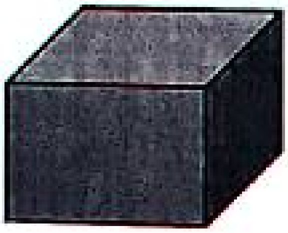
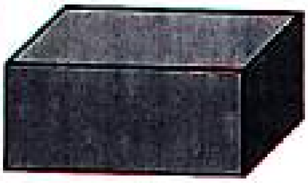
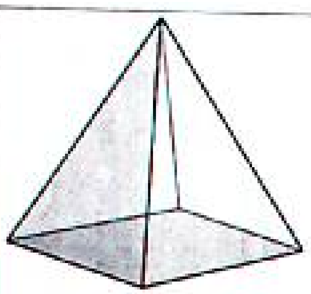


Regular Octagon
8 lines of symmetry

6.2 3Dimensional Objects

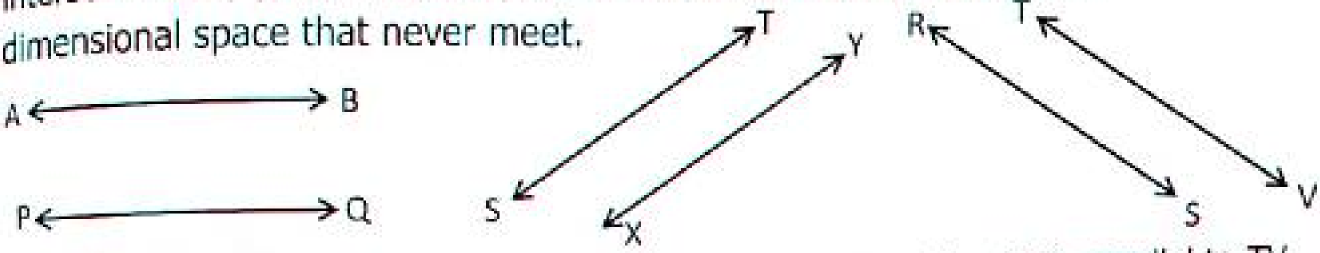
3Dimensional objects or shapes are shapes with dimensions, length, width and height. Examples of 3Dimesnsional objects include cube, cuboid, cylinder, sphere, rectangular prism, triangular prism etc..

Shape	Picture	Faces	Edges	Vertices
sphere		0	0	0

Shape	Picture	Faces	Edges	Vertices
Cube		6	12	8
Cuboid		6	12	8
Pyramid		5	8	5
Cylinder		2	0	0
Cone		1	0	1

6:3 Parallel and Perpendicular Lines

In geometry, parallel lines are coplanar infinite straight lines that do not intersect at any point. Parallel planes are planes in the same three dimensional space that never meet.

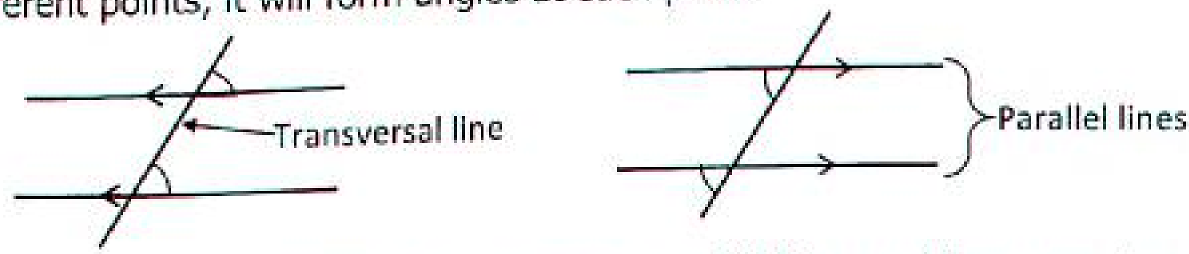


Line AB is parallel to line PQ   Line ST is parallel to line XY   Line RS is parallel to TV

- ✓ Parallel lines have equal distance apart and they cannot meet when prolonged.
- ✓ Two or more lines that lie in the same plane and never intersect each other are termed as parallel line.
- ✓ Examples of parallel lines in our daily life are Zebra crossing, the lines of note books, the opposite edges of a table, railway tracks etc..

Properties of Parallel Lines

- ✓ They are always straight lines with an equal distance between each other.
- ✓ They are coplanar lines.
- ✓ They never intersect, no matter how far you try to extend them in any given direction.
- ✓ If there is a transversal line that intersect two parallel lines at two different points, it will form angles at each point.



The corresponding angles formed by the two parallel lines and transversal are equal.

Drawing accurate parallel lines a geometry set is needed with all the instruments.

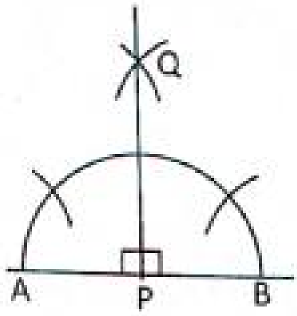
i) Protractor is used for measuring angles

ii) A pair of compasses is used for constructing angles.

iii) A set square is for drawing lines.

iv) A ruler used for drawing straight lines.
v) A pair of divider used for measuring length.

Perpendicular lines are lines that intersect at a right (90 degrees) angle.



Line AB is perpendicular to line PQ.

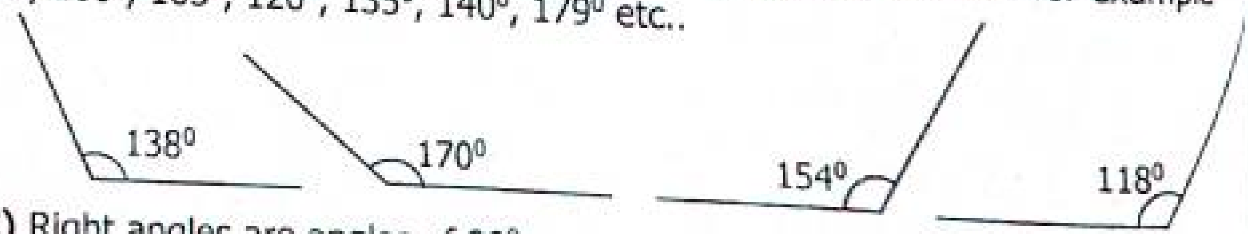
6.4 Types of Angles

There are six types of angles.

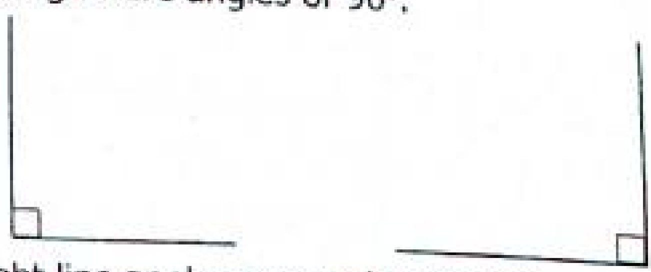
i) Acute angles
ii) Obtuse angles
iii) Right angles
iv) Straight lines angles
v) Reflex angles
vi) Angles at a point

A) Acute angle is an angle less than 90^0 for example 45^0, 30^0, 20^0, 15^0, 80^0, 89^0 etc..

B) Obtuse angle is an angle greater than 90^0 but less than 180^0 for example 92^0, 100^0, 105^0, 120^0, 135^0, 140^0, 179^0 etc..



C) Right angles are angles of 90^0.



D) Straight line angles are angles of 180^0.

E) Reflex angles are angles greater than 180^0 but less than 360^0.
Examples of reflex angles include 195^0, 190^0, 220^0, 230^0, 235^0, 238^0, 350^0, 354^0 etc.

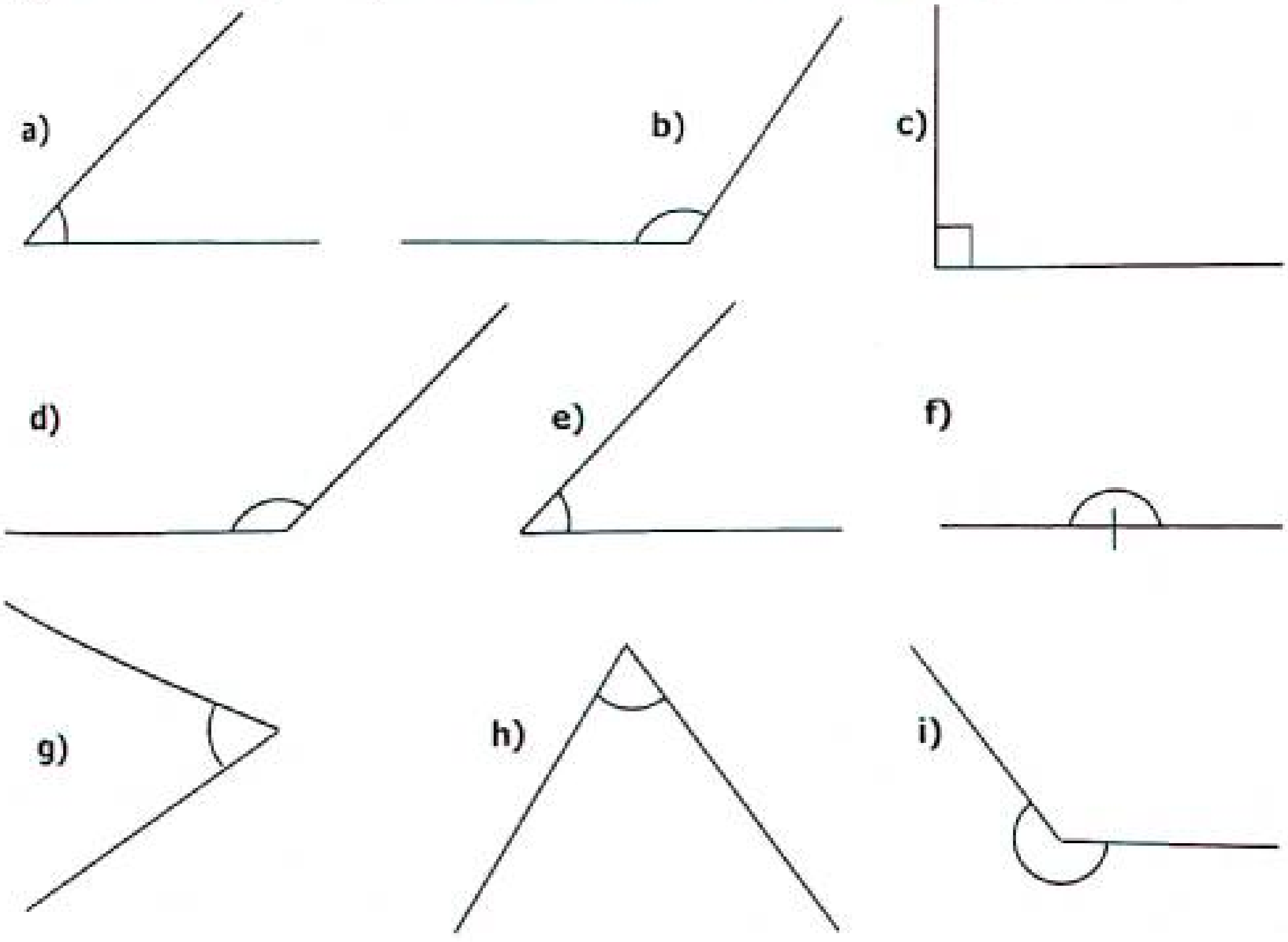
F) Angles at a point are angles of 360^0.

Activity 6:4A

Name the type of angle given below;

1) 128^0 2) 49^0 3) 90^0 4) 10^0 5) 160^0 6) 172^0 7) 54^0
8) 100^0 9) 240^0 10) 56^0 11) 145^0 12) 360^0 13) 247^0 14) 90^0

15) With the help of a protractor measure and describe the type of angle



Activity 6:4B

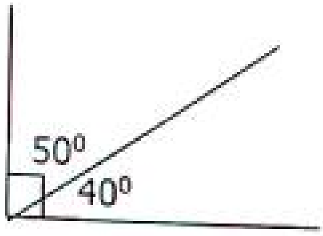
Use a ruler and a protractor to draw the angles below;

1) 50^0 2) 40^0 3) 45^0 4) 65^0 5) 80^0
6) 110^0 7) 130^0 8) 150^0 9) 175^0 10) 180^0
11) 200^0 12) 240^0 13) 270^0 14) 300^0 15) 320^0

6.5 Complementary Angles

Complementary angles are two angles which add up to 90^0.

Example 1

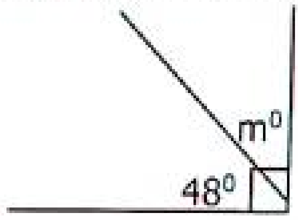


Two complementary angle 50^0 and 40^0 angles)

$50^0 + 40^0 = 90^0$

Example 2

Find the value of m.



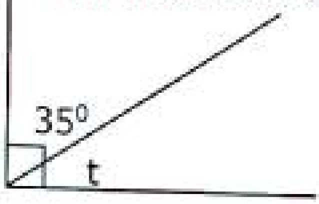
$m + 48^0 = 90^0$(complementary

$m + 48^0 - 48^0 = 90^0 - 48^0$

$m = 42^0$

Example 3

Find the value of angle t.



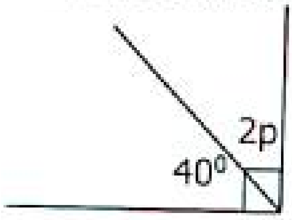
$t + 35^0 = 90^0$

$t + 35^0 - 35^0 = 90^0 - 35^0$

$t = 55^0$

Example 4

Find the value of p.



$2p + 40^0 = 90^0$

$2p + 40^0 - 40^0 = 90^0 - 40^0$

$\frac{\cancel{2}p^1}{\cancel{2}_1} = \frac{\cancel{50}^{0\,25}}{\cancel{2}_1}$

$p = 25^0$

Example 5

What is the complement of 38^0

Let the complement $= x$

$x + 38^0 = 90^0$

$x + 38^0 - 38^0 = 90^0 - 38^0$

$x = 52^0$

Example 6

The two angle of 30^0 and $3y^0$ are complementary angles. Find the value of y.

$3y + 30^0 = 90^0$

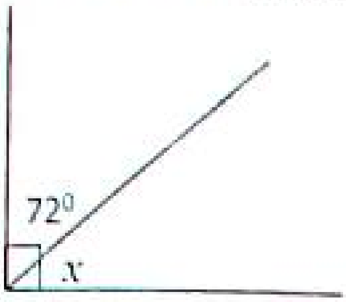
$3y + 30^0 - 30^0 = 90^0 - 30^0$

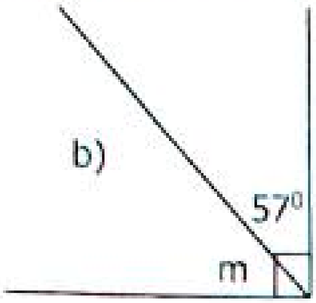
$\frac{\cancel{3}y^1}{\cancel{3}_1} = \frac{\cancel{60}^{20}}{\cancel{3}_1}$

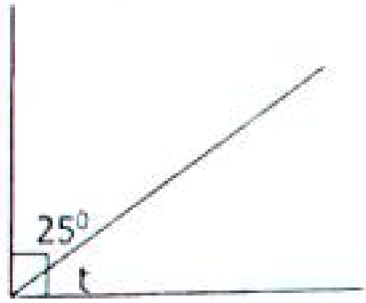
$y = 20^0$

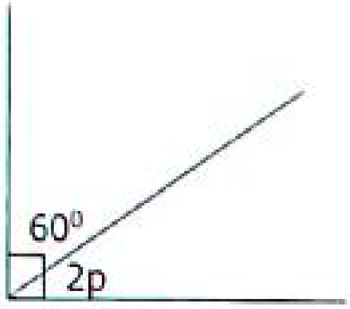
Activity 6:5

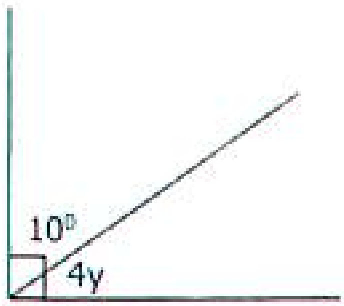
1. Calculate the values of the angles marked with letters below.

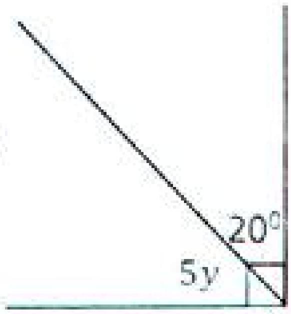
a)


b)


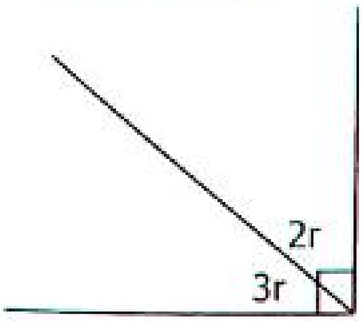
c)


d)


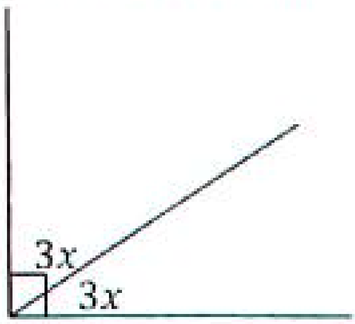
e)


f)


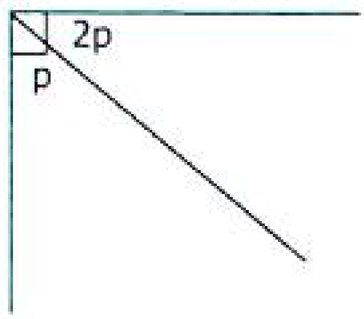
2. Find the value of r.



3. Find the value of x.



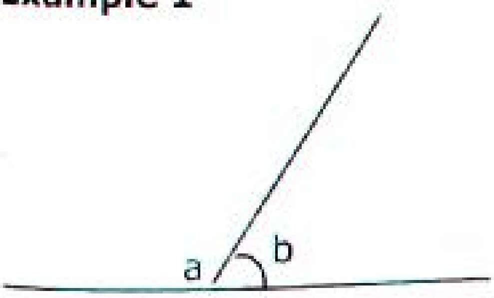
4. Find the value of p.



6.6 Supplementary Angles

Supplementary angles are two angles that add up to 180^0.

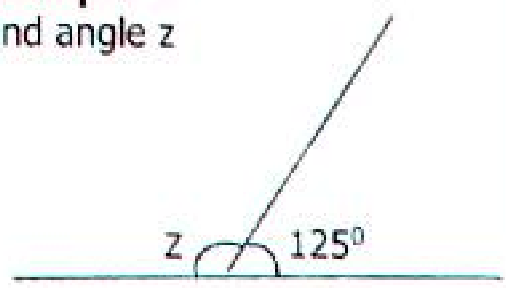
Example 1



$a + < b = 180^0$

Example 2

Find angle z

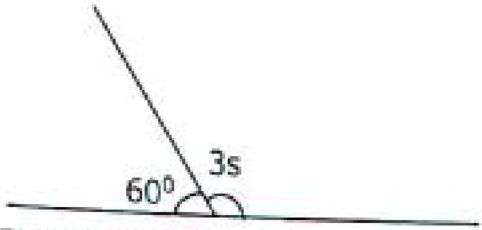


$Z + 125^0 = 180^0$

$Z + 125^0 - 125^0 = 180^0 - 125^0$

$Z = 55^0$

Example 3



Find the value of s.

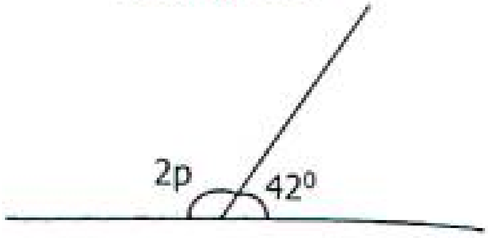
$3s+60^0=180^0$

$3s+60^0-60^0=180^0-60^0$

$\frac{\cancel{3s}^1}{\cancel{3}_1}=\frac{\cancel{120^0}^{40}}{\cancel{3}_1}$

$s=40^0$

Example 4



What is the value of p.

$2p+42^0=180^0$

$2p+42^0-42^0=180^0-42^0$

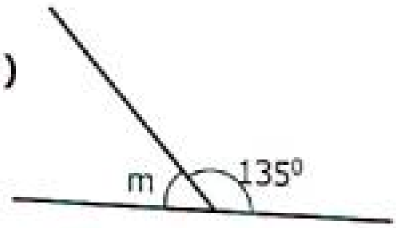
$\frac{\cancel{2p}^1}{\cancel{2}_1}=\frac{\cancel{138^0}^{69}}{\cancel{2}_1}$

$p=69^0$

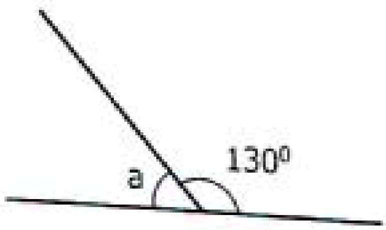
Activity 6:6

1. What is the supplementary of 75^0.
2. Workout the supplementary of 163^0.
3. Find the supplement of 115^0.
4. What is the supplement of 45^0?
5. Workout the value of the unknown letters.

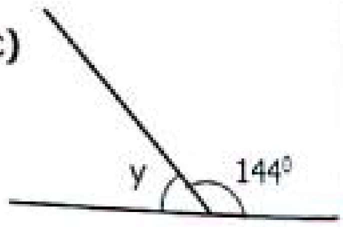
a)



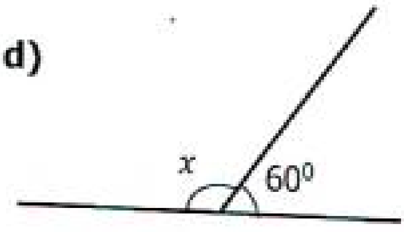
b)



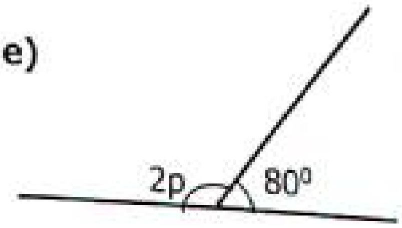
c)



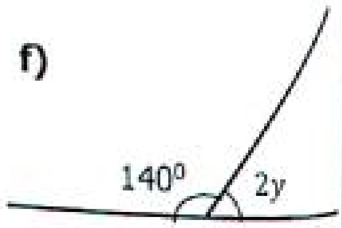
d)



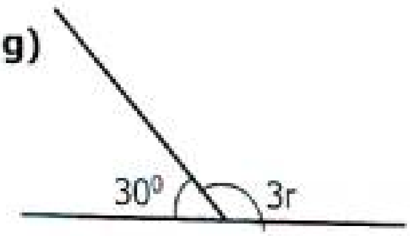
e)



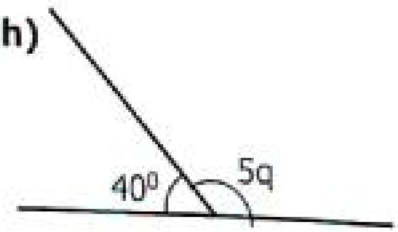
f)



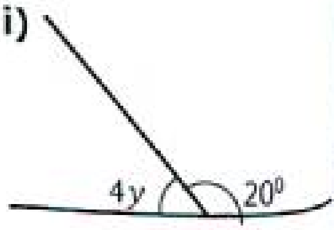
g)



h)



i)

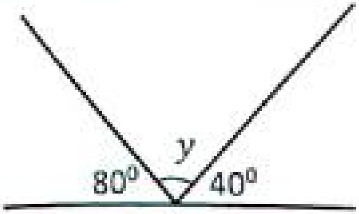


6.7 Angles on Straight Line

Angles on a straight line are two or more which add up to 180^0.

Example 1

What is the value of y?



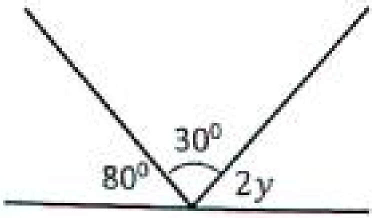
$y+80^0+40^0=180^0$

$y+120^0=180^0$

$y = 60^0$

Example 2

Find the value of y.



$2y+80^0+30^0=180^0$

$2y+110^0=180^0$

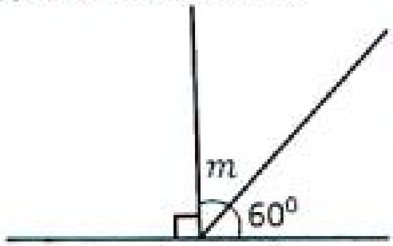
$2y+110^0-10^0=180^0-110^0$

$\frac{2y}{2} = \frac{70^0}{2}$

$y = 35^0$

Example 3

Find the value of m.



$m+90^0+60^0=180^0$

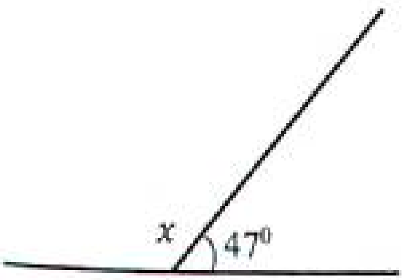
$m+150^0=180^0$

$m+150^0-150^0=180^0-150^0$

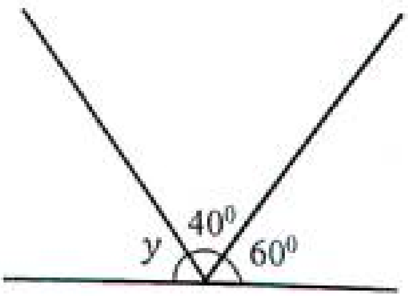
$m = 30^0$

Activity 6:7

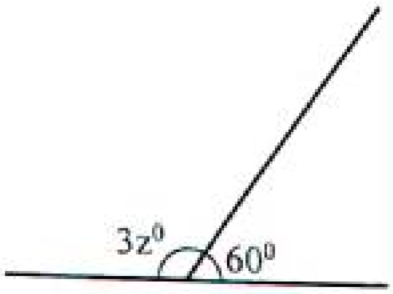
1) Find the value of x.



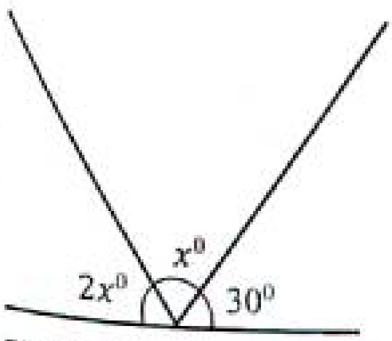
2) What is the value of y



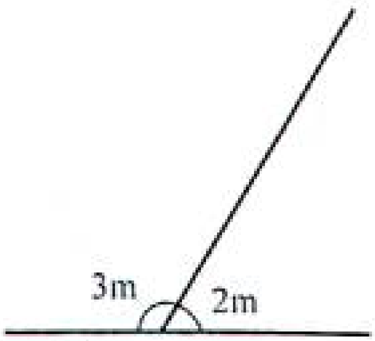
3) Find z.



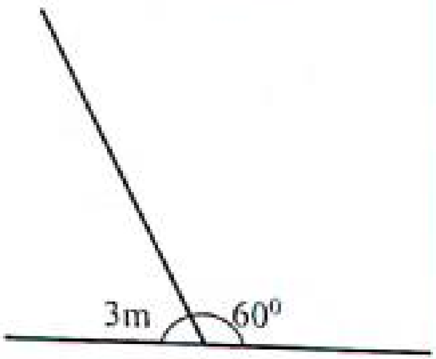
4) Workout the value of x.



5) Find x.

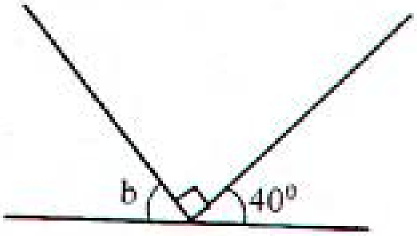


6) What is the value of w?

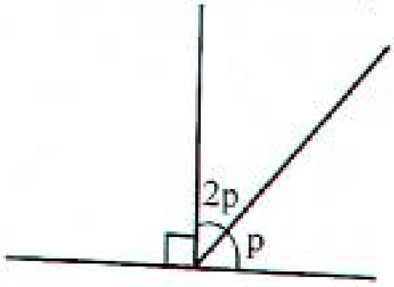


7) The two angles $3m^0$ and $6m^0$ are on a straight line. Find the value of m.

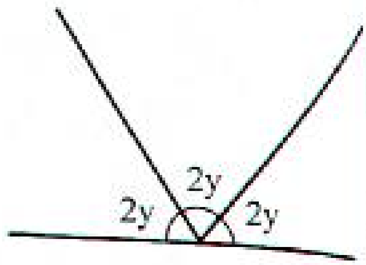
8) What is the value of b.



9) Find the value of p.



10) Find the value of y.



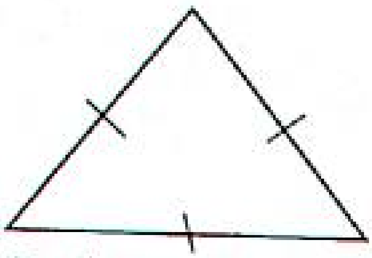
6.8 Polygons

A polygon is a flat or plane, two dimensional closed shape bounded with straight sides. Examples of polygons are Triangles, quadrilaterals, pentagon, hexagon, heptagon, octagon, nonagon, decagon, nuodecagon, duodecagon.

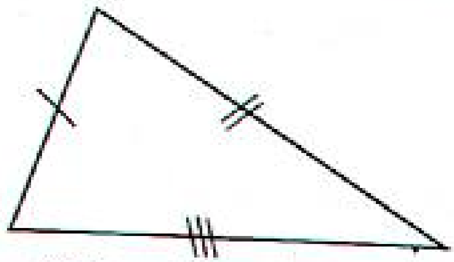
Triangles

A triangle is a three sided polygon.
There are four sided polygons.

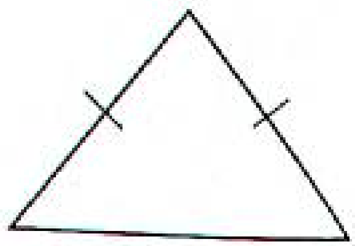
- Equilateral triangle
- Right angled triangle
- Isosceles triangles
- Scalene triangles



Equilateral triangle



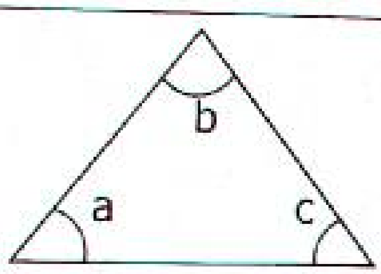
Scalene triangle



Isosceles triangle

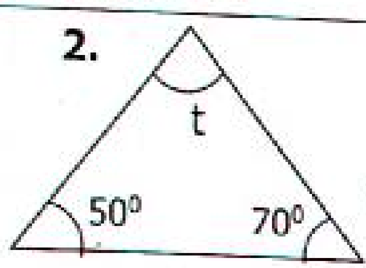
Interior Angles Sum of a Triangle is 180^0

1.



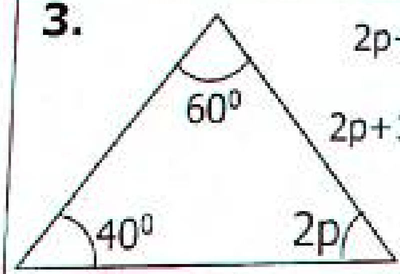
$< a + < b + < c = 180^0$

2.



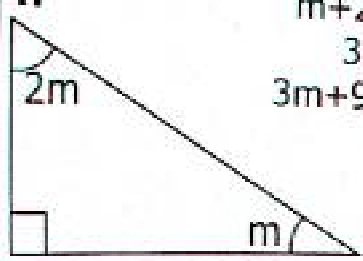
$t + 50^0 + 70^0 = 180^0$
$t + 120^0 = 180^0$
$t + 120^0 - 120^0 = 180^0 + 120^0$
$t = 60^0$

3.



$2p + 40^0 + 60^0 = 180^0$
$2p + 100^0 = 180^0$
$2p + 100^0 - 100^0 = 180^0 - 100^0$
$\frac{2p}{2} = \frac{80^0}{2}$
$p = 40^0$

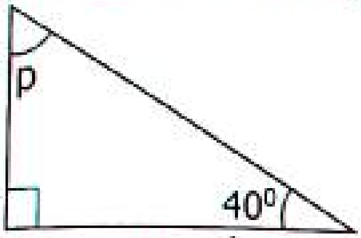
4.

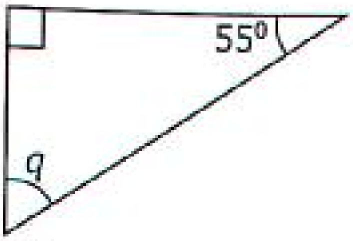


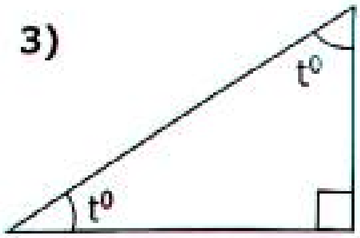
$m + 2m + 90^0 = 180^0$
$3m + 90^0 = 180^0$
$3m + 90^0 - 90^0 = 180^0 - 90^0$
$\frac{\cancel{3m}^1}{\cancel{3}_1} = \frac{\cancel{90^0}^{30}}{\cancel{3}_1}$
$m = 30^0$

Activity 6:8

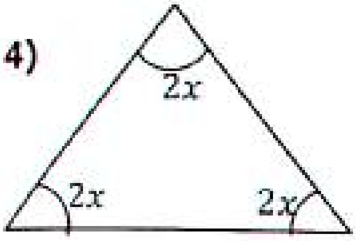
Find the size of the missing angle.

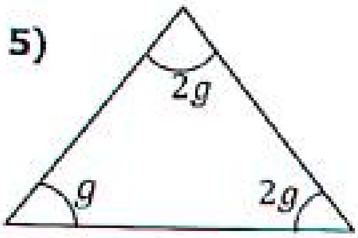
1)


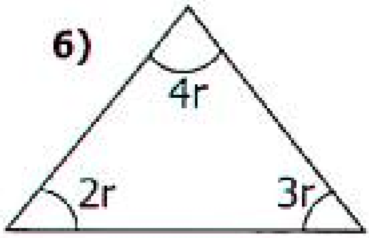
2)


3)


Find the value of the marked letters.

4)


5)


6)


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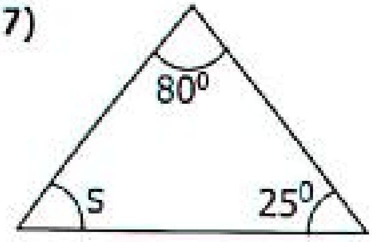

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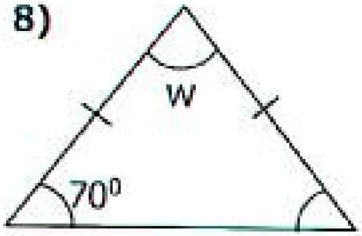
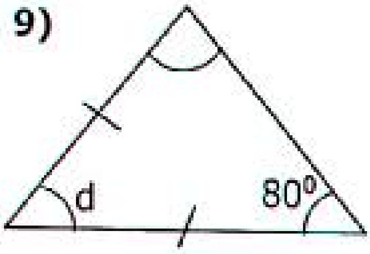


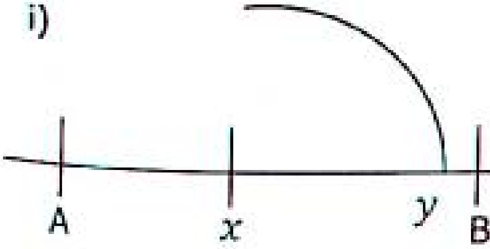
9)


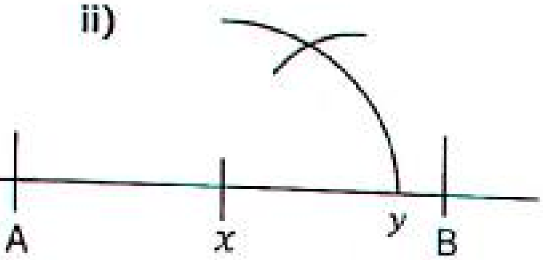
6.9 Constructing Angle 60^0 and 120^0

We need a pair of compasses, a ruler and a pencil.

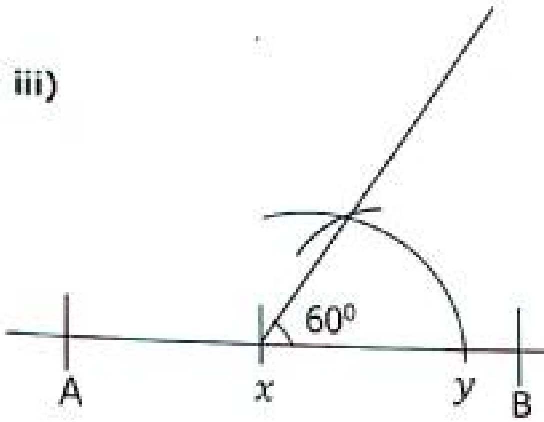
Procedure

- ✓ Draw a straight line AB and mark a point on the line segment at x.
- ✓ Place a compass needle at x and draw an arc to meet line AB at y.
- ✓ Take y as the centre to draw another arc to bisect the first arc.
- ✓ Draw a line through to complete angle 60^0.
- ✓ Using the same procedure, the opposite of angle 60^0 is 120^0

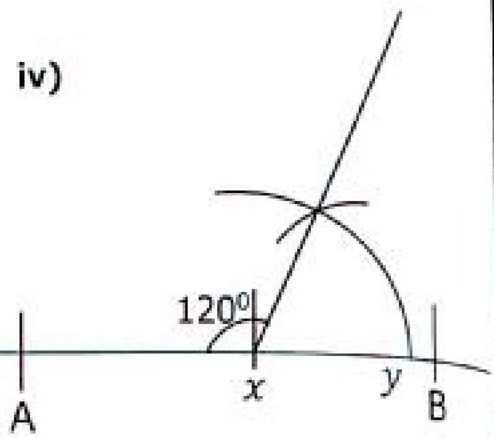
i)


ii)


iii)



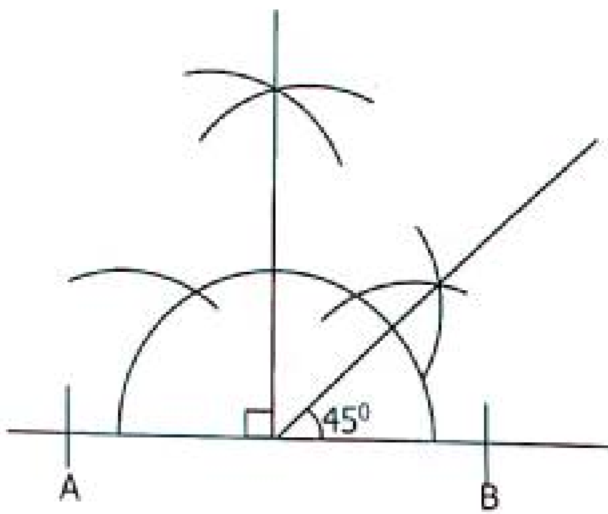
iv)

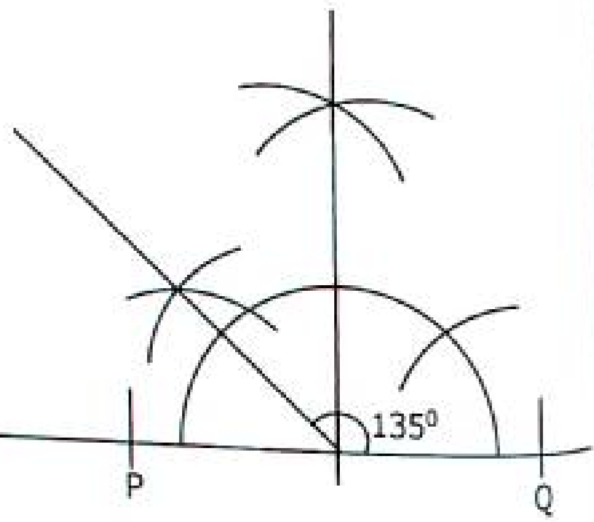


6.10 Construction of 90^0, 45^0 and 135^0

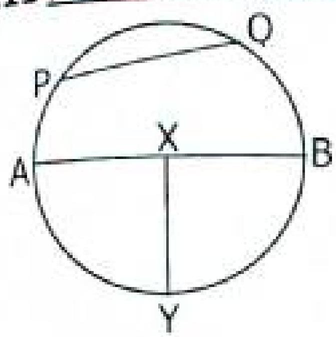
Procedure

- ✓ Use a ruler and a pair of compasses
- ✓ Construct angle 90^0
- ✓ Bisect angle 90^0 to get 45^0.
- ✓ Since angle 45^0 and angle 135^0 are supplementary.
- ✓ Angle $(90^0+45^0)=135^0$





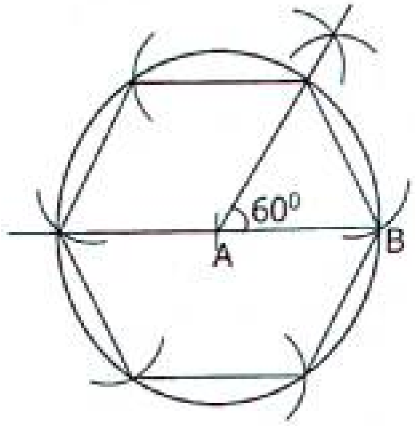
6.11 Parts of the circle



X is the center.
AB = Diameter
XY = Radius
PQ = Chord
Angle AXY = sector

Construction of a Hexagon

- ✓ Construct a hexagon of radius 3cm.
- ✓ Construct angle 60^0 on a base line.
- ✓ Draw a circle around it using a radius of 3cm



Activity 6:11

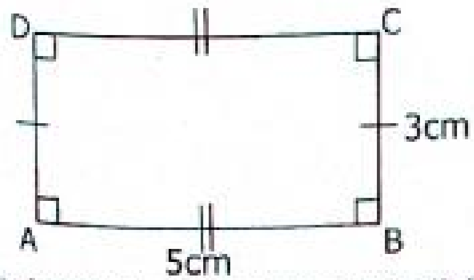
1. Use a ruler, a pair of compasses and a pencil to construct a regular hexagon of radius 4cm.
2. Use a ruler, a pencil and a pair of compasses only, construct a hexagon of radius 4cm using an angle of 60^0.
3. Construct a circle a hexagon using a radius of 2.8cm
4. Use angle of 60^0 to construct a hexagon of radius 3.2cm.

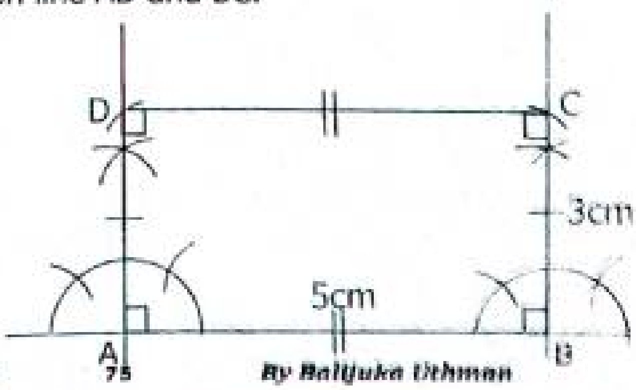
6.12 Construction of Rectangles

Using a ruler, a pencil and a pair of compasses only construct a rectangle ABCD where AB=5cm and BC=3cm (Measure the diagonal)

Procedure

- ✓ Sketch the diagram of a rectangle
- ✓ Draw a line AB=5cm
- ✓ Construct an angle of 90^0 at point A and B respectively.
- ✓ Measure the length 3cm and mark on line AD and BC.
- ✓ Use a ruler to join D to C.





Activity 6:12

1. Using a ruler, a pair of compasses and a pencil only, construct a rectangle whose length is 5cm and width 3cm.
2. With the help of a ruler, a pair of compasses and a pencil only, construct a rectangle whose length is 6cm and width 4.2cm.
3. With the help of a pencil, a ruler and pair of compasses only, construct a rectangle PQRS where line PQ=6cm and line QR=3.5cm.
4. Using a ruler, a pencil and a pair of compasses only construct a rectangle PQRS where PQ=5.5cm, QR=3.5cm. measure the length of the diagonal PR.
5. With the help of a ruler, a pencil and a pair of compasses only construct a rectangle BCDE where BC=4.8cm and BE=3.2cm. Measure the length of the diagonal LN.
6. With the help of a pair of compasses, a pencil and a ruler only, construct a rectangle PQRS where line PQ=6cm and line QR=4.5cm.

6.13 Construction of Squares

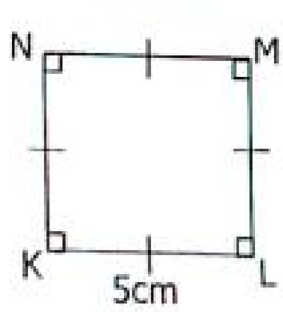
Example

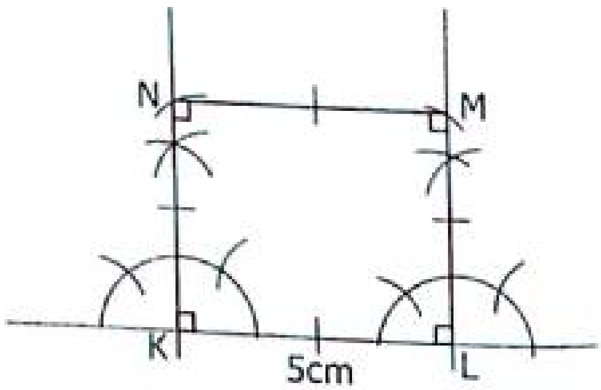
With the help of a pencil, a ruler and a pair of compasses only, construct a square KLMN whose sides measure 5cm.

Procedure

- ✓ Sketch the diagram of a square KLMN.
- ✓ Draw a line KL=5cm
- ✓ Construct an angle of 90^0 at point K and L respectively.
- ✓ Measure the length 5cm and mark on line KN and LM.
- ✓ Use a ruler to join M to N.

Sketch





Activity 6:13

1. With the help of a pencil, a ruler and a pair of compasses only, construct a square KLMN whose sides measure 5cm.
2. Use a sharp pencil, a ruler and a pair of compasses, construct a figure PQRS where PQ=QR=4cm.

3. With the help of a ruler, a pencil and a pair of compasses only, construct a square MNOP of sides 4.8cm each.
4. Construct a square ABCD where the sides measure 4.2cm. use a ruler, a pencil and a pair of compasses only.
5. Using a ruler, a pencil and a pair of compasses only, construct a square PQRS whose sides measure 5.3cm.
6. With the help of a ruler, a pencil and a pair of compasses only, construct a square whose side is 4.5cm.

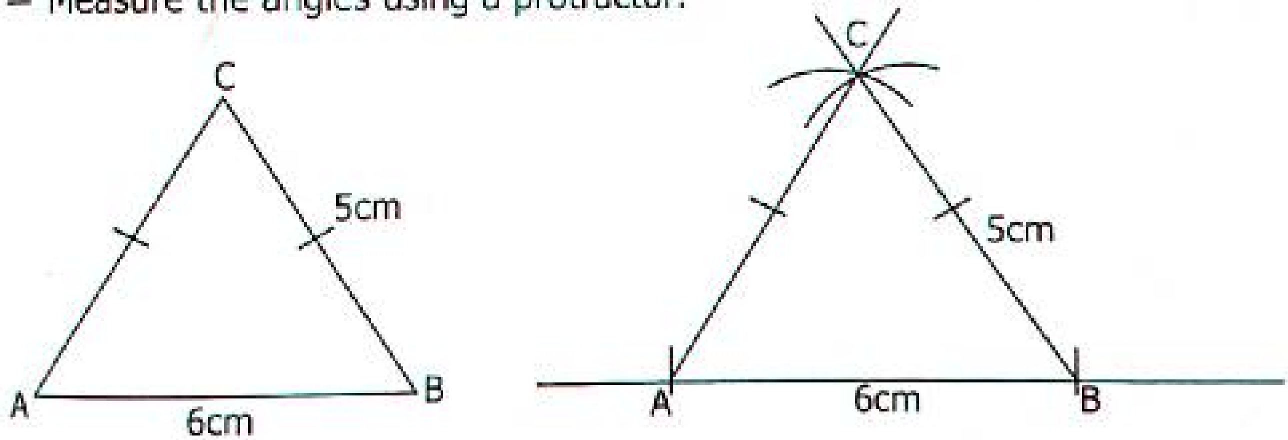
6.14 Constructing Triangles When Given the Three Sides (S.S.S)

Construct a triangle ABC where AB=6cm, BC=5cm and AC=5cm.

(i) Measure angle A or Angle BAC
Measure and B or Angle ABC

Procedure

- Sketch the triangle
- Draw a line a mark off AB=6cm
- Measure 5cm using a pair of compasses and aim at A to draw an arc above then aim at B to draw another arc to cut the first arc.
- Measure the angles using a protractor.



Activity 6:14

Use a ruler, a pencil and a pair of compasses only construct the following triangles.

1. Triangle ABC, where AB=6cm, BC=7cm, AC=6cm measure angles.
   (i) ABC (ii) CAB
2. Triangle PQR where PQ=6cm, PR=4cm, QR=5cm. Measure angles.
   (i) PQR (ii) RPQ
3. Triangle LMN, where LM=8cm, MN=7.2cm, LN=7cm. Measure angles.
   (i) LMN (ii) NLM

4. Triangle PQR where PQ=4.2cm, PR=5cm and QR=5.3cm. Measure angles (i) PQR (ii) RPQ.
5. Triangle XYZ where XY=4cm, YZ=5cm, XZ=6cm. Measure angle XYZ.
6. (a) With the help of a sharp pencil, a ruler and a pair of compasses only, construct an equilateral triangle MTN such that MT=TN=MN=6cm.
(b) Find the perimeter of the triangle above.

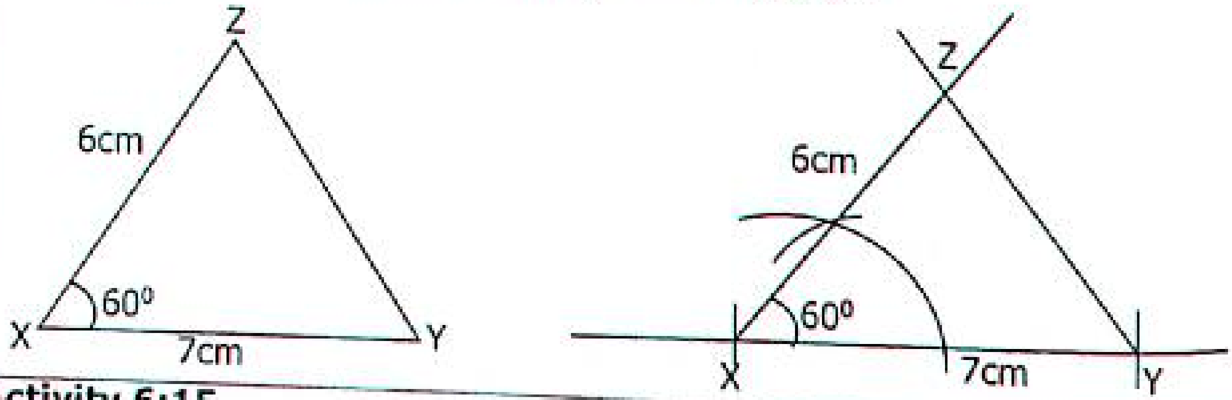
6.15 Construct Triangles When One Angle and Two Sides Given (SAS)

Example

Construct a triangle XYZ where XY=7cm, angle ZXY=60^0 AND XZ=5cm. Measure line YZ.

Procedure

- Prepare a pencil, ruler and a pair of compasses.
- Draw a sketch of a triangle XYZ.
- Draw a line and a mark off XY=7cm.
- Construct an angle of 60^0 at X.
- Measure the length 5cm from X to Z.
- Draw a line from Y to Z to complete the triangle.

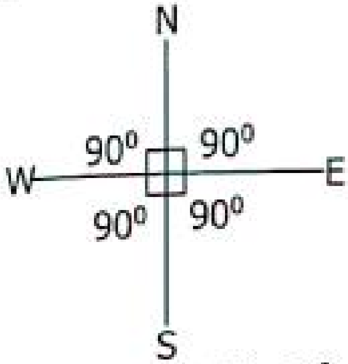


Activity 6:15

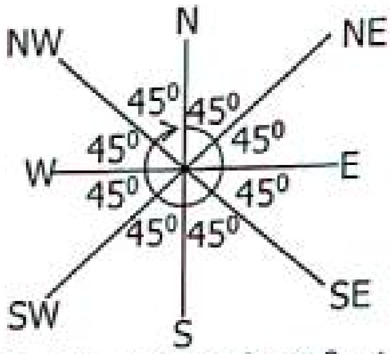
1. Using a pair of compasses, a ruler and a pencil, construct a triangle MOA such that MO=5cm, MA=6cm and angle AMO=90^0.
2.(a) with the help of a pencil, a ruler and a pair of compasses only, construct a triangle ABC where AB=7cm and BAC=90^0 and line AC=5cm.
(b) Measure line BC
3. (a) Use a ruler, a pencil and a pair of compasses only, construct a triangle PQR where PQ=5cm, angle PQR=90^0 and PR=6CM.
(b) Measure the length QR.

6.16 Compass Direction and the Angles Around or Magnetic Compass

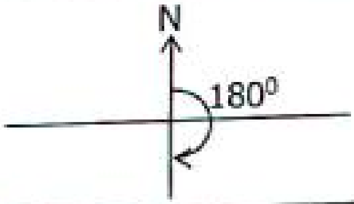
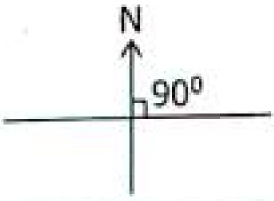
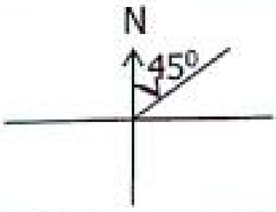
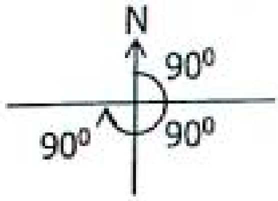
- A compass direction or magnetic compass is an instrument used to show direction.
- An angle is a measure of turn.
- Rotation is a clockwise turn through an angle of 360^0
- One revolution when using a magnetic compass makes 360^0.



$90^0+90^0+90^0+90^0$
$= 360^0$



$45^0+45^0+45^0+45^0+45^0+45^0+45^0+45^0$
$= 360^0$

A half turn	$= \left(\frac{360}{2}\right)^0 = 180^0$	
A quarter turn	$= \left(\frac{360}{4}\right) = 90^0$	
$\frac{1}{8}$ of a turn	$= \left(\frac{360}{8}\right) = 45^0$	
$\frac{3}{4}$ turn	$= \frac{3}{4_1} \times \cancel{360}^{90^0}$ $(3\times90)^0 = 270^0$	

A revolution has 4 quarters

Example 1
John made ½ turn clockwise direction. What angle did he turn?
$\frac{1}{2} \times 360^0$
$\frac{1}{\cancel{2}_1} \times \cancel{360}^{0\,180^0} = 180^0$
He turned 180^0

Example 2
If John turns $\frac{2}{3}$. What angle does he make?
$= \frac{2}{\cancel{3}_1} \times \cancel{360}^{0\,120}$
$= 2 \times 120$
$= 240^0$
He turned 240^0

Example 3
What turn is made in an angle of 40^0?
$360^0 = 1$ full turn
$40^0 = \frac{1}{\cancel{360}_9} \times \cancel{40}^1$
$= \frac{1}{9}$ *of a turn*

Example 4
Find the turn made by an angle of 240^0.
$360^0 = 1$ complete turn
$240^0 = \frac{1}{\cancel{360}_9} \times \cancel{240}^4$
$240^0 = \frac{4}{9}$ *of a turn.*

Activity 6:16
Work out the following
1. How many degrees are in $\frac{1}{10}$ turn?
2. How many degrees make up $\frac{5}{6}$ turn?
3. Work out the angle represented by $\frac{3}{4}$ turn.
4. Find the angle represented by $\frac{2}{5}$ turn.
5. How many degrees are in $\frac{3}{8}$ turn?
6. Workout the angle represented by $\frac{1}{4}$ turn.
7. How many degrees are in $\frac{5}{8}$ turn?
8. How many degrees make up $\frac{7}{12}$ turn?

Work out the fraction represented by the angles below;
a) 120^0 b) 90^0 c) 45^0 d) 60^0 e) 150^0 f) 180^0 g) 270^0 h) 240^0

6.17 <u>Angles Made by a Minute Hand</u>
A minute hand covers 60minutes to make a complete turn.
One complete turn is equal to 360^0 making 60 minutes.
How many degrees make up a minute?
60 minutes $\rightarrow 360^0$
1 minute $\rightarrow \left(\frac{360}{60}\right)^0$
Therefore 1 minute $\rightarrow 6^0$

Example 1

Calculate the angle made by the minute hand in the diagram.

15minutes

1minute $\rightarrow 6^0$

15minutes $\rightarrow (15\times6)^0$

15minutes $\rightarrow 90^0$

Example 2

Workout the angle made by the minute hand on the clock face below.

40minutes

1minute $\rightarrow 6^0$

40minutes $\rightarrow (40\times6)^0$

40minutes $\rightarrow 240^0$

Activity 6:17

Calculate the value of the angle represented by the minute hand on the clock face.

1. 

2. 

3. 

4. 

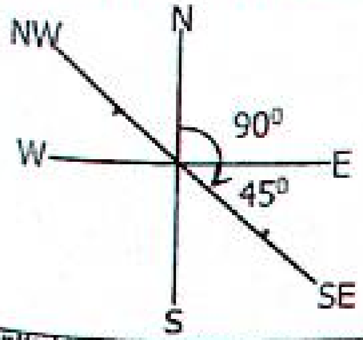
5. 

6. 

6.18 Finding the Angles on a Compass

Example 1

Find the value of the angle between North and south East clockwise direction.

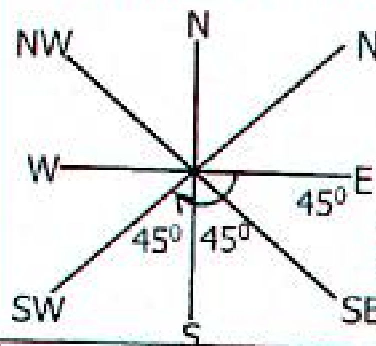


90^0+45^0

135^0

Example 2

Find the angle between East and South West clockwise direction.



$= 45^0+45^0+45^0$

$= 135^0$

Activity 6:18

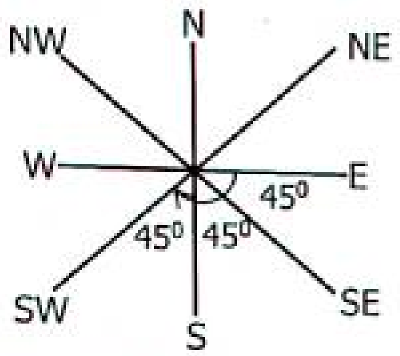
Workout the angles formed by the given directions.

1. From North to East clockwise.
2. From North to West clockwise.
3. From South East to West clockwise.
4. From East to North clockwise.
5. From North West to East clockwise.
6. From North to West anti-clockwise.
7. From Southwest to South East anti-clockwise direction.
8. From South East to North anti-clockwise direction.

6.19 Clockwise and Anti-clockwise Direction

Example 1

Joseph was facing in the East, he turned clockwise direction to face southwest. What angle did he turn through?



Clockwise direction

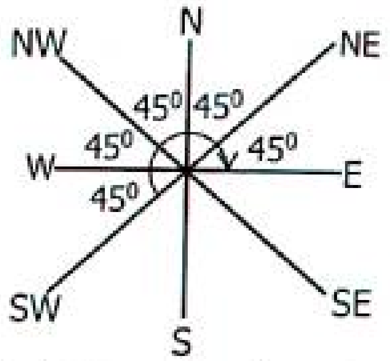
$45^0+45^0+45^0$

135^0

Joseph turned an angle of 135^0

Example 2

Opio was facing South west. He turned Anticlockwise direction to face East. What angle did he turn through?



Anti-clockwise direction

$45^0+45^0+45^0+45^0+45^0$

$90^0+90^0+45^0$

Opio turned an angle of 225^0

Activity 6:19

Workout the numbers below;

1. Through what angle does one turn from North East to South East?
2. Through what angle does one turn from South West to North?
3. Omario was facing North West, he turned clockwise direction to South East. What angle did he turn through?
4. Byaruhanga faced North East, he turned to West clockwise direction. What angle did he turn through?
5. Kabatooro was facing North, she turned anticlockwise direction to West. What angle did he turn through?
6. What angle did Jacob turn through from South to North West anti-clockwise direction?
7. A man was facing West. He turned an angle of 135^0 in a clockwise direction. Where is he facing now?
8. In what direction will Apollo be facing if he turned 225^0 from South anticlockwise?

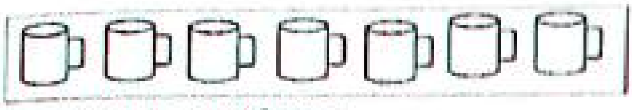
TOPIC 7: INTERPRETATION OF DATA AND GRAPHS

7.1 Pictograph

A Pictograph can be also called picture graph. It's called picture graph because the graphs are represented by pictures i.e trees, balls, apples, people, animals, oranges etc.

Example 1

Scale  represents 10cups. How many cups are represented below?



1cup represents 10cups

7cups represent (10×7)cups

7cups represent 70cups

Example 2

The graph below shows the number of balls picked by six pupils in primary five.

Name of pupils	Balls picked
Joseph	
Ibrahim	
Suzan	
Amos	
Abdallah	

Given that  = 20 balls

a) How many balls did Suzan pick?

1 ball →20 balls

7 balls → (20×7) balls

7 balls → 140 balls

Suzan picked 140 balls

b) How many balls did Amos pick?

1 ball → 20 balls

5 balls → (20×5) balls

5 balls → 100 balls

Amos picked 100 balls

c) Find the total number of balls picked. by Joesph, Suzan and Abdallah.

1 ball → 20 balls

14 balls → (20×14) balls

14 balls → 280 balls

d) Calculate the total number of balls picked altogether.

1 ball → 20 balls

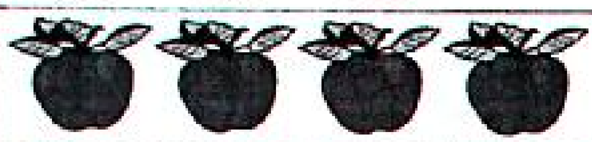
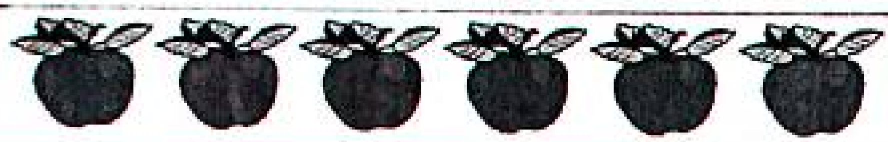
22 balls → (20×22) balls

22 balls → 440 balls

TOPIC 7: Interpretation of Data and Graphs

Activity 7:1

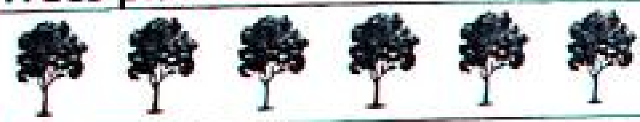
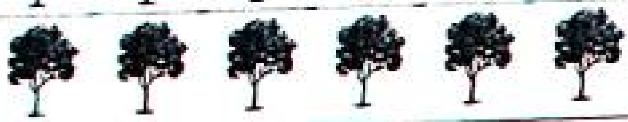
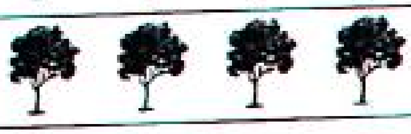
1. The pictograph below shows the number of apples harvested at Mr. Odongo's farm. Study it and answer questions that follow.

Month	Apples harvested
June	
July	
August	
September	
October	
November	

Scale  = 80 apples

a) How many apples were harvested in August?
b) How many apples were harvested in July?
c) When did Mr. Odong harvest the same number of apples?
d) How many more apples were harvested in July than in October?
e) When did Odong harvest the least number of apples?
f) What type of graph is this?
g) What is the graph about?

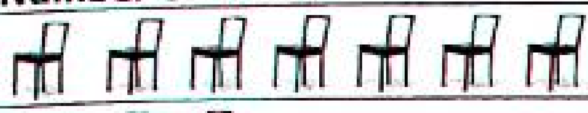
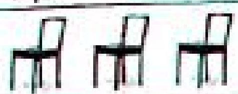
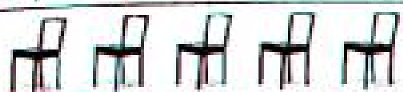
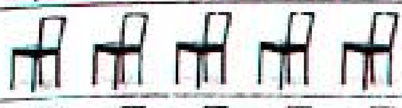
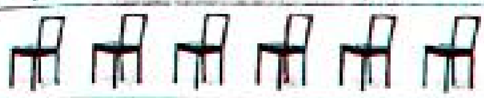
2. The pictograph below shows the number of trees planted in the first six months of the year.

Months	Trees planted
January	
February	
March	
April	
May	
June	

Scale  = 15 trees

a) What is the graph about?
b) How many trees were planted in March?
c) When was the least number of trees planted?
d) Find the total number of trees planted in the last three months.
e) Calculate the total number of trees planted in the six months.

3. The pictograph below sjows the number of chairs supplied to different school a by the district Education Officer.

School Name	Number of chairs
Motherwell Junior School	
Namirembe Parents School	
Sir Apolllo Kaggwa Mengo	
Nakasero Primary school	
Kyaggwe Road Primary School	

Scale  = 8 chairs

a) How many chairs did Kaggwe Road P/s get?
b) Which school got the biggest number of chairs?
c) How many chairs did two schools Sir Apollo primary school and Namirembe parents school get?
d) Calculate the total number of chairs supplied to the five schools.

4. The table below shows the number of stars counted by different children.

Name of the Child	Stars counted
Barungi Ismail	☆☆☆☆☆
Kirungi Majidu	☆☆☆☆☆☆☆
Karungi Saidah	☆☆☆☆☆☆
Birungi Sudat	☆☆☆☆☆☆☆☆☆
Murungi Imran	☆☆☆
Kanyunyuzi	☆☆☆☆☆

Scale ☆ = 50stars

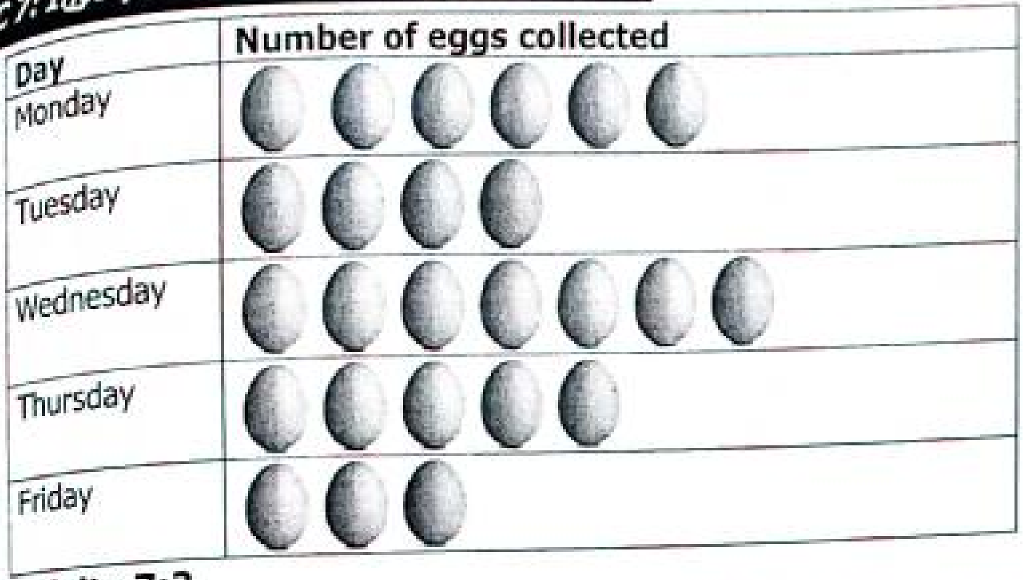
a) How many stars did Karungi Saidah count?
b) Name the child who counted the highest number of stars.
c) Find the total number of stars counted by Saidah and Imran.
d) How many stars did Kirundi Majidu count?

7.2 Drawing Pictographs

Example

Draw a pictograph and show the number of eggs collected by Miss Jazira in 5 days. Monday 30 eggs, Tuesday 20 eggs, Wednesday 35eggs, Thursday 25 eggs and 15 eggs on Friday. Use a scale = 5 days.

Day	Number of eggs	Representation
Monday	30	$\frac{30}{5} = 6$
Tuesday	20	$\frac{20}{5} = 4$
Wednesday	35	$^{35}/_{5} = 7$
Thursday	25	$^{25}/_{5} = 5$
Friday	15	$^{15}/_{5} = 3$



Activity 7:2

1. Draw a pictogram showing the number of patients who visited different hospitals in Kampala, Kawempe in a week. Mulago hospital 450 patients, Kawempe 350, Kiruddu 200 and Nagulu 400 patients.

Scale  = 50 patients

2. Draw a pictograph and show the number of rabbits kept by different schools in Lira town. Central primary school 70 rabbits, St. Kizito primary school 40 rabbits, Lira town school 30 and City Junior school 25 rabbits.

Scale  = 10 rabbits

3. Uganda Taxi operators and Drivers' Association recorded passengers who travelled from Kampala to Mbale using taxis as follows. January 80, February 120, March 200, April 160, May 100 and June 150. Use the information to draw a pictograph.

7.3 Tally Charts

A tally chart is a simple way of recording and counting frequencies. We draw tally marks and use them to count in groups of five.

Number	3	4	5	6	7	12	14	19
Tallies	𝍷𝍷𝍷	𝍷𝍷𝍷𝍷	𝍸	𝍸 𝍷	𝍸 𝍷𝍷	𝍸 𝍸 𝍷𝍷	𝍸 𝍸 𝍷𝍷𝍷𝍷	𝍸 𝍸 𝍸 𝍷𝍷𝍷𝍷

Activity 7:3

1. The tally chart shows the food items sold during the weekend.

Food items	Tallies	Number of people
Ice cream	卌 卌 卌 |||	____
Burger	卌 卌 ||	____
Fried rice	卌 卌 卌 ||||	____
Pizza	卌 卌 卌 |	____
Cake	卌 卌 卌 卌 ||	____

a) How many people bought fried rice on the weekend?
b) How many people ate pizza?
c) Find the total number of people who enjoyed ice cream and cakes.
d) How many more people ate ice cream than burger?

2. The table below shows the number of pupils who play different games.

Soccer	Tennis	Football	Baseball	Basket ball
15	23	17	34	29

Complete the tally table to represent the information above

Game	Number	Tallies
Soccer	15	____
Tennis	23	____
Food ball	17	____
Base ball	34	____
Basket ball	29	____

3. The tally chart shows primary five pupils who like different subjects.

Subject	Number of pupils
English	卌 卌 卌 ||||
Religious education	卌 卌 ||
Mathematics	卌 卌 卌 卌 |||
Science	卌 卌 卌 卌
Social studies	卌 卌 |||
Computer science	卌 |||

a) How many pupils like Mathematics?
b) How many pupils like science?
c) Find the total number of pupils who like Religious education?
d) Calculate the total number of pupils in primary five.

4. The tally chart below represents the number of votes got by different people.
a) Fill the blank spaces in chart

Name	Tally marks	Number of votes
Emmy	卌 ///	8
Amelia	______	10
David	______	9
Gary	卌 //	______
Ham	卌 卌 ////	______

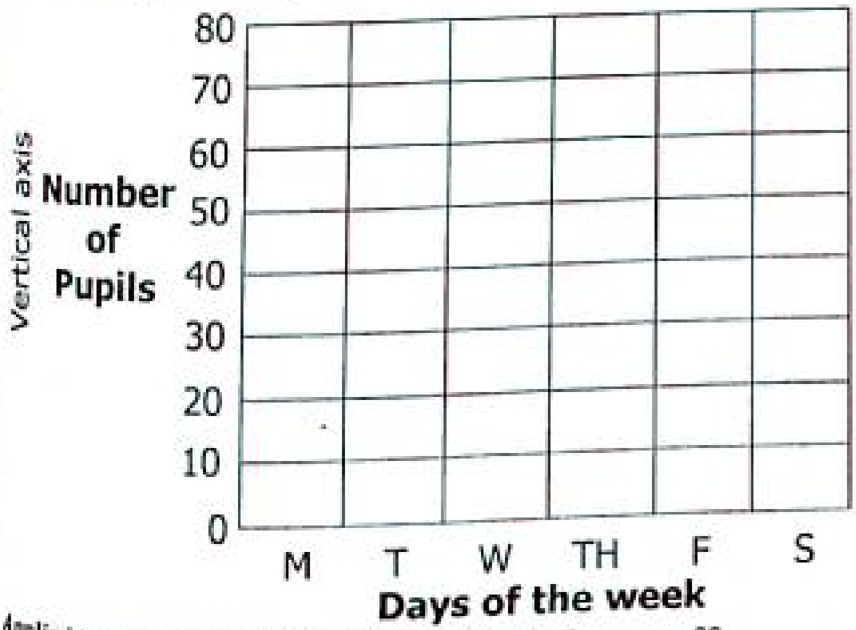
b) How many votes were cast altogether?

5. Primary five class scored the following marks in a mathematics test.
62, 40, 60, 63, 62, 80, 53, 60, 90, 82, 80, 62, 63, 53, 90, 82, 80, 40, 63, 53, 60, 62, 40, 80, 63, 40, 63, 60, 40, 60, 80, 53, 90, 63
Represent the information above on the tally chart.

Marks	Frequency	Tally
40	______	______
53	______	______
60	______	______
62	______	______
63	______	______
80	______	______
82	______	______
90	______	______

7.4 Bar graphs

Finding the Horizontal Scale and Vertical Scale (axes)



✓ Horizontal axis is the line running from left to right, in the above graph, it represents days of the week.
✓ The vertical axis shows the number of pupils.
✓ Finding the scale on the vertical axis

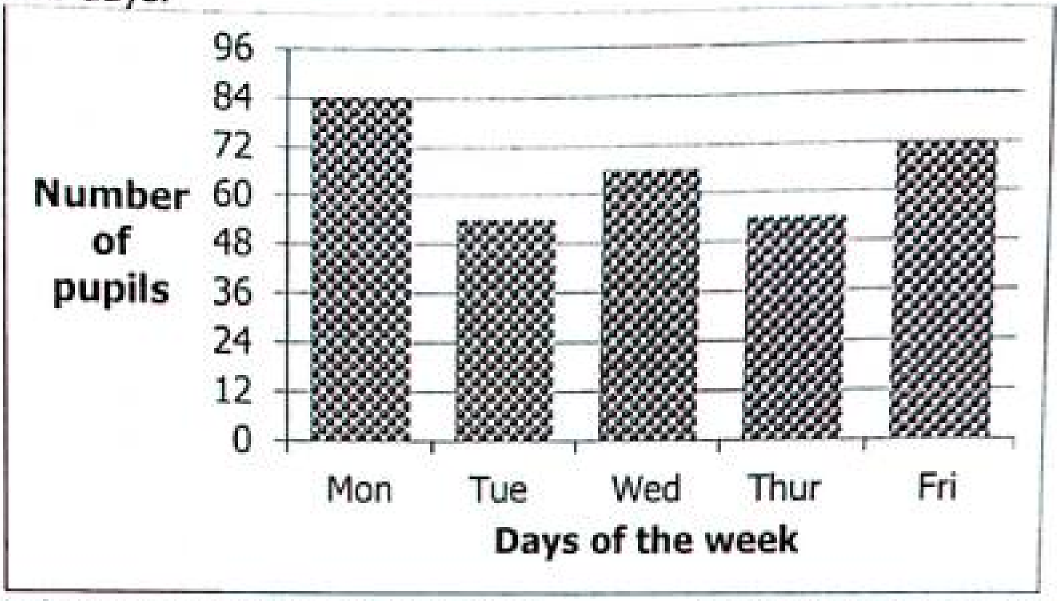
Scale = $\left(\frac{80-20}{6}\right)$

$= \left(\frac{60}{6}\right)$

1square represents 10 pupil

1 square = 1 day

Study the bar graph below and answer questions that follow;
The graph below shows the number of pupils who attended classes in the five days.



a) What is the scale on the vertical axis?

2 squares represent 12 pupils

1 square represent $\left(\frac{12}{2}\right)$ pupils

1 square represent 6 pupils

b) What is the scale on the horizontal axis?

1 square represent 1 day

1 square = 1day

c) How many pupils attended on Tuesday?

(48+12) pupils

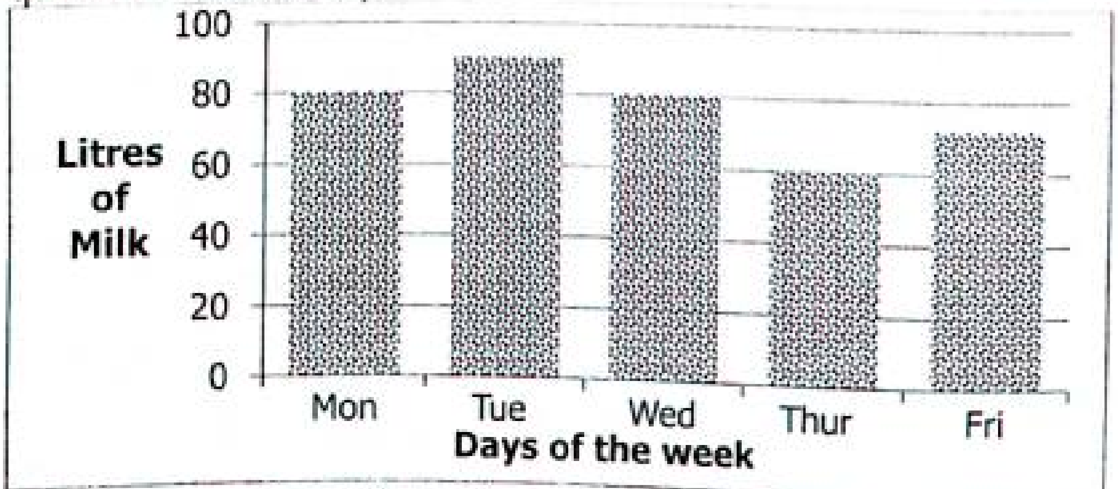
60 pupils attended on Tuesday

d) Which two days had the same number of attendance?

Tuesday and Thursday had the same attendance

Activity 7:4

1. The graph below shows the litres of milk sold in a week. Use it to answer questions that follow;



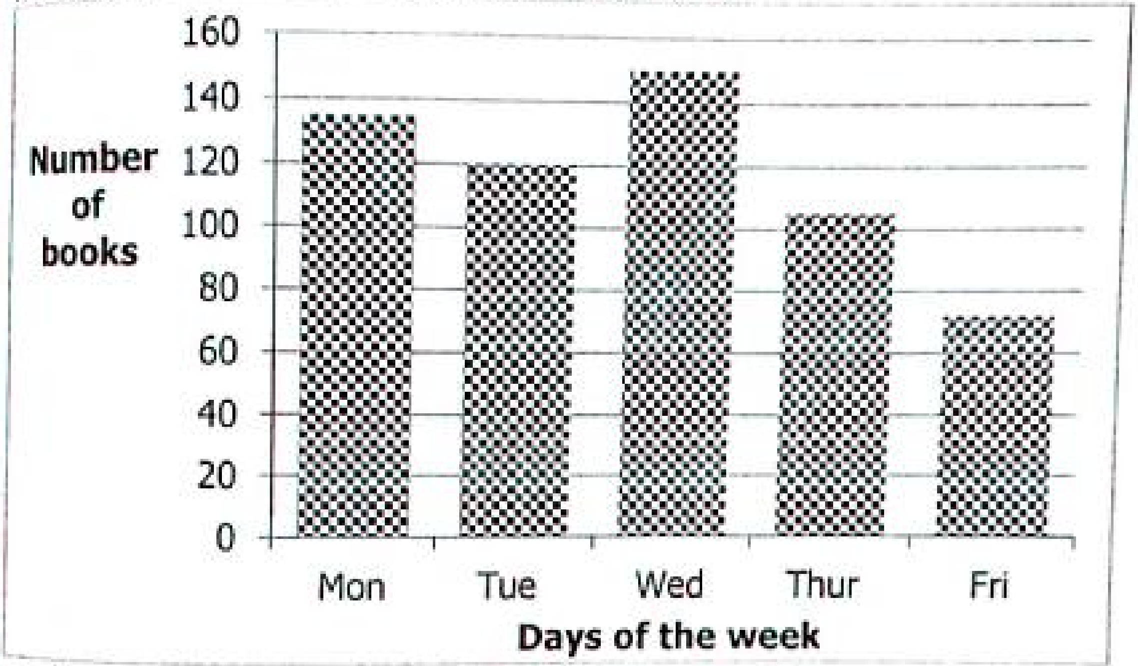
a) How many litres of milk were sold on Friday?

b) How many litres of milk were sold on Tuesday and Thursday?

c) On which day was the highest number of litres of milk sold?

d) Which two days of the week had the same number of litres sold?

2. Use the bar graph below and answer the questions that follow. The graph is about the number of books sold in a week.

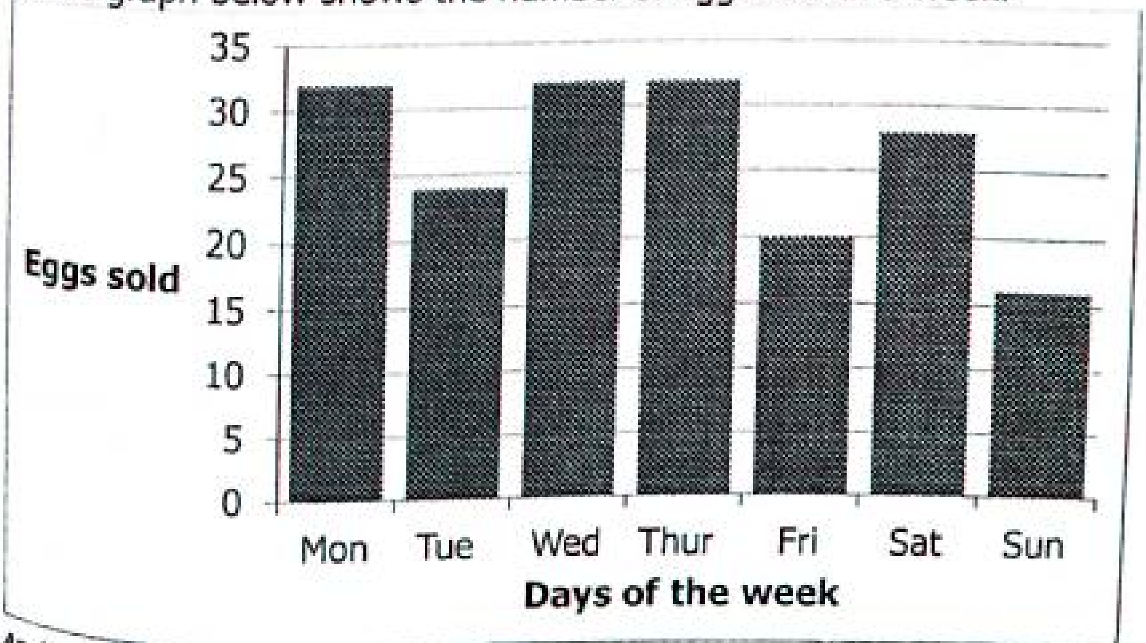


a) How many books were sold on Tuesday?

b) How many more books were sold on Wednesday than Friday?

c) Find the total number of books sold during the week.

3. The graph below shows the number of eggs sold in a week.



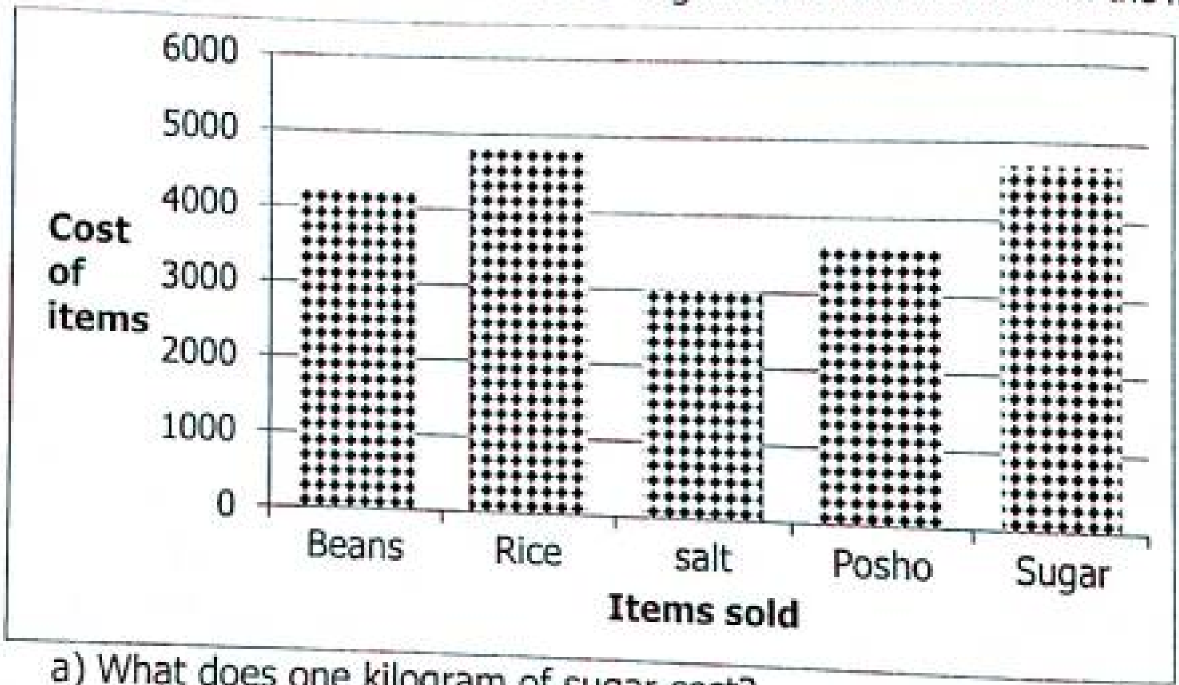
a) What does one small square represent on the vertical axis?

b) How many eggs were sold on Wednesday?

c) How many eggs were sold on Thursday?

d) Calculate the total number of eggs sold on Monday, Tuesday and Wednesday.

4. The graph below shows the cost of 1kg for different items from the market.



a) What does one kilogram of sugar cost?

b) Find the cost of a kilogram of salt.

c) Find the cost of 20kgs of beans.

d) Work out the cost of 3kg of posho.

e) If Mr. Lule wants to buy 5kg of rice, how much money can he pay?

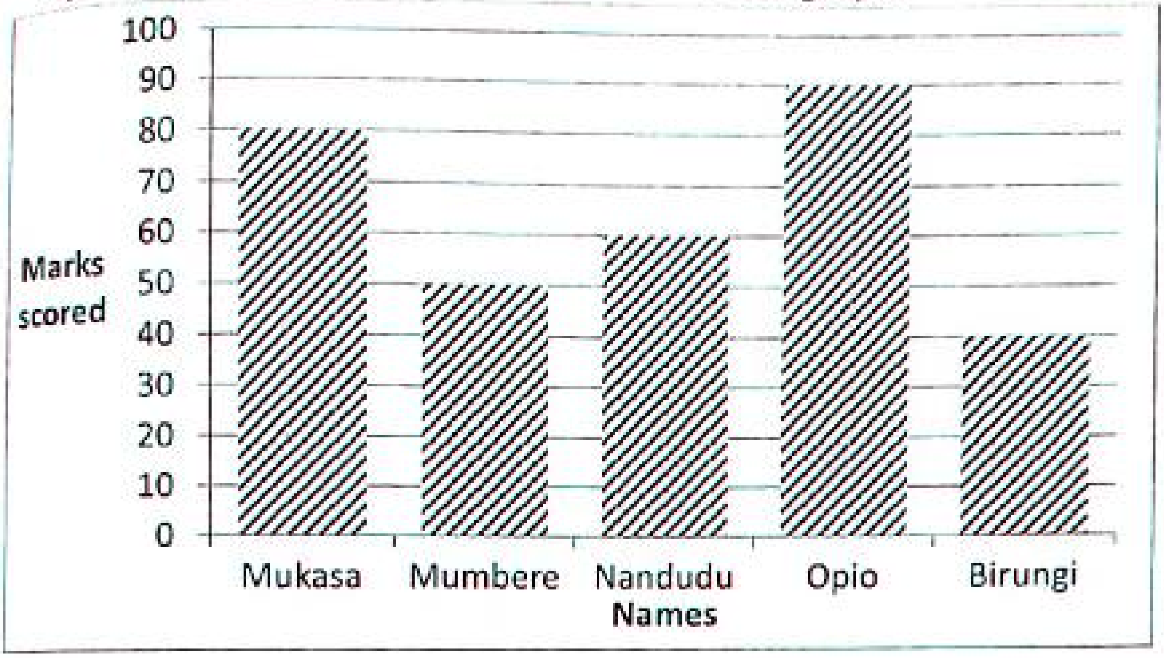
7.5 Representing Information on a Bar Graph

Example 1

The table below shows the marks scored by different pupils in a mathematics test.

Mukasa	Mumbere	Nandudu	Opio	Birungi
80	50	60	90	40

a) Represent the above information on a bar graph.



b) Who got the highest marks in mathematics?
Opio got the highest marks

c) Calculate the range.
Range = Highest – Lowest/(90 – 40)marks

d) Calculate the scale on the vertical axis.

$$\text{Scale} = \left(\frac{60-40}{2}\right) marks$$

$$= \left(\frac{20}{2}\right) marks$$

1 square = 10marks

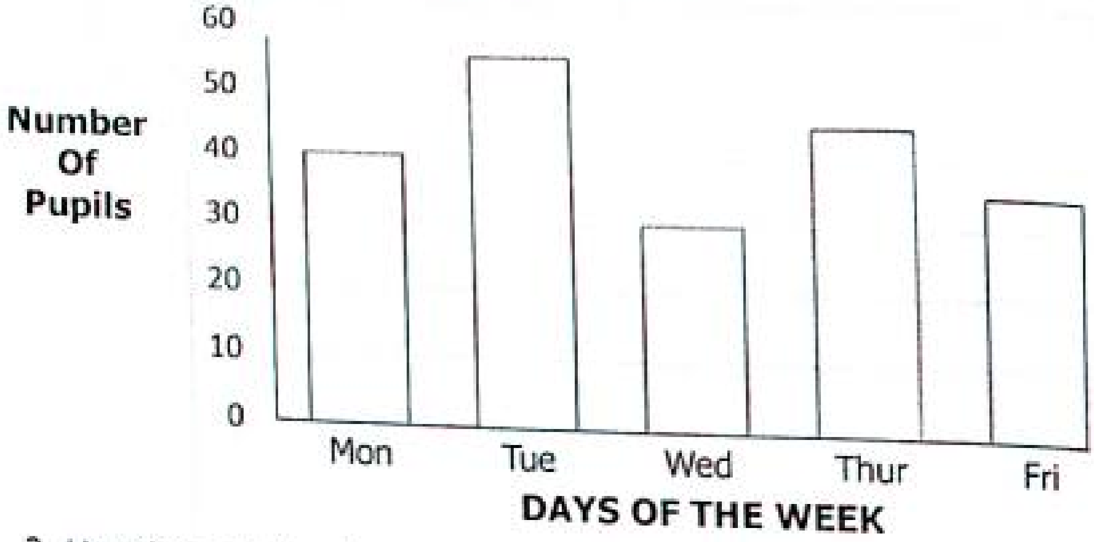
e) Calculate the total marks scored by all the five.

Total = (80+50+60+90+40)marks
(130+150+40)marks
320marks

Activity 7:5

1. The table below shows daily attendance for a week. Use the information to draw a bar graph.

Day	Monday	Tuesday	Wednesday	Thursday	Friday
Number of pupils	40	55	30	45	35



2. Use the table to draw a bar graph.

Days of the week	Mon	Tue	Wed	Thu	Fri	Sat
Number of apples sold	35	40	50	45	35	25

Scale 1:5 apples

3. The table below shows the number of oranges distributed to 5 classes.

Class	P.1	P.2	P.3	P.4	P.5
Number of pupils	90	40	55	60	80

Scale 1:10 pupils

4. The table below shows the number of pupils in primary five who attended classes on Friday.

Class	P.1	P.2	P.3	P.4	P.5	P.6
Number of pupils	35	45	50	30	45	25

Take the scale 1square = 5pupils on vertical axis.

a) Represent the above information on a bar graph.
b) Name the two classes which had the same number of pupils.
c) Find the total attendance that day.

5. The table below shows Odur's mid-term performance.

Subject	English	Science	SST	Mathematics
Score	85	70	95	90

a) How many subjects did Odur write?
b) Which subject did Odur get the highest score?
c) Workout the range of his score.
d) Find the mark in all the subjects.

7.6 Finding the mode and Modal Frequency

Mode is the number which appears most in the given data.

Example 1

Find the mode of 7.

Number	3	4	5	6	7
Frequency	1	1	1	1	3

The mode is 7.

Example 2

What is the mode 72, 80, 72, 90, 72, 80, 72, 68, 72, 65

Mark	65	68	72	80	90
Frequency	1	1	5	2	1

Mode = 72

Example 3

What is the modal frequency of 4,3,5,6,4,3,2,4,5,3,4,5,4,6

Mark	2	3	4	5	6
Frequency	1	3	5	3	2

Modal frequency is 5

Example 4

Kanyunyuzi scored the following marks in English home work; 4, 6, 9, 8, 7, 6, 5. What was her model mark?

Mark	4	5	6	7	8	9
Frequency	1	1	2	1	1	1

Model mark = 6

Activity 7:6

1. Obua scored the following marks in home work exercises 3, 2, 8, 5, 11, 7, 4, and 10.
   (a) Find his mode (b) Find his model frequency
2. Ali scored 5, 7, 6, 9, 4, 8, 6 in daily exercises last week.
   (a) What was his model mark? (b) Find his model frequency
3. Below are the marks scored by the pupils in a mathematics test; 70, 40, 30, 50, 40, 30, 40, 70, 40 and 70.
   (a) How many pupils did test? (b) Find the modal mark.
   (c) What was their model frequency?

7.7 Average/Mean

Average is the number you get after dividing the total scores by the number of scores.

$$Average = \frac{total\ scores}{number\ of\ score} \quad \text{OR} \quad \frac{sum\ of\ score}{number\ of\ scores}$$

Example 1	Example 2
Find the average of 8, 7, 9, 10, 6.	Calculate the average of 20, 16, 24 and 4.
$Average = \frac{sum}{no. of\ items}$	$Average = \frac{sum}{no. of\ scores}$
$Average = \frac{8+7+9+10+6}{5}$	$Average = \frac{20+16+24+4}{4}$
$Average = \left(\frac{40}{5}\right)$	$Average = \left(\frac{64}{4}\right)$
Average = 8	Average = 16

Activity 7:7

1. Find the average 2, 4, 8, 10 and 6.
2. Calculate the average of 3, 7, 5, 1 and 9.
3. Joseph scored the following marks in weekend home work; 4, 5, 6, 4, 7 and 4. Find his average mark.
4. Joan scored the following marks in End of term exams; 90, 80, 93 and 85. Calculate his average score.
5. Ali scored the following marks in End of year exams; 90, 94, 96, 97 and 93. Calculate his average mark.
6. Workout the average of 16, 32, 8 and 4.
7. A fruit seller sold the following number of apples in 6days 42, 40, 60, 35, 28, 35. Workout the mean number of apples sold.
8. Find the mean of 8, 10, 4, 1, 6 and 7.
9. Seven children had the following ages; 2, 3, 7, 6, 5, 1 and 4. Find the mean age.
10. The mean age of 6pupils is 13years. Find their total age.
11. Find the total mass of 8 students whose average mass is 47kg.
12. What is the average mark of 5pupils is 82marks. Find their total marks.
13. Abooki scored the following marks in the end of term one exams, 80, 60, 70 and 90.
    (a) Calculate Abooki's average mark.
    (b) what was the range of Abooki's score?
14. Kompany scored the following marks in end of year examinations. Calculate his average mark 96, 90, 93, 94 and 97.
15. Alice scored the following marks in homework 4, 7, 4, 5, 6, 4. Find his average mark.
16. Find the average of 8, 4, 0, 12.

7.8 More About Average/Mean

Example 1	Example 2
The average of 24, 30 and x is 28. Find the value of x.	The mean of 6, 10 and p is 14. Find the value of p.
$average = \frac{sum}{no.of\ items}$	$mean = \frac{sum\ of\ data}{no.of\ data}$
$\frac{28}{1} = \frac{24+30+x}{3}$	$\frac{14}{1} = \frac{6+10+p}{3}$
✓ The L.C.D of 1 and 3 is 3	✓ The L.C.D of 1 and 3 is 3
✓ Multiply by 3 on both sides	✓ Multiply by 3 on both sides
$3 \times \frac{28}{1} = \left(\frac{24+30+x}{\cancel{3}_1}\right) \times \cancel{3}^1$	$3 \times \frac{14}{1} = \left(\frac{6+10+p}{\cancel{3}_1}\right) \times \cancel{3}^1$
$84 = 24+30+x$	$42 = 6+10+p$
$84 = 54+x$	$42 = 16+p$
$84 - 54 = 54 - 54+x$	$42 - 16 = 16 - 16+p$
$30 = x$	$26 = p$
$x = 30$	$P = 26$

Activity 7:8

1. The mean of two numbers is 42, if one of the numbers is 45, find the second number.
2. The average of two numbers is 24, if the two numbers are p and 30, find the value of p.
3. T+18 and 2t have an average of 24. Find the value of t.
4. The average age of two boys is 12 years. If one of the boys is 15years, find the age of the second boy.
5. The mean of 25, 30 and m is 20. Find the value of m.
6. The mean mass of three people is 30kg. Two of them are 28 and 24. Find the mass of the third person.
7. The average of 2y, 3y and y is 24. What is the value of y?
8. The average of 2p, 72and 60 is 64. Find the value of p.
9. The mean age of 4 pupils in primary five is 11 years. Three of the pupils are 12 years 6years and 7 years. Find the age of the fourth pupil.
10. The average mass of 4 men is 62kg. The mass of three of the men are 42kg, 72kg and 54kg. What is the mass of the fourth man?

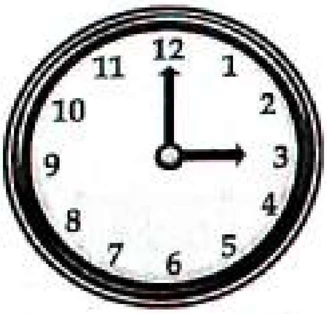
TOPIC 8: TIME

Time is a continued sequence of existence and events that occurs in an irreversible succession from past through the present to the future.
Time is the progression of events from the past to the present into the future.
Time is familiar to everyone.
Clocks are based on seconds, minutes, hours, day, week, month, year, decade or century.

8.1 Description of Time

1 year = 365days
1 year = 12months
1month = 30/31 or 29/28 days
1month = 4 weeks
We use a clock face to tell time

1 day = 24hours
1 hour = 60minutes
1 minute = 60 seconds
10 years = 1 decade
100 years = 1 century



Time can be shown on the clock face, hand watch and on a cell phone.

Changing Time from Years to Months

Example 1

Changing time from years to months
Express 3years to months
1year = 12months
3years = (12×3) months
3years = 36months

Example 2

Change $5\frac{1}{2} years$ to months
1year = 12months
$5\frac{1}{2}\, years = \left(\frac{11}{2} \times 12\right) months$
$5\frac{1}{2}\, years = \frac{11}{\cancel{2}_1} \times \cancel{12}^6 months$
$5\frac{1}{2}\, years = 66 months$

Example 3

Convert $1\frac{1}{4} years$ to months
1 years = 12months
$1\frac{1}{4}\, years = \left(\frac{5}{\cancel{4}_1} \times \cancel{12}^3\right) months$
$1\frac{1}{4}\, years = (5 \times 3) months$
$1\frac{1}{4}\, years = 15 months$

Example 4

How many months are $10\frac{1}{4} years$?
1 year = 12months
$10\frac{1}{4}\, years = \left(\frac{41}{\cancel{4}_1} \times \cancel{12}^3\right) months$
$10\frac{1}{4}\, years = (41 \times 3) months$
$10\frac{1}{4}\, years = 123 months$

Activity 8:1

1. Express the following hours to minutes

a) 4 hours	b) 5hours	c) 7hours	d) 11hours	e) 12hours
f) $3\frac{1}{2}hours$	g) $2\frac{1}{2}hours$	h) $6\frac{1}{2}hours$	i) $8\frac{1}{2}hours$	j) $12\frac{1}{2}hours$
k) $2\frac{1}{4}hours$	L) $4\frac{1}{4}hours$	m) $\frac{1}{4}\ hours$	n) $5\frac{1}{4}hours$	o) $7\frac{1}{4}hours$

2. A football match took $3\frac{1}{2}$ hours. How many minutes did it take?
3. A mathematics lesson took $1\frac{1}{2}hours$. Express the time taken in minutes.
4. A military parade took $4\frac{1}{2}hours$ waiting for the president to address them. How long did the parade last in minutes?
5. Kipsiro took $1\frac{3}{4}hours$ to complete a marathon race. Express the time he took in minutes.
6. A parents' meeting at Motherwell Junior school took $2\frac{2}{5}hours$. Express the time taken in minutes.

8.2 Express Minutes to Hours

- ✓ 60minutes = 1 hour
- ✓ For one to change minutes to hours, divide the given minutes by 60 minutes.

Example 1 Change 120minutes to hours 60minutes = 1hour 120minutes = $\left(\frac{120}{60}\right)hours$ 120minutes = 2hours	**Example 2** Express 180minutes to hours. 60minutes = 1hour 180minutes = $\left(\frac{180}{60}\right)hours$ 180minutes = 3hours
Example 3 How many hours are in 150minutes 270minutes? 60minutes = 1hour 150minutes = $\left(\frac{150}{60}\right)hours$ 150minutes = $\left(\frac{15}{6}\right)hours$ 150minutes = $2\frac{3}{6}\ or\ 2\frac{1}{2}hours$	**Example 4** How many hours are in 60minutes = 1hour 270minutes = $\left(\frac{270}{60}\right)$ 270minutes = $\frac{27}{6}$ 270minutes = $4\frac{3}{6}$ 270minutes = $4\frac{1}{2}hours$

Activity 8:2

Express the following from minutes to hours

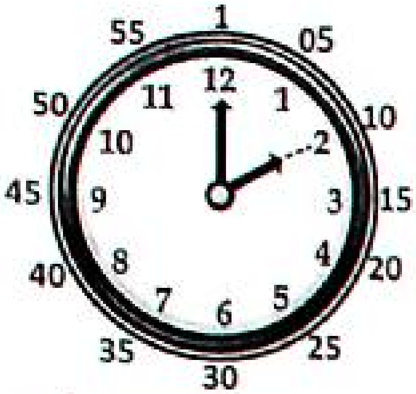
1) 240minutes	2) 300minutes	3) 60minutes	4) 360minutes
5) 420minutes	6) 90minutes	7) 210minutes	8) 75minutes
9) 135minutes	10) 165minutes	11) 255minutes	12) 330minutes

8.3 Telling Time in the Morning and Afternoon

A day begins at 12:00(mid-night)

"a.m" mean Anti-meridian, it is the time before mid-day.

"p.m" means post-meridian which means afternoon.



Change $\frac{1}{2}hour$ to minutes.

$\frac{1}{2}h = \frac{1}{2} \times 60minutes$

$= \frac{1}{\cancel{2}_1} \times \cancel{60}^{30}$minutes

$\frac{1}{2}hour = (1 \times 30)minutes$

$\frac{1}{2}hour = 30minutes$

Change $\frac{1}{4}$ hour to minutes

$\frac{1}{4}hour{=}(\frac{1}{4} \times 60)minutes$

$\frac{1}{4}hours{=}\frac{1}{\cancel{4}^1} \times \cancel{60}^5$ minutes

= (1×15)minutes

$\frac{1}{4}hours{=}$ 15minutes

Express $\frac{2}{5}$ hours to minutes

$\frac{2}{5}hours = \left(\frac{2}{5} \times 60\right) minutes$

$\frac{2}{5}hours = \frac{2}{\cancel{5}_1} \times \cancel{60}^{12}$

$\frac{2}{5}hours = (2 \times 12)minutes$

$\frac{2}{5}hours = 24minutes$

Telling the Time By the Clock face

2 O'clock

8 O'clock

11 O'clock

A half past 1 O'clock
30minutes past 1 O'clock

A half past 3 O'clock
30minutes past 3 O'clock

A half past 9 O'clock
30minutes past 9 O'clock

TOPIC 8: Time

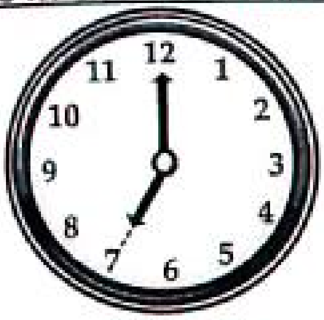
Tell Time by the Clock Face in the Morning

7:00a.m

4:00a.m

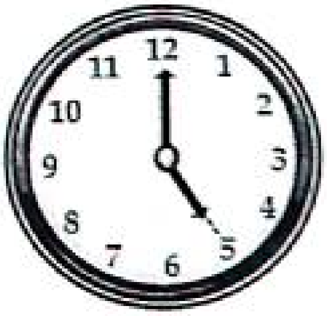
10:00a.m

Time in the Afternoon and Evening



7:00p.m

2:00p.m



5:00p.m

A half past and hour/30minutes past

A half past 7 o'clock
30minutes past 7 o'clock

A half past 4 o'clock
30minutes past 4 o'clock

A half past 11 o'clock
30minutes past 11 o'clock

Activity 8:3

Use a.m/p.m to tell time shown below;

Morning

Morning

Morning

Afternoon

Afternoon

Evening

8.4 A Quarter past or 15 Minutes past an Hour

A quarter past 1 O'clock
15minutes past 1 O'clock
1:15 O'clock

A quarter past 5 O'clock
15minutes past 5 O'clock
5:15 O'clock

A quarter past 9 O'clock
15minutes past 9 O'clock
9:15 O'clock

Activity 8:4

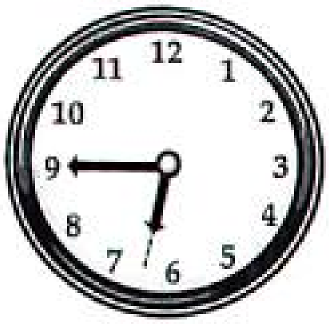
Describe the time shown on the clock faces below;

8.5 A quarter to (15minutes to an hour)

Morning

5:45a.m

Morning



6:45a.m

Evening

11:45p.m

Activity 8:5

1. Show the following time on the clock faces below;

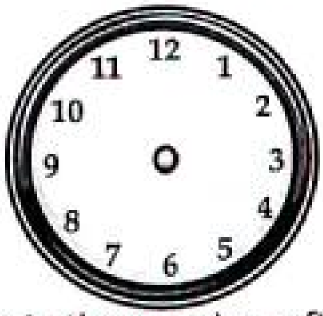
12:45p.m

7:45p.m

10:45p.m

2:45a.m

10:45a.m



3:45a.m

2. Use the clock face to tell time in the morning, afternoon or evening.

8.6 Some Minutes Past and Minutes to an Hour

11:25 O'clock
25minutes past 11 O'clock

10:10 O'clock
10minutes past 10 O'clock

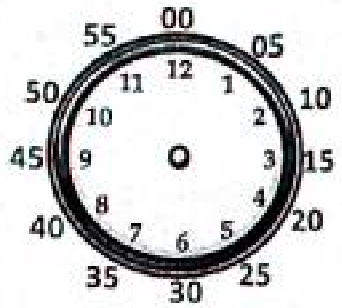
11:35 O'clock
35minutes past 11 O'clock

TOPIC 8: Time

Activity 8:6

Indicate the time given on the clock face

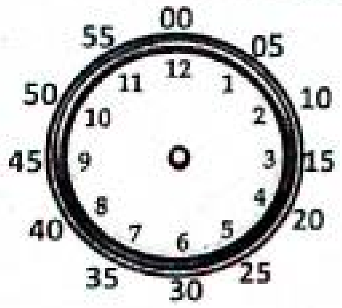
7:30a.m



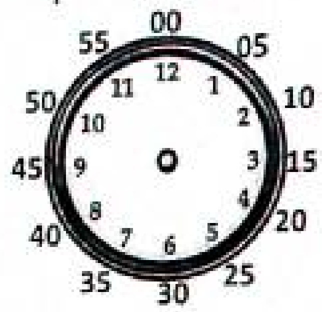
9:20p.m

11:15p.m

25minutes to 6 O'clock

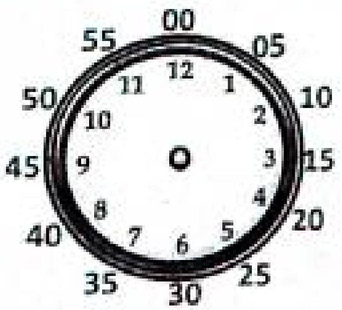


A quarter to 1 O'clock



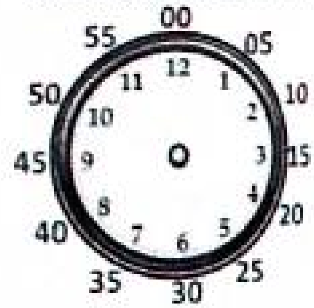
5minutes to 8 O'clock

20minutes to 5 O'clock



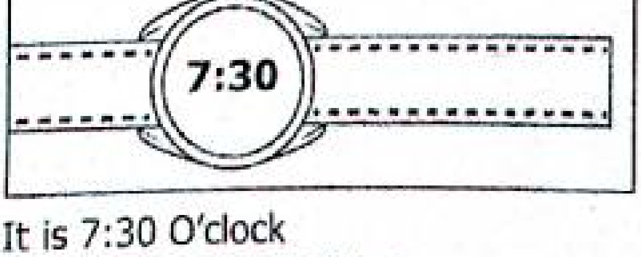
25minutes past 8 O'clock

10minutes to 12 noon



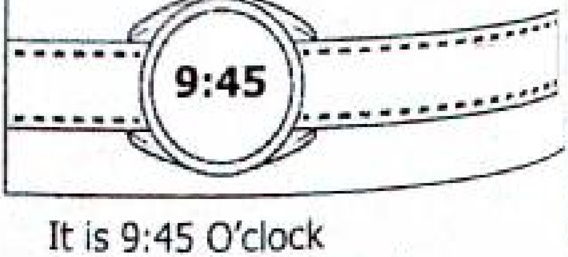
8.7 Telling time Using Digital Watch

Example 1



It is 7:30 O'clock
30minutes past 7 O'clock
30minutes to 8 O'clock

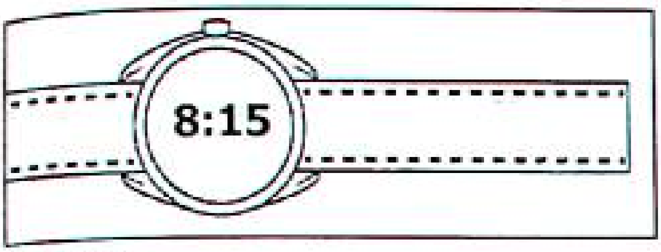
Example 2



It is 9:45 O'clock
A quarter to 10 O'clock
15minutes to 10 O'clock

TOPIC 8: Time

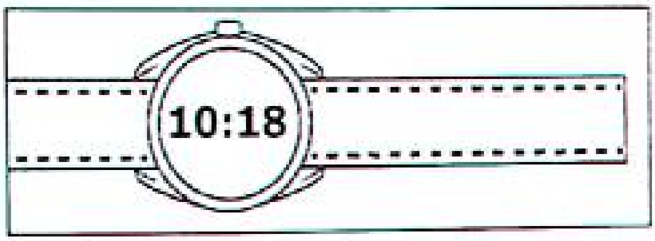
Morning



8:15a.m

A quarter past 8 in the morning

Afternoon

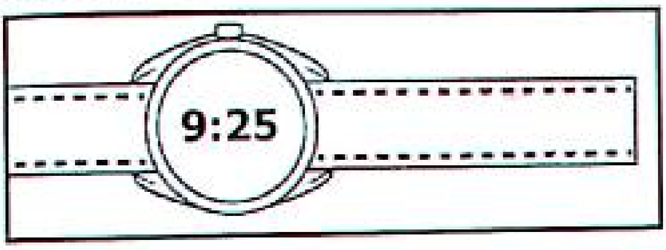


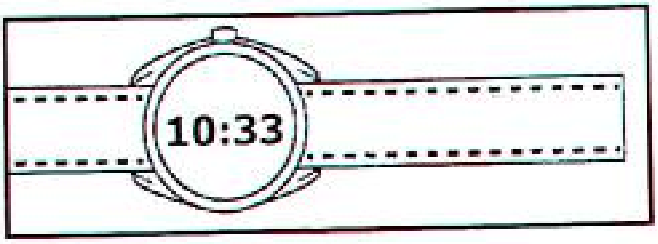
10:18p.m

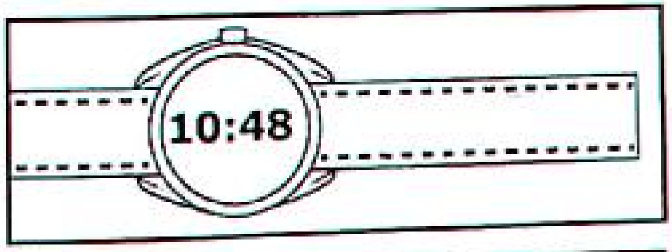
18minutes past 10 O'clock in the evening

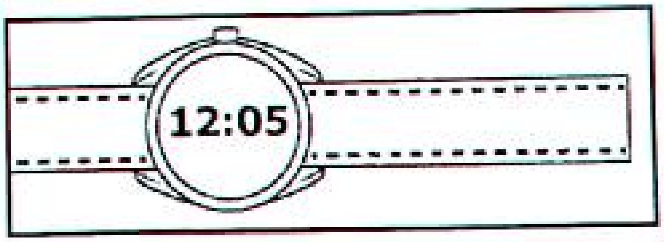
Activity 8:7

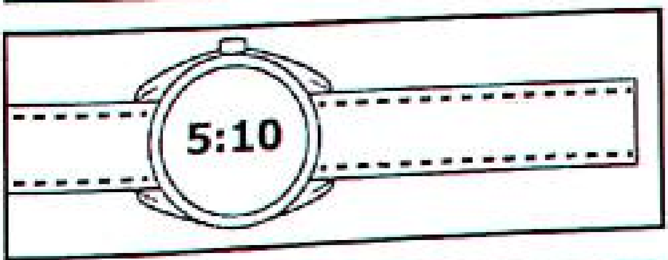
Write down the time shown on the digital watch.

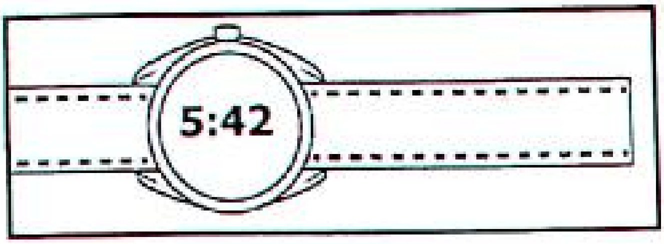


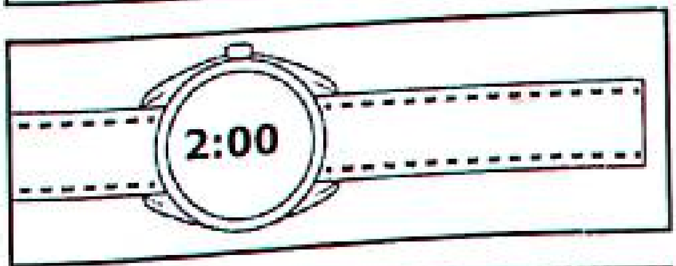


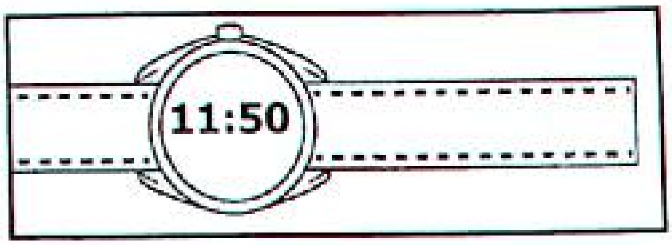


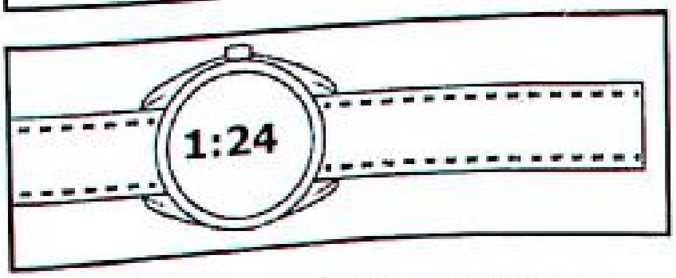


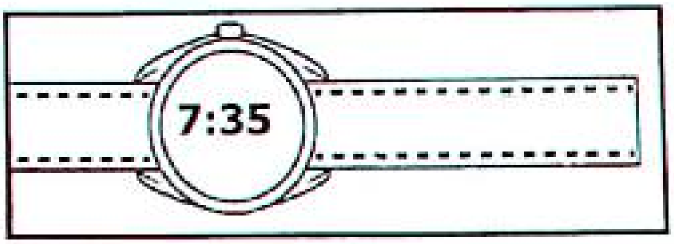












8.8 Addition of Hours and Minutes

Example 1

Add

Hours	Min
2	25
+ 3	15
5	40

25+15=40
2+3=5

Example 2

Add

Hours	Min
4	35
+3	40
8	15

35+40=75
75÷60=1r15
1+4+3=8hours

Activity 8:8

Add the following numbers

1) Hours 6 +2; Min 25 30
2) Hours 3 +5; Min 12 42
3) Hours 2 +1; Min 30 40
4) Hours 2 +4; Min 45 35
5) Hours 1 +1; Min 24 48
6) Hours 3 +2; Mins 15 45
7) Hours 7 +8; Mins 25 30
8) Hours 1 +4; Min 45 45

10) An examination started at 9.00am and took 2hours 30minutes. At what time did it end?

8.9 Subtraction of Hours and Minutes

Example 1

Hours	Min
2	40
- 1	25
1	15

Example 2

Hours	Min
5	15
-2	25
3	50

60 +15 = 75
75 − 25 = 50

Activity

1) Hours 6 -1; Min 20 10
2) Hours 12 -10; Min 55 30
3) Hours 8 -3; Min 48 15
4) Hours 4 -1; Min 45 50
5) Hours 7 -2; Min 25 40
6) Hours 6 -2; Min 15 30
7) Hours 10 -4; Min 05 40
8) Hours 9 -3; Min 10 50

8:10 Duration (Time Span)

Example 1

A football match started at 3.30pm and ended at 5:20pm. How long did the match take?

Hours	Min	
$\cancel{5}^4$	20	(20 + 60) = 80
- 3	30	80 – 30 = 50
1	50	

It took 1 hour 50minutes

Example 2

A parents' meeting at Golden Treasure school started at 8:30am and ended at10:25am. How long was the meeting?

Hours	Min	
$\cancel{10}^9$	25	(25 + 60) = 85
-8	30	85 – 30 = 55
1	55	

Example 3

Joseph left Mbarara at 10:30am and arrived Kigali Rwanda at 4:20pm. How long did the journey take?

- ✓ The two indicated time are in different units a.m and p.m
- ✓ Work out the remaining time before noon.
- ✓ Then add to get the total time taken.

$\cancel{12}^{11}$. 00		(60 – 30) = 30
-10	30	
1 :	30	

1 . 30	80÷60 = 1h rem 20min
+4 : 50	
6 : 20	

The journey took 4hours 50minutes.

Activity 8:10

1. A lesson started at 9.00am and ended at 11:00am. How long did the lesson take?
2. Odong slept at 10:00pm and wake up at 4:20am. For long did Odong sleep?
3. A bus left Kotido at 5:30am and arrived Kampala at 4:45pm. How long did the journey take?
4. Owere rode his bicycle from Tororo at 11:40am and arrived Iganga at 2:45pm. How long did he take?
5. An aeroplane left Entebbe airport at 9:30am and landed at Heathrow airport in London at 2:15pm. How long did the journey take?
6. A motorist left Adjuman at 3:00pm and arrived Kampala at 2:00am. How long did he take to Kampala?
7. An examination started at 8:30am and ended at 10:15am. How long was the exam?

8. A church service started at 7.15am and ended at 10:30am. How long was the service?

9. Bolingo started ploughing his garden at 8.50am and completed at 12:20pm. How long did he take ploughing?

8.11 <u>Days and Weeks</u>

Expressing Weeks to Days

Example 1

How many days are in 4weeks?

1 week = 7days

4 weeks = (7×4) days

4 weeks = 28 days

Example 2

Change 12 weeks in days.

1 week = 7days

12 weeks = (12×7)days

12 weeks = 84 days

Expressing Days to Weeks

Example 1

Express 77days to weeks.

7 days = 1 week

1day = $\left(\frac{1}{7}\right)$ *week*

77days = $\left(\frac{1}{\cancel{7}_1} \times \cancel{77}^{11}\right)$ *weeks*

77days = 11weeks

Example 2

How many weeks are in 35days?

7days = 1 week

1 day = $\left(\frac{1}{7}\right)$ *week*

35days = $\left(\frac{1}{\cancel{7}_1} \times \cancel{35}^{5}\right)$ *week*

35days = (1×5)weeks

35days = 5weeks

Activity 8:11

1. Complete the table below.

Week	2	5	6	____	____	10	15
Days	14	____	____	56	49	____	____

2. How many days make up 3 weeks?
3. How many days are in 9weeks?
5. How many days make up 13 weeks?
6. Express 63 days in weeks.
6. Convert 21 days to weeks.
7. How many weeks are in 28 days?

8.12 <u>Adding of Weeks and Days</u>

1.

Weeks	Days
4	3
+2	2
____	____

2.

Weeks	Days
2	2
+3	2
____	____

3.

Weeks	Days
6	3
+2	4
____	____

4.

Weeks	Days
1	4
+2	3
____	____

5.

Weeks	Days
3	4
+2	5
____	____

6.

Weeks	Days
2	3
+ 5	6
____	____

7.

Weeks	Days
2	1
+2	6
____	____

8.

Weeks	Days
3	3
+5	5
____	____

TOPIC 8: Time

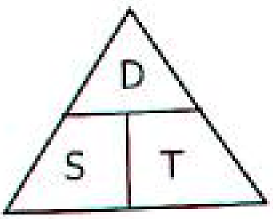
8.13 Subtraction of Weeks and Days

1. Weeks	Days	2. Weeks	Days	3. Weeks	Days	4. Weeks	Days
8	4	4	5	6	4	5	3
-4	2	-2	3	-4	1	-1	6

5. Weeks	Days	6. Weeks	Days	7. Weeks	Days	8. Weeks	Days
3	2	5	1	6	2	10	3
-2	3	-3	6	-2	4	-8	4

9. Weeks	Days	10. Weeks	Days	11. Weeks	Days	12. Weeks	Days
12	2	4	3	5	1	10	2
-1	5	-2	4	-2	4	-5	6

Distance, Time and Speed



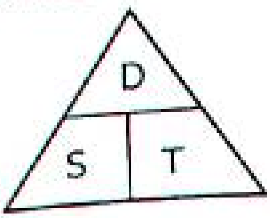
- ✓ Distance is the lengths between two points.
- ✓ Distance is measured in Metres and Kilometres.
- ✓ Speed in kilometres per hour or metrics per second.
- ✓ Time in hours, minutes or seconds

8.14 Finding Distance

Distance = Speed × Time

Example 1

Oding travelled from Lira for 3hours at a speed of 64km/hr. calculate the distance it covered .



Distance = Speed × Time

Distance = $\frac{64km}{hr} \times 3hr$

Distance = 64km × 3

Distance = 192km

Example 2

It took $2\frac{1}{2}$ hours for a train to travel from Mombasa to Nairobi at a speed of 60km/hr. What is the distance from Mombasa to Nairobi?

Distance = Speed × Time

Distance = $\frac{60km}{hr} \times 2\frac{1}{2}hr$

Distance = $\frac{60km}{hr} \times 2\frac{1}{2}hr$

Distance = $\frac{60km^{30}}{hr_1} \times \frac{5hr^1}{2_1}$

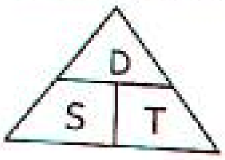
Distance = 30km×5

Distance = 150km

Activity 8:14

1) Global coach left Kampala for Mbarara at 80km/hr and took $3\frac{1}{2}$ hours to cover the journey. Calculate the distance from Kampala to Mbarara.
2) A motor rally driver drove at a speed of 110km/hr for 1½hours. What distance did he cover?
3) A train travelling at 55km/hr took 6hours to move from Nairobi to Jinja. Calculate the distance between the two towns.
4) A lorry carrying sacks of maize travelled from Kigumba to Jinja at an average speed of 45km/hr for 5 hours. What distance did it cover?
5) It takes 3½hours for YYbus to move from Lira city to Kampala travelling at 78km/hr. What is the distance from Lira to Kampala?
6) It takes 1½hours for Bombadia aeroplane to fly from Johannesburg to Accra in Ghana at aspeed of180km/hr. Find the distance from Johannesburg to Accra.
7) A soldier walked for 2½hours at a steady speed 6km/hr. What distance did he cover?
8) Kakise bus travelled at a speed of 80km/hr for 4hours. What distance did it cover?

8.15 <u>Finding Speed</u>



$$Speed = \frac{Distance}{Time}$$

Example 1

The distance from Abim to jinja is 248km. If the journey took 4hours, calculate the average speed for the journey.

Speed $= \frac{distance}{time}$

Speed $= \frac{240km}{3hours}$

Speed $= \frac{\cancel{240km}^{80}}{\cancel{3hrs}_1}$

Speed = 80km/hr

Example 2

A taxi left Kitgum at 8:00am for Luwero a distance of 240km. It arrived at 12:00noon. What was its average speed?

Speed $= \frac{distance}{time}$

Speed $= \frac{240km}{4hrs}$

Speed $= \frac{\cancel{240km}^{60}}{\cancel{4hrs}_1}$

Speed = 60km/hr

Time taken

$$\begin{array}{r} 12:00 \\ -\ 8:00 \\ \hline 4:00 \\ \hline \end{array}$$

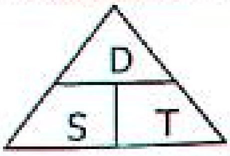
= 4hrs

Activity 8:15

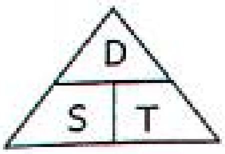
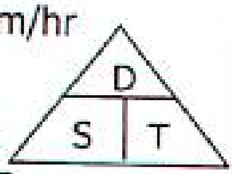
Workout the following numbers;

1. A car covered a distance of 180km in 3hours. What was the cars' average speed?
2. Global bus covered a journey of360km in 4hours. At what speed was it travelling?
3. A cyclist rode 75km in 3hours. At what speed was he cycling?
4. A train covered a distance of 480km in 6hours. At what speed was it travelling?
5. What distance does a driver cover at a speed of 60km/hr for 3hours?
6. A cyclist takes 2½hours to cover a distance of 200km.Calculate his average speed.
7. A motorist travelled 72km in 3 hours. What speed was it moving?
8. A speeding boat covered 280km on Lake Victoria in 4hours. Find its speed.
9. the distance from Busia via Bugiri to Kampala is 320km. Omollo covered it in 4hours, at what speed was he travelling?
10. At what speed should a Gateway bus travel in order to cover 150km in 2½hours?
11. A cyclist covered 100km in 2 hours 30minutes. Calculate his speed in km/hr.

8.16 <u>Calculating Time</u>



Time = $\frac{Distance}{Speed}$

Example 1	Example 2
Example 1 Calculate the time taken by a speeding vehicle at 72km/hr to cover a distance of 360km.  $Time = \frac{Distance}{speed}$ $Time = \frac{360km}{72km/hr}$ $Time = \frac{\cancel{360km}^{5}}{\cancel{72km/hr}_{1}}$ Time = 5hours	**Example 2** The distance from Kabale to Masaka is 350km. Trinity bus left Kabale at a speed of 70km/hr. How long did it take to cover the journey? D= 350km, S= 70km/hr  $T = \frac{Distance}{Speed}$ $T = \frac{350km}{70km/hr} = \frac{\cancel{350km}^{5}}{\cancel{70km/hr}_{1}}$ T = 5hours

Activity 8:16

1. Opandi covered a distance of 240km at a speed of 80km/hr. How long did he take?
2. Muzee Zedde drove a distance of 240km from Jinja to Gulu at 60km/hr. How long did he take driving?
3. A motorist travelled from Arua to Soroti a distance of 480km at 60km/hr. Calculate the time taken by the motorist.
4. Kamya drives his Ipsam from Kotido to Soroti at a speed of 80km/hr covering a distance of 120km. What time does he take to cover the distance?
5. A car covered a distance of 720km from Kabale to Soroti at a speed of 60km/hr. Find the time it took to cover the journey.
6. What time is needed to cover a distance of 440km at a speed of 44kn/hr?
7. A train left Nairobi for Kampala a distance of 840km at 70km/hr. Calculate the time a train took to cover the journey.
8. A bus travelling at 65km/hr covered 390km. Calculate the time it took.
9. An aeroplane flew from Entebbe airport to Accra-Ghana a distance of 18,000km. If it travelled at 360km/hr, how long did it take to complete the journey?
10. Its only 150km from Lira to Arua. If Zawadi bus travelled at 60km/hr, how long will it take to cover that distance?

11. A lorry carrying sacks of charcoal from Kitgum to Kampala covered a distance of 441km at 63km/hr. How long did it take to reach Kampala?

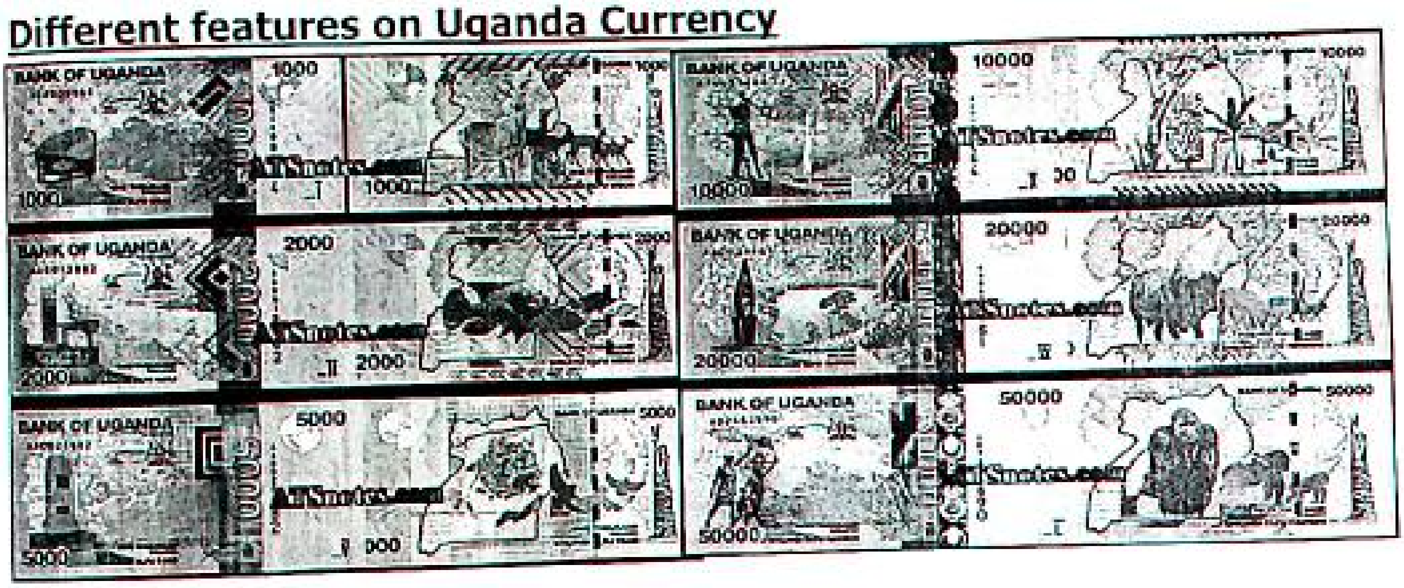
TOPIC 9: MONEY

Money is a medium of exchange for goods and services. Uganda we use two types of currency;

(i) Coins (ii) Notes (Paper money)

Coins	Notes
Sh.50	1000 sh. notes
Sh.100	2000 sh. notes
Sh.200	5000 sh. notes
Sh.500	10,000 sh. notes
Sh.1000	20,000 sh. notes
	50,000 sh. notes

Different features on Uganda Currency



9.1 Exchanging Money

Example 1 John had to exchange a sh.50,000 notes for Sh.1000. How many notes did he get? Sh.50,000 ÷ sh.1000 $sh\left(\frac{50000}{1000}\right) = 50notes$	**Example 2** How many sh.500 coins can be got from sh.10,000? Sh.10000÷sh.500 $\left(\frac{\cancel{sh.10000}^{20}}{\cancel{sh.500}}\right) = 20\ coins$
Example 3 Byaruhanga had 6otes of sh.2000. How Much money did he have? 1 note → sh.2000 6 notes → sh.(2000×6) 6 notes → sh.12000	**Example 4** How much money are is in 12 coins of sh.500 each? 1 coin → sh.500 12 coins → sh.(500×12) 12 coins → sh.6000

TOPIC 9: Money

Activity 9:1

1. How many coins of sh.500 can be got from ten-thousand-shilling note?
2. How many 200 shilling coins that can be obtained from five thousand shilling note?
3. How many sh.2000 notes can be obtained from sh.80,000?
4. Joseph has 10 coins of sh.500 and 8 coins of sh.200. How much money does he have?
5. Juma paid for a shirt using 4notes of sh.5000. How much did he pay for the shirt?
6. Applied mathematics text costs 10 notes of sh.2000. Calculate the total cost of the text book.
7. Ogwang Joseph went to speek Hotel to have a good lunch and paid 7coins of sh.500 and 2 notes of sh.10,000. How much did he pay for lunch?
8. Latipha went to the market to buy a bunch of bananas, she paid 6 coins of sh.1000 and 4 notes of sh.5000. How much did she pay for a bunch of bananas?
9. How many sh.200 coins are in sh.10,000 note?

9.2 <u>Simple Direct Proportion</u>

Example 1 The cost of a pen is sh.700. Find the cost of 5 similar pens. 1 pen costs sh.700 5 pens cost sh.(700×5) 5 pens cost sh.3500	**Example 2** Given that the cost of 3 loaves of bread is sh.12000, what is the cost of 1 loaf? 3 loaves cost sh.12000 1 loaf costs sh.$\left(\frac{12000}{3}\right)$ 1 loaf costs sh.$\frac{12000}{3}$ 1 loaf costs sh.4,000
Example 3 Two text books cost sh.18000. Find the cost of 3 similar text books. 2 books cost sh.18000 1 book costs $sh.\left(\frac{18000}{2}\right)$ 1 book costs sh.9000 3 books cost sh.(9000×3) 3 books cost sh.27000.	**Example 4** 4 caps cost sh.8000. Find the cost of 7caps. 4 caps cost sh.8000 1 cap costs $sh.\left(\frac{8000}{4}\right)$ 1 cap costs sh.2000 7 caps cost sh.(2000×7) 7 caps cost sh14000

Activity 9:2

Work out the numbers below.

1. A tray of eggs cost sh.8000. Find the cost of 3 trays.
2. One books costs sh.800. What is the cost of 3 similar books?

3. A mango costs sh.1200. Calculate the cost of 5 mangoes.
4. A litre of milk costs sh.800. What is the cost of 2 litres?
5. One kilogram of salt costs sh.1200. Find the cost of 8 kilograms.
6. 3cups cost sh.12,000. Find the cost of 5 similar cups.
7. Mary bought 2 dresses at sh.8000. How much did each dress cost?
8. The cost of 6 books is sh.1800. What is the cost of 9books?
9. 2bottles of soda cost sh.2000. Find the cost of 5 bottles of sodas.
10. 3mangoes cost sh.3000. Find the cost of 6mangoes.
11. Akello bought 2kg of sugar. If each kilogram costs sh.3000, how much did she pay for sugar?
12. John bought a pen at sh.500 and a book at sh.3200. How much change did he get back if he had sh.5000?
13. Kipsiro bought a pair of trousers at sh.38000.Kipsiro had sh.50000 note, how much was his change?
14. Kiplangat had sh.80,000 and bought a suit at sh.49500. How much was his change?

9.3 Working out Shopping Bills

Example 1

Pauline went shopping and bought the items;

2 pens at sh.1500 each

4 books at sh.500 each

2 geometry sets at sh.2400 each

a) How much did she pay for all the items?

Items	Method	Amount
Pens	Sh.(1500×2)	Sh.3000
Books	Sh.(500×4)	Sh.2000
Sets	Sh.(2400×2)	Sh.4800
	Total cost	Sh.9800

b) If he had sh.10000, what was his change?

Sh.10000 – sh.9800

Sh.200

Activity 9:3

1. Use the price list below to answer questions that follow;

1kg of salt costs sh.1200

1kg of beans costs sh.2400

A bar of soap costs sh.3000

A kg of sugar cost sh.4500

a) What is the cost of 3kg of salt?

b) Find the cost of 2kg of beans and 2kg of sugar.

c) What is the cost of 4kg of salt and 2bars of soap?

2. Use the price list to answer questions that follow;
   1kg of sugar at sh.3500
   1packet of Nomi at sh.500
   1kg of salt at sh.500
   A bar of soap at sh.3200
   a) What is the most expensive item?
   b) Calculate the cost of 3bars of soap.
   c) What is the cost of 3packets of Nomi and 4kg of salt?
   d) How much would one pay for 5bars of soap?
3. Faith went to the super market and bought the following items;
   3kg of rice at sh.3000 per kg
   2bars of soap at sh.3000 each
   4loaves of bread at sh.4000 each
   a) How much did she pay for all the items?
   b) If she had sh.50,000 note, how much change did she get?
4. Ronaldo Christiano went to the market and bought the following items;
   2kg of sugar at sh.3000 each
   3rulers at sh.1200each
   2packets of tea leaves at sh.2500 each
   3shopping bags at sh.1000 each bag
   a) How much was his total expenditure?
   b) If he had twenty thousand shilling note, what was his change?
5. Mugisha went to the shop and bought the following items;
   A pen at sh.1200
   1kg of salt at sh.1500
   3books at sh.1500 each book
   2kg of sugar at sh.3500 per kg
   a) Calculate his total expenditure.
   b) If he was given a change of sh.1500, how much did he have at the beginning?
6. A teacher went to the market and bought the following items;
   3pens at sh.800 each
   2rulers at sh.1500 each
   4books at sh.1200 each
   3rubbers at sh.500 each
   a) How much did he pay for all the items?
   b) If the teacher had sh.15,000, how much was his change after buying all the items?

9.4 Completing Shopping Bills

Finding the amount = Unit cost ×Quantity

Finding the quantity = $\frac{Amount}{Quantity}$

Finding the amount using

Amount/Total = Unit cost × Quantity

Example 1

Atmandu bought 6 dresses at sh.3000 each, how much did she spend altogether?

Amount = unit × quantity

Amount = sh.(3000×6)

Amount = sh.18000

She spent sh.18000

Example 2

The table below shows Witney's shopping bill, complete it;

Items	Quantity	Unit cost	Total/Amount
Sugar	3kg	Sh.3200	Sh.9600
Rice	2kg	Sh.4300	Sh.8600
Salt	4kg	Sh.1200	Sh.4800
Bread	2loaves	Sh.2800	Sh.5600
		Total	Sh.28600

Sugar	Rice	Salt	Bread
Sh.3 2 0 0	Sh.4 3 0 0	Sh.1 2 0 0	Sh.2 8 0 0
× 3	× 2	× 4	× 2
Sh.9 6 0 0	Sh.8 6 0 0	Sh.4 8 0 0	Sh.5 6 0 0

Activity 9:4

Complete the shopping bills below.

1. a) A house wife had sh.30,000 and bought the items as shown in the shopping bill below. Complete the table.

Items	Quantity	Unit cost	Amount
Milk	2litres	Sh.1400	______
Bread	3loaves	Sh.2400	______
Sugar	2kg	Sh.4400	______
Tea leaves	3packets	Sh.800	______
		Total	______

b) How much change did the house wife get?

2. Miss Atim went shopping with sh.50,000 and bought the following items as shown in the table.

a) Complete the table below;

Items	Quantity	Unit cost	Amount
beans	3kg	Sh.2000	sh.______
Rice	3kg	Sh.3000	sh.______
Salt	2kg	Sh.1200	sh.______
Meat	4kg	Sh.1500	sh.______
		Total	sh.______

b) How much did she remain with after buying all the items?

3.The table below shows Mawanda' shopping bill use it to answer questions that follow;

a) Complete the table below

Items	Quantity	Unit cost	Amount
Sugar	2kg	Sh.4500	sh.______
Salt	3kg	Sh.2000	sh.______
Books	3 books	Sh.3000	sh.______
Pens	2 pens	Sh.500	sh.______
		Total	sh.______

b) If he had sh.30,000, what was his balance?

4. a) Juma bought the items as shown in the table below. Complete the table correctly.

Items	Quantity	Unit cost	Amount
Rice	3kg	Sh.3200	sh.______
Soap	2bars	Sh.5500	sh.______
Bread	4 loaves	Sh.1800	sh.______
		Total	sh.______

b) If he went with a fifty thousand shilling note, how much change did he get?

5. Daniella went to the supermarket and bought the items as shown in the table. a) Complete the table correctly

Items	Quantity	Unit cost	Amount
Omo	2packets	Sh.4500	sh.______
Salt	3packets	Sh.3500	sh.______
Books	2½litres	Sh.2400	sh.______
Pens	500g	Sh.4000	sh.______
		Total	sh.______

b) Daniella had sh.30,000, how much money did she remain with?

9. 5 Finding the Unit Cost/Unit price

Example 1

Unit cost = $\frac{Amount}{Quantity}$

Isabirye bought 4kg of posho at sh.24,000. Calculate the unit cost (the cost of 1kg of posho).

Unit cost = $\frac{Amount}{Quantity}$

Unit cost = $sh.\left(\frac{24000}{4}\right)$

Unit cost = sh.6000

1kg of posho costs sh.6000

Example 2

1. (a)The table below shows how Maswanku spends his money. Study it carefully and fill the blank spaces.

Items	Quantity	Unit cost	Amount
Sugar	2kg	Sh.3600	sh.7,200
Meat	3kg	Sh.12000	sh.36,000
Rice	4kg	Sh.3000	sh.12,000
		Total	sh.55,200

Sugar	Meat	Rice
$sh.\left(\frac{7200}{2}\right) = 3600$ $sh.3600$	$sh.\left(\frac{36000}{3}\right)$ $sh.12{,}000$	$sh.\left(\frac{12000}{4}\right)$ $sh.3000$

(b) If Mr. Maswanku had sh.60,000, how much money did he remain with?

He remained with sh(60,000 – 55,200)

= sh.4800

Activity 9:5

1. Jacob went shopping and bought the items as shown in the table below;

Items	Quantity	Unit cost	Amount
Fountain pen	3 pens	Sh.______	sh.1500
Books	4 books	Sh.______	sh.3200
Pencils	6 pencils	Sh.______	sh.1800
Geometry set	2 geometry sets	Sh.______	Sh.5000
		Total	sh.______

Jacob had 20,000 shilling note, how much did he remain with?

2. Complete the table below;

Items	Quantity	Unit cost	Amount
Books	4 books	Sh._____	sh.4800
Tooth paste	3 tubes	Sh._____	sh.6000
Nido milk	2 tins	Sh._____	sh.4800
Omo	1 tin	Sh._____	sh.5200
		Total	sh._____

3. Complete the shopping below.

Items	Quantity	Unit cost	Amount
Bread	2 loaves	Sh._____	sh.4800
Blue band	1 tin	Sh._____	sh.3200
Sugar	1½kg	Sh._____	sh.4500
Tea leaves	2kg	Sh._____	sh.20,000
		Total	sh_____

4. Wafula primary candidate gave a shopping list to his father as shown below;

Items	Quantity	Unit cost	Amount
Biscuit	2 packets	Sh._____	Sh.7200
Ground nuts	3kg	Sh._____	Sh.21000
Books	2 dozens	Sh._____	Sh.24000
Pens	1 dozen	Sh._____	Sh.18000
		Total	Sh._____

9.6 Finding Quantity of Items

Quantity $= \frac{Amount}{Unit\ cost}$

Example 1

Kalibbala bought beans at sh.24000, if each kilogram of beans costed sh.3000, how many kilograms did he buy?

Quantity $= \frac{Amount}{unit\ cost}$

$= \frac{\cancel{sh.24000}^{8}}{\cancel{sh.3000}_{1}} = 8kg$

2. Study and complete the table below

Items	Quantity	Unit cost	Amount
Maize flour	4 kg	Sh.1200	Sh.4800
Millet flour	3 kg	Sh.1500	Sh.4500
Rice flour	1½ kg	Sh.8000	Sh.12000
Wheat flour	2 kg	Sh.4000	Sh.8000
		Total	

Maize flour	Millet flour	Rice flour	Wheat flour
$\frac{sh.4800}{sh.1200} = 4kg$	$\frac{sh.4500}{sh.1500} = 3kg$	$\frac{sh.12000}{sh.8000} = 1\frac{1}{2}kg$	$\frac{sh.8000}{sh.4000} = 2kg$

Activity 9:6

1. Complete the shopping table below;

Items	Quantity	Unit cost	Amount
Books	________	Sh.2000	Sh.10,000
Pencils	________	Sh.800	Sh.40000
Pens	________	Sh.500	Sh.1500
Reams	________	Sh.9000	Sh.18000
		Total	Sh.______

2. (a) Mr. Ndabada Moses bought the items in the table below. Study it carefully and fill the blank spaces

Items	Quantity	Unit cost	Amount
Salt	______kg	Sh.800	Sh.1200
Sugar	______kg	Sh.3600	Sh.10800
Beans	______kg	Sh.2500	Sh.10,000
Milk	______kg	Sh.1400	Sh.4200
		Total	Sh.______

b) If Mr. Ndabada had sh.30,000. Ho much was his balance?

3. The table below shows how Mr. Kimpanze spent his money on different items. (a) Complete the shopping bill.

Items	Quantity	Unit cost	Amount
Cloth	______kg	Sh.10000	Sh.60000
Dress	______	Sh.30000	Sh.90000
Shirt	______	Sh.15000	Sh.60000
Trousers	______	Sh.40000	Sh.80000
		Total	Sh.______

b) How much did spend altogether?

4. General questions about shopping. Study the table and fill the blank spaces.

Items	Quantity	Unit cost	Amount
Beans	______	Sh.3000	Sh.______
Cow peas	3kg	Sh.______	Sh.8400
Soya beans	5kg	Sh.3500	Sh.13500
Meat	______	Sh.14000	Sh.______
		Amount	Sh.______

5.(a) An Indian went to Nakasero market and bought the items as shown below. He had sh.90,000 for shopping.

Items	Quantity	Unit cost	Amount
Beans	10kg	Sh.3000	Sh.______
Sugar	2kg	Sh.______	Sh.840
Rice	______kg	Sh.4500	Sh.13500
G.nuts	2kg	Sh.______	Sh.______
		Total	Sh.______

b) How much money did he remain with after spending to all items?

9.7 Finding Profit

Profit = Selling price – Buying price

Example 1

A shop keeper bought a cell phone at a whole price of sh.49,000 and sold it at sh.63000.

He sold it at high price that means he made a profit.

Buying price = sh.49000,

Selling price = sh.63000

Profit = Selling price – buying price

Profit = S.P – B.P

Profit = sh.(63000 – 49000)

Profit = sh.14000

Example 2

Mr. Odur bought a dining at sh.88,000 and sold it sh.96,000. How much profit did he make?

Buying price = sh.88,000, Selling price = sh.96000

Profit = S.P – B.P

Profit = sh.(96000 – 88,000)

Profit = sh.8000

Activity 9:7

1. Mr. Kalibata bought a phone at sh.89000 and sold it at sh.100,000. What was his profit?
2. Sh.48000 was spent on buying a small radio which was later sold at sh.57000. How much was the profit?
3. Nakandi bought a dress at sh.14500 and sold it at sh.15,900. What was her profit?
4. A shop keeper bought a sack of sugar at sh.112500 and sold it at sh.135000. How much did he gain?

5. Alice bought a sheep at sh.175000 and sold it at sh.187000. Calculate her profit.
6. Majorine bought a source pan at sh.12500. she then sold it at sh.13500. What profit did she make?
7. Joseline bought a jerrican of water at sh.7000 and sold it at sh.9700. How much profit did she get?
8. Obadia bought a radio at sh.93000 and sold it at sh.105000. What was his profit?

9.8 Finding the Selling Price When Given Buying Price and Profit

Selling Price = B.P + Profit

Example 1 Messi bought a pair of shoes at sh.190,000 and sold it making a profit of sh.12700. How much did Messi sell the pair of shoes? Selling price = Buying price + Profit S.P = sh.(190,000 + 12700) S.P = sh.202,700	**Example 2** A shop keeper bought a sack of rice at sh.215000. He sold it making a profit of sh.18700. How much did he sell the sack of rice? S.P = B.P + Profit S.P = sh.215000 + sh.18700 = sh.233700

When the profit is got, we calculate selling price by adding buying price (B.P) and profit.

Activity 9:8

1. Erianah bought a bucket at sh.25000 and sold it later making a profit of 7,000. What was his seling price?
2. A man bought television set at sh.545000. He later sold it making a profit of sh.55,000. Calculate the selling price of the television.
3. Fatuma bought 3 saucepans at sh.12000. He sold them at a profit of sh.4500. What washer selling price?
4. A trader bought a carton of soda at sh.10,500 and sold it maiking a profit of sh.1500. Calculate his selling price.
5. A shopkeeper bought a pair of shoes at sh.37500 and sold it at a profit of 5500. Find his selling price.
6. Kauthar bought 6 cakes at sh.3000 and sold each cake making a profit of sh.300.
   a) How much profit did she get? b) Calculate his selling price generally.
7. Obongi bought a dozen of cups at sh.60000. He sold each cup at sh.2000.
   a) How much profit did he realize?
   b) What was his selling price?

8. Kaswabuli bought a goat at sh.144000. At What price must he ell it to get a profit of sh.48000?

9.9 Finding Loss

Loss = Buying price – Selling price

A loss is realized when the buying price is higher than the selling price.

Example 1	Example 2
A shirt which was bought at sh.2400 was sold at sh.21500. Find the loss realized Loss = B.P – S.P Loss = sh.(24000 – 21500) Loss = sh.2500	Joseph bought a bag at sh.38000 and sold it at ah.36500. Find his loss. Loss = B.P – SP Loss = sh.(38000 – 36500) Loss = sh.1500

Activity 9.9

1. Allan bought a bicycle at sh.275000 and sold it at sh.235000. What was his loss?
2. Calculate the loss on a sack of sugar which was bought at sh.210,000 and sold at sh.187,000.
3. Kidimbo bought a shopping bag at sh.47000 and sold it at sh.45800. Calculate his loss.
4. Calculate the loss on a side board which was bought at sh.455,000 and sold for 447,000.
5. A man bought a calf at sh.980,000 and sold it sh.720,000. What was his loss?
6. A mobile phone set was bought at sh.275000 and sold at sh.269000. What was the loss?
7. A tin of cooking oil costs sh.180000, a shop keeper sold it at sh.196,000. Calculate his loss.
8. Calculate the loss on a table which was bought at sh.156000 and sold for sh.137,000.

9.10 Finding the buying Price When Given Loss

Buying price = Selling price + Loss

Example 1	Example 2
Obur sold a shirt at sh.6000. What was his buying price? Buying price = S.P + Loss B.P =sh.(24000 + 6000) B.P = sh.30,000	Selling a stool at sh.17000, Mr. Byaruhanga made a loss of sh.3000. At how much did he buy the stool? Buying price = S.P + Loss B.P = sh.(17000 + 3000) B.P is sh.20,000

Activity 9:10

1. Mr. Wafula sold a side board at sh.52000 making a loss of sh.3000. how much did he buy it?
2. Josephat sold his plot of land at sh.12,500,000 making a loss of sh.300,000. How much did he buy the plot of land?
3. Muhammad sold an audio system at 875000 making a loss of sh.25000. How much did he buy the plot of land?
4. Okware sold a gross of pens at sh.240,000 making a loss of sh.36000. At what price did he buy the pens?
5. When business lay sold a bicycle at 280,000, she made a loss of sh.45,000. How much did she buy the bicycle?
6. Namukadde sold a cock at sh.47,000 making a loss of sh.8000. How much did she buy the cock.
7. Birungi Sudat sold a school bag at sh.37,000 and made a loss of sh.5000. How much did Sudat buy the school bag?
8. A head teacher sold a girl's uniform at sh.28,000 making a loss of sh.4000. How much did the head teacher buy the uniform?
9. Lady Sweetner sold a packet of sweets at sh.15500 and made a loss of 3500. How much did he buy the packet of sweets?
10. Mujomba sold a calculator at sh.62000 making a loss of sh.8000. What was the buying price of the calculator?
11. A sauce pan is sold at sh.24000 making a loss of sh.6000. What was the buying price of the sauce pan?
12. Madam Babandana sold a dress for sh.28,000. She made a loss of sh.7500. What was the buying price of the dress?

9.11 Transport Fares

Example 1

The table below shows train fare from Tororo to Kampala.

Towns		Fares
Tororo	Busembatya	Sh.9000
Busembatya	Iganga	Sh.5000
Iganga	Jinja	Sh.8000
Jinja	Kawolo	Sh.4000
Kawolo	Kampala	Sh.6000

a) Mr. Lumbuye travelled from Jinja to Kampala. How much did he pay as transport fares?

Jinja – Kawolo	sh.4 0 0 0
Kawolo – Kampala	+sh.6 0 0 0
	Sh.1 0 0 0 0

Lumbuye paid sh.10,000

c) What is the fare from Busembatya to Jinja?

Busembatya – Iganga	sh.5 0 0 0
Iganga – Jinja	sh.8 0 0 0
	Sh.1 3 0 0 0

b) Calculate the fare from Tororo to Jinja.

Tororo – Busembatya	sh.9 0 0 0
Busembatya – Iganga	sh.5 0 0 0
Iganga – Jinja	sh. 4 0 0 0

d) How much should one pay from Busembatya to Kawolo?

Busembatya – Iganga	sh.5 0 0 0
Iganga – Jinja	sh.8 0 0 0
Jinja – Kawolo	sh.4 0 0 0
	Sh.1 7 0 0 0

Activity 9:11

1. The table below shows the fare from Kampala to Fort portal towns.

Towns	Bus fares
Kampala – Mityana	Sh.10000
Mityana – Kasanda	Sh.5000
Kasanda – Mubende	Sh.8000
Mubende – Kyegegwa	Sh.6000
Kyegegwa – Fortportal	Sh.6000

a) What is the fares from Kampala to Mubende?
b) Nyangoma travelled from Kasanda to Fort portal, how much did he pay for transport?
c) How much is from Mubende to Fort portal?

2. Use the table below to work out the fares charged from different towns to Gulu to Kampala.

Different towns	Bus fares
Gulu – Kigumba	Sh.12000
Kigumba – Nakasongola	Sh.8000
Nakasongola – Luweero	Sh.5000
Luweero – Kampala	Sh.7000

a) How much is from Kampala to Luweero?
b) How much does one pay from Kampala to Kigumba?
c) Mr. Okello and his wife travelled from Gulu to Kampala. How much did they pay altogether?
d) How much is from kigumba to Luweero?
e) Three school children travelled from Nakasongola to Kampala, how much did they pay altogether, if they were half ticket pay?

TOPIC 10: LENGTH, MASS AND CAPACITY

LENGTH

Length is the distance between two points
1kilometre = 1000metres (1KM = 1000m)
1hetometre = 100metres (1HM = 100m)
1 decametre = 100 metres (1DM = 10m)
1 metre = 1000cm (1M = 1000cm)
1 decimetre = 100cm (1dm = 100cm)
1 centimetre = 10mm (1cm =10mm)

10.1 Express Kilometres to Metres

KM	HM	DM	M
1	0	0	0

1Km = 1000 metres

Example 1
Change 4km to metres
1km = 1000m
4km = (1000×4)m
4km = 4000m

Example 2
Express 3.6km to metres.
1km = 1000m
$3.6km = \left(\frac{36}{\cancel{10}_1} \times 100\cancel{0}\right)m$
3.6km = 36 ×100m
3.6km = 3600m

Example 3
Convert 2½km to metres
1km = 1000m
$2½km = \left(\frac{5}{\cancel{2}_1} \times \cancel{1000}^{500}\right)m$
2½km = (5×500)m
2½km = 2500m

Example 4
Express 0.125km to metres
1km = 1000m
$0.125km = \left(\frac{125}{\cancel{1000}_1} \times \cancel{1000}^1\right)m$
0.125km = 125m

Activity 10:1
1. Express 8km to metres.
2. Convert 9km to metres.
3. Change 12km to metres.
4. Express 15km to metres.
5. Convert 25km to metres.
6. Express 4.2km to metres.
7. Express 8.25km to metres.
8. Convert 12.5km to metres.
9. Change 2.18km to metres
10. Convert 10.125km to m.
11. Convert $10\frac{1}{4}km$ to metres
12. Change $8\frac{1}{4}km$ to metres.
13. Express 2½km to metres.
14. Change $7\frac{3}{10}km$ to metres.
15. Convert 0.45km to m.
16. Express 0.255km to m.
17. Express 0.85km to m.
18. What is 0.75km to m.

10:2 Express Metres to Kilometres

M	Dm	Hm	Km
0	0	0	1

1 metre = $\left(\frac{1}{1000}\right) km$ or 0.001km

Example 1

Express 4000 metres to km.

1 metre = $\left(\frac{1}{1000}\right) km$

4000m = $\left(\frac{1}{\cancel{1000}} \times 4\cancel{000}\right) km$

4000m = (1×4)km

4000m = 4km

Example 2

Convert 2500metres to km

1 metre = $\left(\frac{1}{1000}\right) km$

2500m = $\left(\frac{1}{\cancel{1000}} \times 25\cancel{00}\right) km$

2500m = $\left(\frac{25}{10}\right) km$

2500m = 2.5km

Example 3

Change 450 metres to km.

1 metre = $\left(\frac{1}{1000}\right)$km

450m = $\left(\frac{1}{\cancel{1000}} \times 45\cancel{0}\right) km$

450m = $\left(\frac{45}{100}\right) km$

450m = 0.45km

Example 4

Express 75metres to km.

1 metre = $\left(\frac{1}{1000}\right) km$

75m = $\left(\frac{1}{1000} \times 75\right) km$

75m = $\left(\frac{75}{1000}\right) km$

75m = 0.075km

Activity 10:2

1. Express 7000 metres to km.
2. Convert 2000 meters to km.
3. Convert 14000 metres to km.
4. How many km are in 5000m.
5. Express 750 metres to km.
6. Express 270 metres to km.
7. Change 350 metres to km.
8. Change 820 metres to km.
9. Convert 3500 metres to km.
10. Express 1500 metres to km.
11. How many km are in 1350 metres?
12. Express 56 metres to km.
13. Change 84 metres to km.
14. Convert 96 metres to km.

10:3 Express Metres to Centimetres

M	dm	cm
1	0	0

1 metre = 100 centimetres

Example 1

Change 8metres to cm.

1 metre = 100cm

8metres = (100×8)cm

8 metres = 800cm

Example 2

Change $7\frac{1}{2}$ metres to cm.

1 metre = 100cm

$7\frac{1}{2}$ metres = $\left(\frac{15}{\cancel{2}_1} \times \cancel{100}^{50}\right) cm$

$7\frac{1}{2}$ metres = (15×50)cm

$7\frac{1}{2}$ metres = 750cm

Example 3	Example 4
Express 2.4metres to cm. 1 metre = 100cm 2.4metres = $\left(\frac{24}{10} \times 10\cancel{0}\right) cm$ 2.4metres = (24×10)cm 2.4metres = 240cm	Express 0.24metres to cm. 1 metre = 100cm 0.24metres = $\left(\frac{24}{1\cancel{00}} \times 1\cancel{00}\right) cm$ 0.24metres =(24×1)cm 0.24 metres = 24cm

Activity 10:3

Convert the following to centimetres.

1. Express 5metres to cm.
2. Convert 13 metres to cm.
3. Change 15 metres to cm.
4. What is 24 metres to cm.
5. Change 120 metres to cm.
6. Express 1.8metres to cm.
7. Express 9.2metres to cm.
8. Convert 3.45m to cm.
9. Express 7.15mtrs to cm.
10. Change 4.8metres to cm.
11. Express $\frac{1}{4}$mtrs to cm.
12. Express $1\frac{1}{2}$metres to cm.
13. What is $8\frac{1}{2}$metres to cm.
14. Express $12\frac{1}{4}$metres to cm.
15. Express $4\frac{1}{5}$metres to cm.
16. Express $\frac{7}{10}$metres to cm.
17. Change 0.148m to cm.
18. Convert 0.256m to cm.
19. Convert 0.65metres to cm.
20. Express 0.25metres to cm.

10:4 Express Centimetres to Metres

cm	dm	m
0	0	1

1cm = $\left(\frac{1}{100}\right) metres$

1cm = (0.01)metres

Example 1	Example 2
Express 280cm to metres. 1cm = $\left(\frac{1}{100}\right) metres$ 280cm = $\left(\frac{1}{10\cancel{0}} \times 28\cancel{0}\right) metres$ 280cm = $\left(\frac{28}{10}\right) metres$ 280cm = 2.8metres	Convert 40cm to metres. 1cm = $\left(\frac{1}{100}\right) metres$ 40cm = $\left(\frac{1}{10\cancel{0}} \times 4\cancel{0}\right) metres$ 40cm = $\left(\frac{4}{10}\right) metres$ 40cm = 0.4metres

Activity 10:4

1. Express 400cm to metres.
2. Convert 1200cm to metres.
3. Change 700cm to metres.
4. Express 560cm to metres.
5. Express 880cm to metres.
6. Convert 70cm to metres.
7. Express 52cm to metres.
8. Express 99cm to metres.

9. Change 12cm to metres.
10. Express 144cm to metres.
11. Convert 650cm to metres.
12. Change 250cm to m.

10.5 Expressing Square Metres to Square Centimetres

1 metre = 100cm

1sq metre = (100cm×100cm)

1 square metre = 10000 sq cm or $1000cm^2$

Example 1	Example 2
Express 3 sq metre to square centimetres. 1metre = 100cm 1 sq metre = 100cm×100cm 1 sq metre = 10000 sq cm 3 sq metres = (3×10000)sq cm 3 sq metres = 30000 sq cm or $30000cm^2$	Change 5.4 square metres to square centimetres. 1 metre = 100cm 1 square metre = 100cm×100cm 5.4sq metre $=\left(\frac{54}{10}\times 10000\right) sqcm$ 5.4sq metre = (54×1000)sq cm 5.4sq metres = 54000 sq cm

Activity 10:5

1. Express $8m^2$ as square cm.
2. Express $5m^2$ as square cm.
3. Express $12m^2$ as square cm.
4. Convert $2.5m^2$ to square cm.
5. Convert $7.4m^2$ to Square cm.
6. Convert $9.2m^2$ to square cm.
7. Change $2\frac{1}{2}m^2$ to square cm.
8. Change $12\frac{1}{2}m^2$ to square cm.
9. Change $4\frac{1}{4}m^2$ to square cm.
10. Express $12.5m^2$ to square cm.
11. The area of a rectangle is $0.8m^2$. express the area in square centimetres.
12. The area of a school garden is $72m^2$. What is its area in square centimetres?
13. A table surface measures 3m by 2m. calculate its area in square centimetres.

10.6 Expressing From Square Centimetres to Square Metres

Example 1	Example 2
Express 12000 sq cm to square metres 10000 sq cm = 1sq metre 12000 sq cm = $\frac{12000\ \cancel{sq\ cm}}{10000\ \cancel{sq\ cm}}$ 12000 sq cm = $\left(\frac{12}{10}\right) sq\ m$ 12000 sq cm = 1.2 sq m.	Convert 48000 square centimetres to square metres. 10,000sq cm = 1 sq m 1 sq cm = $\left(\frac{1}{10000}\right) sq\ m$ 48000sq m = $\frac{1}{\cancel{10000}_1}\times 480\cancel{00}$ 48000 sq cm = $(1\times\frac{48}{10})$ sq m 48000 sq cm = 4.8 sq m

Activity 10.6

Express the following to square metres.

1) 80000 sq cm. 2) 8000 sq cm. 3) 800 sq cm. 4) 24000 sq cm.
5) 5200 sq cm. 6) 1900 sq cm. 7) 640000 sq cm. 8) 4600 sq cm.
9) 3600 sq cm. 10) 880000 sq cm. 11) 198000 sq cm. 12) 360000sq cm.
13) A rectangle measures 30cm by 50cm. Calculate its area in square metres.
14) The area of a rectangular flower garden is 48500 sq cm. Express its area in square metres.
15) Express the area of a rectangle below in square metres.

10.7 Expressing Square Kilometres to Square Metres

1km = 1000metres
1 sq km = 1000m×1000m
1 sq km = 1000,000m²

Example 1	Example 2
Express 8 sq km to sq metres	Change 2.5sq km to sq m.
1km = 1000m	1km = 1000m
1 sq km = 1000m×1000m	1 sq km = 1000m×1000m
1 sq km = 1000000m²	1 sq km = 1000,000m²
8 sq km = (8×100000)m²	2.5sq km = $\frac{25}{10}$ × 1000000m²
Sq km = 8000000m²	2.5sq km = 2500000m²

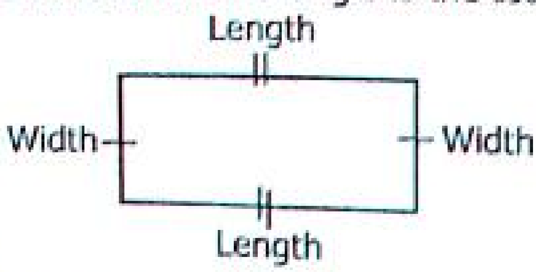
Activity 10:7

Express the following to square metres.

1) 3 sq km 2) 5 sq km 3) 9 sq km 4) 12 sq km 5) 15 sq km
6) 1 sq km 7) 0.4sq km 8) 0.8 sq km 9) 0.15sq km 10) $1\frac{1}{2}$sq km
11) $5\frac{1}{4}$sq km 12) $8\frac{1}{7}$sq km 13) 1.8 sq km 14) 2.4 sq km 15) 4.8 sq km

10.8 Perimeter of Rectangles

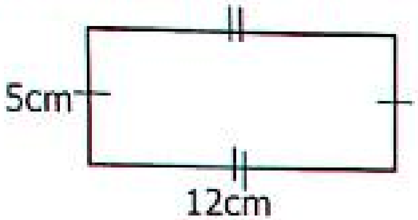
Perimeter of a rectangle is the total distance or length around the figure.



Perimeter = L+W+L+W
Perimeter = L+L+W+W
Perimeter = 2L+2W
Perimeter = 2(L+W)

Example 1

Find the perimeter of the figure below.



Perimeter = 2(L+W)
Perimeter = 2(12cm+5cm)
Perimeter = 2(17cm)
Perimeter = 2×17cm
Perimeter = 34cm

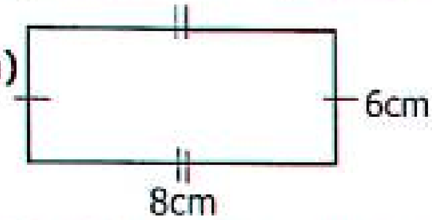
Example 2

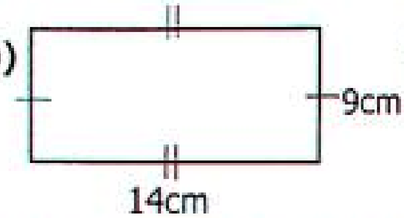
A football field measures length 100m and width 80m.
Find its perimeter.

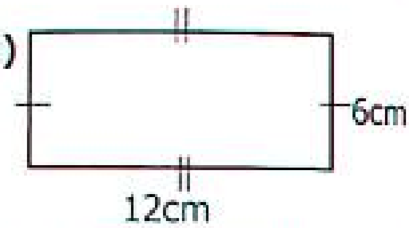
Perimeter = 2(L+W)
Perimeter = 2(100m+80m)
Perimeter = 2(100+80)m
Perimeter = 2×180m
Perimeter = 360m

Activity 10:8

1) Calculate the perimeter of the figures below;

a)


b)


c)


2) Calculate the perimeter of a rectangle whose length is 40mm and width 32mm.
3) A classroom measures 6metres long and 4metres wide. Find the perimeter of the classroom.
4) A rectangular garden measures 12metres by 9metres. Find its perimeter.
5) A rectangular swimming pool measures 300cm by 150cm. What is its perimeter?
6) Find the perimeter of a rectangular flower garden whose length is 80metres and width 50 metres.

10.9 Finding the Length or Width When Given Perimeter

Example 1

The perimeter of a rectangle below is 36cm.
Find the value of p.

12(L+p) = p
2(14cm+p) = 36
28+2p = 36
28 − 28+2p = 36 − 28

$$\frac{\cancel{2}p^1}{\cancel{2}_1} = \frac{\cancel{8}^4}{\cancel{2}_1}$$

p = 4cm

Example 2

Given that, the perimeter of a rectangle is 42cm and the width is 8cm. What is its length?

2(L+W) = p
2(L+8) = 42
2L+16 = 42
2L+16 − 16 = 42

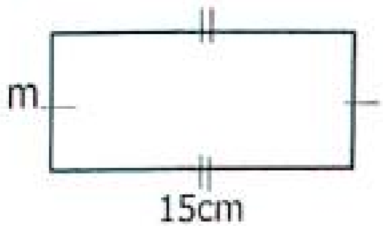
$$\frac{\cancel{2}L^1}{\cancel{2}_1} = \frac{\cancel{26}^{13}}{\cancel{2}_1}$$

L = 13cm

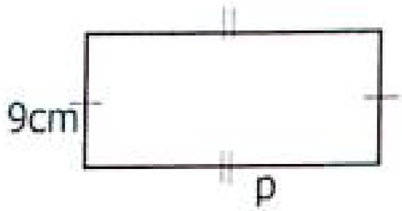
Activity 10:9

1. Find the length of a rectangle whose perimeter is 24cm and width 5cm.
2. If the perimeter of a rectangle is 30cm, Find the width when the length is 8cm.
3. Find the length of a rectangle whose width is 11cm and perimeter 56cm.
4. The perimeter of a rectangle is 20cm. Find its width if the length is 6cm.
5. Given the perimeter of a rectangles below are 48cm. Find the value of the unknown sides.

(i)



(ii)



6. The perimeter of a rectangle is 28mm. Find its length if the width is 4mm.
7. The length of a rectangle is 15cm and the perimeter is 72cm. What is its width?
8. The perimeter of a rectangle is 40cm. Its length is 5cm. Find the width.
9. The perimeter of a rectangle 28cm. Find the width if the length is 8cm.
10. The perimeter of a rectangle is 50cm. Find its length if the width is 14cm.

10.10 Finding the Perimeter of Squares

A square is four sided polygons with all for sides equal.

Perimeter = side + side + side + side

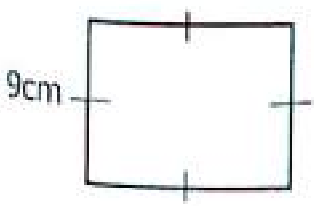
Perimeter = s+ s + s + s

Perimeter = 2s + 2s

Perimeter = 4s

Example 1

Use the diagram below to find its perimeter.



Perimeter = 4s

Perimeter = (4×9)cm

Perimeter = 36cm

Example 2

Find the perimeter of a square whose sides are 12cm.

Perimeter = 4s

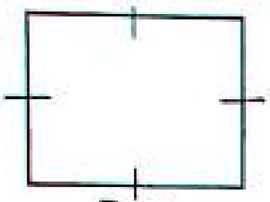
Perimeter = 4×12cm

Perimeter = 48cm

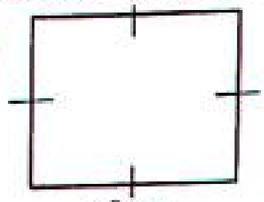
Activity 10:10

1) Calculate the perimeter of the squares below;

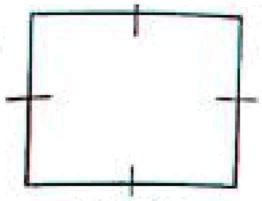
a)



b)



c)

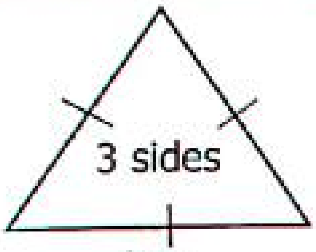


2) Calculate the perimeter of a square whose sides are 12cm.
3) What is the perimeter of a square whose sides is 16cm.
4) A square garden has a length of 18cm. Calculate its perimeter.
5) Find the perimeter of a square whose sides are 13cm.
6) The perimeter of a square is 36cm. Find the length of each side.
7) If the perimeter of a square is 84cm. Find the length of each side.
8) Calculate the length of a square floor whose perimeter is 28metres.
9) Calculate the length of each side of a square whose perimeter is 60cm.
10) Find the length of each side of a square whose perimeter is 112cm.

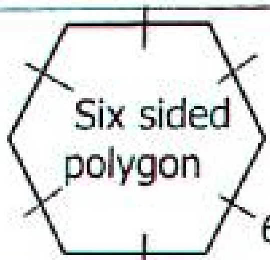
10.11 Perimeter of Different Figures

Remember perimeter is the total length around any given figure

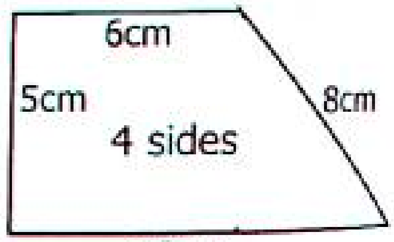
Perimeter of Polygons



P = S + S + S
P = 9cm+9cm+9cm
P = (9+9+9)cm
P = 27cm



P = S+S+S+S+S+S
P=6cm+6cm+6cm+6cm+6cm+6cm
P = (6+6+6+6+6+6)cm
P = 36cm

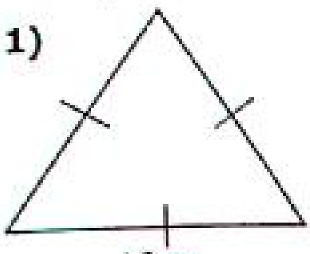


P = S+S+S+S
P=12cm+5cm+6cm+8cm
P = (12+5+6+8)cm
P = 31cm

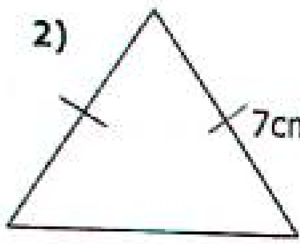
Activity 10:11

Find the perimeter of the figures below;

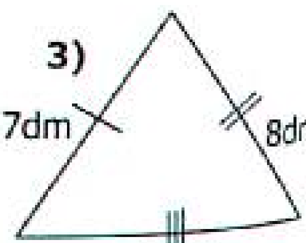
1)



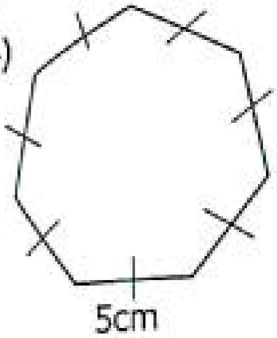
2)



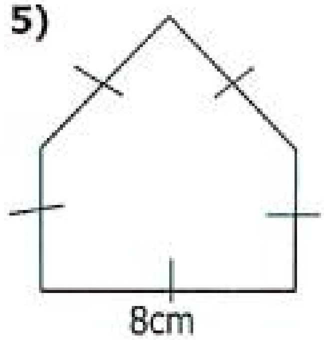
3)



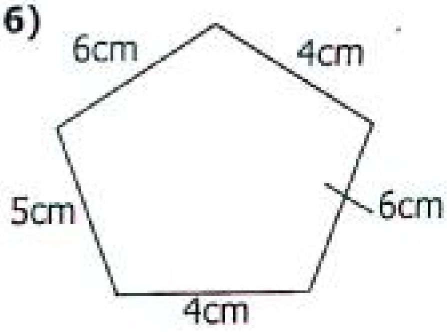
4)



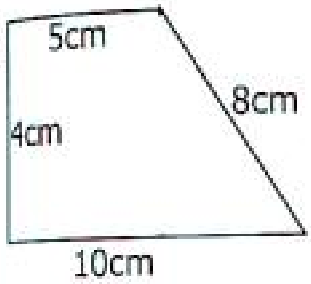
5)



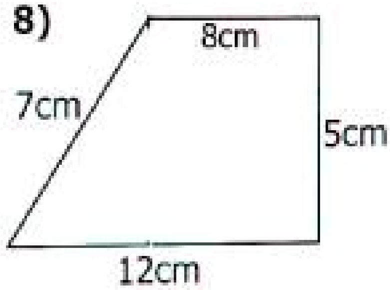
6)



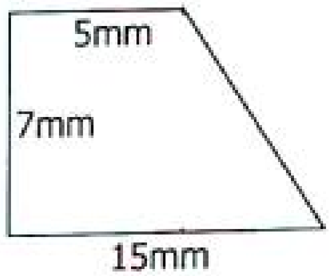
7)



8)

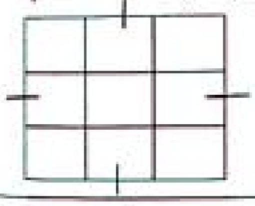


9)



10.12 <u>Area of Squares</u>

A square is a polygon with four sides equal.

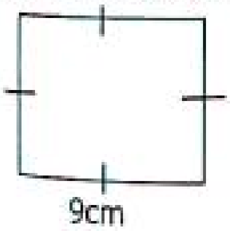


Area = 3units×3units
Area = 9 sq units

Area of a square = side × side
Area = S × S
Area = S^2

Example 1

Calculate the area of a square below

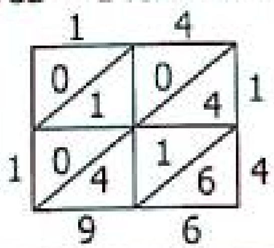


Area = S × S
Area = 9cm×9cm
Area = (9×9)sq cm
Area = 81sq cm

Example 2

Calculate the area of a square garden of sides of 14cm.

Area = S × S
Area = 14cm×14cm



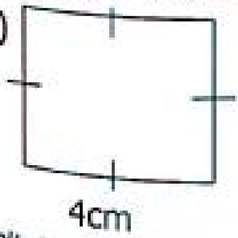
$$\begin{array}{r} 14 \\ \times 14 \\ \hline +56 \\ 14 \\ \hline 196 \\ \hline \end{array}$$

Area = 196sq cm

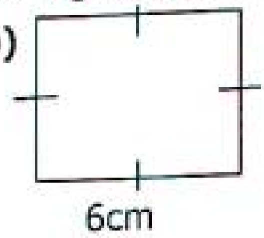
Activity 10:12

1. Calculate the area of figures below

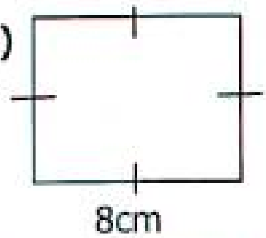
a)



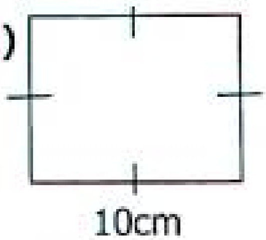
b)



c)



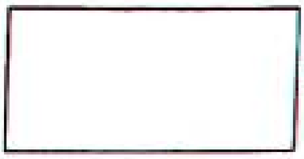
d)



2. Calculate the area of a square flower garden measuring 11metres.
3. Find the area of a square room of length 15 metres.
4. The side of a square piece of cloth 120cm. Calculate the area of the coth.
5. Workout the area of s square whose sides is 8cm.
6. A square compound has a length of 22metres. Find its area.

10:13 Area of Rectangles

✓ Area is the number of square units needed to cover a surface:



✓ The area of a rectangle is the product of its length and breadth (width)
Area=Length × breadth.

✓ Area is measured in square units e.g square centimetres (cm^2), square metres (m^2), square millimetres(mm^2) and square kilometres(km^2).

✓ We can count the number squares to represent the area of the figure.

Example 1

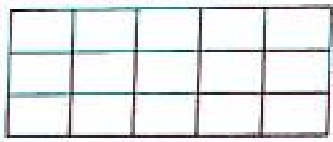
Calculate the area of the figure below.

- *By counting the squares*

1	2	3	4	5
10	9	8	7	6
11	12	13	14	15

Area = 15 sq units

Use of formula; A = L × W



Area = Length × Width
Area = 5units × 3units
Area = (5×3)sq units
Area = 15sq units

Example 2

Calculate the area of the rectangle below.

Area=Length×Width
Area=14cm×8cm
Area=(14×8)sq cm
Area=112 sq cm

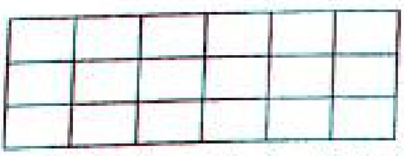
Example 3

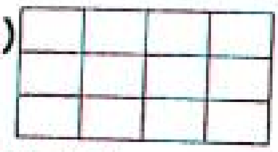
Calculate the area of a rectangle whose length is 12cm and width 9cm

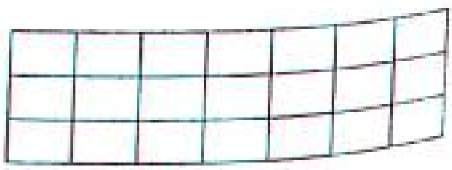
Area = Length × Width
Area = 12cm × 9cm
Area = (12×9)sq cm.
Area = 108 sq cm

Activity 10:13

1. Calculate the area of the figures below;

a) 

b) 

c) 

d) 

e) 

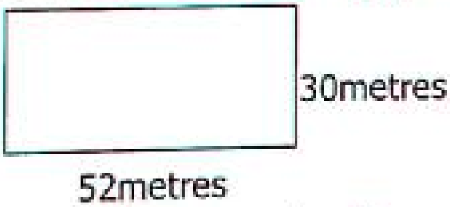
f) 

g) 

h) 

k) 

2. Calculate the area of a rectangular floor whose length is 8metres and width 6metres.
3. Find the of a rectangle whose length is 12cm and width 8cm.
4. Calculate the area of a flower garden measuring 10metres by 6metres.
5. The of a rectangular carpet is 12metres and its width is 5metres. Calculate its area.
6. Calculate the area of a netball field measuring 24metres by 12metres.
7. Calculate the area of a rectangle whose length is 15dm and width 10dm.
8. The diagram below shows a swimming pool. Find the area of the swimming pod.



9. A textbook book measures 26cm by 12cm. calculate its area.
10. Find the area of a piece of paper measuring 18cm length and 9cm width.

10.14 Finding one Side of a Rectangle When Given the Perimeter

Example 1

The area of a rectangle is $36cm^2$. Find the width if the length is 9cm.

$L \times W = A$

$9cm \times w = 36cm^2$

$$\frac{\cancel{9cm} \times w^1}{\cancel{9cm}_1} = \frac{\cancel{36cm}^4 \times \cancel{cm}}{\cancel{9cm}_1}$$

$w = 4cm$

Example 2

Find the length of a rectangle whose area is 72sq cm and width 8cm.

$L \times w = A$

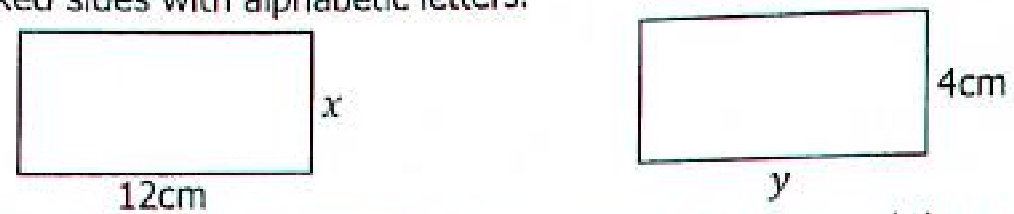
$L \times 8cm = 72$ sq cm

$$\frac{L \times \cancel{8cm}^1}{\cancel{8cm}_1} = \frac{\cancel{72}^9 \times cm \times \cancel{cm}}{\cancel{8cm}}$$

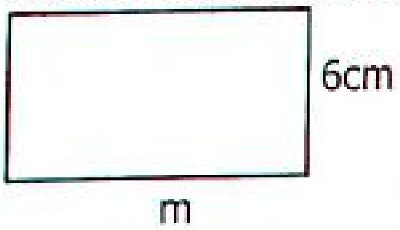
$L = 9cm$

Activity 10:14

1. The area of a rectangle is $84cm^2$. Work out the length of a rectangle.
2. The area of a rectangular piece of paper is $48cm^2$. Find its width if the length is 12cm.
3. The area of a rectangle is $108cm^2$. Find its length if the width is 9cm.
4. Calculate the width of a rectangle whose area is $144cm^2$ when the length is 16cm.
5. Given that the area of the rectangles below is $72cm^2$. Find the value of the marked sides with alphabetic letters.

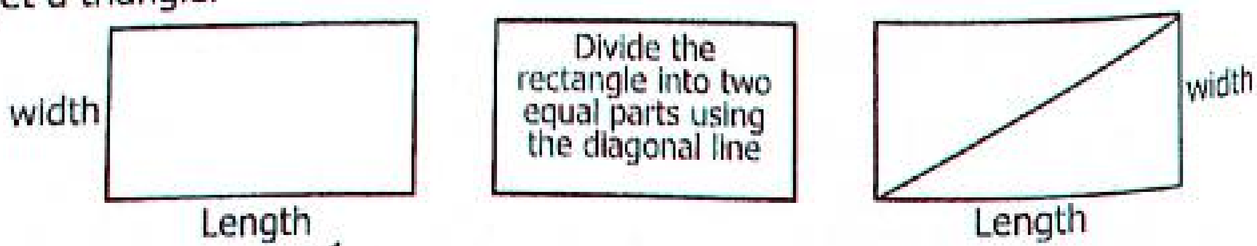


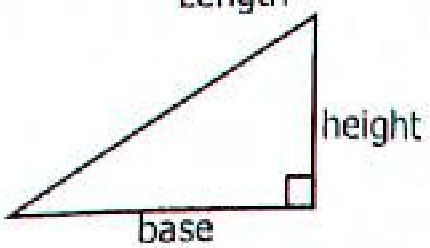
7. Find the width of a rectangle whose length is 11cm and the area of $99cm^2$.
8. The area of a rectangular field is $144m^2$. If its length is 14m. find the width.
9. Find the width of a rectangle whose area is $108cm^2$ if the length is 18cm.
10. Given that the area of a rectangle below is $80cm^2$. Find the value of m.



10.15 Area of Triangles

A triangle is got from a rectangle. Divide a rectangle into two equal sides to get a triangle.



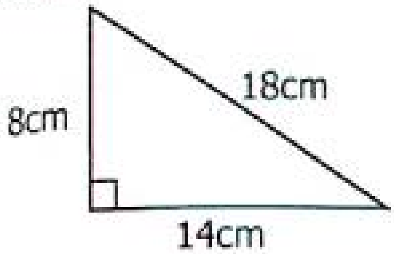


$$Area = \frac{Length \times width}{2}$$

$$Area = \frac{Base \times height}{2}$$

Example 1

Calculate the area of a triangle below.



Area = ½×b×h

Area = ½×14cm×8cm

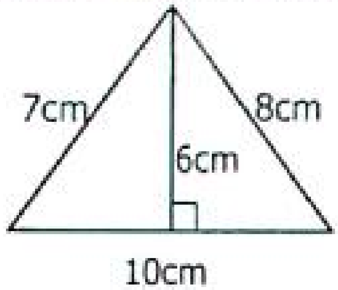
$Area = \frac{1}{\cancel{2}_1} \times \cancel{14}^7 \times 8\ sqcm$

Area = 1×7×8 sq cm.

Area = $56cm^2$

Example 2

Find the area of the triangle below.



Area = ½×b×h

Area =½×10cm×6cm

$Area = \frac{1}{\cancel{2}_1} \times \cancel{10}^5 \times 6\ sq\ cm$

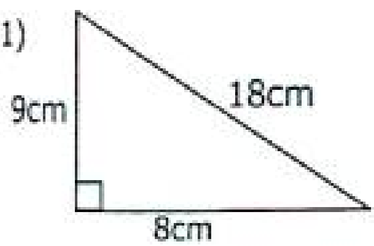
Area = 1×5×6 sq cm

Area = 30sq cm

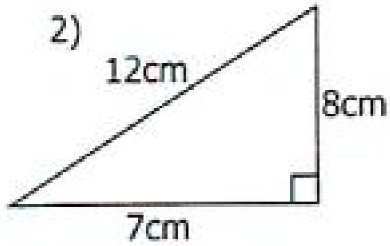
Activity 10:15

Calculate the area of triangles below.

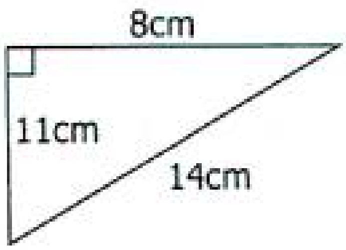
1)



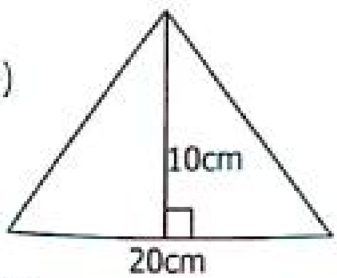
2)



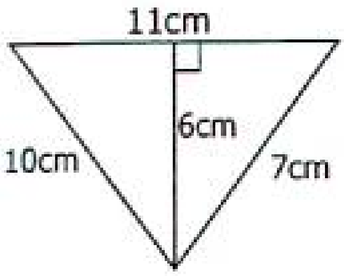
3)



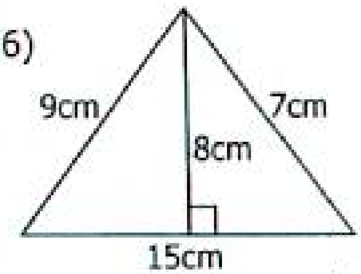
4)



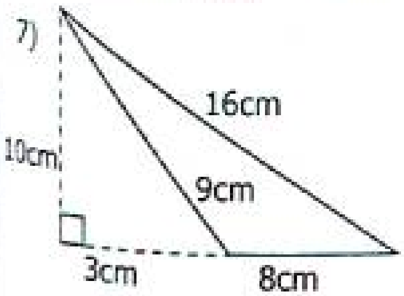
5)



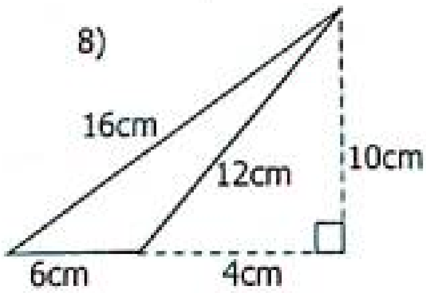
6)



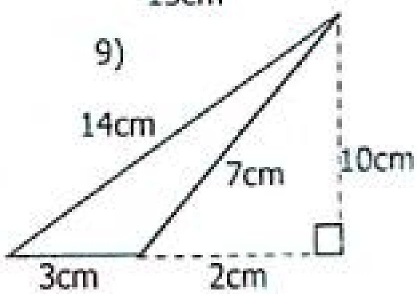
7)



8)



9)

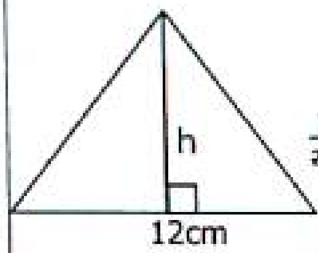


10. Calculate the area of a triangle whose base is 8cm and height 3cm.
11. Find the area of a triangle measuring base 10cm and height 8cm.
12. Workout the area of a triangle whose base is 24cm and height 16cm.
13. Calculate the area of a triangle whose base is 20cm and height 9cm.

10.16 Find the Height or Base of Triangles When Given the Area

Example 1

The area of a triangle below is $48cm^2$.
Find its height.



$$\frac{1}{2}bh = A$$

$$\frac{1}{\cancel{2}_1} \times \cancel{12cm}^6 \times h = 48cm^2$$

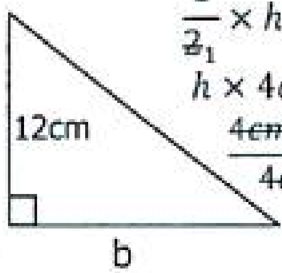
$$6cm \times h = 48cm^2$$

$$\frac{\cancel{6cm} \times h^1}{\cancel{6cm}_1} = \frac{48cm^8 \times \cancel{cm}}{\cancel{6cm}_1}$$

$$h = 8cm$$

Example 2

The area of a triangle is $36cm^2$. Find its base if the height 8cm.



$$\frac{1}{2}bh = A$$

$$\frac{1}{\cancel{2}_1} \times h \times \cancel{8}cm^4 = 36cm^2$$

$$h \times 4cm = 36cm \times cm$$

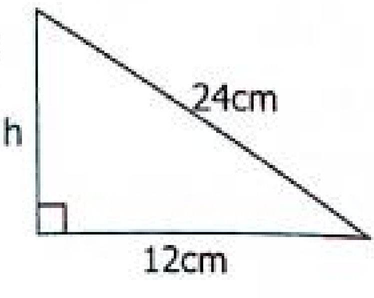
$$\frac{\cancel{4cm}^1 \times h}{\cancel{4cm}_1} = \frac{36cm^9 \times cm}{\cancel{4cm}_1}$$

$$h = 9cm$$

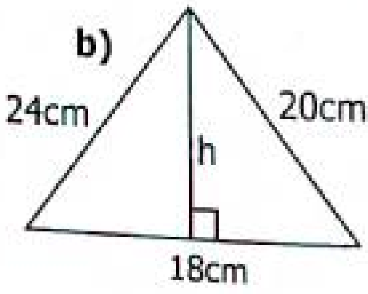
Activity 10:16

1. The area of a triangle is $24cm^2$. Find its height if the base is 8cm.
2. Given that the area of a triangle is $28cm^2$. What is its base if the height is 14cm.
3. The length of a triangle is 12cm. its area is $48cm^2$.Find the width.
4. The area of a triangle is $24cm^2$. Find the base if its height is 8cm.
5. Find the height of a triangle whose base is 15cm and area $60cm^2$.
6. The areas of triangles below is $72cm^2$. Find the heights of the triangles.

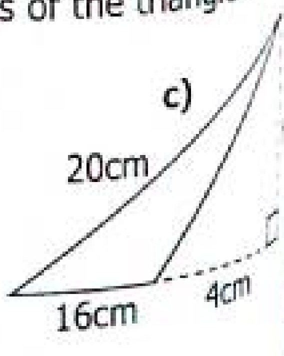
a)



b)



c)

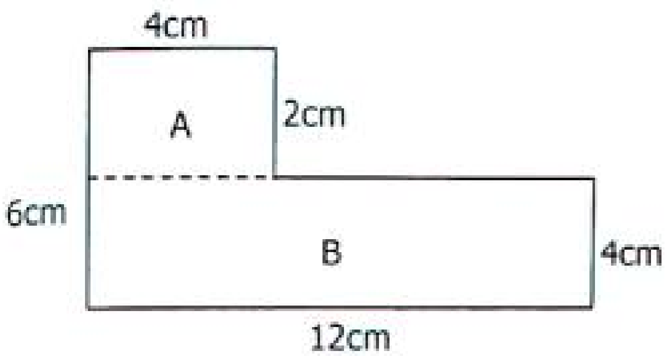


10.17 Area of Combined Figures

Find the area of the figure below.

Example 1

Divide the figure into two manageable figures A and B.



Find area of A

Area = L×W

Area = 4cm×2cm

Area = (4×2)sq cm

Area = 8 sq cm

Area of B

Area =L×W

Area =12cm×4cm

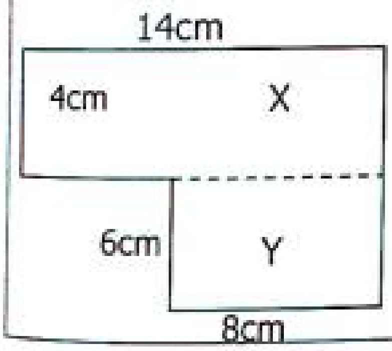
Area = (12×4)sq cm

Area = 48sq cm

Total area of the figure

$$\begin{array}{r} 8 \text{ sq cm} \\ +\ 48 \text{ sq cm} \\ \hline 56 \text{ sq cm} \\ \hline \end{array}$$

Example 2



Area of figure X

Area=L × W

Area=14cm×4cm

Area=(14×4)sq cm

Area=56 sq cm

Area of figure Y

Area = L × W

Area = 8cm×6cm

Area = (8×6)sq cm

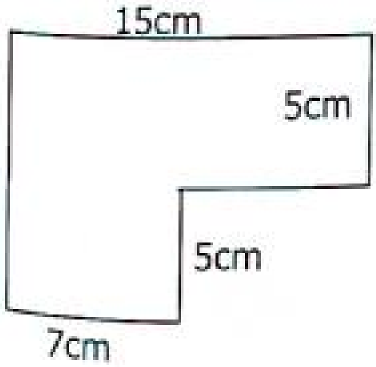
Area = 48 sq cm

Total area of the figure

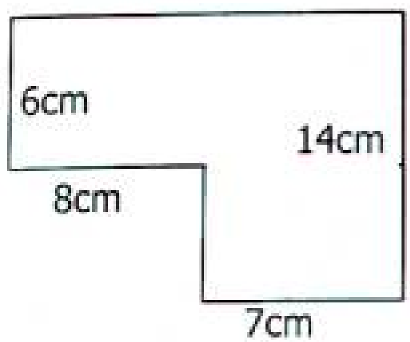
$$\begin{array}{r} 56 \text{ sq cm} \\ +\ 48 \text{ sq cm} \\ \hline 104 \text{ sq cm} \\ \hline \end{array}$$

Activity 10:17

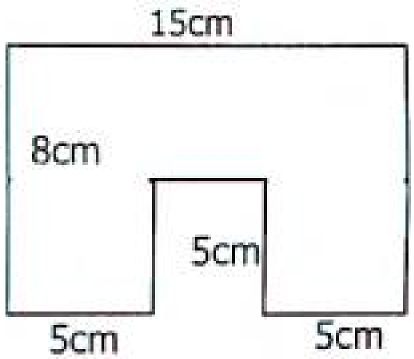
1)



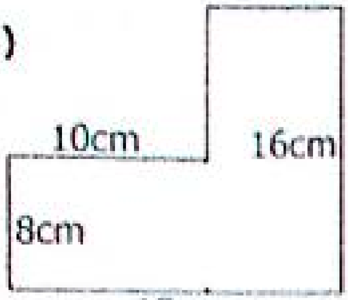
2)



3)

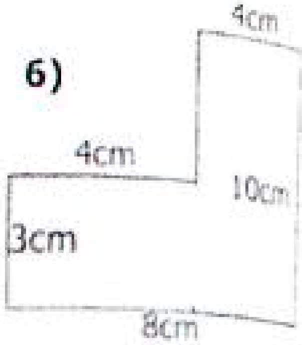


4)

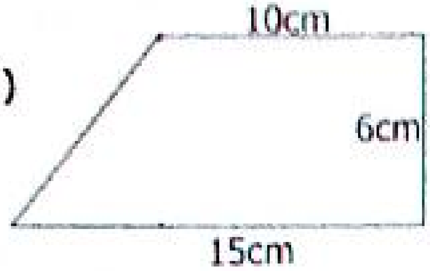


5)

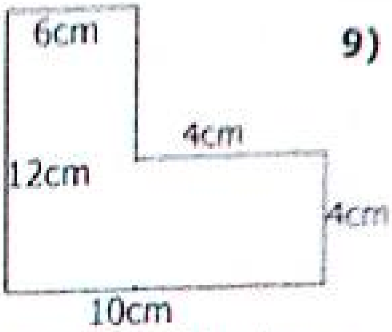
6)



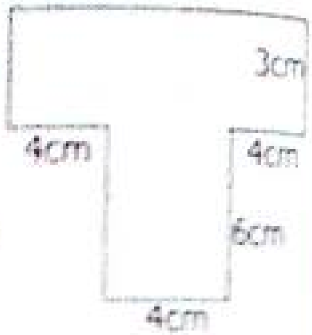
7)



8)

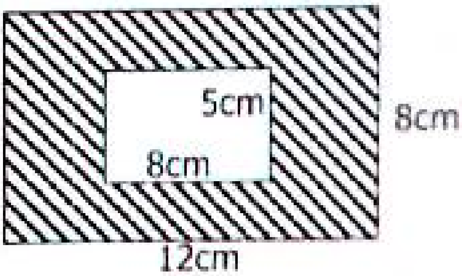


9)



10.18 Calculate the Area of the Shaded Regions

Work out the area of the shaded region.

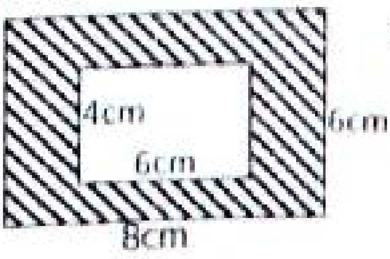


Area of the outer figure	Area of the inner figure	Area of shaded region
Area = 12cm, W = 8cm	L = 8cm, W = 5cm	9 6 sq cm
Area = L × W	Area = L × W	- 4 0 sq cm
Area = 12cm×8cm	Area = 8cm×5cm	5 6 sq cm
Area = (12×8)sq cm	Area = (8×5) sq cm	
Area =96 sq cm	Area = 40 sq cm	

Activity 10:18

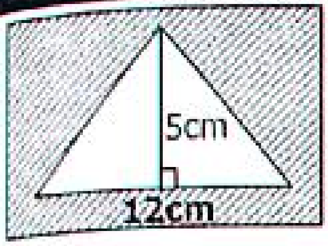
Calculate the area of the shaded regions in the numbers below.

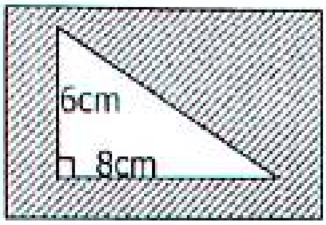
1)

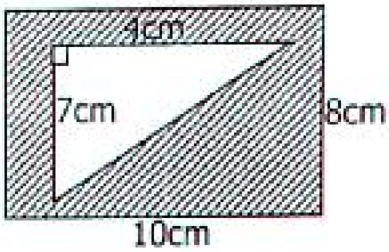


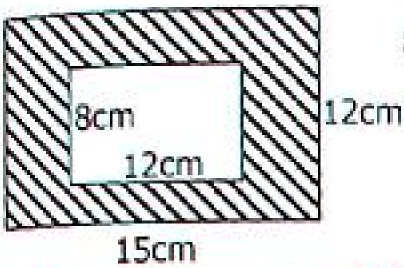
2)

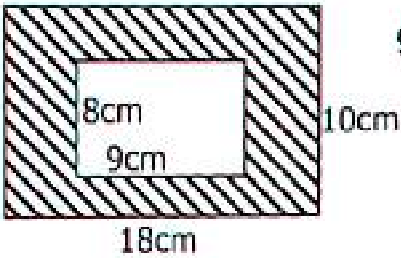
3)

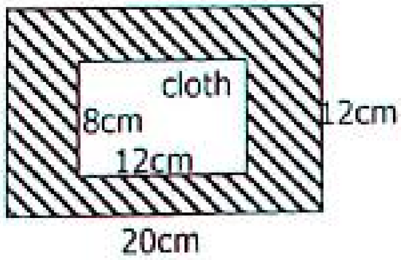
4)


5)


6)


7)


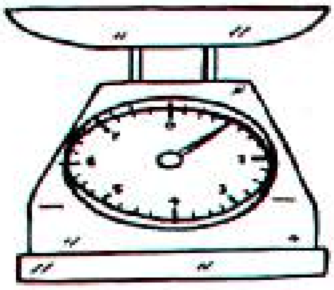
8)


9)


10.19 Metric System (Mass)

Mass is how much matter is in an object. The basic unit of mass is grammes

A weighing scales



1 kilogram = 1000grams (1kg = 1000g)

1 Hectogram = 100 grams (1Hg = 100g)

1 Decagram = 10 grams (1Dg = 10g)

1 Gram = 1000 centigrams (1dg = 100cg)

1 Centigram = 10 milligrams (1cg = 10mg)

Expressing Kilograms to Grams

kg	Hg	Dg	g
1	0	0	0

1kg = 1000g

Example 1

Express 5kg to grams

1kg = 1000g

5kg = (1000×5)g

5kg = 5000g

Example 2

Change 4.8kg to grams.

1kg = 1000g

4.8km = $\left(\frac{48}{\cancel{10}_1} \times 100\cancel{0}\right) g$

4.8kg =(48×100)g

4.8kg = 4800g

Example 4

Change 0.142kg to grams.

1kg = 1000g

0.142kg = $\left(\frac{142}{\cancel{1000}} \times \cancel{1000}^1\right) g$

0.142kg = 142×1g

0.142kg = 142g

Activity 10:19

1. Express 8kg to grams.
2. Convert 12kg to grams.
3. Convert 15kg to grams.
4. Express 5½kg to grams.
5. Change $5\frac{1}{2}$kg to grams.
6 Express 2¼ to grams.
7. Convert 4.2kg to grams.
8. Express 7.25kg to grams.
9. Change 12.5kg to grams.
10. How many grams are in 15.2kg?
11. Express 0.75kg to grams.
12. Express 0.256kg to grams.
13. Change ¼kg to grams

10.20 <u>Expressing Grams to Kilograms</u>

G	Dg	Hg	Kg
0	0	0	1

$1g = \left(\frac{1}{1000}\right) kg$

1kg = 0.001kg

Example 1	**Example 2**
Express 500g to kilograms	Change 1420g to kilograms.
$1g = \left(\frac{1}{10\cancel{00}} \times 5\cancel{00}\right) kg$	$1g = \left(\frac{1}{1000}\right) kg$
$500g = \left(\frac{5}{10}\right) kg$	$1420g = \left(\frac{1}{100\cancel{0}} \times 142\cancel{0}\right) kg$
500g = 0.5kg	$1420g = \left(\frac{142}{100}\right) kg$
	1420g = 1.42kg

Activity 10:20

Express the following to grammes.

1) 300g	2) 400g	3) 700g	4) 800g	5) 1200g
6) 1400g	7) 9800g	8) 7500g	9) 17000g	10) 24000g
11) 56000g	12) 175000g	13) 12500g	14) 1100g	15)1850g

16) A shopkeeper bought 178000grams of rice. How many kilograms of rice did he buy?
17) The mass of a bag of beans is 12000grams. How many kilograms are in 3750grams.
18) Owor bought 41250grams of groundnuts. Express it in kilograms.

10.21 <u>Capacity in Litres and Millilitres</u>

1litres = 1000millilitres (1L = 1000ML)
1 decalitres = 100millilitres (1DL = 100ML)
1 centilitre = 10millilitres (1CL = 10ML)
1 millilitre = $\left(\frac{1}{1000}\right) litres$ (1MM = $\left(\frac{1}{1000}\right) L$

Expressing Litres to Millilitres

Converting from litres to milliliter multiply by 100ml

L	DL	CL	ML
1	0	0	0

1Litre = 1000ml

Example 1

Express 7litres to millilitres

1litre = 1000ml

7litres = (1000×7)ml

7litres = 7000ml

Example 2

Convert 4¼litres to ml.

1litre = 1000ml

4¼litres = $\left(\frac{17}{\cancel{4}_1}\times \cancel{1000}^{250}\right)ml$

4¼litres = (17×250)ml

4¼litres = 4250ml

Activity 10.21

Express the following numbers from litres to millilitres.

1) 5litres 2) 8litres 3) 9litres 4) 10litres 5) 2.4litres

6) 7.2litres 7) 12.4litres 8) 15.8litres 9) 5½litres 10) 8½litres

11) 24¼litres 12) 4½litres 13) 0.124litres 14) 0.75litres 15)0.257litres

16) A drone taxi travelled from Soroti to Kampala and used 18litresof petrol. Express the amount of petrol in cubic units.

17) A bucket of paint contain 12litres. Express the amount of amount of paint in millimetres.

10.22 Expressing Millilitres to litres

✓ When expressing millitres to litres we divide by 1000ml

ML	CL	DL	L
0	0	0	1

1000ml = 1litre

1ML = $\left(\frac{1}{1000}\right) litres$

Example 1

Express 600ml to litres

1000ml = 1litre

$1ml\left(\frac{1}{1000}\right) litres$

6000ml = $\left(\frac{1}{1\cancel{000}}\times 6\cancel{000}\right)L$

6000ml = (1×6)L

6000ml = 6litres

Example 2

Change 8500 ml to litres

1000ml = litres

$1ml = \left(\frac{1}{1000}\right) litres$

$8500ml = \left(\frac{1}{100\cancel{0}}\times 850\cancel{0}\right) l$

$8500ml = \left(\frac{85}{10}\right) l$

$8500ml = 8.5 litres$

Example 3

Express 2560millilitres to litres.

1000ml = 1litres

$1ml = \left(\frac{1}{1000}\right) litres$

$\left(\frac{1}{100\cancel{0}}\times 256\cancel{0}\right) litres$

$2560ml = \left(\frac{256}{100}\right) litres$

2560ml = 2.56litres

Activity 10:22

Change the following to litres

1) 3000ml 2) 5000ml 3) 7000ml 4) 8000ml 5) 7500ml
6) 4800ml 7) 9800ml 8) 2400ml 9) 480ml 10) 550ml
11) 880ml 12) 320ml 13) 200ml 14) 300ml 15) 700ml

17) Adam bought 6500litres of milk. How many millilitres did he buy?

18) Baby Latifah drinks 750millilitre of milk every morning. How much milk does the baby drink in litres?

19) A house wife uses 2750millilitres of water to prepare rice. How many litres of water does he use?

10.23 Addition and Subtraction in Litres and Millilitres

Example 1		Example 2		Example 3	
Litres	ML	Litre	ML	Litre	ML
7	240	2	540	17	690
+8	380	+4	720	− 12	430
15	620	7	260	5	260

Activity 10:23

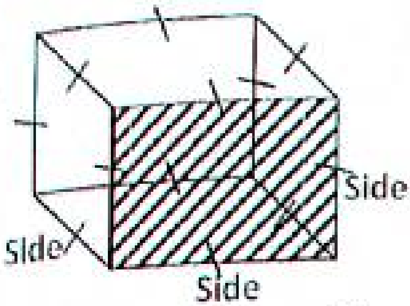
Add the following

1) L	ML	2) L	ML	3) L	ML	4) L	ML
9	146	12	270	3	400	144	350
+7	562	+14	830	+ 5	600	+141	650

5) L	ML	6) L	ML	7) L	ML	8) L	ML
14	288	27	754	10	450	19	675
+16	222	+32	646	- 7	070	-16	160

9) L	ML	10) L	ML	11) L	ML	12) L	M
24	110	34	024	47	175	88	250
-20	149	-28	086	-19	175	-54	750

10.24 Volume of cubes



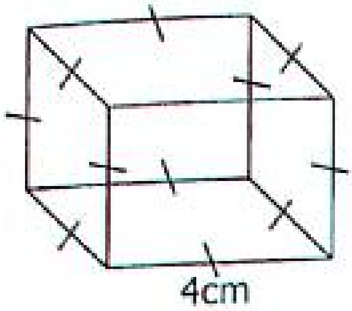
Volume is the space occupied by an object. Volume can be measured using bottles, jerrycans, tanks, buckets etc..

Volume = side × side × side

Volume = (S×S)×S

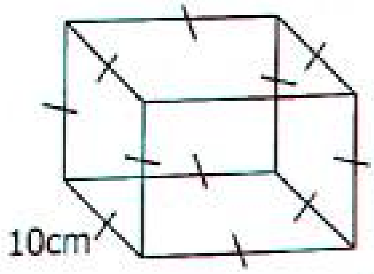
Calculate the volume of the cubes below

Example 1



V = (S × S) × S
V = base area × h
V = (4cm ×4cm)×4cm
V = 16cm^2× 4cm
V = 64cm^3

Example 2



Volume = Base area × h
Volume = (10cm×10cm)×10cm
Volume = 100cm^2×10cm
Volume = 1000cm^3

Example 3

Find the volume of a cube whose Side is 12cm.

Volume = Base area × h
Volume = (12cm×12cm)×12cm
Volume = 144cm^2×12cm
Volume = 1728cm^3

Example 4

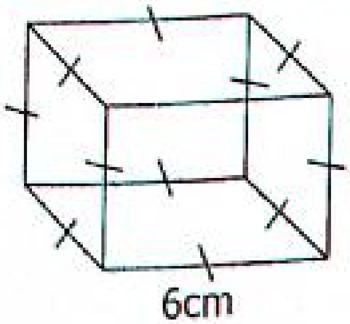
Calculate the volume of a cube whose side is 15cm.

Volume = Base area × h
Volume = (15cm×15cm)×15cm
Volume = 225cm^2×15cm
Volume = 3375cm^3

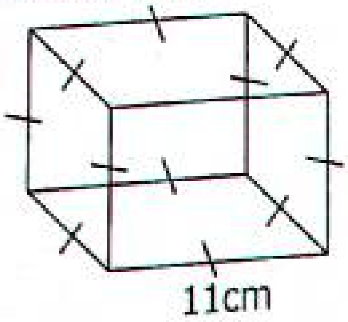
Activity 10:24

1. Calculate the volume of the cubes below.

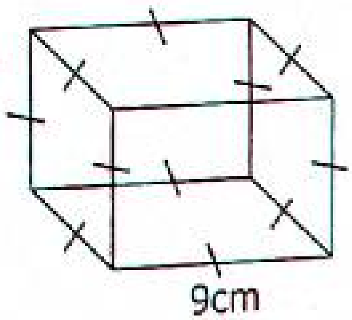
a)



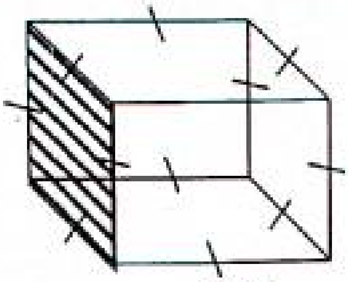
b)



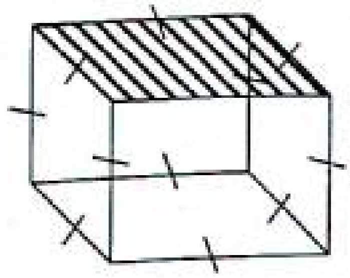
c)



2. Calculate the volume of a cube whose side is 4cm.
3. The sides of a cube is 5cm. Calculate its volume.
4. What is the volume of a cube with sides 7cm.
5. Use the cube below to answer questions that follow.



a) Calculate the area of the shaded region.
b) Work out its volume.

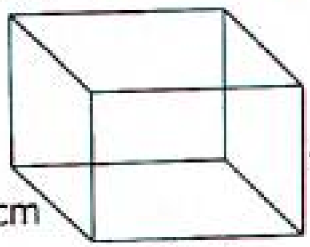


a) What is the area of the shaded part?
b) Calculate the volume of the cube.

10.25 Calculating Volume of Cuboids

Volume of cuboids = Base area × height
Volume of cuboids = (L × W)×h

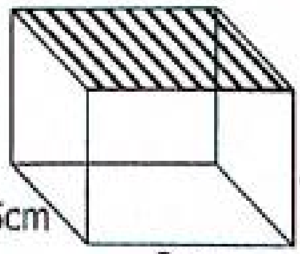
Example 1
Calculate the volume of the cuboid below



Volume = Base area × h
Volume = (L × W) × h
Volume = 5cm×6cm×4cm
Volume = $30cm^2$×4cm
Volume = $120cm^3$

Example 2
Calculate the volume of a Cuboid whose base area is $36cm^2$ and height 5cm.
Volume base area×5cm
Volume = $36cm^2$×5cm
Volume = $(36×5)cm^3$
Volume = $180cm^3$

Example 3
a) Calculate the area of the shaded region.

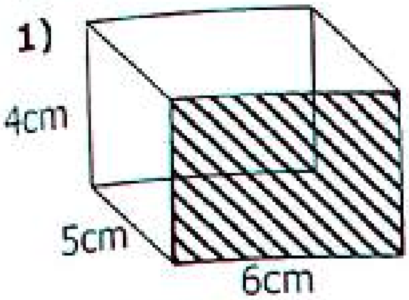


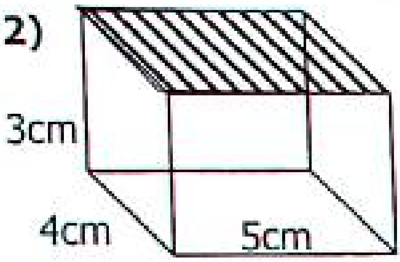
Area = L × W
Area = 8cm×5cm
Area = $(8×5)cm^2$
Area = $40cm^2$

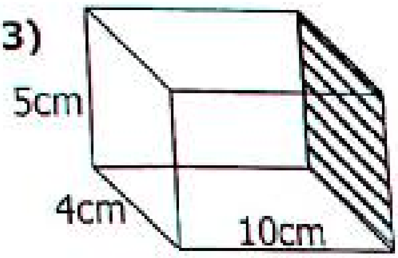
b) Find the volume of the cuboid.
Volume = L × W × h
Volume = 8cm×5cm×6cm
Volume = $40cm^2$×6cm
Volume = $240cm^3$

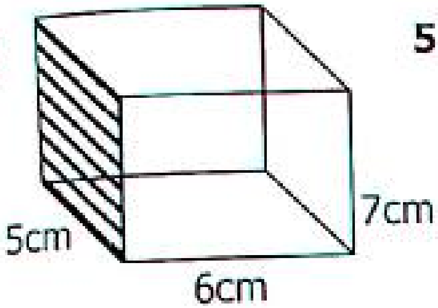
Activity 10:25

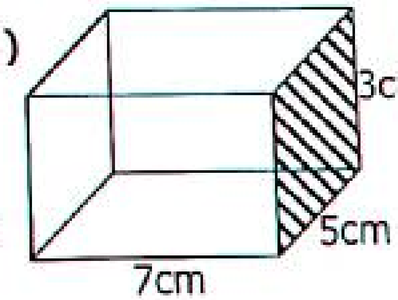
Calculate the volume of the cuboids and the shaded parts.

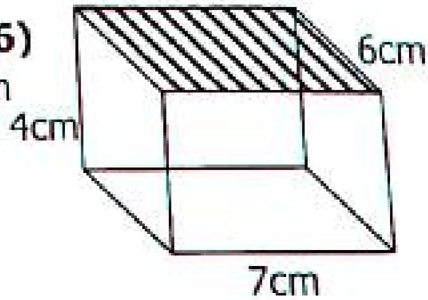
1)


2)


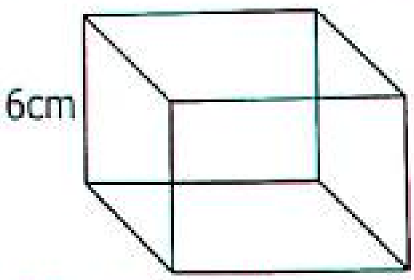
3)


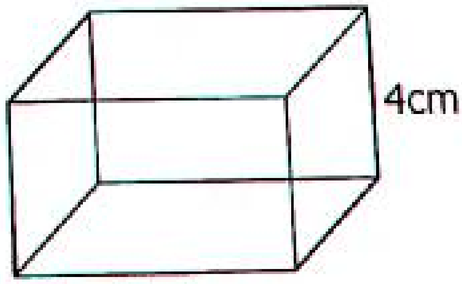
4)


5)


6)


7. Calculate the volume of a cuboid measuring length 10cm, width 5cm and height 8cm.
8. A cuboid measures 8cm by 6cm by 5cm. Calculate its volume.
9. The base area of a cuboid is $72cm^2$. Calculate its volume if the height is 5cm.
10. Find the volume of a cuboid whose base area is $24cm^2$ and height 8cm.
11. Workout the volume of the cuboids below whose base area are $45cm^2$.

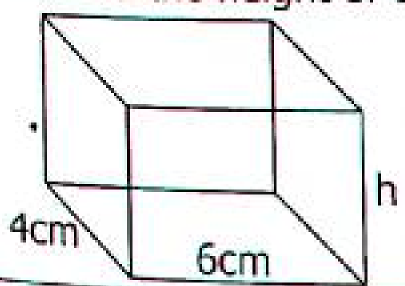
a)


b)


10.26 Find one side of a Cuboid When Given the Volume

Example 1

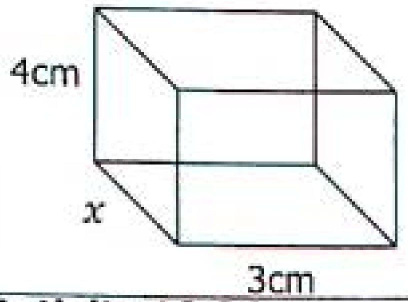
The volume of the cuboid below is $120cm^3$. Find the height of the cuboid.



$$L \times W \times h = \text{volume}$$
$$6cm \times 4cm \times h = 120cm^3$$
$$24cm^2 \times h = 120cm^3$$
$$\frac{\cancel{24cm}^1 \times \cancel{cm} \times h}{\cancel{24cm}_1 \times \cancel{cm} \times \cancel{cm}} = \frac{\cancel{120cm}^5 \times \cancel{cm} \times \cancel{cm}}{\cancel{24cm}_1 \times \cancel{cm}}$$
$$h = 5cm$$

Example 2

The volume of a cuboid below is $48cm^3$



L×w×h = volume

$x \times 3cm \times 4cm = 48cm^3$

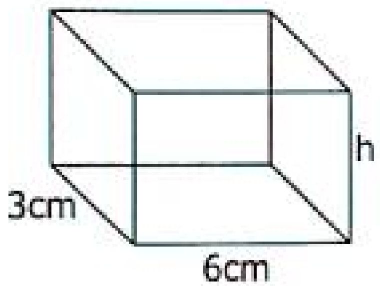
$12cm^2 \times x = 48cm^3$

$$\frac{\cancel{12cm} \times \cancel{cm} \times x}{\cancel{12cm} \times \cancel{cm} \times \cancel{cm}} = \frac{\cancel{48cm^4} \times \cancel{cm} \times \cancel{cm}}{12\cancel{cm} \times \cancel{cm}}$$

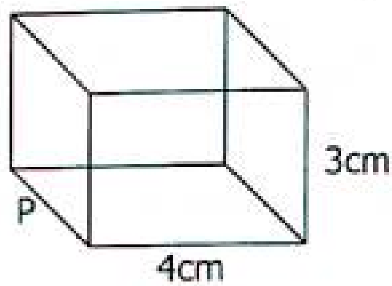
$x = 4cm$

Activity 10:26

1. Calculate for the value of h if the Volume is $36cm^3$.



2. The volume of a cuboid below is $60cm^3$. Find the value of p.



3. The volume of a cuboid is $60cm^3$.Find its width if the length is 5cm and height 4cm.
4. The volume of a box is $64cm^3$. Find the length if the width is 4cm and height 4cm.
5. The length of a cuboid is 4cm, width 5cm and volume $100cm^3$. What is the height of the cuboid?
6. The volume of a box is $24cm^3$. Its length is 3cm, width 4cm.Find its height.

10.27 <u>Finding Capacity in Litres</u>

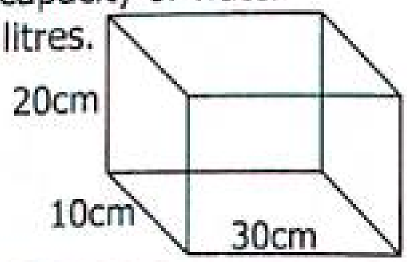
Volume = Base area × height

1litre = $1000cm^3$

1 litre = $1000cm^3$

Example 1

The tank below contain water.
Calculate the capacity of water in the tank in litres.



Volume = L×w×h

Volume = 30cm×10cm×20cm

Volume = $(30\times10\times20)cm^3$

Volume = $6000cm^3$

$1000cm^3$ = 1litre

$6000cm^3 = \frac{6000cm^3}{1000cm^3}$

$6000cm^3$ = 6litres

Activity 10:27

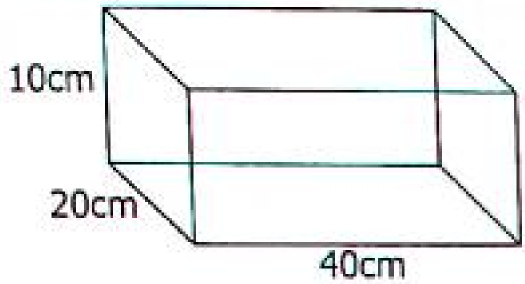
Express the following to litres;

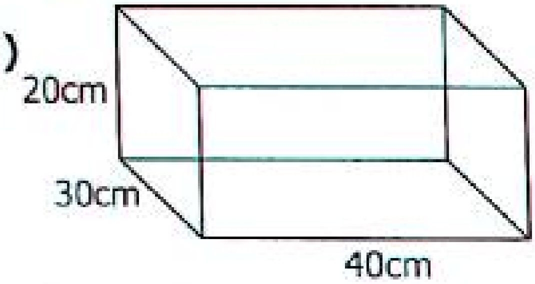
1) 14000ml	2) 5000ml	3) 9000ml	4) 10000ml	5) 15400ml
6) 24700ml	7) 5800ml	8) 7200ml	9) 850cm^3	10) 700cm^3
11) 450cm^3	12) 280ml	13) 1500cm^3	14) 9700cm^3	15) 1850cm^3

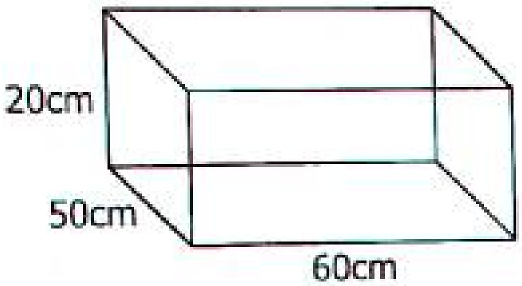
16) The mother to Letifa bought 15780millilitrs of milk from a diary farm. How many litres of milk did she buy?

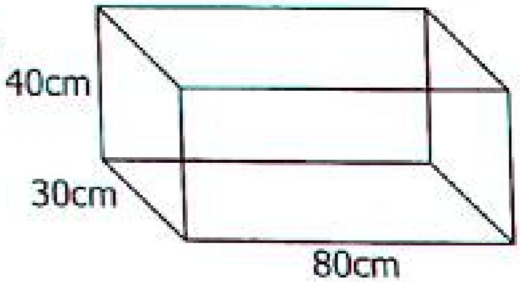
17) On a rainy day, Okware collected 187500cm^3 of rain water. Express the amount of water in litres.

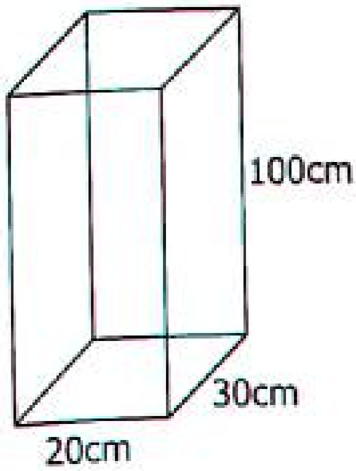
18) Calculate the capacity of water the tanks below can hold.

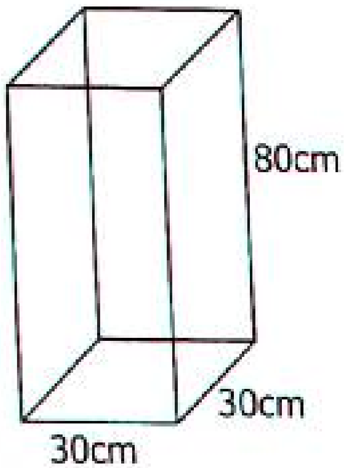
a) 

b) 

c) 

d) 

e) 

f) 

19) Calculate the volume of a tank which measures 40cm by 30cm by 60cm. Give your answer in litres.

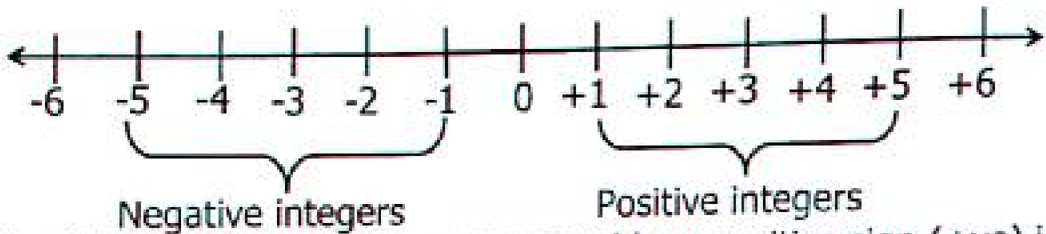
TOPIC .11: INTERGERS

Integers are whole numbers or negative numbers including zero that don't include fractions or decimals.

- ✓ Positive integers {+1, +2, +3, +4, +5, +6.................}
- ✓ Negative integers {-1, -2, -3, -4, -5, -6.......................}

Zero is neither a positive nor negative.

Integers can be shown on a number line.

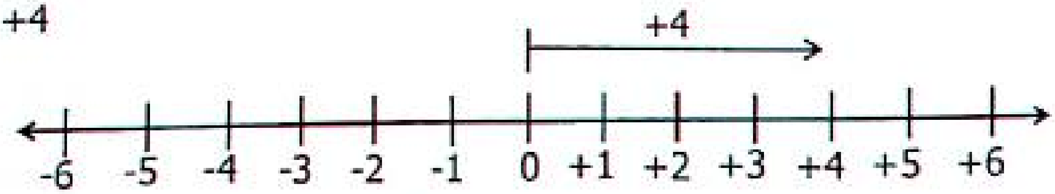


Positive integers are numbers represented by a positive sign (+ve) i.e +2, +8, +20.....

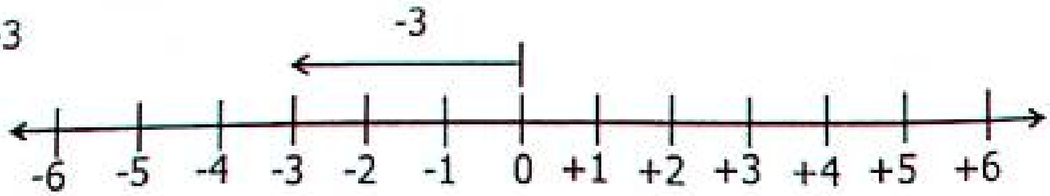
Negative integers are represented by a negative sign (-ve) i.e -1, -2, -3, -8 etc

11.1 Representing Integers on a Number Line

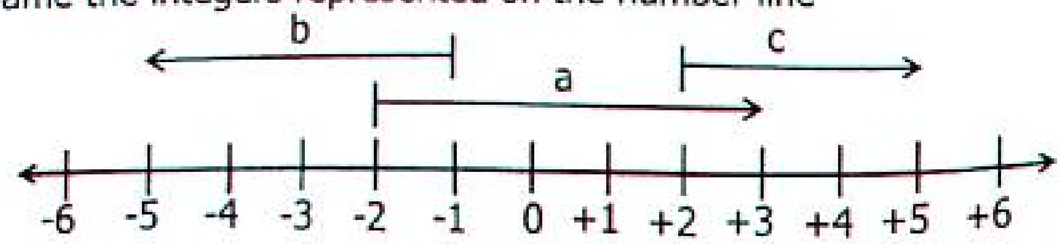
i) +4



ii) -3



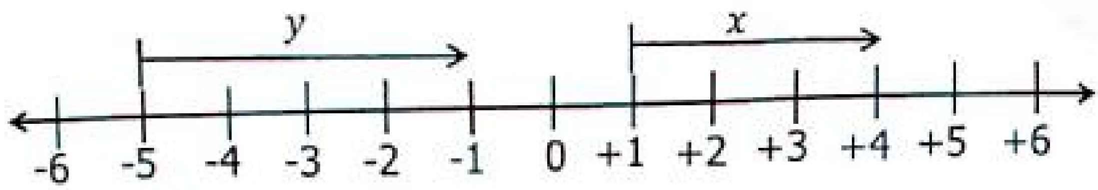
iii) Name the integers represented on the number line

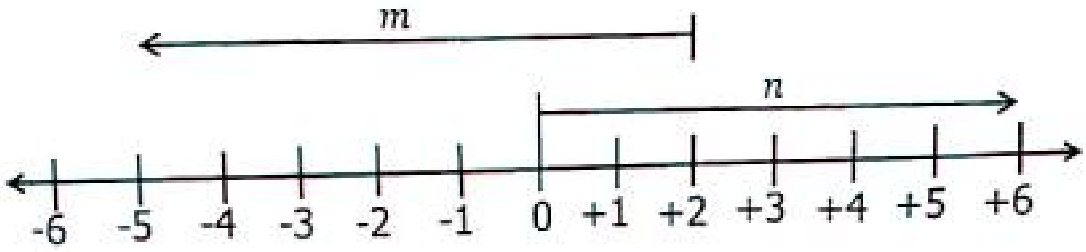


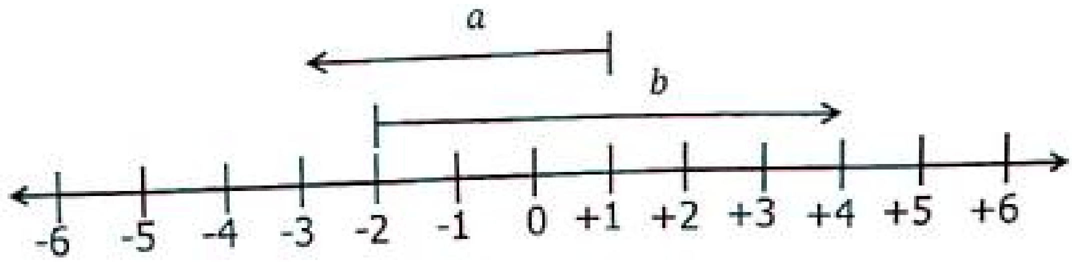
a = +5 b = -4 c = +3

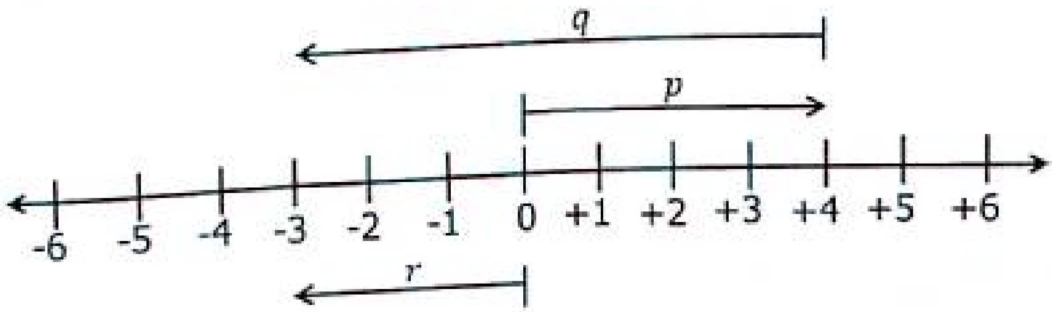
Activity 11:1

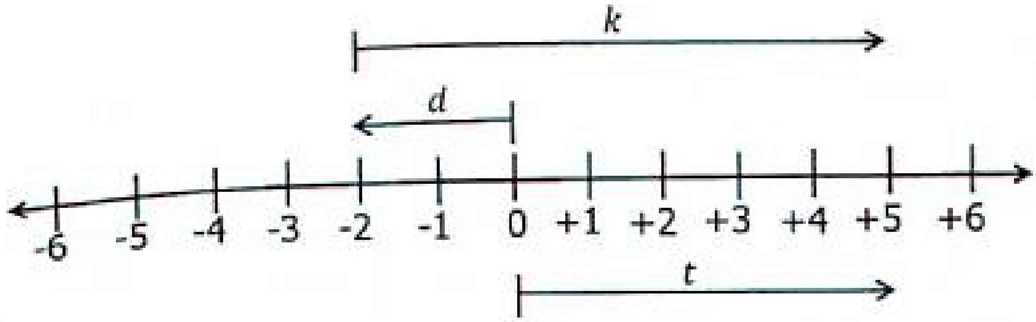
Name the arrows on the number line

1. 

2. 

3. 

4. 

5. 

11:2 Arranging Integers in Ascending Order or Descending Order

- ✓ Integers on a number line are arranged in ascending order from left to right.
- ✓ The best method to use when ordering integers is using a number line.
- ✓ Ascending order mean arranging numbers from the smallest to the biggest number.
- ✓ Descending order arranging from the biggest to the smallest.

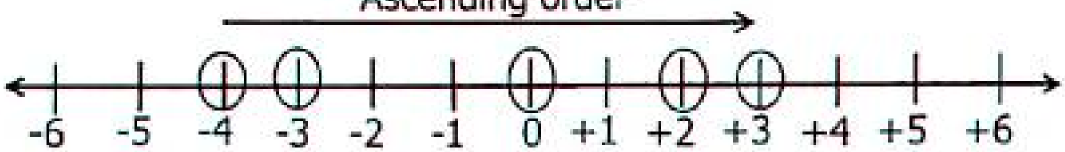
Example 1

Arrange 3, -4, 2, -3, 0 in ascending order.

Procedure

- ✓ Draw a number line to possess all the given integers.
- ✓ Mark the integers given on the number line.
- ✓ Record the integers from the left to the right hand.

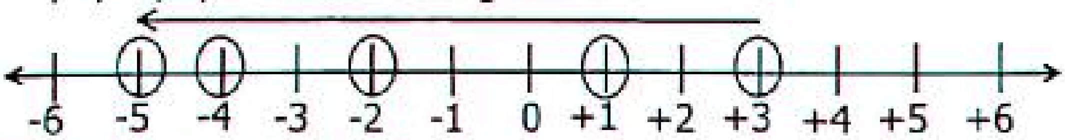
Ascending order



Arrangement -4, -3, 0, +2, +3

Example 2

Arrange -2, 1, -4, 3, -5 in descending order.



Arrangement +3, +1, -2, -4, -5

Activity 11:2

1. Arrange the following integers in ascending order;

a) 0, -4, -1, +3, +4, -2 b) +2, -2, 5, -3, 3 c) +2, 0, -3,+1, -2, -5

d) +4, -3, +1,+2, -2 e) +4, -3, +1, +2, -1 f) -5, +5, -3, +2, -1

g) -10,+9,+4, -5, -2 h) -3, -4, +4,+3,-2 i) 0, -2, +3, -5, +1

2. Arrange the following integers in descending order.

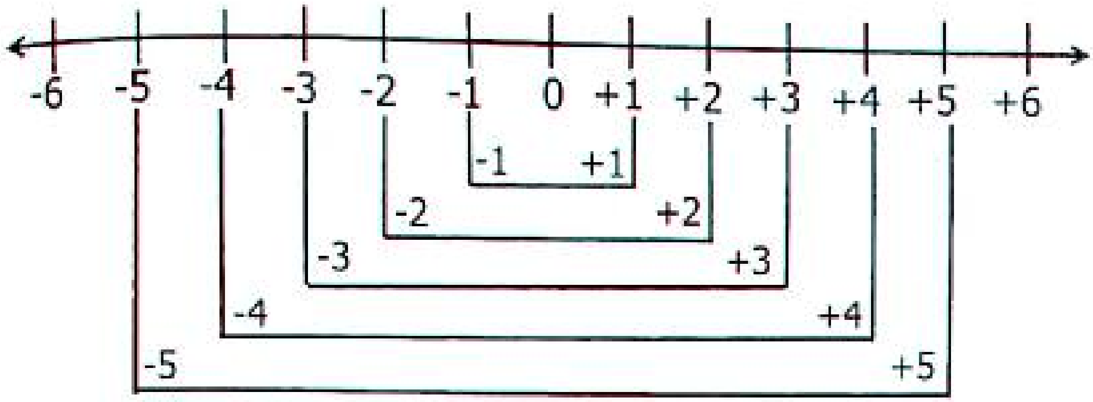
a) -2, +2, +3, -4, 0 b) -5, 4, -3, 5, -2, +1 c) +3, -4, 0, -2, 5, 2

d) -10, +7, -8, 5, -3 e) -7, +6, -5, +4, +3 f) +8, -2, +4, 0, -5

11.3 Additive Inverse

Additive inverse is a number which gives zero when added to a number, inverse of integers refers to the opposite of an integer.

Additive inverse as shown on a number line.



Finding the additive inverse by calculation.

Example 1

What is the additive inverse of -3?

Let the additive inverse of -3 = p

P + -3 = 0

P +-3+3 = 0+3

P = +3

The additive inverse of -3 is +3

Example 2

Calculate the additive inverse of +24.

Let the additive inverse of +24 = y

$y + +24 = 0$

$y + 24 - 24 = 0 - 24$

$y = -24$

The additive inverse of -24 is +24

The additive inverse property states that a number added to its inverse the result is zero (0).

Activity 11:3

Work out the additive inverse of the following integers;

1) +4 2) -5 3) +6 4) -7 5) -12 6) -9 7) +15 8) +18
9) -20 10) -25 11)+17 12)+8 13)+120 14)-100 15)+50 16) -40

11.4 Addition of Integers Without Using a Number Line

Rules of Integers

Positive × Positive = Positive (+ × +) = +

Positive × Negative = Negative (+ × –) = –

Negative × Positive = Negative (– × +) = –

Negative × Negative = Positive (– × –) = +

Example 1

+3 + +4

+3 +(+4)

+3 + 4

+7

+	+ + +++++
–	

Example 2

-5 + +2

-5 +(+2)

-5 + 2

-3

–	– – – – –
+	+ +

Example 3

+6 + -4

+6 +(-4)

+6 – 4

+2

+	+ + + + + +
–	– – – –

Activity 11:4

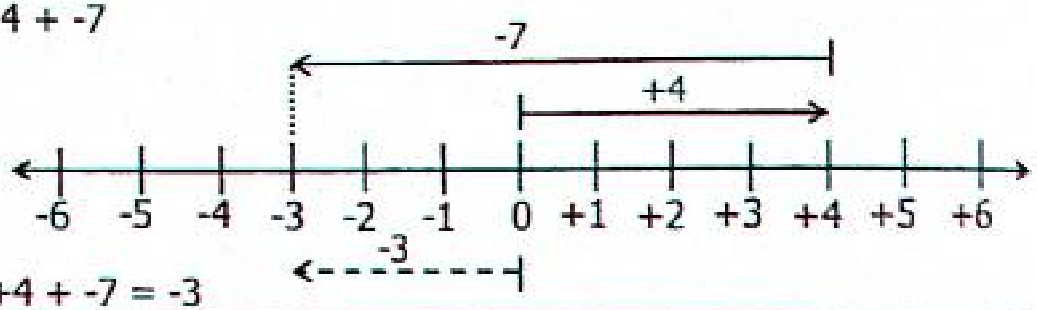
Add the following integers without using a number line.

1) +4 + -1	2) -4 + -1	3) -4 + +1	4) +1 + -4	5) -3 + -4
6) -3 + +4	7) +3 + -7	8) -4 + +3	9) +5 + -4	10) +3 + -7
11) -3 + +7	12) -6 + +3	13) +7 + -3	14) -8 + -6	15) +4 + -8
16) -1 + -3	17) -8 + -3	18) -6 + -2	19) +6 + -4	20) -2 + -5

11:5 Addition of Integers Using a Number Line

Example 1

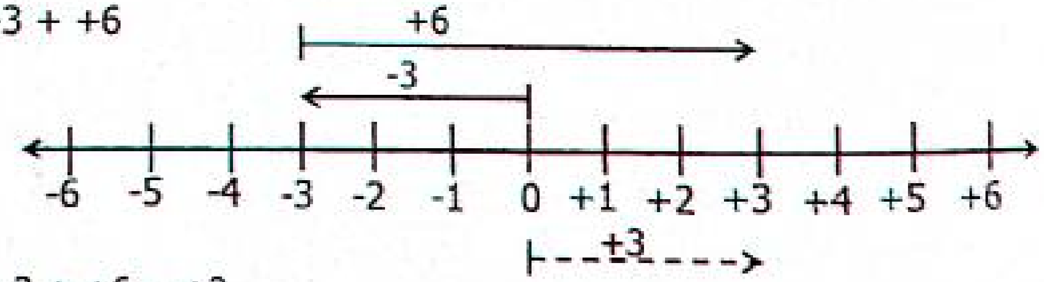
Work out +4 + -7



Therefore +4 + -7 = -3

Example 2

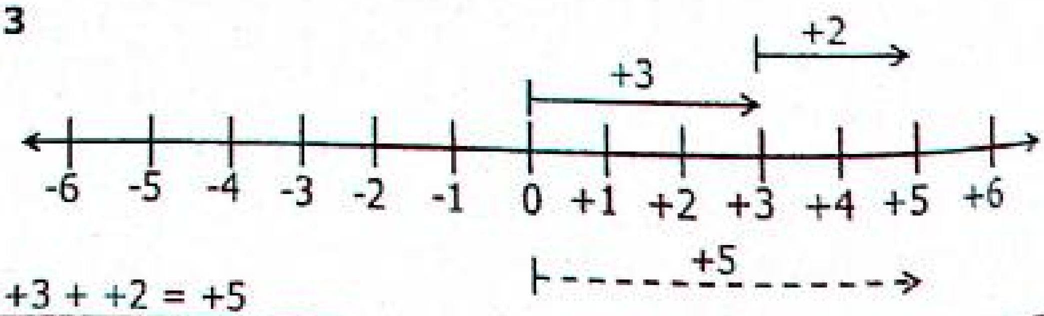
Work out -3 + +6



Therefore -3 + +6 = +3

Example 3

+3 + +2



Therefore +3 + +2 = +5

Activity 11:5

Represent the numbers below on a number line and write their mathematical statements.

1) +3 + +4	2) +2 + +4	3) +3 + +2	4) +3 + -2
5) +4 + -2	6) +2 + -5	7) -4 + -1	8) -5 + +2
9) -3 + -5	10) -2 + -3	11)-4 + -3	12) -1 + -6

11.6 Subtraction of Integers Without Using a Number Line

Work out the following

Example 1	Example 2	Example 3
-3 - +5	-4 – 5	+3 – +5
-3 – (+5)	-4 – (-5)	+3 – (+5)
-3 – 5	-4 + 5	+3 – 5
-8	+1	-2

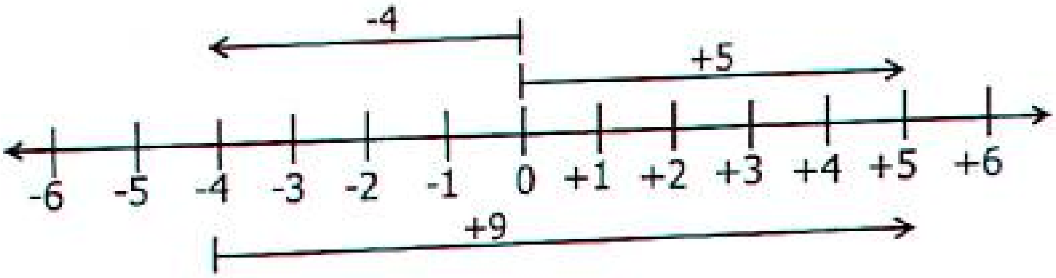
Activity 11:6

1) +6 – +2
2) +3 – +5
3) +12 – +7
4) -15 – +3
5) +3 – -3
6) -4 – +7
7) -24 – -28
8) +6 – +3
9) -3 – +2
10) -8 – +5
11) +6 – -4
12) +10 – +6
13) -3 – -4
14) -5 – -7
15) -6 – +10
16) -1 – -3

11.7 Subtraction of Integers Using Number Lines

Example 1

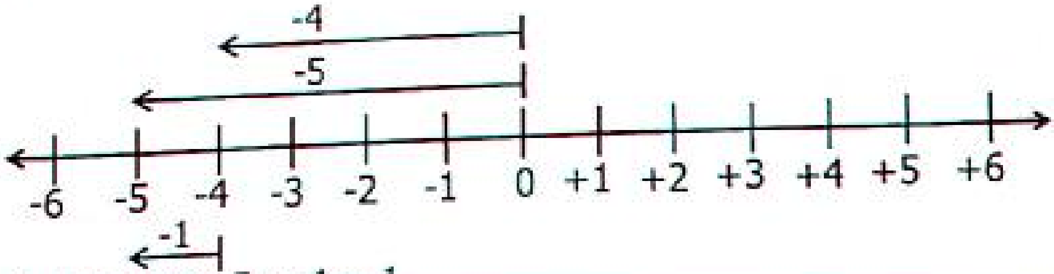
+5 – -4



Mathematical statement +5 – -4 = +9

Example 2

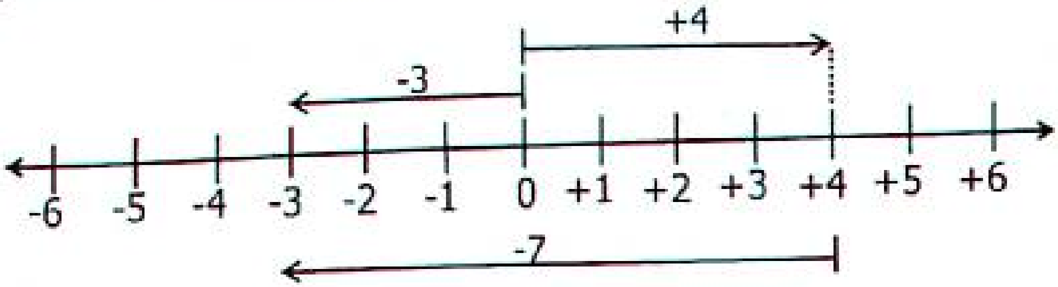
-5 – -4



Mathematical statement -5 – -4 =-1

Example 3

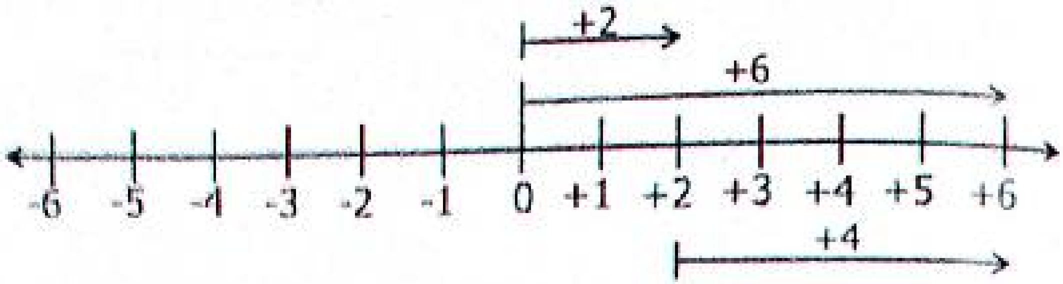
-3 – +4



Mathematical statement -3 – +4 = -7

Example 4

+6 – +2



Mathematical statement +6 - +3 = +4

Activity 11:7

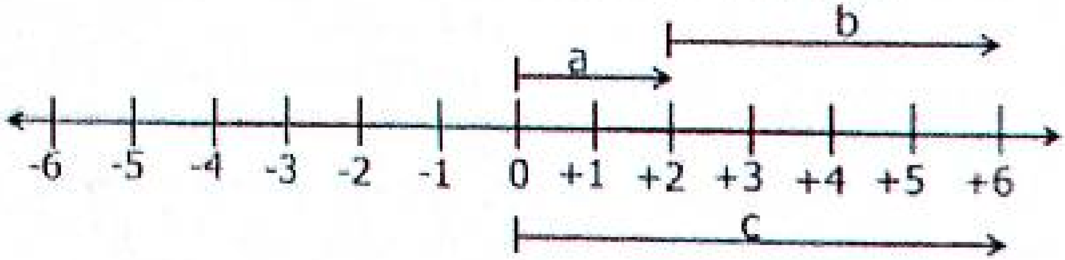
Work out the following using Number Lines

1) -3 – -7 2) -2 – -4 3) -3 – -5 4) +3 – -7 5) +2 – -4
6) +3 – -5 7) -4 – +2 8) -2 – +3 9) -1 – +5 10) +7 – +3
11) +5 – +3 12) -2 – -6 13) -3 – -4 14) +4 – -2 15) -5 – +3

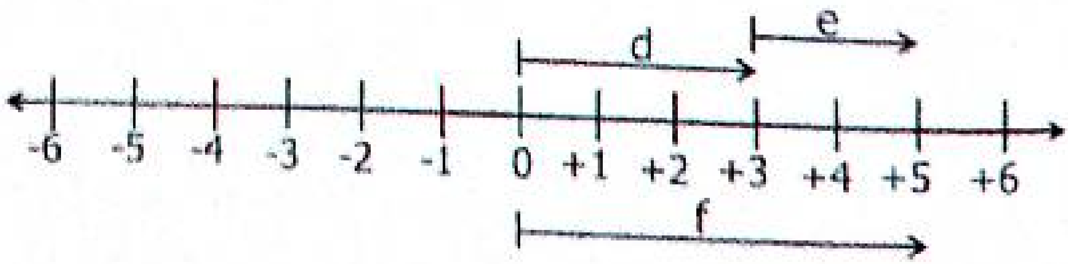
Activity 11:8

Write a mathematical statement for each number line below.

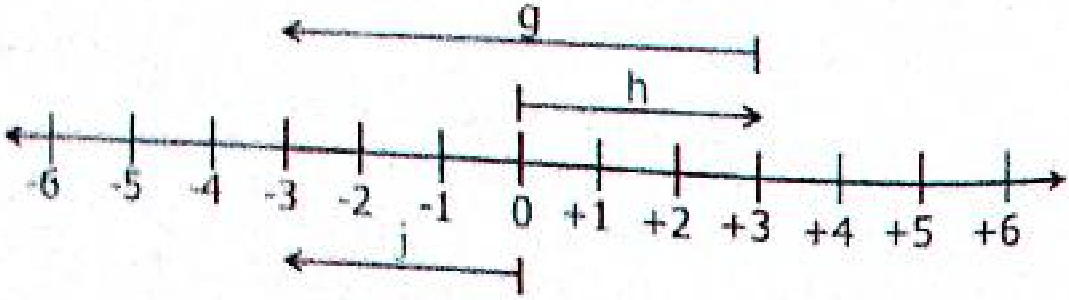
1)



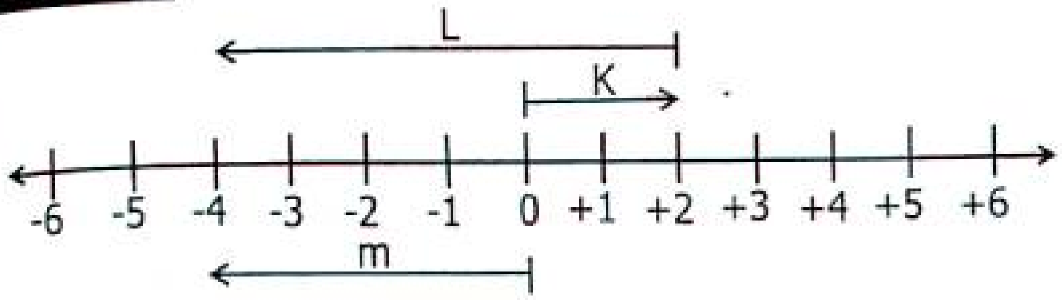
2)



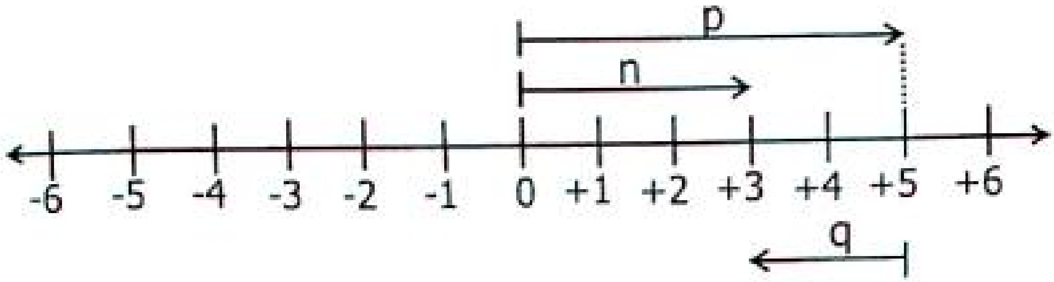
3)



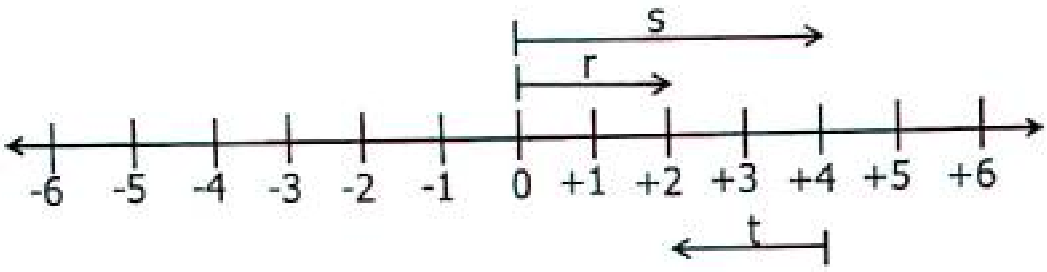
4)


L
K
-6 -5 -4 -3 -2 -1 0 +1 +2 +3 +4 +5 +6
m

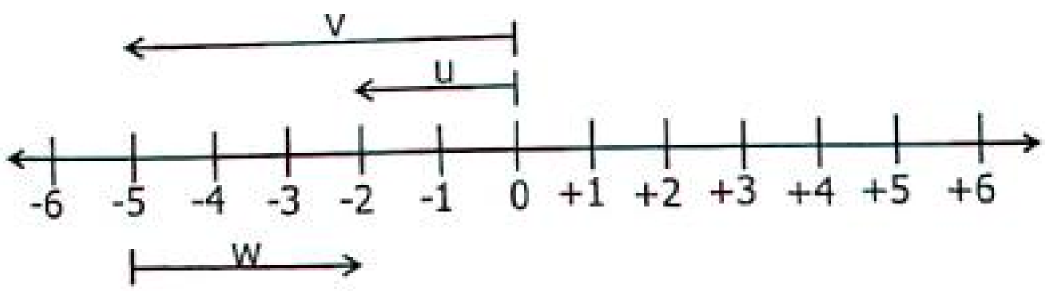
5)


p
n
-6 -5 -4 -3 -2 -1 0 +1 +2 +3 +4 +5 +6
q

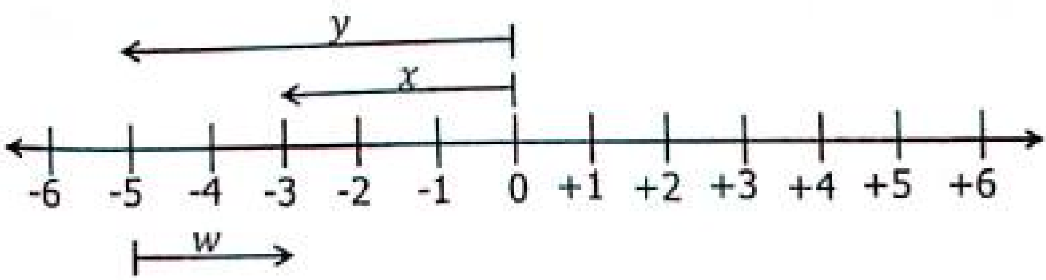
6)

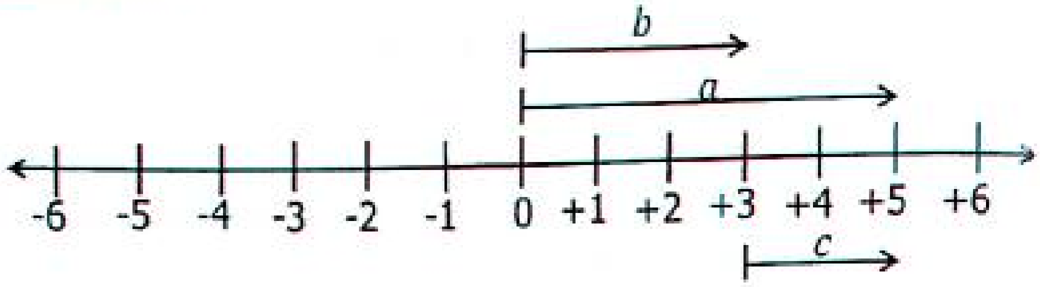

s
r
-6 -5 -4 -3 -2 -1 0 +1 +2 +3 +4 +5 +6
t

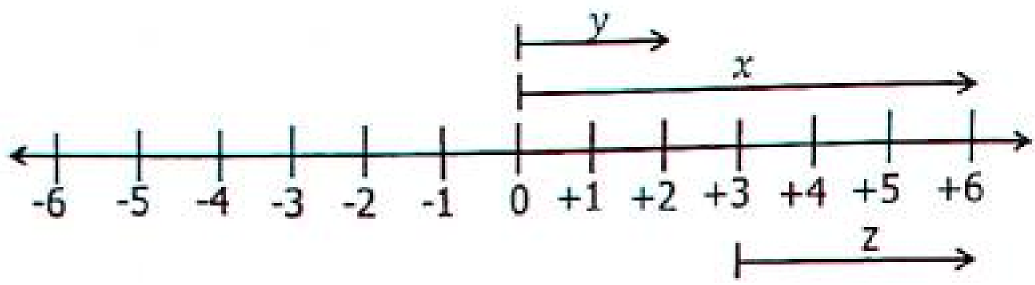
7)

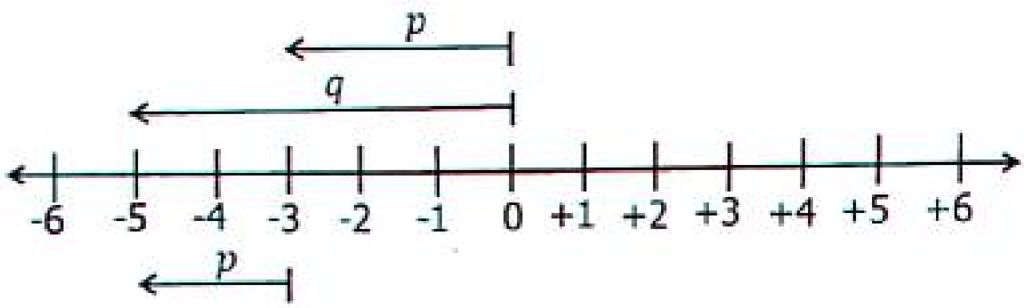

v
u
-6 -5 -4 -3 -2 -1 0 +1 +2 +3 +4 +5 +6
w

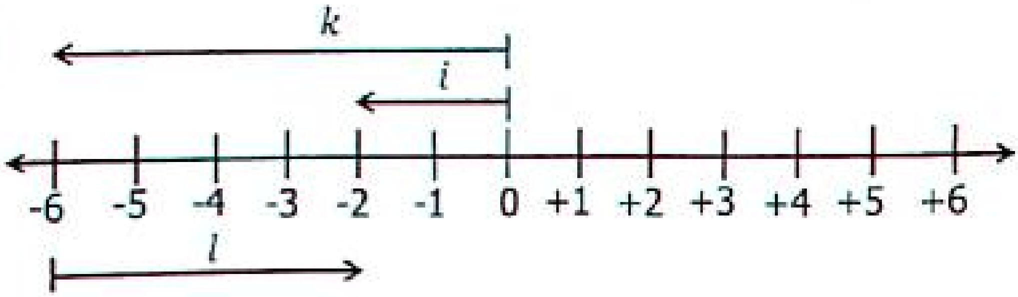
8)


y
x
-6 -5 -4 -3 -2 -1 0 +1 +2 +3 +4 +5 +6
w

9) 

10) 

11) 

12) 

11.8 Multiplication of Integers Without Using a Number Line

Positive × Positive = Positive
Positive × Negative = Negative
Negative × Positive = Negative

Example 1	Example 2	Example 3
$+2 \times +3$	$+2 \times -3$	-2×-3
$+ \times + = +$	$+ \times - = -$	$- \times - = +$
$+2 \times +3 = +6$	$+2 \times -3 = -6$	$-2 \times -3 = +6$

Activity 11:8

Workout the numbers below

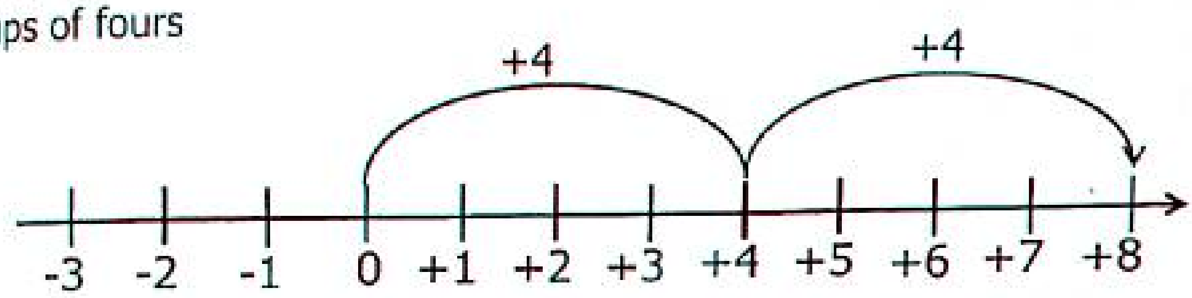
1) +3 × +4 2) +5 × +3 3) +6 × +2 4) +6 × +5
5) -5 × +4 6) -2 × +3 7) -4 × +3 8) -7 × +3
9) -5 × -7 10) -6 × -4 11) -12 × -4 12) -9 × -4
13) +7 × -6 14)+10 × -3 15)+8 × -6 16) +15 × -4

11.9 Multiplication of Integers Using a Number Line

Example 1

+2 × +4

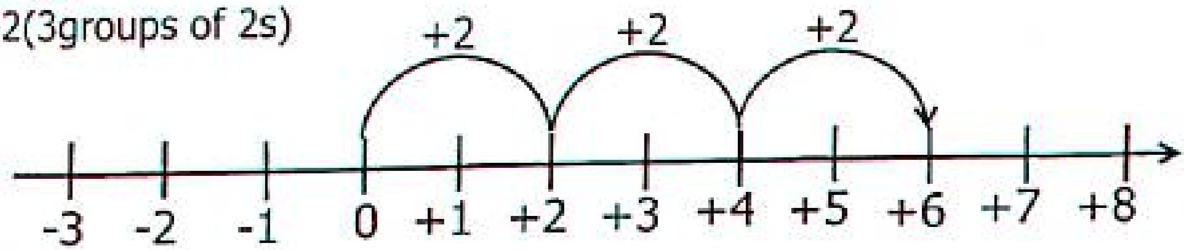
2groups of fours



Therefore +2 × +4 = 8

Example 2

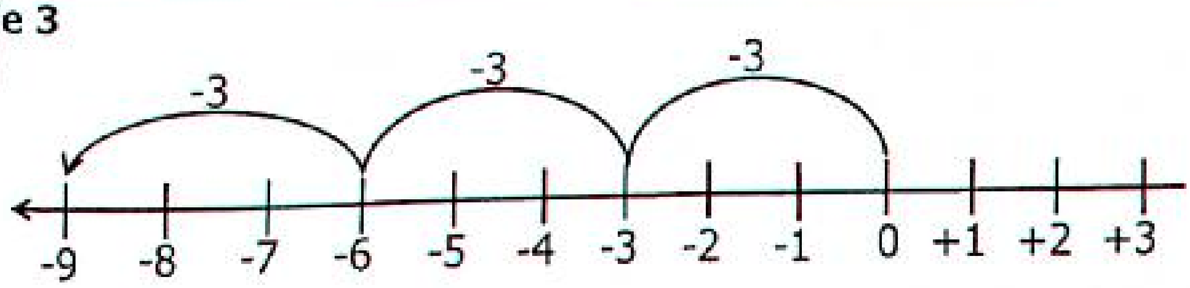
+3 × +2(3groups of 2s)



Therefore +3 × +2 = +6

Example 3

+3 × -3



+3 × -3 = -9

Activity 11:9

Represent the integers on number lines

1) +3 × +2 2) +2 × +3 3) +4 × +2 4) +2 × -3 5) +3 × -2
6) +2 × -4 7) +2 × +5 8) +5 × +2 9) 3 × -3 10) +2 × -2

11.10 Division of Integers

Positive ÷ Positive = Positive
Positive ÷ Negative = Negative
Negative ÷ Negative = Positive
Negative ÷ Positive = Negative

Example 1	Example 2	Example 3
+4 ÷ +2	-8 ÷ +2	+24 ÷ -6
+ ÷ + = +	- ÷ + = −	+ ÷ − = −
+4 ÷ +2 = +2	-8 ÷ +2 = -4	+24 ÷ -6 = -4

Activity 11:10

1) -4 ÷ +2 2) -4 ÷ -2 3) +8 ÷ -2 4) -9 ÷ -3 5) +10 ÷ +2
6) +8 ÷ -4 7) -12 ÷ -3 8) +12 ÷ -4 9) -10 ÷ +5 10) +18 ÷ -6

11.11 Application of Integers

Example 1	Example 2
The temperature at the top of Mt. Elgon Was 10^0c. it dropped by 5^0c at night. Find the new temperature of the mountain. When we talk about dropping it's a negative 10c – 5^0c = 5^0c The temperature is 5^0c	Aristotal a scientist was born in 42BC and died in 54AD. How old was he when he died? The scientist live from 42BC to 54AD. BC is in negative. (54 – -42) years (54 – (-42) (54 + 42) years

Activity 11:11

1. Obura moved 12steps forward then 7steps backward. Where did he end?
2. A butterfly moved 6 steps backward and 3 steps forward in that order. Find how many steps is it from the starting point?
3. Ali borrowed sh.2500 from Joseph and then another sh.350 from Juma, how much does Ali owe the two people?
4. A frog jumped 4 steps forward and then turned and jumped 7 steps backward.
5. The temperature reading of a place in the morning was 10^0c. If it rose by 15^0c by mid-day, what was the temperature of the place by mid-day?

11.12 Finite System

Finite system is a system of recording the remainder (modular system).
Possible numbers in different finite system

(Finite 4) = {0, 1, 2, 3}
Finite 5 = {0, 1, 2, 3, 4}
Finite = 6 = {0, 1, 2, 3, 4, 5}
Finite = 7 = {0, 1, 2, 3, 4, 5, 6}
Finite = 8 = {0, 1, 2, 3, 4, 5, 6, 7}

TOPIC 11: Integers

Addition in Finite System

2+3 = ____(mode 6)

2+3 = 5 (mode 6)

2+1(finite 5)

2+1=3(finite 5)

All numbers got after adding and they are less than the finite, the result is written direct but if its equal or more than the finite divide and record the remainder.

Example 1

Add 2 +5 = ____(finite 6)

2+5 = ____(finite 6)

7 is more than 6, we divide by the finite and record the remainder.

7 ÷ 6 = 1 rem (1)

1 is the answer

$\therefore 2 + 5 = 1(finite\ 6)$

Example 2

Add 3 + 4 = ____(finite 5)

3 + 4 = 7(finite 5)

Divide 7 by 5 = 1 rem (2)

7 ÷ 5 = 1 rem (2)

2 is the answer

$\therefore 3 + 4 = \ 2\ (finite\ 5)$

Example 3

3 + 3 = x (finite 5)

$x = 3 + 3(finite\ 5)$

$x = 6(finite\ 5)$

$x = 6 \div 5 = 1rem\ 1(finite\ 5)$

$x = 1(finite\ 5)$

Example 4

$2 + 3 + 4 = y(finite\ 5)$

$y = 2 + 4(finite\ 5)$

$y = 9(finite\ 5)$

9 ÷ 5 = 1 rem 4

$y = 4(finite\ 5)$

Activity 11:12

Workout the following numbers without using a dial or clock face

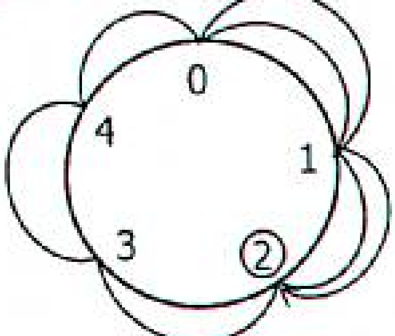
1. 2+2 = ____(finite 5)
2. 2+3 = ____(finite 5)
3. 3+4 = ____(finite 5)
4. 4+4 = ____(finite 5)
5. 3+2 = ____(finite 6)
6. 4+5 = ____(finite 6)
7. 3+5 = ____(finite 6)
8. 2+4 = ____(finite 6)
9. $6 + 4 = y(finite\ 7)$
10. 5 + 4 = m(finite 7)
11. 3 + 2 + 1 = p(finite 6)
12. $3 + 3 + 2 = x(finite\ 5)$
13. 4 + 4 + 4 = w (finite 8)
14. 3 + 4 = z (finite 7)
15. 2 + 3 =.r (finite 6)
16. 3 + 1 = t (finite 4)

11.13 Addition in Finite Using a Dial or Clock Faces

Example 1

Add 4+3 = ____(finite 5) using a clock face.

- ✓ Possible digits in finite 5 {0,1, 2, 3, 4}
- ✓ Represent the digits on the clock face
- ✓ Begin from zero and move 4 steps and then continue 3 steps clock wise direction.
- ✓ Where you end is the answer.



4 +3 = 2 (finite 5)

Activity 11:13

Add the following using a dial/clock face.

1) 2 + 2 = ___(mod 5)
2) 2 + 4 = ___(mod 5
3) 2 + 1 = ___(mod 5)
4) 4 + 2 = ___(mod 5)
5) 1 +1+2 = ___(mod 5)
6) 2 +2 + 2 = ___(mod 5
7) 2 + 4 = ___(mod 6)
8) 3 + 4 = ___(mod 6)
9) 5 + 5 = ___(mod 6)
10) 2 + 3 +1 = ___(mod 6)
11) 3 + 3 + 2 = ___(mod 6)
12) 4 + 3 = ___(mod 7)
13) 5 + 5 = ___(finite 7)
14) 5 + 2 +1 = ___(finite 8)
16) 3 + 2 + 2 = ___(finite 9)
17) 2 + 2 = ___(finite 9)
18) 6 + 2 = ___(finite 9)

11.14 Subtraction in Finite System

Example 1 **(Method 1)**

2 – 3 = ___(finite 5)

2 – 3 is impossible, therefore add 5 to 2

2 + 5 – 3 = __(finite 5)

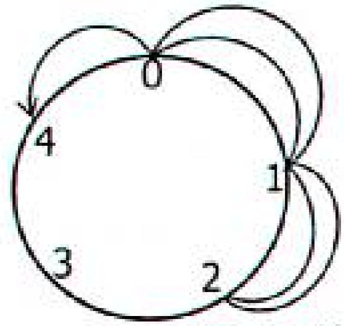
7 – 3 = 4 (finite 5)

2 – 3 = 4 (finite 5)

(Method 2)

Finite 5 = 0, 1, 2, 3, 4

Move 2 steps from zero clockwise direction then 3 steps anti-clockwise direction.



2 – 3 = 4(finite 5)

Activity 11:14

Workout the following without using a dial.

1. 3 – 4 = ____(finite 5)
2. 2 – 4 = ____(finite 5)
3. 4 – 3 = ____(finite 5)
4. 5 – 2 = ____(finite 6)
5. 3 – 5 = ____(finite 7)
6. 2 – 5 = ____(mode 7)
7. 3 – 6 = ____(mod 8)
8. 4 – 7 = ____(mode 8)

Workout the following using a clock face/dial.

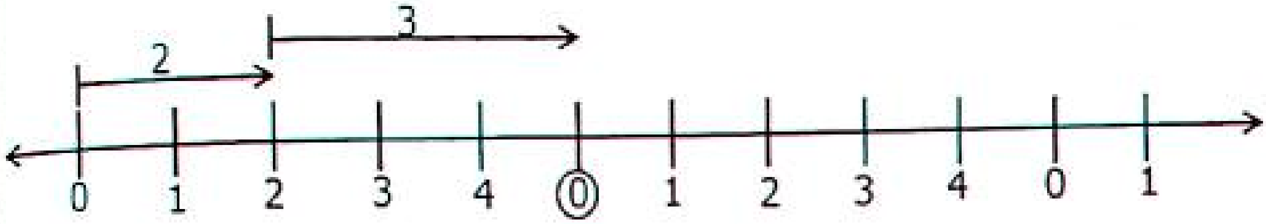
1. 4 – 5 = (mod 6)
2. 1 – 3 = (mod 6)
3. 2 – 6 = (mod 7)
4. 4 – 1 = (mod 7)
5. 4 – 5 = (mod 7)
6. 1 -2 = (mod 12)
7. 1 – 4 = (mod 12)
8. 2 – 3 = (mod 12)

11.15 Addition in Finite System using a Number line

Example 1

Add 2 + 3 =_____(finite 5) using a number line.

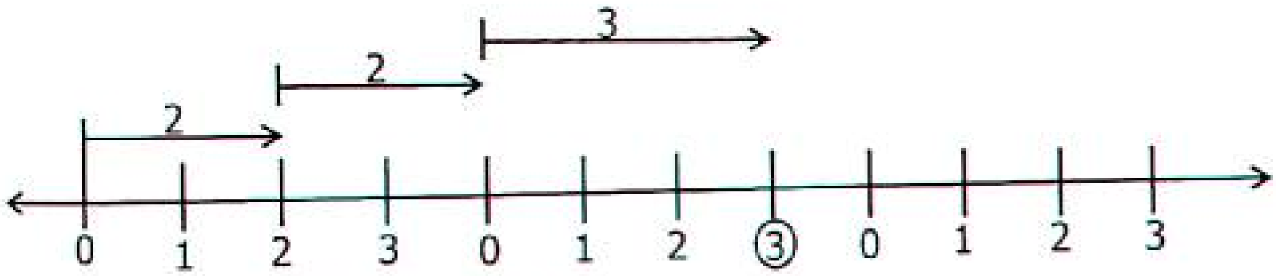
Draw a number line and represent 2 + 3 (finite 5)



2 + 3 = 0 (finite 5)

Example 2

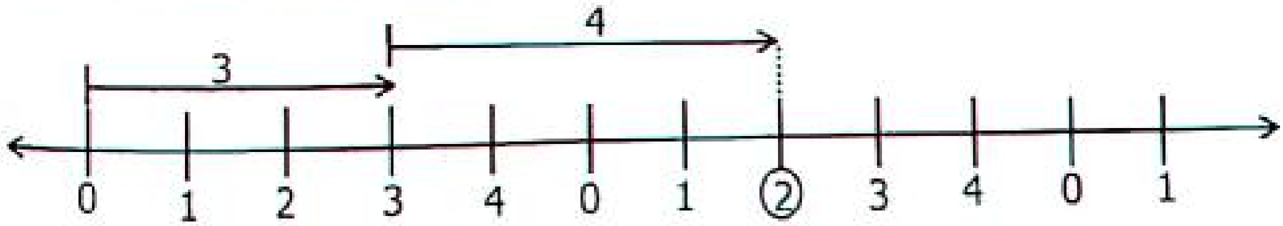
Add 2 + 2 + 3 = (mod 4) using a number line



2 + 2 + 3 = 3(mod 4)

Example 3

Add 3 + 4 = ___(finite 5) using a number line



3 + 4 = 2(finite 5)

Activity 11:16

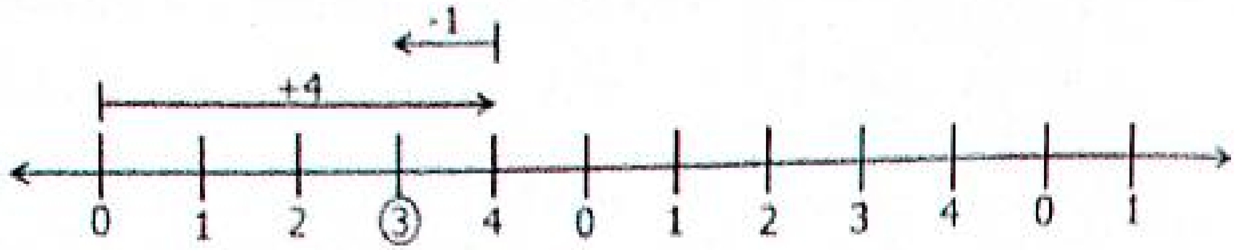
Work out the following using a number line.

1) 2+2=___(finite 5) 2) 2+1=___(finite 5) 3) 3+2=___(finite 5)
4) 3+3=__(mod 5) 5) 4+4=__(mod 5) 6) 2+4=__(mod 5)
7) 4+2+1=___(mod 5) 8) 2+2+3=__(mod 5) 9) 2+3+4=__(mod 5)
10) 4+3+3=__(finite 5) 11) 3+3+3=__(finite 5) 12) 4+4+2=__(finite 5)

11.17 Subtraction in Finite System Using a Number Line

Example 1

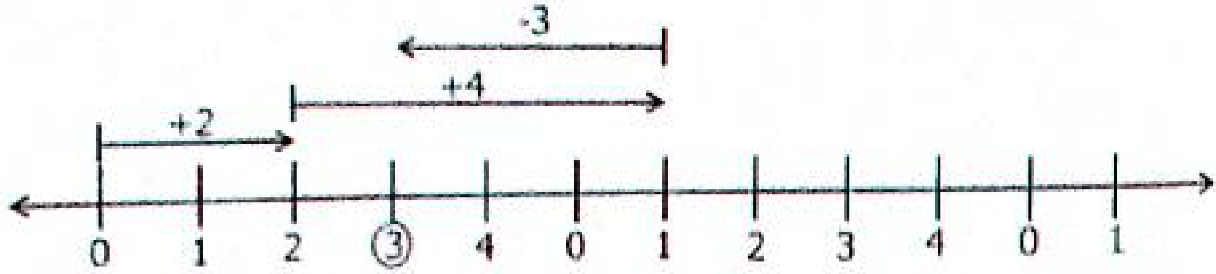
4 – 1 = ____(finite 5)



4 – 1 = 3(Finite 5)

Example 2

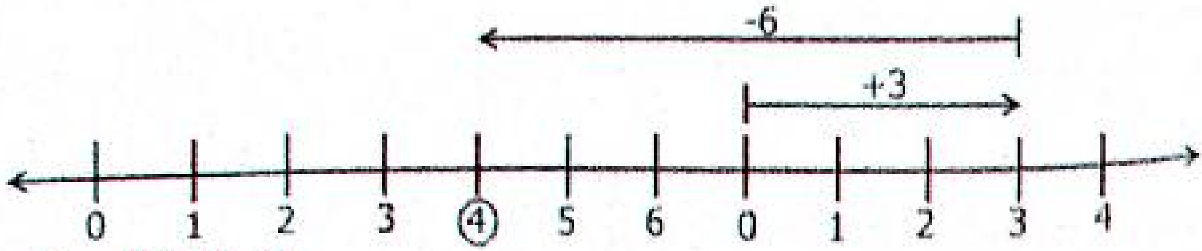
2 + 4 – 3 = (mod 5)



2 + 4 – 3 = 3(finite 5)

Example 3

3 – 6 = ____(mod 7)



3 – 6 = 4(finite 7)

Activity 11:17

1) 3 – 1=____(mod 5) 2) 4 – 2=__(mod 5) 3) 3 – 2=__(mod 5)
4) 1 – 2=____(mod 5) 5) 2 – 3=__(mod 5) 6) 2 – 4=__(finite 5)
7) 1 – 3=__(finite 5) 8) 2 – 3=__(finite 7) 9) 1 – 4 =__(finite 7)
10) 0 – 3=_(finite 7) 11) 5 – 4=_(mod 12) 12) 1 – 4=_(mod 12)

TOPIC 12: ALGEBRA

In groups, collect different items used in a classroom that is to say books, pens, pencils, rubbers, rulers, pair of compasses, protractors, chalk etc, put them in groups accordingly, when put in different groups means collecting like terms.

12.1 Collecting Like Terms

Example 1	**Example 2**
2books + 2pens + 4books + 7pens Collect like terms 2books + 4books + 2pens + 7pens 6books + 9pens	The same statement can be represented as below 2books + 2pens +4books + 7pens 2b + 2p + 4b + 7p 2b + 4 b +2p + 7p 6b + 9p

Activity 12.1

Collect like terms

1) a+2a+3a
2) 3p+4p+6p
3) 9m+m – 2m
4) 5z – 2z+6z+3z
5) 14q-5q +2q
6) 6b+4b-5b
7) 4b+8a+7b+2a
8) 4m+2p-m+4p
9) 3b+3b-2b+7b
10) 9p+10+3p-3
11) 8g+3g-4g +g
12) 7g+5c+8g-2c
13) 6y+9y-2x-3x
14) 5d+3d-d
15) 6b+4b+10+b
16) 8a – 5a+2b+b

Use the suitable letters instead of words to collect like terms.

1) 4apples + 8mangoes +3apples – 3mangoes
2) 8cups + 3plates – 5cups – 2cups
3) 2books + 3pencils + 13books – 2pencils
4) 3jugs + 6spoons +4jugs – 4spoons
5) 5tins + 7forks – 2tins – 3falks
6) 7rulers + 8rubbers – 2rulers + 3rubbers
7) 12bulls + 8bulls – 3bulls +7cows
8) 2ducks + 18hens + 7ducks – 7hens

12.2 Algebraic Expression Involves all Operations ($+, -, \times, \div$)

Addition/Subtraction, Multiplication and Division

Substitution

Example 1 Given that a=4, b=2 and c=5 Workout $2a + bc$ $2 \times a + b \times c$ $(2 \times 4) + (2 \times 5)$ $8 + 10$ $= 18$	**Example 3** If p=3, y=5, z=7. Work out $2p^2+2y+z$. $2 \times p \times p + 2 \times y + z$ $2 \times 3 \times 3 + 2 \times 5 + 7$ $18 + 10 + 7$ 35
Example 2 Find $3c^2 - ab$ $(3 \times c \times c) - (a \times b)$ $(3 \times 5 \times S) - (4 \times 2)$ $75 - 8$ 67	**Example 4** $p^2 + y^2 - z^2$ $(p \times p) + (y \times y) - (z \times z)$ $(8 \times 8) + (5 \times 5) - (7 \times 7)$ $64 + 25 = 49$ $89 - 49$ 50

Activity 12:2

Given that a = 2, b = 3, c = 4, p = 5. Work out the following using the values above.

1) abc 2) $2bc + 4a$ 3) $3p + a^2$ 4) $4b + 2p$ 5) $3c + ab$

6) $a^2 + b2$ 7) $(a+b)- c$ 8) $2(ab+c)$ 9) $\frac{2abc}{c}$ 10) $3pa + 2b$

11) $abc - 2b$ 12) $6bc - 3p$ 13) $\frac{abcp}{2b}$ 14) $\frac{bp}{2}$ 15) $\frac{4ab}{2c}$

12.3 Solving Simple Equations by Subtracting from Both Sides

Example 1 $P + 4 = 16$ Subtract from both sides $P + 4 - 4 = 16 - 4$ $P = 16 - 4$ $P = 16 - 4$ $P = 12$	**Example 2** $2y + 6 = 20$ Subtract 6 from both sides $2y + 6 - 6 = 20 - 6$ $\frac{2y}{2} = \frac{14}{2}$ $y = 7$

Activity 12.3

Solving the following;

1) $y + 3 = 10$	2) $p + 12 = 18$	3) $m + 8 = 10$	4) $x + 7 = 16$
5) $b + 9 = 17$	6) $a + 3 = 6$	7) $7 + w = 15$	8) $12 + a = 20$
9) $24 + y = 27$	10) $3 + z = 17$	11) $8 + p = 11$	12) $15 + x = 22$
13) $2y + 4 = 20$	14) $7p+1 = 22$	15) $6y + 14 = 32$	16) $3m + 6 = 18$
17) $5x + 2 = 22$	18) $9z + 15 = 42$	19) $8 + 2y = 18$	20) $4b+3=19$
21) $12 + 2x = 20$	22) $30 + 4y = 50$	23) $24 + 8g = 40$	24) $15 + 9y = 51$

12.4 Word Problems Forming and Solving Equations

Example 1

The sum of two number is 18, if one of the numbers is 24. Find the other number.

Let the number = t

1st Number	2nd No.	Sum
t	18	24

$t + 18 = 24$

$t + 18 - 18 = 27 - 18$

The other number is 4.

Example 2

The cost of a book and a pen is sh.2400 If the book costs sh.1800. Find the cost of the pen.

Let the cost of a pen = p

Book	Pen	Sum
Sh.1800	P	Sh.2400

sh.1800 + p = sh.2400

sh(1800 − 1800)+p=sh.(2400 − 1800)

P = sh.600

The pen costs sh.600.

Example 3

Joseph is twice as old as Mary.

Their total age is 27years.

What is their age?

Mary	Joseph	Total
m	$2 \times m$	

$m + 2m = 27$

$$\frac{\cancel{3}m^1}{\cancel{3}_1} = \frac{\cancel{27}^9}{\cancel{3}_1}$$

$m = 9$

Mary is 9years

Joseph is $(2 \times m)$

(2×9) years

Joseph is 18 years.

Example 4

A television costs three times the cost a radio. Their total cost is sh.360000.

a) Find the cost of a radio.

Let the cost of a radio = r

$n + 3r = 360000$

$4r = 360000$

$$\frac{\cancel{4}r^1}{\cancel{1}^{4r}} = \frac{\cancel{360000}^{90000}}{\cancel{4}_1}$$

$r = 90000$

Therefore the radio costs sh.90000

b) what is the cost of the T.V?

Tv costs 3r

$= 3 \times r$

Sh.(3×90000)

Sh.270000

The T.V costs sh.270000.

Activity 12:4

1. The cost of a plate is twice the cost of a cup. Their total cost is sh.18000. calculate the cost of each of them.
2. In a class of 40pupils, 24pupils are boys and the rest are girls. How many girls are in the class?
3. Allan is 5years older than Abdallah. If their total age is 35years. How old is Abdallah?
4. The sum of two numbers is 27. If one of the numbers is 7 more than the other, workout the value of each number.
5. Mr. Odong is 8 years older than his wife Apio and their total age is 56years. What are their ages?
6. A father is 18 years older than his son. Their total age is 58 years. How old is each?
7. Malembo is three times as old as Mumbere. Their total age is 48years. How old is each?
8. Wafula is 15 years older than Mafabi. If their total age is 40 years, how old is each?
9. A jerrycan costs twice as much as a bucket. Calculate the cost of each if their total cost is sh.18000.
10. The cost of a table is 3p and that of a chair is 2p. Their total cost is sh.360,000, Workout the cost of each.
11. A door costs sh.24000 more than a window, their total cost 150,000. Find the cost of the door.
12. Swengere and the wife shared some mangoes. The wife got 12 mangoes more than Swengere. If they shared a total of 50 mangoes, how many mangoes did each get?
13. I think of a number, double it and add 7, the result is 23. Find the number.
14. What number when added to 12, the result is 32?

12.5 More About Solving Equations

Example 1	Example 2
$2y + y + 6 = 24$ $3y + 6 = 24$ $3y + 6 - 6 = 24 - 6$ $\frac{\cancel{3}y^1}{\cancel{3}_1} = \frac{\cancel{18}^6}{\cancel{3}_1}$ $y = 6$	$3p + 4 + 4p = 46$ $3p + 4p + 4 = 46$ $7p + 4 = 46$ $7p + 4 - 4 = 46 - 4$ $\frac{\cancel{7}p^1}{\cancel{7}_1} = \frac{\cancel{42}^6}{\cancel{7}_1}$ $P = 6$

Activity 12:5

Solve the equations below

1) $2x + 3x + 2 = 17$
2) $4m + 2m + 1 = 19$
3) $8y + 2y = 80$
4) $4m + 3m + 3m = 31$
5) $p + p + 2p + 4 = 24$
6) $4y + y + 2 = 32$
7) $7x + x = 56$
8) $3q = 28$
9) $5n + 2n + 4 = 53$
10) $3b + 3b = 24$

12.6 Solving Equations by Adding on Both Sides and Dividing on Both Sides

Example 1

Solve for y: $y - 8 = 11$

Add 8 on both sides

$y - 8 + 8 = 11 - 8$

$y = 3$

Example 2

Solve for p: $3p + 2p - 6 = 24$

$5p - 6 = 24$

$5p - 6 + 6 = 24 + 6$

$\frac{5p^1}{5_1} = \frac{30^6}{5_1}$

$P = 6$

Example 3

I think of a number subtract 8, the result is 10. Find the number.

Let the number = m

$m - 8 = 10$

$m - 8 - 8 = 10 + 8$

$m + 0 = 18$

$m = 18$

Example 4

What is number when doubled and 6 subtracted the result is 40?

Let the number = g

$g \times 2 - 6 = 40$

$2g - 6 = 40$

$2g - 6 + 6 = 40 - 6$

$\frac{2y^1}{2_1} = \frac{46^{13}}{2_1}$

$y = 13$

Activity 12:6

Solve the following equations from 1 – 15;

1) $x - 7 = 10$
2) $y - 3 = 7$
3) $p - 12 = 8$
4) $m - 15 = 27$
5) $b - 8 = 2$
6) $2w - 8 = 10$
7) $3x - 10 = 14$
8) $5y - 9 = 11$
9) $6y - 13 = 11$
10) $8z - 12 = 44$
11) $4r = 28$
12) $5y = 45$
13) $3y = 36$
14) $7m = 56$
15) $12n = 132$
16) I think of a number subtract 12 from it, the result is 17. Find the number.
17) I think of a number subtract 15 the result is 6. What is the number?
18) I thinks of a number, double then subtract 6 the answer is 14. Find the number.
19) The difference between two numbers is 12. If the bigger number is 20, find the second number.

20) The range of two numbers is 8. If the bigger number is 20, find the second number.

21) John had some cows on his farm. He sold 18 cows and remained with 92 cows. How many cows did he have at first?

12.7 Forming and Solving Equations Involving Area of Rectangles

Example 1	Example 2
The area of a rectangle is 42cm. If the length is 7cm. Find its width $L \times W = A$ $7 \times W = 42$ $\frac{7w^1}{7_1} = \frac{42^6}{7_1}$ $W = 6cm$	Find the length of the rectangle whose area is $48cm^2$ and width 6cm. $L \times W = A$ $6 \times L = 48$ $\frac{6L^1}{6_1} = \frac{48^8}{6_1}$ $L = 8cm$

Activity 12:7

1. The area of rectangle is $36cm^2$. Find its width if the length is 9cm.
2. The area of a rectangle is $42mm^2$. Calculate its length if the width is $6cm^2$.
3. The length of a rectangle is twice its width. If its perimeter is 60cm. Find the length and the width of the rectangle.
4. Find the length of a flower garden whose width is 7m with an area of $63m^2$
5. The area of a rectangle is $12cm^2$. Find its length if the width is 3cm.
6. The area of rectangular garden is $120cm^2$. Given that the width is 10cm, find its length.
7. Given that the area of a rectangle is $72cm^2$. Find the width if the length is 12cm.
8. The width of a rectangle is 7cm and its area is $8cm^2$. Work out the length.

12.8 Opening the Brackets and Solving Simple Equations

Example 1	Example 2
$2(t + 4) = 12$ Open the brackets by multiplying by 2 all the numbers in the brackets $2 \times t + 2 \times 4 = 12$ $2t + 8 = 12$ $2t + 8 - 8 = 12 - 8$ $\frac{2t^1}{2_1} = \frac{4^2}{2_1}$ $t = 2$	$3(p - 4) = 12$ $3 \times p - 4 \times 3 = 12$ $3p - 12 = 12$ $3p - 12 + 12 = 12 + 12$ $\frac{3p^1}{3_1} = \frac{24^8}{3_1}$ $p = 8$

Activity 12:8

Open the brackets and solve the following equations;

1) $2(t + 3) = 12$	2) $2(p + 5) = 20$	3) $(x + 6) = 18$
4) $3(x + 2) = 24$	5) $3(a + 2) = 24$	6) $3(3x + 6) = 48$
7) $4(y + 7) = 36$	8) $4(r + 2) = 48$	9) $5(s + 2) = 26$
10) $5(w + 3) = 30$	11) $6(x - 2) = 6$	12) $6(y - 3) = 12$
13) $7(y - 1) = 42$	14) $7(m - 2) = 21$	15) $8(n - 6) = 56$
16) $8(n - 4) = 0$	17) $2(p - 8) = 4$	18) $3(x - 3) = 12$
19) $3(x - 4) = 15$	20) $4(2x - 3) = 20$	21) $5(2x - 3) = 15$
22) $6(2y - 10 = 18$	23) $7(y - 1) = 35$	24) $8(x - 2) = 48$

12.9 Solving Fractional Equations

Example 1

$\frac{t}{5} = 10$

✓ Make 10 a fraction by $\frac{t}{5} = \frac{10}{1}$ dividing by one.

✓ Find the L.C.D of 5 and 5 is 5

✓ Multiply each side by the L.C.D $5 \times \frac{t}{5} = \frac{10}{1} \times 5$

✓ Cancel by 5 on both sides where applicable $t = 50$

Example 2

$\frac{3x}{4} = 6$

✓ Make 6 fraction by dividing by one $\frac{3x}{4} = \frac{6}{1}$ $L.C.D = 4$

✓ Find the L.C.D of 4 and 1 $4 \times \frac{3x}{4} = 6 \times 4$

✓ Multiply each side by the L.C.M $\frac{\cancel{3}x^1}{\cancel{3}_1} = \frac{\cancel{24}^8}{\cancel{3}_1}$

✓ Cancel where applicable $x = 8$

TOPIC 12: Algebra

Activity 12:9

Solve the following

1) $\frac{p}{3} = 6$ 2) $\frac{t}{4} = 3$ 3) $\frac{m}{2} = 4$ 4) $\frac{x}{2} = 0$ 5) $\frac{y}{3} = 2$ 6) $\frac{m}{4} = 6$

7) $\frac{x}{5} = 1$ 8) $\frac{p}{6} = 3$ 9) $\frac{a}{6} = 2$ 10) $\frac{b}{7} = 12$ 11) $\frac{c}{8} = 15$ 12) $\frac{d}{9} = 10$

13) $\frac{e}{10} = 3$ 14) $\frac{f}{11} = 8$ 15) $\frac{g}{12} = 7$ 16) $\frac{h}{15} = 6$ 17) $\frac{2y}{3} = 6$ 18) $\frac{3x}{4} = 12$

17) What number when divided by 6 the result is 4?

18) I think of a number divide it by 3 the result is 9.

19) Some mangoes were divided amongst 6 pupils, if each pupils got 12. How many mangoes did they share?

20) What number when divided by 5 gives 24?

12.10 Application of Algebra

Example 1	**Example 2**
The sum of two numbers is 24. If one of the numbers is 8. Find the other number.	The sum of 3 numbers is 33. Two of the numbers are 12 and 14, find the third number.
Let the other number = p	Let the third number = m
P + 8 = 24	m + 12 + 14 = 33
P + 8 − 8 = 24 − 8	m + 26 = 33
P = 24 − 8	m + 26 − 26 = 33 − 26
P = 16	m = 7
The other number is 16	

Example 3

Antony is 8 years older than Abas. If their total age is 24 years, how Old is Abas?

Let Abas's age = b

Abas	Anthony	Total
b	b + 8	24

$b + b + 8 = 24$

$2b + 8 = 24$

$2b + 8 - 8 = 24 - 8$

$\frac{\cancel{2}b^1}{\cancel{2}_1} = \frac{\cancel{16}^8}{\cancel{2}_1}$

$b = 8$

Abas is 8 years

Example 4

The product of two numbers is 24. Find the other number if one of them is 4.

Let the other number = t

1st No	2nd No	product
t	4	24

$t \times 4 = 24$

$4t = 24$

$\frac{\cancel{4}t^1}{\cancel{4}_1} = \frac{\cancel{24}^6}{\cancel{4}_1}$

$t = 6$

The other number is 6

Activity 12:10

1. The sum of two numbers is 48. If one of the numbers is 30. Find the second number.
2. If two numbers sum up to 72. Find the second number if the first number is 24.
3. The sum of three numbers is 60. Find the third number if two of the numbers are 21 and 27.
4. The sum of 24, 30, 35 and x is 100. Find the value of x.
5. The sum of p, 20, 24 and 30 is 90. What is the value of p?
6. Amooti is 12years older than Akiiki, their total age is 47. How old is each of them?
7. Adur is 9kg heavier than Odur. Their mass is 59kg. Find Adur's mass.
8. Byaruhanga is twice as old as Kusiima, their total age is 42 years. How old is each of them?
9. The product of two numbers is 72. If one of the numbers is 12, find the second number.
10. The product of x and $3x$ is 108. Find the value of x.

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Applied Mathematics For Today is a primary five course book written for pupils and teachers.

This book has provided the required items, clear explanations of re-al Mathematical concepts and exercises which help in the building of different concepts. The exercises are modulated and good for pupils. This book is prepared to meet all the twelve major topics in Mathematics as outlined in the new Uganda Primary School Curric-ulum.
It has answered all your questions you have been looking for.